

Policy Entanglement in Active Inference

A Coupling-Parameter Deformation Framework for Multi-Stream Policy Posterior Distributions,
Machine-Checked and Simulated with a Typed Float Boundary

Daniel Ari Friedman
Active Inference Institute
`daniel@activeinference.institute`
ORCID: 0000-0001-6232-9096
DOI: 10.5281/zenodo.20418904
Version 1.0
Active Inference Journal (2026)
License: CC-BY-4.0
[Policy Entanglement source repository](#)

May 27, 2026

Contents

Abstract	6
Part I — Introduction	6
Motivation and Position: Why Multi-Stream Active Inference Needs Parametric Policy Entanglement	7
The problem in plain language	7
Current coping strategies and their limits	8
Significance	9
Position relative to alternatives	10
How the framework is developed: four parallel tracks	10
Reading guide	11
Section dependencies (forward signposts)	11
Background and Prior Work: Antecedents in Active Inference, Variational Coupling, Information Geometry, and Tensor Networks	11
Discrete-time POMDP active inference	12
Mean-field variational families and their limits	12
Hierarchical, deep, and sophisticated active inference	13
Information geometry	13
Tensor networks and entanglement entropy	13
KL / path-integral control and reinforcement learning	14
Multi-agent inference, Markov blankets, and renormalization	14
Computational psychiatry and neural implementation	14
Lean 4, Mathlib, and formal verification of probabilistic systems	15
Reproducible scientific software	15
Part II — Theory	15
Setup and Assumptions: Finite-Horizon Discrete POMDPs, Multi-Stream Policy Factorization, and the Mean-Field Baseline	16
Single-stream POMDP active inference	16
The multi-stream extension	17
The mean-field baseline	17
Lambda Deformation: Definition and Properties	17
Coupling potentials	17
Definition: lambda-entangled policy posterior	18
The coupling parameter lambda as a precision-like coupling weight	18
What J and K _c look like in practice	18
Entanglement Decomposition Theorem: Marginals, Coupling, and Total Correlation	19
Statement	19
Proof sketch	20
What the decomposition says	20
Optimal coupling: existence of lambda*	22
Takeaways	23
Worked Examples: Bernoulli Toy, Motor-Attention Coupling, and Multi-Timescale Coupling	23
The K=2 Bernoulli toy: full closed form	23
Two-stream motor + attention with realistic EFE	24
Multi-timescale coupling	27
Information Geometry: Geodesics, Flatness, and Pythagorean Decomposition	27
Dual coordinates and the mean-field submanifold	27
The lambda-family is an e-geodesic	30
Connection to alpha-projections and curved exponential families	33
Connection to escort distributions and deformed exponential families	33
Takeaways	33

Spectral Structure: Schmidt Rank, Archetypal Decomposition, and Tensor-Train Bond Profiles	33
Bipartite Schmidt decomposition	33
The leading singular vectors as archetypal eigenvectors	34
Multi-stream tensor decomposition	37
Birth and death of archetypes	39
Takeaways	39
Heterogeneous Inference: Mixed VFE/EFE Ensembles and the Coupling-Tax Bound	39
Three-level update hierarchy	39
Coupled marginal updates	39
The VFE-only suboptimality bound	39
Updating lambda: precision-like learning on coupling	41
Habit accumulation and revertibility	42
Takeaways	42
Phase Structure: Disordered, Mixed, and Frozen Coupling Regimes	42
Phase diagram	42
Order parameters	44
Model predictions and behavioral hypotheses	44
Comparative Statics: Coupling Payoff, Two-Parameter Generalization, and Sensitivity Surfaces	44
The pay-off structure	44
Two-parameter habit/EFE generalization	45
Sensitivity to potential structure	45
Part III — Formal Verification	47
What is proved	47
How the verification stack composes	47
Where to read the per-row content	47
Lean 4 Formalization: Current Boundary, Witness Contracts, and Mathlib Scope	48
Current Lean artifact	48
What Mathlib4 is used for	49
Mathlib4 analytic targets	50
Live-source injection	50
What the Lean track proves today	50
The verification stack	51
Validation gates	51
Part IV — Empirical Grounding	52
Empirical Simulation Suite: Bernoulli Validation, Coupling-Tax Envelope, Phase Diagram, and Spectral Structure	52
Two-stream Bernoulli ($K=2$) closed-form validation	53
Heterogeneous VFE / EFE coupling-tax bound	54
Phase Structure of the $K = 2$ Ising Toy: Disordered, Mixed, and Frozen Regimes	54
Spectral structure (Schmidt rank, archetypes)	58
Information-geometric structure (e-geodesic)	58
Multi-stream coupling graph	58
$K > 2$ ensembles	58
Long-horizon rollouts and trajectory stationarity	66
m -projection revertibility witness	70
Robustness and ablation stress tests	70
Software and numerical stack	73
Head-to-head BTAI baseline and adversarial-perturbation harnesses	76
Anchored figure index	78
pymdp 1.0.1 POMDP Harness: Architecture, Lambda Sweep, and Deterministic Rollout	78
Architecture: layered, not duplicated	78
The static λ -sweep	79
Deterministic coupled rollout	84

Anchored figure index for the pymdp harness	84
pymdp Free-Energy Bundle Observables and Auto-Injected Summary Statistics	84
Statistical contracts	85
Summary statistics across the sweep	85
Visualizations	85
Theorem 5.1 Numerical Witness	85
Anchored figure index for the pymdp free-energy bundle	93
pymdp Validation: Three-Tier CI Gates, Structured JSONL Run Log, and Bit-Reproducibility Contract	93
Three-tier validation	93
Structured run log (JSONL)	94
Software and source provenance	94
Reproducibility contract	94
Test budget snapshot	95
Part V — Connections to Existing Frameworks	95
Framework Connections I: pymdp/SPM Baseline, Hierarchical/Deep AIF, Sophisticated Inference, and Branching-Time AIF	96
Relationship calculus for AIF variants	96
Relation to recent unification efforts	96
Mean-field active inference (pymdp, SPM, ActiveInference.jl)	97
Hierarchical / Deep Active Inference	97
Sophisticated Inference (recursive EFE)	97
Branching-Time Active Inference	99
Synthesis: How Every Connection Lands on Lambda, J, or Kc	99
Framework Connections II: KL/Path-Integral Control, Options Frameworks, Products-of-Experts, and Copula Variational Inference	100
KL / Path-Integral Stochastic Optimal Control	100
Options framework / Hierarchical RL	101
Product of Experts and Mixture of Experts	101
Copula variational inference	101
Recovery dictionary	101
Framework Connections III: Multi-Agent Inference, RG-AIF Coarse-Graining, Markov Blankets, and CEREBRUM Case Grammar	102
Interactive Inference / Multi-Agent Active Inference	102
Renormalization-group active inference (structural analogy)	103
Markov Blankets and Bayesian Mechanics	103
CEREBRUM and case-grammar approaches	103
Part VI — Discussion and Outlook	105
Open Theoretical Questions: Analytical Gaps, Identifiability, Empirical Frontiers, and Practical Research Directions	105
Where to start	105
How to read each question	105
Analytical questions	106
Identifiability and inference questions	106
Empirical questions	107
Conceptual / cross-disciplinary questions	107
Practical / formalization questions	108
Discussion and Outlook: Worldview, Live Artifact State, and Limitations	108
What the framework commits to	108
What the framework declines to commit to	108
The parametric-entanglement worldview	108
Relation to the Free Energy Principle and active-inference process theory	109

Six load-bearing properties	109
Live state of the artifact	110
Alignment and Dysregulation Hypotheses	111
Limitations	111
Threats to validity	112
Verification ledger	112
Open directions	113
Proof of the Entanglement Decomposition	113
Definitions	114
Expanding the EFE expectation	114
Expanding the prior log-expectation	114
Combining	114
Interpretation (sign of I)	114
Closed exponential-family form (collapsed identity)	115
Convexity of Free Energy in Lambda: Conditions and Counter-Example	115
The closed exponential-family form	115
The convexity ledger	115
Sufficient condition for global convexity on $[0, \infty)$	116
$K = 2$ Symmetric Ising specialization	116
Numerical verification	116
Complete Derivation: Two-Stream Bernoulli Toy	116
Setup	116
Joint posterior	116
Marginals (uniform by symmetry)	117
Joint and marginal entropies	117
Closed-form mutual information	117
Coupling from a target alignment (alignment-inversion, <i>not</i> a VFE optimum)	117
Free-energy curve	118
Numerical cross-check	118
Tensor-Train Inference Algorithm: Bond-Dimension Sweep, MPS Contraction, and Sparsity-Rank Tradeoff	119
Tensor-train representation of J	119
The exponential map	119
Inference cost	119
Sampling	119
Marginal extraction	119
Bond-dimension recursion	119
Approximate compression	120
Connection to the sparsity-rank tradeoff	120
Lean 4 Boundary Fragment: Live ActinfPolicyEntanglement Source Excerpts and Validation Wiring	120
Package structure	120
Mathlib-free boundary	120
Status snapshot	121
Building locally	121
FepSketches Re-exports for the Sibling fep_lean Layout	121
Where each manuscript theorem lives in Lean	121
The twenty live Lean companions	122
Decomposition fragment	122
Geometry fragment	124
Spectral fragment	125
Convexity fragment	126
Heterogeneous fragment	126
Connections fragment	127
Markov-blanket fragment	127
How drift is prevented	128

How a witness is consumed today	128
Notation Concordance: Symbol Registry Across Manuscript, LaTeX, Python, and Lean	128
How to read this concordance — one symbol, four tracks	129
Sign conventions	129
Sets, indices, and basic objects	130
POMDP generative-model symbols	130
Distributions	130
Coupling potentials and parameters	131
Information-theoretic quantities	133
Free energies	134
Manifolds, projections, and dual coordinates	134
Spectral and tensor-network symbols	134
Heterogeneous-ensemble quantities	135
KL-control, copula, and product-of-experts symbols	135
Relationship classes and claim-strength labels	136
Hyperparameters (figure / sweep / simulation)	136
LaTeX preamble macros	137
Lean type abbreviations	137
Phase / verdict labels (manuscript-level)	137
Status notes	137
Conventions	138
Common pitfalls when editing across tracks	138
GNN fifth-track concordance (structural-and-numerical)	138
Reference Tables: Claim Strength, Variant Recovery, Lean Inventory, pymdp Bundle Statistics, and JSONL Run-Log Schema	139
Claim-Strength Legend	139
Active-Inference Variant Recovery Ledger	139
Lean 4 Boundary-Fragment Module Inventory	140
MathlibProofs $\mathbb{R}$ Analytic Discharge Package: Inventory	142
pymdp Free-Energy Bundle: Auto-Injected Summary Statistics	143
Robustness and Ablation Stress-Test Ledger	143
Evidence Ladder and Claim Provenance	143
pymdp JSONL Run-Log Field Schema	145
GNN as a Shipped Fifth Track: Triple-Play Mapping, a Verified K=2 Bernoulli Round-Trip, and a Lean Typed-Contract Emitter	146
Scope and Claim-Strength: a Shipped Bridge, with an Explicit Non-Claim About Proof	146
GNN Background: the Triple Play and the Active Inference Ontology	147
Acronym Disambiguation: GNN (Notation) vs. GNN (Graph Neural Network)	147
Mapping the $(A, B, C, D, E, J, \lambda)$ Tuple to a GNN Spec	148
Worked Example: the K=2 Bernoulli Toy as a Shipped GNN Round-Trip	148
What GNN Preserves and What It Abstracts Away	149
Shipped Elaboration: GNN to a Lean Typed-Structure Contract	150
Shipped Round-Trip: GNN to pymdp Harness Configuration	150
Downstream Use, Open Directions, and Open Questions	151
Bibliography	152
Active inference and the free-energy principle	152
Branching-time and structured AIF	155
Information geometry	155
Information theory and multi-information	155
Variational and copula inference	156
Variational inference robustness and Lipschitz bounds	156
Tensor networks and quantum-inspired ML	156
KL / path-integral control duality	157
Control / RL as probabilistic inference	157
Hierarchical RL / options	157
Markov blankets and particular physics	158

Computational psychiatry	158
Integrated information / consciousness	158
Multi-agent / interactive inference	158
Lean / formalization	159
Software dependencies (companion code)	160
Scientific visualization	160

Abstract

Active inference models often need to choose among several policy streams at once, for example streams tied to different effectors, sensory channels, agents, agents within a group, or planning horizons. Standard discrete active-inference implementations keep this manageable by treating those streams as independent, but that simplification removes the dependencies that make coordinated action possible. This manuscript introduces policy entanglement: a controlled deformation of the usual independent policy posterior by a scalar coupling strength and explicit compatibility and preference potentials. The construction preserves the finite active inference setting while making cross-stream dependence a first-class modeling object rather than an implicit artifact of the chosen factorization. The framework keeps a claim-strength ledger that distinguishes exact recoveries, parameterized embeddings, numerical witnesses, and structural analogies. Mean-field active inference is the exact independent case. Products of experts, copula variational inference, options, hierarchical and sophisticated inference, branching-time active inference, renormalization-style compression, and Markov-blanket multi-agent views are connected as special cases through their stated posterior-factorization maps.

The central result is a free-energy decomposition that separates ordinary per-stream free energy, coupling preference terms, the coupling normalizer, and the information cost of leaving independence. The decomposition makes multi-information the explicit surcharge paid by a non-factorized policy posterior and shows how coupling strength, compatibility structure, and off-diagonal preference costs enter the same accounting identity. This result supplies the organizing principle for the rest of the paper. It supports an information-geometric reading of the coupled policy family as a path away from the mean-field submanifold, a projection identity that returns the coupled posterior to its independent marginals, a spectral and tensor-train view of dominant coordinated policy modes, a heterogeneous-ensemble coupling-tax bound, and a phase vocabulary for under-coupled, mixed, and highly concentrated policy posteriors. These interpretations are intentionally limited: the manuscript does not claim a neural, clinical, biological, or quantum implementation, and Markov-blanket and tensor-network language is used as scoped modeling analogy unless a specific theorem row or generated artifact supports a stronger statement.

The main decomposition analytic identity is machine-checked in $\mathbb{R}$ in the Mathlib-backed Lean layer with an axiom audit and negative controls. A separate stock-Lean boundary fragment remains Mathlib-free and exposes the theorem surface as typed contracts for the Python simulation layer and the manuscript registry, including witness-consuming rows where analytic payloads are deliberately supplied at the boundary. The executable numerical layer remains a Float pipeline, so a verified $\text{Float} \leftrightarrow \mathbb{R}$ error bridge is still an explicitly open interface rather than an implied proof; conservative interval brackets on the $K=2$ decomposition sweep certify Float residuals within a widened high-precision envelope (`output/reports/float_real_residual.json`) without promoting the registry row to `proved`. The empirical companion uses a partially observable Markov decision process (POMDP) simulation layer built with `pymdp` and NumPy to sweep coupled policy ensembles, run short and long rollouts, check the projection identity to round-off precision, produce free-energy, entropy, total-correlation, action-distribution, robustness, and adversarial sidecars, and render figures from those artifacts. The manuscript, figures, theorem map, citation registry, notation glossary (§S6), bibliography, and PDF are regenerated from the same source-owned pipeline, so prose claims are tied to Lean sources, Python witnesses, output metadata, and validation gates rather than maintained by hand.

All manuscript methods, tests, and documentation are available as open-source software at https://github.com/ActiveInferenceInstitute/policy_entanglement (DOI: <https://doi.org/10.5281/zenodo.20418904>).

Part I — Introduction

This part frames the multi-stream policy-inference problem and surveys the prior work on which the framework builds. Two chapters:

The Free Energy Principle enters this manuscript as a disciplined variational framing rather than as a slogan: adaptive systems can be modeled as maintaining tractable beliefs that bound sensory surprise, while active inference turns that variational bound into a process theory of perception, action, learning, and policy selection [Friston et al.,

2006, Friston, 2010, Buckley et al., 2017, Parr et al., 2022, Pezzulo et al., 2024]. That framing has also drawn careful scope critiques, especially around Markov blankets and the temptation to treat FEP as a theory of everything rather than as a modeling principle with explicit assumptions [Aguilera et al., 2022, Raja et al., 2021, Menary and Gillett, 2022]. The finite POMDP setting used here is therefore intentionally narrow: expected free energy supplies the policy objective, pymdp supplies a reproducible implementation substrate, and the contribution of this manuscript is to ask what happens when the *policy posterior itself* is allowed to carry explicit cross-stream dependence [Friston et al., 2017a, Parr and Friston, 2017a, Kaplan and Friston, 2018, Da Costa et al., 2020, Smith et al., 2022, Heins et al., 2022].

- **§1** — *why* parametric coupling is needed. Real agents juggle many concurrent policy streams; strict mean-field factorization can discard cross-stream structure that is directly visible in the motivating examples of §1. A single tunable coupling parameter recovers strict factorization at one limit and arbitrary joint structure at the other.
- **§2** — *what already exists* in active inference, control theory, information geometry, and tensor networks that this framework relates to exactly, parametrically, or analogically. Eight axes of antecedents are surveyed (discrete-time POMDP AIF; mean-field variational families; hierarchical / deep / sophisticated AIF; information geometry; tensor networks; KL / path-integral control and RL; multi-agent inference and renormalization; formal verification and reproducible scientific software). The later recovery ledger separates exact recoveries (for example mean-field AIF and single-stream KL control) from parametric embeddings (for example explicit factor-graph policy factors, options, copula VI, and multi-agent coupling) and structural analogies (for example hierarchical, sophisticated, branching-time, renormalization-group, and Markov-blanket readings). The specific mappings are cataloged in §17 through §19.

After Part I a reader has the motivating examples, the literature context, and the conceptual frame within which the analytical content of §3 onward is developed. Every symbol used in this manuscript is cataloged in §S6 across all four tracks (manuscript, LaTeX preamble, Python, Lean), with status notes flagging where a track diverges (e.g. unicode-vs-ASCII identifier conventions). The Lean track ships in two complementary packages: `MathlibProofs/`, which machine-checks the central S01 free-energy identity (Theorem 5.1) in $\mathbb{R}$ with foundational-only `#print axioms` (`propext`, `Classical.choice`, `Quot.sound`; two independent negative controls audited on every build), and `lean/ActinfPolicyEntanglement/`, a stock-Lean v4.29.0 boundary fragment that ships the 21-row theorem surface as a typed API for the Python computational layer and the manuscript registry; §12 carries the per-row content table and `docs/reference/veridical_status.md` the running audit.

Repository artifact. All four tracks (prose, equations, Python / pymdp, Lean 4) are maintained as a single working repository artifact [Friedman, 2026a]. The public Zenodo DOI (10.5281/zenodo.20418904) and source repository are recorded in `manuscript/config.yaml` and `CITATION.cff`. The entire build — rendered PDF, pymdp 1.0.1 numerical sidecars, dashboard invariants, and the MathlibProofs $\mathbb{R}$ -level discharge — regenerates from a single command (`./run.sh --pipeline` or `python scripts/run_all.py`), and the repository’s machine-readable `CITATION.cff` provides the preferred citation form.

Motivation and Position: Why Multi-Stream Active Inference Needs Parametric Policy Entanglement

The problem in plain language

Active inference furnishes a normative theory of a perceiving-and-acting unit by minimizing variational free energy across perception, action, and learning [Friston et al., 2006, Friston, 2010, Buckley et al., 2017, Friston et al., 2017a, Da Costa et al., 2020, Parr et al., 2022]. In the discrete-time POMDP formulation, a unit is specified by an observation likelihood $A : p(o \mid s)$, a transition tensor $B : p(s' \mid s, a)$, prior preferences over outcomes C , a prior over hidden states D , and a habit / prior over policies $E(\pi)$, where policies π live on a space of action-affordances replicated across a (possibly flexible) time horizon [Smith et al., 2022, Heins et al., 2022]. Expected free energy then gives each policy posterior both an epistemic term (seek observations that resolve uncertainty) and a pragmatic term (seek preferred outcomes), so action selection is not bolted onto perception but derived from the same variational calculus [Friston et al., 2017a, Sajid et al., 2021, Parr and Friston, 2019, Friston, 2018]. The formalism is mathematically compact, biologically motivated, and increasingly used across perception, decision-making, robotics, and computational psychiatry [Adams et al., 2013, Schwartenbeck and Friston, 2016, Lanillos et al., 2021], while its direct empirical status remains an active comparative-modeling question [Hodson et al., 2024].

An agent of any non-trivial complexity, however, is not best described by a *single* policy variable. A real agent

juggles **many concurrent policies**: motor and attentional, fast and slow, modality-specific, with some streams treating the next move as a one-shot gradient descent on variational free energy (VFE) and others engaging in deep counterfactual planning under expected free energy (EFE). The standard discrete POMDP treatment in `pymdp` [Heins et al., 2022] and the continuous-state SPM-DEM lineage [Friston et al., 2008] split inference across independent *factors* and policies — that is, they rely on strict mean-field factorizations across hidden-state factors, observation modalities, and, in the multi-stream construction studied here, policy variables [Da Costa et al., 2020, Smith et al., 2022].

The mean-field separation is computationally cheap and biologically suggestive — it mirrors functional segregation, sparse cortical connectivity, and the modular character of much of the cortex — but it discards a structure that is visible in natural behavior and engineered agents: actions taken on one stream are routinely contingent on, anticipatory of, and instrumentally coupled to actions on others. A drummer’s left hand does not freely sample its policy distribution while the right hand plans the next fill; an autonomous vehicle’s lateral and longitudinal controllers share predictive structure with the symbolic route planner [Lanillos et al., 2021]; a human reaching for a cup while reading does not factor reach-policy and saccade-policy independently. Natural-task eye-tracking studies find that gaze typically arrives at the next task object before manipulation begins [Land and Hayhoe, 2001], and object-manipulation studies show hand shaping and grip forces coupled to visual sampling of the object [Johansson et al., 2001]. Contemporary world-model RL agents such as DreamerV3 [Hafner et al., 2023] give an adjacent machine-learning example: cross-modal predictive state is useful, but the architecture does not expose a scalar coupling parameter that the modeler can tune, infer, or validate.

This document develops a tunable, analytically tractable, machine-checkable formalism for the *parametric entanglement* of concurrent policies in a single agent. The key move is structural rather than architectural: we do not commit to translation layers between symbolic planners and motor controllers (which would make the modeler responsible for the entire bridge), nor to fully joint enumeration of cross-modal policies (which combinatorially explodes), nor to hand-designed attention-gating that re-introduces the mean-field assumption at a higher level. We instead introduce a *single tunable coupling parameter* that interpolates between strict factorization and arbitrary joint structure, with a closed-form decomposition of free energy that exposes precisely what the entanglement buys. In FEP language, λ does not replace precision, preferences, or expected-free-energy minimization; it makes the strength of cross-stream policy dependence an explicit object that can be proved about, simulated, visualized, and validated.

The phrase *policy-posterior coupling* is used narrowly. Standard active-inference notation already distinguishes posteriors over hidden states, policies, and parameters [Friston et al., 2017a, Smith et al., 2022]. This manuscript changes only the variational family for the policy posterior: the policy prior factors E_k , the expected-free-energy weights $\gamma_k G_k$, and the agent’s generative-model tensors remain visible, while λJ adds an explicit cross-stream factor on joint policy space. Thus λ is not neural precision, not a new preference distribution, and not a biological measurement by itself; it is the scalar strength of a specified coupling potential inside the finite policy posterior.

Current coping strategies and their limits

The active-inference and probabilistic-modeling literatures have developed several strategies for handling cross-stream dependence on top of a mean-field backbone. Each is genuinely useful in its regime; each leaves a structural gap that the parametric construction below is designed to close.

- **Architectural translation layers.** A common engineering pattern is to factor the agent into independent per-stream modules and hand-design a translation layer between them — for example, a symbolic-planner outputs a goal that a motor-controller consumes as a target. This decouples the modules at design time but moves the entire alignment problem into the translation layer, which is typically non-differentiable, non-Bayesian, and the implicit source of brittleness when the modules’ assumptions drift apart.
- **Hierarchical / deep active inference.** Hierarchical AIF [Friston et al., 2017a, Pezzulo et al., 2018] organizes the agent into a stack of generative models with explicit message passing across levels. Within each level the mean-field assumption is re-imposed; cross-stream dependence is captured only through the bottleneck of the inter-level messages. When two streams at the *same* level depend on each other directly (left hand and right hand; saccade and reach), hierarchical AIF cannot express the dependence without inflating the level structure to a degree that loses biological plausibility.
- **Sophisticated inference.** Sophisticated inference [Friston et al., 2021] introduces beliefs about beliefs through recursive expected free energy, capturing how an agent’s plan depends on what it will *come to believe* once an action is taken. This is a powerful generalization along the temporal axis but it does not address concurrent cross-stream coupling at a single decision point; the recursion is over future selves, not over peer streams.
- **Branching-time active inference.** Branching-time AIF [Champion et al., 2022] expands the policy tree and prunes it with Monte Carlo tree search, allowing a much richer enumeration of joint futures. The price

is computational: the policy space scales exponentially in horizon and arity, and the agent must spend search budget the framework does not give it any principled way to allocate.

- **Attention-gating and precision modulation.** Dopaminergic precision learning [Friston et al., 2014, Schwartenbeck et al., 2015], the broader precision-in-action literature [Limanowski et al., 2024], and structured attention mechanisms [Hoffman and Blei, 2015] gate which streams influence the posterior at a given moment. This is well-suited to coarse on / off coupling but provides no continuous parameter for how *strongly* coupled the streams should be, nor any analytical decomposition of the cost paid for coupling.
- **Copula variational inference.** Copula VI [Tran et al., 2015, Han et al., 2016] retains mean-field marginals and adds a parametric copula on top to model dependence. The construction is elegant in static settings but does not natively integrate with the EFE / planning side of active inference, and the choice of copula family is a non-trivial modeling commitment that does not interpolate smoothly to or from the mean-field baseline.
- **Tensor-network parameterized variational families.** Tensor networks — matrix product states, tree-tensor networks, hierarchical Tucker decompositions [Han et al., 2018, Glasser et al., 2020, Verstraete et al., 2008, Eisert et al., 2010] — give compact structured joint distributions with controlled bond dimensions. We are not aware of a prior active-inference treatment that uses this exact tensor-network family as the policy posterior; one contribution of this manuscript is to show that the parametric coupling below *is* a tensor-network family in disguise, recovering MPS-style algorithmic structure as a side effect.

Across these strategies, the diagnostic pattern is consistent: each one solves a piece of the cross-stream-coupling problem and leaves the remaining pieces to the modeler. A unified framework needs a single parameter that interpolates between strict mean-field and arbitrary joint structure, a closed-form bookkeeping for the cost of leaving the mean-field manifold, and a *principled* way to classify each neighboring strategy as an exact slice, a parametric modeling route, or a structural analogy.

This placement also clarifies what the manuscript is *not* claiming. The framework is not a new general account of the Free Energy Principle, nor a replacement for the process theory of active inference. It is a focused extension of the variational family used for policy inference: the usual mean-field policy posterior remains the $\lambda = 0$ anchor, while finite λ represents a controlled departure from that anchor. The Markov-blanket and particular-physics literatures motivate why boundaries and conditional independences matter for self-organizing systems [Kirchhoff et al., 2018, Ramstead et al., 2018], and critical reviews warn that those ideas do not automatically license unrestricted biological or cognitive scope claims [Aguilera et al., 2022, Raja et al., 2021]. The claim tested here is narrower: within a finite-POMDP agent whose modeling boundary is already fixed, cross-stream policy dependence can be made a transparent mathematical and computational degree of freedom.

Significance

Six load-bearing properties of the framework:

(i) **Theoretical economy.** The current AIF literature handles cross-stream coordination through a proliferation of architectural devices: branching-time tree search [Champion et al., 2022], hierarchical / deep AIF with explicit message passing [Pezzulo et al., 2018, Friston et al., 2017a,b], sophisticated inference with recursive EFE [Friston et al., 2021], and factor-graph forward-backward over multi-modal generative models [de Vries and Friston, 2017, Friston and Parr, 2017, Parr et al., 2019, van de Laar, 2019]. Part V classifies these relationships instead of flattening them into one claim: mean-field AIF is the exact $\lambda = 0$ anchor; factor-graph message passing is a parametric implementation route whose inserted policy factor has exact semantics only inside the chosen graph; hierarchical, sophisticated, and branching-time AIF remain structural analogies unless their extra temporal or recursive machinery is explicitly built into J and the generative model.

(ii) **A clean decomposition theorem.** §5 below gives a free-energy decomposition (Theorem 5.1) into (a) per-stream marginal free energies, (b) a coupling / partition-function bundle ($\gamma\lambda\langle K_c \rangle$, $\log Z_E(\lambda)$, and $-\lambda\langle J \rangle$), and (c) the multi-information (total-correlation) term $I(q_\lambda) \geq 0$ — see Eq. (5.2). The trade-off between coupling and correlation makes precise when “joint structure pays for itself” relative to mean-field baselines, where comparable accounts in the structured-VI literature [Saul and Jordan, 1995, Wainwright and Jordan, 2008, Hoffman and Blei, 2015] state the trade-off only implicitly. Three corollaries follow immediately: Corollary 5.2 (coupling-pays-for-itself verdict), Corollary 5.3 (mean-field reduction at $\lambda = 0$), and Corollary 5.4 (strict gain when q is non-mean-field).

(iii) **Information geometry.** The family $\{q_\lambda\}$ is an *exponential geodesic* (e-geodesic) departing the mean-field submanifold, which is itself e-flat. This places the framework on the well-developed scaffolding of dually-flat statistical manifolds [Amari and Nagaoka, 2000, Amari, 2016, Nielsen, 2020], and connects revertibility (collapse to mean-field) to the canonical *m-projection* (taking marginals). Non-extensive analogs via escort / ϕ -deformed exponential families [Naudts, 2011] generalize the construction to systems with non-extensive statistics.

(iv) **Spectral interpretation.** For bipartite ($K = 2$) couplings, the joint posterior $q_\lambda(\pi^1, \pi^2)$ has a Schmidt decomposition; for $K > 2$, a Tucker or tensor-train decomposition. The leading singular vectors / Tucker factors

are **archetypal eigenvectors** — the small number of dominant cross-stream behavioral modes that recur across instances of a habit. This is a direct probabilistic-graphical-model homologue of bipartite entanglement entropy in quantum many-body systems [Verstraete et al., 2008, Eisert et al., 2010, Han et al., 2018], and inherits all the favorable algorithmic structure of low-rank tensor decomposition.

(v) **Heterogeneous VFE/EFE ensembles.** Many real agents are mixed: some streams are reflexive (one-step VFE descent) and others plan (EFE counterfactuals). §9 derives an $O(\lambda^2)$ suboptimality bound (Theorem 9.1) for VFE-only streams operating inside coupled ensembles. This formalizes a long-standing intuition: coupled reflexive controllers can ride along with planners *up to a bounded coupling-norm tax* (Corollary 9.2), providing a quantitative answer to “how reflexive can my low-level controllers be while still benefiting from higher-level planning?”

(vi) **Lean formalizability.** The framework is built from objects (finite probability distributions, KL divergence, Shannon entropy, Bregman divergences) that are natural targets for Lean’s Mathlib [Community, 2020]. A prior line of work shipped a 50-topic Mathlib-checked FEP catalog and companion monograph [Friedman, 2026c]; here, §12 documents this manuscript’s two-package Lean architecture — a Mathlib4-backed MathlibProofs/ library that machine-checks the central decomposition (Theorem 5.1) in $\mathbb{R}$ with foundational-only `#print axioms`, and a stock-Lean v4.29.0 `ActinfPolicyEntanglement` boundary fragment that ships the 21-row theorem surface as a typed API — reusing the same toolchain pin [de Moura and Ullrich, 2021] and re-export layout as that codebase. The boundary fragment is `sorry / axiom / Mathlib-free` and source-extracted into the manuscript prose; the per-row content table (§12) and the running audit at `docs/reference/veridical_status.md` document what each row certifies at the boundary versus what the Mathlib4 discharge layer establishes.

Position relative to alternatives

We make a deliberate choice not to take the *embodiment-as-architecture* route of solving the multi-policy problem by partitioning into hardware / software modules with explicit translation layers between symbolic reasoning and low-level control. That route has well-known costs: the translation layer becomes the entire alignment problem in microcosm, and it is non-differentiable. Instead, we take the *parametric entanglement* route: tunably couple the policy distributions themselves, leaving the structural distinction between modalities / timescales intact at the level of marginals while permitting arbitrary joint structure to emerge through a learned coupling potential.

This is closest in spirit to: copula variational inference [Tran et al., 2015, Han et al., 2016], structured VI [Saul and Jordan, 1995, Wainwright and Jordan, 2008, Hoffman and Blei, 2015], product-of-experts compositions [Hinton, 2002], and tensor-network parameterized variational families [Han et al., 2018, Glasser et al., 2020] — but specialized to the policy-inference setting of active inference with explicit habit (J), expected-free-energy (G), and sophistication (γ). It also connects to KL / path-integral control [Kappen, 2005, Theodorou and Todorov, 2012], options frameworks [Sutton et al., 1999], multi-agent active inference [Maisto et al., 2024], and renormalization-group AIF [Friston et al., 2025]. Part V assigns each connection an exact, parametric, or analogical relationship class, so “connection” never has to mean “full recovery.”

How the framework is developed: four parallel tracks

The manuscript is unusual in that the same mathematical object is developed simultaneously on **four parallel tracks** (prose, equations, Python, Lean), each with its own primary literature, its own quality criteria, and its own role in the overall argument. The tracks are cross-referenced via a registry of injection tokens; a reader can enter on any track and land on any other in a single hop, and a CI validator fails the build on dangling tokens, hardcoded grid counts, seeds, or rollout horizons.

- **Prose (§3 onward).** The natural-language presentation: definitions, theorem statements, proof sketches, worked examples, interpretive commentary. Every chapter in Part II opens with the objects under discussion and closes with a three-bullet *Takeaways* box that surfaces the load-bearing claims. Section numbering is owned by `manuscript/refs/labels.yaml` so that renumbering propagates everywhere automatically.
- **Formalism in equations (§4, §5, §7, §8, §9).** Every load-bearing identity is a registered equation in the same labels file, auto-numbered as `S.K` and resolved through registry tokens so the manuscript’s mathematical structure can be traversed without ever encountering a hand-written equation number. The four foundational identities — the entanglement decomposition (Eq. (5.2)), the e-geodesic identity (Eq. (7.3)), the Pythagorean decomposition (Eq. (7.4)), and the $O(\lambda^2)$ coupling-tax bound (Eq. (9.2)) — anchor the rest.
- **Simulations on real pymdp agents (§14, §15).** The empirical layer is *not* a re-implementation of pymdp’s mathematics; it instantiates real `pymdp.agent.Agent` objects [Heins et al., 2022] with the JAX backend [Bradbury et al., 2018], runs deterministic fixed-point-iteration inference, and reads off the resulting policy posteriors, expected free energies, and free-energy bundles for every λ on the coupling sweep. The analytical λ -coupling layer takes those pymdp outputs and adds the cross-stream term — so the simulation tracks exactly

the deformation the theorems describe, not a stylized stand-in. At $\lambda = 0$ the coupled joint is *exactly* the outer product of the pymdp per-stream posteriors; at finite λ the joint concentrates on cross-stream archetypes that the bare pymdp run never sees.

- **Lean 4 companions in a real lake build environment (§12, §E).** The formal track is a Lean 4 boundary fragment under `../lean/` that compiles against stock Lean v4.29.0 [de Moura and Ullrich, 2021] with **no Mathlib dependency**, zero strict `sorry`s, zero axioms beyond stock Lean, and no `unsafe` / `partial` / `noncomputable` declarations. Every theorem in the body that admits a witness-form statement (including Theorem 5.1) has a Lean companion that *actually compiles* in the project’s lake build; its source is auto-extracted in the theorem sections and appendix via registry Lean tokens. The witness-form discipline makes the analytic-content boundary explicit: Mathlib4 is used in the manuscript as the named library context for those analytic obligations, while the current source blocks remain exactly the code that builds today.

The four-track contract is not aspirational; it is CI-gated. The cross-reference and hyperlink test suites under `../tests/test_token_resolution.py` and `../tests/test_project_wide_hyperlinks.py` fail the build if any `[[SECREP:]]`, `[[THMREF:]]`, `[[LEAN:]]`, `[[VAR:]]`, `[[FIG:]]`, or markdown link does not resolve, so a reader who finds a dangling reference has discovered a bug rather than an omission.

Reading guide

The manuscript has three logical layers; readers can profitably enter at any of the three depending on background. Choose the *conceptual* path if the question driving you is “why does multi-stream coupling matter and what does it look like when an agent does it well?”; choose the *analytical* path if you want to verify the load-bearing mathematical claims and follow the proofs; choose the *empirical* path if you want to reproduce every numerical result against the pymdp-grounded harness and inspect the JSONL run log.

- **Conceptual / qualitative path.** This section (§1) → §3 (POMDP recap) → §4 (the parametric deformation) → §6 (worked toys) → §10 (phase regimes) → §21 (worldview). No proofs required; the closed-form $K=2$ Bernoulli derivation is the fastest entry point to the mathematics.
- **Analytical / theorem path.** §5 (Theorem 5.1, the load-bearing identity) → §7 (e/m-flatness, e-geodesic, Pythagorean) → §8 (Schmidt rank, archetypes, tensor-train) → §9 (the $O(\lambda^2)$ coupling-tax bound, Theorem 9.1) → §11 (comparative statics).
- **Empirical / reproducibility path.** §13 (closed-form Bernoulli + heterogeneous tax + phase + spectral) → §14 (1.0.1 architecture) → §15 (the free-energy bundle and derived summary statistics) → §16 (validation gates + JSONL run log + reproducibility contract).

The Lean formalization plan (§12, §E) is orthogonal to all three paths and can be read at any point; every registered theorem in this manuscript names its Lean companion so a reader can navigate prose → theorem-statement → Lean-source in one hop. A single notation glossary (§S6) consolidates every symbol — including manuscript / LaTeX / Python / Lean counterparts — so prose that introduces new symbols can always be cross-referenced without ambiguity.

Section dependencies (forward signposts)

Builds on	Used by
§3	every later section
§4	§5, §6, §7, §8
§5 (Theorem 5.1)	§6, §11, §10
§7 (Theorem 7.4, Proposition 7.5)	§8, §9, §17
§8 (Proposition 8.1)	§10, §19
§9 (Theorem 9.1)	§11, §13
§13	§14, §15, §16
§14	§15, §16

Background and Prior Work: Antecedents in Active Inference, Variational Coupling, Information Geometry, and Tensor Networks

This chapter catalogs the lines of work the framework builds on and points to where each one returns in the body. Detailed comparisons live in Part V (§17, §18, §19). This chapter provides the substantive orientation required

for the body of the paper. The eight subsections below mirror eight axes of structural commitment: a Markov-decision substrate, a variational family, hierarchical / sophisticated extensions, an information-geometric ambient, a low-rank algorithmic backend, a control-theoretic and reinforcement-learning dual, multi-agent / renormalization analogies, and a formal-verification plus reproducible-software scaffold. Clinical and neural language is treated here as hypothesis-generating context, not as diagnostic evidence.

Discrete-time POMDP active inference

The discrete-time POMDP formulation of active inference is the home setting throughout. The standard treatment combines a generative model (A, B, C, D, E) with planning-as-inference under expected free energy [Friston, 2010, Friston et al., 2014, 2017a, Kaplan and Friston, 2018, Smith et al., 2022, Sajid et al., 2021, Parr et al., 2022], introducing policies as random variables over which the agent maintains a posterior shaped jointly by past observations and anticipated outcomes. Recent unification and agency-formalization work further clarifies how multiple expected-free-energy formulations can be derived from shared roots, how EFE-based planning may be recast as variational inference, and how active inference can be used as a model-comparison language for agency [Champion et al., 2024, de Vries et al., 2025, Nuijten and Lukashchuk, 2025, Da Costa et al., 2024]. That literature is not purely consolidating: Millidge, Tschantz, and Buckley analyze where EFE derivations diverge, so this manuscript treats each per-stream G_k as a specified modeling choice rather than a unique consequence of FEP [Millidge et al., 2021]. Implementations now span several software lineages: `pymdp` targets open-source discrete state-space active inference in Python [Heins et al., 2022], `RxInfer.jl` supplies reactive variational message passing for real-time Bayesian inference in Julia [Bagaev and Podusenko, 2023], `ActiveInference.jl` provides Julia tooling for POMDP active-inference simulation and parameter estimation [Nehrer et al., 2025], and SPM-DEM remains the continuous-state dynamic-expectation-maximization ancestor [Friston et al., 2008]. The present manuscript’s empirical harness is built directly against `pymdp` 1.0.1 (§14), while the mathematical recap in §3 and the Lean recap in §12 keep the same generative-model schema explicit.

Within that lineage, expected free energy is the point at which epistemic and pragmatic control meet: policies are preferred when they both reduce expected uncertainty and realize prior preferences. Cognitive-consistency interpretations make the same split visible in psychological language: epistemic and motivational closure are coupled but not identical objectives [Friston, 2018]. This is why policy coupling is not a cosmetic addition. If two policy streams share epistemic opportunities or pragmatic costs, strict factorization can erase precisely the joint structure through which the process theory expresses coordinated action; epistemic-value and navigation examples make that risk concrete in ordinary active inference before any coupling extension is added [Parr and Friston, 2017a, Kaplan and Friston, 2018]. The action-oriented-model-learning literature makes the same pressure visible from another angle: agents need models that are parsimonious because they are oriented toward possible action [Tschantz et al., 2020]. Here the parsimonious object is the coupling potential J and its scalar strength λ , rather than an additional hidden-state factor or hand-designed controller interface. Factor-graph and graphical-brain accounts show how generative-model structure induces message-passing schedules [de Vries and Friston, 2017, Friston and Parr, 2017], and approximation-family work distinguishes mean-field, Bethe, and marginal message passing as different commitments about how local beliefs compose [Parr et al., 2019, Yedidia et al., 2005]. The word “free energy” is doing different work across those lineages: Bethe / Kikuchi free energies govern approximate marginal inference on factor graphs, while active inference adds policy selection, preferences, and epistemic terms. In the present manuscript, an additional policy factor can implement the λ -coupling inside a chosen graph; the exactness belongs to that chosen graphical model, not to active inference in the abstract.

The standard discrete-POMDP treatment commits to *one* policy random variable per agent, or to a factorized family of per-factor policy posteriors when the model is decomposed. That design is tractable, but multi-stream coordination is then carried by the modeling factorization rather than by an explicit coupling object [Da Costa et al., 2020, Smith et al., 2022]. This commitment is the entry point for the parametric coupling introduced in §4.

Mean-field variational families and their limits

Mean-field factorization is the workhorse approximation of variational inference and its active-inference instantiation: factorize the joint posterior across hidden-state factors, observation modalities, and policy variables to make the per-factor updates tractable [Wainwright and Jordan, 2008, Blei et al., 2017, Da Costa et al., 2020, Smith et al., 2022]. The cost is well known and well-studied: Hoffman and Blei note that the independence approximation “limits the fidelity” of the approximation [Hoffman and Blei, 2015], while Blei, Kucukelbir, and McAuliffe present mean-field and structured variational families as choices in an optimization problem rather than as exact posterior semantics [Blei et al., 2017]. In the present setting, any genuine cross-factor dependence is discarded, and the magnitude of the resulting bias is measured by the multi-information between factors — the same quantity that appears as the surcharge in Theorem 5.1. The point is not that mean-field is wrong; it is that mean-field lacks an internal observable

for the dependence it has projected away.

Three lines of prior work attempt to relax mean-field while preserving tractability:

- **Structured VI** preserves selected dependency substructures while approximating the remaining interactions [Saul and Jordan, 1995, Wainwright and Jordan, 2008, Blei et al., 2017], and structured stochastic VI [Hoffman and Blei, 2015] introduces dependence between subsets of variables via amortized sub-models with shared parameters, capturing coarse-grained dependencies at the cost of additional architectural commitment.
- **Copula variational inference** [Tran et al., 2015, Han et al., 2016] inserts a parametric copula on top of the mean-field marginals, preserving the per-stream factor structure and confining dependence to the copula density in the classical copula sense [Nelsen, 2006].
- **Tensor-network parameterized variational families** [Biamonte and Bergholm, 2017, Cichocki et al., 2016, Han et al., 2018, Glasser et al., 2020] use matrix-product-state and tree-tensor-network factorizations as compact joint distributions, importing algorithmic infrastructure from tensor-network physics and numerical multilinear algebra [Verstraete et al., 2008, Eisert et al., 2010, Orus, 2014].

The framework here is a parametric interpolation across all three: $\lambda = 0$ recovers strict mean-field; finite λ with appropriate J realizes products of experts exactly, copula VI parametrically, and tensor-network variational structure as the algorithmic backend for low-rank coupling (§18). Crucially, these are structural correspondences, not benchmark claims: at each exact or parametric specialization the entanglement decomposition of Theorem 5.1 reduces by substitution to the relevant variational-free-energy expression. Where the relationship is only analogical, such as renormalization-group AIF, the recovery dictionary marks it as analogical rather than exact.

Hierarchical, deep, and sophisticated active inference

Hierarchical / deep active inference [Friston et al., 2017a,b, Pezzulo et al., 2018] decomposes the agent into a hierarchy of generative models with explicit message passing across levels — a treatment that captures temporally-extended structure and cortically-plausible scale separation [Friston et al., 2025], but that retains the mean-field assumption *within* each level. Sophisticated inference [Friston et al., 2021] introduces beliefs about beliefs through recursive expected free energy, modeling how an agent’s plan depends on the beliefs it anticipates having after taking an action. Branching-time AIF [Champion et al., 2022] addresses the exponential cost of policy enumeration via Monte Carlo tree search applied to the EFE tree.

Each motivates a different structural choice, but none should be read as a full exact recovery in the current artifact. Hierarchical AIF corresponds to block-bidiagonal J as a cross-level concentration witness (Theorem 17.1), while temporal-scale separation and directed top-down / bottom-up message passing remain part of the source process theory. Sophisticated inference embeds with recursion-depth-many streams and a belief-about-belief J (Proposition 17.2), while the recursive observation-conditioning is a constraint on how that J is supplied. Branching-time AIF suggests tensor-train or prefix-tree compression of q_λ , but the present manuscript does not run a head-to-head BTAI algorithmic comparison. The detailed mappings are tabulated in §17 and the appendix recovery ledger.

Information geometry

Dually-flat statistical manifolds, e - and m -coordinates, α -projections, and Bregman divergences are the natural language for parametric extensions of variational families [Amari and Nagaoka, 2000, Amari, 2016, Nielsen, 2020, Ay et al., 2017]. The λ -deformation lives on this scaffolding: $\{q_\lambda\}$ is an e -geodesic (Theorem 7.4), the mean-field submanifold is e -flat (Proposition 7.1), and revertibility (collapse to mean-field marginals) is the canonical m -projection (Proposition 7.2). The Pythagorean decomposition (Proposition 7.5) provides an exact split of the KL from the entangled posterior to any reference distribution into a multi-information term and a marginal-to-reference KL.

Non-extensive generalizations via Tsallis / Rényi entropies and escort distributions [Naudts, 2011] yield ϕ -deformed exponential families on which the same construction goes through with quantitatively different geometry; we flag the q -deformed analog as a natural extension (§7.4) without developing it here. Information geometry also supplies vocabulary used in integrated-information studies of consciousness [Tononi, 2008], but the present framework makes only a policy-space analogy: total correlation and low-rank structure quantify dependence among policy streams after the modeling boundary is fixed. It does not claim to solve consciousness, define phenomenology, or validate an integrated-information theory.

Tensor networks and entanglement entropy

Schmidt decomposition, matrix-product states, and the area law of entanglement entropy [Verstraete et al., 2008, Eisert et al., 2010] supply both the algorithmic backend and the conceptual vocabulary for the spectral structure

of entangled posteriors. General tensor-network surveys and machine-learning treatments supply the computational bridge from that physics vocabulary to low-rank tensor representations of high-dimensional arrays [Biamonte and Bergholm, 2017, Cichocki et al., 2016, Orus, 2014]. In the bipartite ($K = 2$) case the joint $q_\lambda(\pi^1, \pi^2)$ admits a Schmidt decomposition whose leading singular vectors are the *archetypal eigenvectors* — the few dominant cross-stream behavioral modes (§8). For $K > 2$ a tensor-train factorization with bond-dimension profile $(r_1, \dots, r_{K-1})$ controls representational capacity (Proposition 8.1, §D).

The tensor-network perspective also gives the framework its sharpest computational claim: low-rank coupling potentials produce low-rank posteriors (Theorem 8.3, the *sparsity-rank tradeoff*), making structured-coupling inference scalable in the bond dimension rather than in the raw policy-space size. The tensor-network literature has independently developed efficient inference algorithms — DMRG, TEBD, MERA [Han et al., 2018, Glasser et al., 2020, Schollwöck, 2011] — that become available once a coupling potential is represented in a tensor-network form; the present artifact demonstrates the rank/bond-profile side of that claim through the multi- K sweep and tensor-train figures, while full tensor-network inference is left as an algorithmic extension. Categorical-systems-theory formulations of structured active inference [Smithe, 2024] further suggest that the same compositional vocabulary admits formal-verification-amenable presentations of structured policies. All tensor-network language in the manuscript is therefore a probability-tensor and linear-algebra analogy. It is not a claim that policy streams are quantum subsystems, or that quantum entanglement semantics are being imported into active inference.

KL / path-integral control and reinforcement learning

KL / path-integral control duality [Kappen, 2005, Todorov, 2006, Theodorou and Todorov, 2012, Rawlik et al., 2013] formalizes the equivalence between stochastic optimal control and free-energy minimization: the value function under a controlled stochastic system is the log-partition of an unnormalized path measure weighted by exponentiated cost. The options framework / hierarchical RL [Sutton et al., 1999, Barto and Mahadevan, 2003, Bacon et al., 2017, Doll et al., 2015] decomposes long-horizon RL into reusable temporally-extended sub-policies with explicit initiation / termination structure. Both fit cleanly into the parametric construction when the required factors are supplied: KL control corresponds to the special-purpose choice in which λJ is the negative control-effort cost (single-stream limit); options correspond to an option-like two-stream factorization with J encoding option-membership structure (§18).

A subtler connection is to **products of experts** [Hinton, 2002]: the λ -entangled prior $\mathcal{E}_\lambda(\pi) = \prod_k E_k(\pi^k) e^{\lambda J(\pi)}$ is literally a product of experts on joint policy space, with the coupling potential acting as one additional expert that *only* fires on configurations where the cross-stream relationship holds.

Multi-agent inference, Markov blankets, and renormalization

Interactive / multi-agent active inference [Maisto et al., 2024] models multi-agent cooperation as joint inference under shared generative models, while collective active-inference models show how group-level coordination can arise from surprise minimization under coupled generative models [Heins et al., 2024]. Renormalization-group AIF [Friston et al., 2025] derives hierarchical generative models via repeated application of a coarse-graining operator that preserves dynamics-form. Bayesian-mechanics treatments of Markov blankets [Friston, 2019, Da Costa et al., 2021] identify a system with a partition of states into internal / external / blanket, while Markov-blanket accounts of autonomy emphasize that a system boundary is an inferential and conditional-independence structure rather than just a physical envelope [Kirchhoff et al., 2018, Ramstead et al., 2018]. The critical literature is important here: both technical and philosophical reviews argue that Markov blankets require careful modeling assumptions and do not, by themselves, settle the scope of FEP, active inference, or cognitive boundaries [Aguilera et al., 2022, Raja et al., 2021, Menary and Gillett, 2022].

All three relate to NK -stream entanglement only after the modeling boundary and coupling graph are specified. Multi-agent inference is a parametric joint-policy prior when within-agent and across-agent blocks of J are added to a shared or linked generative model. Renormalization-group AIF is an analogy between coarse-graining and low-rank policy-posterior projection. Markov-blanket separation is a policy-space analog quantified by normalized multi-information (§19), not a recovery of the state-space blanket construction. The manuscript therefore does not infer new biological boundaries from J ; it supplies a way to measure and manipulate policy dependence once a modeling boundary has already been chosen.

Computational psychiatry and neural implementation

Computational-psychiatry programs use AIF to model dysregulation of perception, action, and learning [Adams et al., 2013, Schwartenbeck and Friston, 2016], often associating symptom dimensions with failure modes of the free-energy-minimization machinery (sensory precision dysregulation, prior over-weighting, habit rigidity). Dopaminergic / policy-precision accounts [Friston et al., 2014, Schwartenbeck et al., 2015, Parr and Friston, 2017b] and recent

precision-in-action reviews [Limanowski et al., 2024], together with empirical precision-weighting evidence in psychosis [Haarsma et al., 2021], motivate the precision vocabulary, but the present framework borrows only the *formal* precision analogy for λ (§9.4). It does not identify λ with dopamine, synaptic gain, or any specific neural variable.

The phase-structure analysis in §10 flags a *coupling-drift* failure mode (under- or over-coupling of behavioral streams) that is qualitatively distinct from per-stream prior corruption. We treat the clinical language as a model-generated hypothesis, not a diagnostic claim: if the coupling account is right, controlled tasks would be expected to reveal cross-stream covariance signatures (joint-action repertoire, between-trial mode statistics, coupling-spectrum effective rank) that are not visible in per-stream marginal measures alone. This is consistent with the habit-vs-deliberation balance literature [Doll et al., 2015] while adding an explicit coupling parameter that a follow-up empirical protocol could estimate.

Lean 4, Mathlib, and formal verification of probabilistic systems

Lean 4 [de Moura and Ullrich, 2021, FRO, 2026a] and the Mathlib library [Community, 2020, FRO, 2026b] are the formal-verification backbone of the present work. Lean 4 is a dependently-typed proof assistant with a minimal trusted kernel and a tactic framework expressive enough to mechanize contemporary mathematics; Mathlib provides — among many other things — the measure theory, linear algebra, and analysis targeted by the separate additive discharge of the framework’s witness-form theorems.

A prior line of work shipped a 50-topic Mathlib-checked FEP catalog and companion monograph [Friedman, 2026c]; the present manuscript’s `ActinfPolicyEntanglement` boundary fragment (§12, §E) reuses the same toolchain pin and re-export layout, with witness-form theorems isolating the analytic content for a separate Mathlib4 discharge layer. Every numbered theorem in this manuscript — including the spectral semicontinuity and tensor-train rank claims of §8 (Proposition 8.2, Theorem 8.3) and the hierarchical / sophisticated-inference embeddings of §17 (Theorem 17.1, Proposition 17.2) — now carries a live Lean companion in that fragment. The choice to keep the present fragment **Mathlib-free** is deliberate: it forces every theorem statement to be expressible in the in-house `CommScalar  $\alpha$` typeclass, exposes the analytic content as explicit witness arguments, and makes the boundary’s hygiene budget (zero strict sorry, zero axioms, zero `unsafe` / `partial` / `noncomputable`) easy to verify against stock Lean v4.29.0.

The verified-numerics boundary remains separate from this proof story. Numerical analysis supplies the right vocabulary for round-off, conditioning, and stability [Higham, 2002], while IEEE 754 fixes the floating-point arithmetic standard a formal bridge would have to model [IEEE, 2019], and proof-assistant floating-point libraries such as Flocq show what such a bridge requires [Boldo and Melquiond, 2011]. The present artifact machine-checks the exact $\mathbb{R}$ mathematics and tests the Float pipeline to strict tolerances; it does not claim a formal $\text{Float} \leftrightarrow \mathbb{R}$ theorem.

Reproducible scientific software

The empirical layer is built on NumPy [Harris et al., 2020], SciPy [Virtanen et al., 2020], JAX [Bradbury et al., 2018] (which underpins `pymdp` 1.0.1), and Matplotlib [Hunter, 2007]. Every numeric value in the manuscript flows from a real run via the `[[VAR:<key>]]` token system (§16); every figure carries reproducibility tEXt metadata (source script, function, hyperparameter snapshot, git revision, ISO timestamp, and compact plotted-data summaries); every cross-reference resolves through the same registry that the renderer consumes. This positions the manuscript inside the broader open-science / reproducible-research practice and — concretely — makes the manuscript an executable artifact: running `scripts/run_all.py` regenerates every numeric, every figure, and every theorem-table row from a single deterministic pipeline.

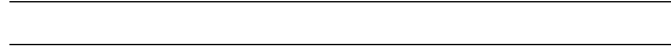
Part II — Theory

The mathematical core of the manuscript. Each chapter introduces the objects, states the load-bearing theorems about them, and immediately gives the proof or proof sketch (full proofs in the supplements). The chain proceeds:

1. §3 — discrete-time POMDP active inference and the mean-field baseline against which everything is measured.
2. §4 — the parametric coupling construction: J (habit) and K_c (preference) potentials, and the λ -entangled posterior they define.
3. §5 — the load-bearing identity (Theorem 5.1) splitting free energy into marginals, coupling, and a non-negative multi-information term.

4. §6 — three worked cases (Bernoulli closed form, motor-attention, multi-timescale) that make the abstract machinery concrete.
5. §7 — dual e/m information geometry; the λ -family is an e-geodesic (Theorem 7.4); Pythagorean decomposition (Proposition 7.5).
6. §8 — Schmidt rank (Proposition 8.1), archetypal eigenvectors, tensor-train bond dimensions (Theorem 8.3).
7. §9 — VFE/EFE mixed ensembles and the $O(\lambda^2)$ coupling-tax bound (Theorem 9.1).
8. §10 — disordered / mixed / frozen coupling regimes and their behavioral signatures.
9. §11 — sensitivity, two-parameter generalization, potential-structure dependence.

Every theorem in this part carries a Lean companion (see the per-row status in [docs/reference/veridical_status.md](#)) and a Python numerical witness (see [docs/reference/_theorem_map.md](#)). The supplements (§A, §B, §C, §D) carry the full derivations.



Setup and Assumptions: Finite-Horizon Discrete POMDPs, Multi-Stream Policy Factorization, and the Mean-Field Baseline

We work throughout with finite, discrete state and action spaces — adequate for the analytical results, computationally tractable, and aligned with the discrete-time POMDP formulation that dominates AIF practice (pymdp, RxInfer, ActiveInference.jl, SPM-DEM) [Heins et al., 2022, Bagaev and Podusenko, 2023, Nehrer et al., 2025, Friston et al., 2008]. Continuous extensions are sketched in §21. Every symbol introduced here also appears in the unified notation glossary (§S6) with its LaTeX, Python, and Lean counterparts.

Single-stream POMDP active inference

A standard discrete POMDP active inference unit:

Symbol	Meaning
$s \in \mathcal{S}$	hidden state (finite)
$o \in \mathcal{O}$	observation (finite)
$a \in \mathcal{A}$	action (finite)
$\pi \in \Pi$	policy: sequence/map of actions over horizon T
$A_{ij} = p(o = i \mid s = j)$	likelihood matrix
$B_{ij}^a = p(s' = i \mid s = j, a)$	transition tensor
$C(o) = \log \tilde{p}(o)$	log-prior preferences
$D(s) = p(s_0)$	initial state prior
$E(\pi)$	habit / policy prior on Π
γ	policy precision

Under planning-as-inference with expected free energy [Friston et al., 2014, Da Costa et al., 2020]:

$$q(\pi) \propto E(\pi) \exp(-\gamma G(\pi)), \quad G(\pi) = \text{Risk}(\pi) + \text{Ambiguity}(\pi) \quad (3.1)$$

with the standard EFE decomposition into pragmatic and epistemic components

$$G(\pi) = \underbrace{\mathbb{E}_{q(o,s|\pi)} [\log q(s \mid \pi) - \log p(s \mid o, \pi)]}_{\text{epistemic value, negated}} + \underbrace{\mathbb{E}_{q(o|\pi)} [D_{\text{KL}}(q(o \mid \pi) \parallel \tilde{p}(o))]}_{\text{pragmatic value, negated}}. \quad (3.2)$$

(Sign conventions for F , G , K_c are cataloged in §S6.1; the “negated” annotations above match the standard EFE convention there.)

VFE-only streams are obtained by treating G as an immediate variational objective (no recursive lookahead), where action selection is the gradient step on F at the current belief; the policy distribution effectively collapses to a delta on the next-action argmin. In planning-as-inference with γ moderate, $q(\pi)$ remains a proper distribution over the policy space.

Having fixed the single-stream POMDP baseline, we now generalize to the multi-stream setting that is the substantive subject of this manuscript.

The multi-stream extension

We posit K concurrent policy variables

$$\pi = (\pi^1, \pi^2, \dots, \pi^K), \quad \pi^k \in \Pi^k, \quad \Pi = \prod_{k=1}^K \Pi^k. \quad (3.3)$$

Generic streams. The K streams are generic in three independent senses, all of which the framework handles uniformly:

- **Modality.** Each stream may correspond to a distinct observation modality (visual, auditory, proprioceptive) and/or distinct hidden-state factor (location, identity, intent).
- **Time horizon.** Streams may differ in planning horizon T_k — a fast attentional policy and a slow navigational policy are simultaneously representable without forcing them onto a shared clock.
- **Inference mode.** Each stream may use VFE-only, EFE-planning, or sophisticated inference (recursive EFE; [Friston et al., 2021]).

Concretely: a drummer’s left-hand timing stream (motor modality, short horizon, reflexive VFE) co-exists with a right-hand fill-planning stream (motor modality, longer horizon, EFE planning); an autonomous vehicle couples a very short-horizon steering controller (proprioceptive, VFE) with a much longer-horizon route planner (symbolic, EFE) and an intermediate-horizon attention allocator (visual, EFE); a reading-while-reaching agent factors immediate saccade policy (visual, VFE) from short-horizon reach policy (proprioceptive, EFE). In each example the streams differ along at least two of the three axes above, and the *coordination* between them is precisely what the entanglement framework formalizes.

Let $\mathcal{V} \subseteq \{1, \dots, K\}$ index VFE-only streams and $\mathcal{P} = \{1, \dots, K\} \setminus \mathcal{V}$ index planning streams. The heterogeneous-ensemble pay-off when these two stream classes are coupled is developed in §9.

The mean-field baseline

Strict mean-field across streams:

$$E_{\text{MF}}(\pi) = \prod_{k=1}^K E_k(\pi^k), \quad G_{\text{MF}}(\pi) = \sum_{k=1}^K G_k(\pi^k), \quad q_{\text{MF}}(\pi) = \prod_{k=1}^K q_k(\pi^k), \quad (3.4)$$

with each $q_k(\pi^k) \propto E_k(\pi^k) \exp(-\gamma_k G_k(\pi^k))$ a single-stream posterior. This is the regime exemplified by `pymdp`’s factorized-A, factorized-B, factorized-policy treatment of multi-factor models [Heins et al., 2022].

Standing assumptions. - Every per-stream policy alphabet Π^k is finite; the joint $\Pi = \prod_k \Pi^k$ is therefore finite. Continuous-policy extensions are conjectured in §20 (Q4) but are out of scope for the analytical core of this manuscript. - The per-stream priors E_k are strictly positive on Π^k (no exact zeros), so KL divergences and log-likelihoods are everywhere finite. - The per-stream EFE G_k and coupling potentials J, K_c are real-valued and bounded on Π , so $\mathbb{E}_{q_\lambda}[\cdot]$ exists for every observable used below and the Gibbs partition function $Z(\lambda)$ is real-analytic in λ on $[0, \infty)$.

Standing notation. - $H(\cdot)$: Shannon entropy. - $D_{\text{KL}}(p\|q)$: Kullback-Leibler divergence. - $I(p) = \sum_k H(p^k) - H(p)$: total correlation (multi-information [McGill, 1954, Watanabe, 1960]) of joint p with respect to its K marginals p^k (see Eq. (5.3)). - $\mathcal{M}$: manifold of joint distributions on Π . - $\mathcal{M}_{\text{MF}} \subset \mathcal{M}$: mean-field submanifold (product distributions); its e-flatness is recorded in Proposition 7.1 and developed in §7.

Lambda Deformation: Definition and Properties

In symbols: $q_\lambda(\pi) \propto q_{\text{MF}}(\pi) e^{\lambda(J(\pi) - \gamma K_c(\pi))}$, with q_{MF} the mean-field baseline introduced in §3.3 and the entangled-prior / entangled-posterior expansion derived below.

The symbols introduced below — $J, K_c, \lambda, \gamma, \mathcal{E}_\lambda, q_\lambda$ — are cataloged in the unified notation glossary (§S6) with their LaTeX, Python, and Lean counterparts.

Coupling potentials

Introduce two real-valued coupling potentials on the joint policy space:

- $J : \Pi \rightarrow \mathbb{R}$ — *habit coupling* (prior-side): structural cross-stream tendencies that the agent has built up and considers a-priori plausible.
- $K_c : \Pi \rightarrow \mathbb{R}$ — *preference coupling* (EFE-side): cross-stream costs/preferences that show up in expected free energy beyond the per-stream EFEs.

These need not be informed by anything outside the agent: they are parameters of the agent’s generative model, learned from experience.

Definition: lambda-entangled policy posterior

Define the λ -entangled prior and λ -entangled posterior as

$$\mathcal{E}_\lambda(\pi) = \frac{1}{Z_E(\lambda)} \prod_{k=1}^K E_k(\pi^k) \exp(\lambda J(\pi)) \quad (4.1)$$

$$G_\lambda(\pi) = \sum_{k=1}^K G_k(\pi^k) + \lambda K_c(\pi) \quad (4.2)$$

$$q_\lambda(\pi) \propto \mathcal{E}_\lambda(\pi) \exp(-\gamma G_\lambda(\pi)) = \frac{1}{Z(\lambda)} [\prod_k E_k(\pi^k)] \exp(\lambda(J(\pi) - \gamma K_c(\pi)) - \gamma \sum_k G_k(\pi^k)) \quad (4.3)$$

Boundary cases. - $\lambda = 0$: $\mathcal{E}_0 = E_{\text{MF}}$, $G_0 = G_{\text{MF}}$, $q_0 = q_{\text{MF}}$. - $\lambda \rightarrow \infty$: q_λ collapses onto the modes of $J - \gamma K_c$ — *pure archetypal joint policies*. (Here “modes” means elements of the argmax set $\text{argmax}_\pi (J(\pi) - \gamma K_c(\pi))$). As $\lambda \rightarrow \infty$, q_λ concentrates on this argmax set, with mass restricted to and re-normalized over those policies weighted by the bare-posterior measure $E_{\text{MF}}(\pi) e^{-\gamma G_{\text{MF}}(\pi)}$; in particular, when E_{MF} and G_{MF} are constant on the argmax set, q_λ concentrates uniformly on the set.) - Intermediate λ : the regime of practical interest.

For analytical separation of habit vs. preference coupling, one can carry two parameters λ_E, λ_G with coupling potentials J, K_c respectively. We use a single λ in the body of the paper for clarity, with the two-parameter generalization appearing in §11 for comparative statics.

The coupling parameter lambda as a precision-like coupling weight

A central conceptual point: λ is *not* merely an arbitrary engineering knob. In the exponential-family representation it behaves as a **precision-like inverse-temperature** on the coupling structure, dual (in the Legendre sense) to the sufficient statistics $\langle J \rangle$ and $\langle K_c \rangle$ under the joint posterior. This is mathematically analogous to the role γ plays for EFE in the standard single-stream model, while remaining a distinct model parameter rather than a claim about any specific neural precision signal. Concretely:

$$\frac{\partial \log Z(\lambda)}{\partial \lambda} = \langle J - \gamma K_c \rangle_{q_\lambda}. \quad (4.4)$$

Updating λ by gradient on free energy (§9) is therefore a formal analog of precision learning [Friston et al., 2014, Schwartenbeck et al., 2015, Limanowski et al., 2024]. We return to this in §9.4. Two consequences are worth flagging now. First, λ can be learned from the same free-energy objective the agent already minimizes — there is no auxiliary regularizer, no separate meta-learner in the model. Second, under the model, coupling strength can vary with task context or learning history. Any biological, arousal, fatigue, or developmental reading is hypothesis-level until an empirical protocol estimates λ from joint-action data; tonic over-coupling and under-coupling are therefore framed as hypotheses about joint-statistical signatures, not as diagnostic claims (§10).

What J and K_c look like in practice

Five archetypal forms span the modeling range, each tuned to a different *kind* of cross-stream dependency. The choice of J is therefore the substantive modeling decision — once J is fixed, the deformation in Eq. (5.2) is determined, and the framework’s theorems hold for any bounded-norm choice. The five archetypes:

- **Sparse pairwise.** $J(\pi) = \sum_{(i,j) \in \mathcal{C}} J_{ij}(\pi^i, \pi^j)$ with $\mathcal{C}$ a small set of edges. Recovers pairwise Markov-random-field structure and is the natural generalization of factor graphs; appropriate when only a few stream pairs (e.g. left/right-hand pairs in motor coordination) carry meaningful joint dependence.

- **Tensor-network low-rank.** $J(\pi) = \log[\text{MPS}(\pi^1, \dots, \pi^K)]$ with bond dimension r . Recovers all the favorable algorithmic structure of matrix product states [Han et al., 2018]; bond dimension is a complexity dial that interpolates between mean-field ($r = 1$) and arbitrary joint structure (full r).
- **Block-bidiagonal hierarchical.** J couples only stream k to $k + 1$. This is the structural analog used by the hierarchical / deep AIF witness-form mapping (§17.2): it captures cross-level coordination in the joint policy posterior, while temporal-scale separation and directed top-down / bottom-up message passing remain part of the source generative model.
- **Symbolic-task biased.** J encodes a logical constraint (“if policy 1 is reach, then policy 2 is open hand”) via large-magnitude penalties on violating configurations. Appropriate when domain knowledge dictates discrete co-occurrences.
- **Learned tabular.** J is a free parameter on Π (size $\prod_k |\Pi^k|$) and is learned end-to-end. Computationally infeasible for large K but conceptually clean and useful as the universal upper bound against which structured choices are benchmarked.

The framework is agnostic to the choice of J, K_c — its theorems hold for any bounded-norm potentials. The art of the modeler is in choosing parsimonious, mechanistically interpretable forms.

Entanglement Decomposition Theorem: Marginals, Coupling, and Total Correlation

Statement

The variational free energy under q_λ admits a clean decomposition. Define

$$F[q_\lambda] = \mathbb{E}_{q_\lambda}[\gamma G_\lambda(\pi) - \log \mathcal{E}_\lambda(\pi)] - H(q_\lambda). \quad (5.1)$$

Let q_λ^k denote the marginal of q_λ on stream k , and $F[q_\lambda^k]$ the *marginal* free energy if stream k were treated in isolation against its own E_k, G_k .

Theorem 5.1 (Entanglement Decomposition).

$$F[q] = \sum_k F[q^k] + \gamma \lambda \mathbb{E}_q[K_c] + \log Z_E(\lambda) - \lambda \mathbb{E}_q[J] + I(q) \quad (5.2)$$

where the total correlation

$$I(q) = \sum_k H(q^k) - H(q) = D_{\text{KL}}(q \parallel \prod_k q^k) \quad (5.3)$$

is non-negative and vanishes iff q_λ is mean-field. (The sign on every term follows §S6.1 in S06 — specifically the **plus- $I(q_\lambda)$** and the $+\gamma\lambda\langle K_c \rangle$ conventions.)

The Lean companion is a boundary statement auto-extracted from the live source:

```
-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:75` [status: **boundary**]
/-- **Theorem 5.1 (Entanglement Decomposition)** - boundary witness
form.
```

```
Given a Mathlib-supplied algebraic split
`F[q] = marginal_part + coupling_expectation + agentic_gain`,
where `coupling_expectation` is the integrated `couplingLogWeight`
against `q`, this theorem certifies the equation in the boundary
fragment. Every coupling parameter `(J, K_c, γ, λ)` is genuinely used
via `couplingExpectationSkeleton`, so the statement is non-vacuous.
```

```
**Typed-API-contract disclaimer.** The Lean body is literally
`hWitness` - the identity term on the algebraic split.
This is not a stand-alone proof of Theorem 5.1; it is a typed-API
contract that forces `(J, K_c, γ, λ)` into the conclusion via
`couplingExpectationSkeleton` and locks the four-term decomposition
shape. The non-vacuous algebraic content of the decomposition
(commutative-ring re-grouping of the four bookkeeping scalars) is
discharged separately by `entanglement_decomposition_four_terms_assoc_skeleton`
and `entanglement_decomposition_four_terms_commute_skeleton` below
(genuine `CommScalar` proofs). The full analytic content (Gibbs
expansion + KL chain rule) is supplied as `hWitness` and discharged
```

by the separate MathlibProofs layer; the numerical realization is in
`src/lean/decomposition.py` and verified at the dashboard invariant
`decomposition\_lhs\_eq\_rhs\_max\_residual` (worst-case `5.55e-16`). -/
theorem entanglement_decomposition {K Pol}
(q : JointDist K Pol) (s : List (PolicySpace K Pol))
(logE G : PolicySpace K Pol → Float)
(J K_c : CouplingPotential Float K Pol) (gamma lam : Float)
(marginal_part agentic_gain : Float)
(hWitness : variationalFreeEnergy q logE G gamma s =
marginal_part
+ couplingExpectationSkeleton q s J K_c gamma lam
+ agentic_gain) :
variationalFreeEnergy q logE G gamma s =
marginal_part
+ couplingExpectationSkeleton q s J K_c gamma lam
+ agentic_gain :=

Honesty note. The *Float* boundary companion above carries `status: boundary` in `manuscript/refs/labels.yaml`, `not status: proved`: it type-checks under stock Lean v4.29.0 and locks the four-term algebraic skeleton, but its analytic content is supplied as a hypothesis. **The analytic content itself is now machine-checked in $\mathbb{R}$:** `MathlibProofs.entanglement_decomposition_generalK` proves, for general K and a general entangled q (positivity + normalization only; never assumed to factorize), the multi-information non-negativity, the KL chain-rule decomposition, and m -projection minimality — depending only on the three standard foundational axioms, with the previously-deferred product-marginalization core discharged from the standing positivity + normalization hypotheses without further structural assumptions, and negative-control-verified as non-vacuous. **The full S01 boxed identity itself** ($F = \sum F[q^k] + \gamma \lambda \langle K_c \rangle + \log Z_E - \lambda \langle J \rangle + I(q)$) is then machine-checked in $\mathbb{R}$ by `MathlibProofs.free_energy_decomposition_full` for the genuine entangled posterior (`log Z_E` the definitional normalizer, foundational-only axioms, two independent negative controls). The *Float* companion is the numerically-corroborated computational image of the $\mathbb{R}$ proof; closing the $\text{Float} \leftrightarrow \mathbb{R}$ formal bridge is the one *verification-stack interface* residual (the open residual discussed at §21; interval brackets on the $K=2$ sweep grid in `output/reports/float_real_residual.json` corroborate *Float* decomposition residuals but do not discharge the bridge); the remaining typed-API rows are the analytic-discharge follow-on tracked at §20, not a separate verification gap). See the *honest substantive / typed-API split* in `docs/reference/veridical_status.md` for the round-by-round audit of which Lean theorems are *substantive machine-checked proofs* versus *typed-API contracts*.

Proof sketch

Expand $F[q_\lambda] = \mathbb{E}_{q_\lambda}[\gamma G_\lambda - \log \mathcal{E}_\lambda] - H(q_\lambda)$ using

$$\gamma G_\lambda(\pi) = \gamma \sum_k G_k(\pi^k) + \gamma \lambda K_c(\pi), \quad \log \mathcal{E}_\lambda(\pi) = \sum_k \log E_k(\pi^k) + \lambda J(\pi) - \log Z_E(\lambda) \quad (5.4)$$

for the normalized entangled prior $\mathcal{E}_\lambda \propto (\prod_k E_k) e^{\lambda J}$. Linearity of expectation separates $\mathbb{E}[\gamma G_\lambda]$ and $\mathbb{E}[\log \mathcal{E}_\lambda]$ into per-stream pieces plus the $\gamma \lambda \langle K_c \rangle$, $\lambda \langle J \rangle$, and $\log Z_E(\lambda)$ terms. The entropies $-H(q_\lambda)$ combine with $\sum_k \mathbb{E}_{q_\lambda^k}[\gamma G_k - \log E_k]$ into $\sum_k F[q_\lambda^k] + (\sum_k H(q_\lambda^k) - H(q_\lambda))$; the parenthesis is multi-information

$$I(q) = \sum_k H(q^k) - H(q) = D_{\text{KL}}(q \parallel \prod_k q^k) \quad (5.3)$$

, i.e. $I(q_\lambda)$. Altogether this yields

$$F[q] = \sum_k F[q^k] + \gamma \lambda \mathbb{E}_q[K_c] + \log Z_E(\lambda) - \lambda \mathbb{E}_q[J] + I(q) \quad (5.2)$$

. Full line-by-line algebra is in §A; the Python record `entanglement_decomposition_rhs` sums the same four grouped terms and matches the Gibbs definition of F on random joints (see `tests/test_decomposition.py`).

What the decomposition says

Three terms, three readings:

(i) $\sum_k F[q_\lambda^k]$. The free energy that would obtain if each stream were optimized in isolation against the *marginal* of q_λ . This is *not* the same as $\sum_k F[q_k^{\text{MF}}]$, because the marginals q_λ^k are themselves shaped by coupling — coupling deforms what a stream considers a-posteriori plausible even before we charge the multi-information cost.

(ii) $\lambda(\gamma\langle K_c \rangle - \langle J \rangle)$. The *coupling cost* — what it costs (or pays) the agent to maintain the joint structure encoded by the potentials. Sign depends on alignment between habit coupling J (which the agent likes) and preference-side coupling K_c (which contributes EFE). When J and K_c point in compatible directions — e.g., when habitual joint policies happen to also minimize joint EFE — this term can be net negative and pays for itself.

(iii) $I(q_\lambda) \geq 0$. The multi-information term appears **with a plus sign** in

$$F[q] = \sum_k F[q^k] + \gamma \lambda \mathbb{E}_q[K_c] + \log Z_E(\lambda) - \lambda \mathbb{E}_q[J] + I(q) \quad (5.2)$$

: holding the marginals fixed, any genuine cross-stream correlation raises the Gibbs variational objective by exactly $I(q_\lambda)$ nats. When joint structure still *pays off* for the agent is a comparison with the coupling / EFE / log-partition bundle in the same identity — not literal subtraction of I from “free energy” in isolation. [Corollary 5.2–Corollary 5.4](#) formalize the sign bookkeeping of that trade-off.

Read together, the three terms expose the central trade-off of policy entanglement. The marginal sum (i) tracks how each stream is faring on its own slice of the deformed posterior; the coupling bundle (ii) is the *price tag* of maintaining the structure encoded in J and K_c , and can be paid in either direction depending on whether habit and preference are aligned or in tension; the multi-information (iii) is the *correlation surcharge* that any non-mean-field joint owes purely for being correlated. Coupling pays for itself precisely when the bundle (ii) is sufficiently negative to absorb both the correlation surcharge (iii) and any worsening of per-stream marginals (i). Stating it this way also clarifies why the framework refuses to call (iii) a “cost” in isolation: $I(q_\lambda)$ is the price of joint structure measured *against* mean-field, but (i) and (ii) measure how much that structure buys back, and only the sum across all three is a meaningful free-energy comparison.

The Lean companions for the three immediate corollaries are auto-extracted below.

[Corollary 5.2](#) (the *coupling-pays-for-itself* verdict) is a tri-state classifier from the sign of the bookkeeping, *now upgraded to a proved theorem* (status: proved) whose Lean form discharges the sign-soundness identity by direct case analysis on the bookkeeping variable:

```
-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:168` [status: **proved**]
/-- **Corollary 5.2 correctness theorem (boundary identity)**: the
`couplingVerdict` Boolean is `true` precisely when the coupling tax is
strictly less than the agentic gain – i.e. the verdict semantics
faithfully decide the “coupling-pays-for-itself” predicate.
```

This pins the verdict’s contract on the boundary fragment: a `true` answer is *not* an oracle assertion but a Lean-level proof that `tax < gain`. Discharged by unfolding the `decide` and forwarding the proof component, no `sorry`. -/

```
theorem couplingVerdict_correct (gain tax : Float) :
  couplingVerdict gain tax = true ↔ tax < gain := by
```

[Corollary 5.3](#) (mean-field reduction at $\lambda = 0$) collapses the four-summand bookkeeping to its sum-of-marginals + total- correlation-gain pair:

```
-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:195` [status: **proved**]
/-- **Corollary 5.3 (Mean-field optimum at  $\lambda = 0$ )**: at ` $\lambda = 0$ `, the
coupling log-weight contribution vanishes pointwise. Polymorphic over
`[CommScalar  $\alpha$ ]`; combine with `entanglement_decomposition` to recover
the pure mean-field statement ` $F[q_0] = \sum_k F[q_0^k] + \text{agentic\_gain}$ `. -/
theorem couplingLogWeight_pointwise_at_zero { $\alpha$  : Type} [CommScalar  $\alpha$ ]
  {K Pol}
  (J K_c : CouplingPotential  $\alpha$  K Pol) (gamma :  $\alpha$ )
  ( $\pi$  : PolicySpace K Pol) :
  couplingLogWeight J K_c gamma 0  $\pi$  = 0 :=
```

[Corollary 5.4](#) (strict gain when q is non-mean-field) is the sign companion of [Eq. \(5.3\)](#). Unlike the two proved corollaries above, its Lean companion is a typed boundary witness (status: boundary, faithfulness: typed-witness): the strict-gain inequality is carried as a typed contract, with the analytic discharge at the Mathlib layer.

```
-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:215` [status: **boundary**]
/-- **Corollary 5.4 (Total-correlation unfolding)**: the boundary
fragment’s `totalCorrelation` is by construction the per-stream
entropy sum minus the joint entropy. This forwarder makes the
definitional unfolding available to downstream modules without
re-importing `FreeEnergy`’s namespace.
```

The strict-positivity claim ` $I(q) > 0$  iff  $q$  is not mean-field` is the ` $Iq > 0 \leftrightarrow \neg \text{IsMeanField } q$ ` companion that requires Mathlib’s KL-non-negativity (see `IsNonNegMultiInformation` in `[FreeEnergy.lean]`(`FreeEnergy.lean`)) and is exposed by the `MathlibProofs` extension. Stock-Lean, zero-`sorry`. -/

```
theorem totalCorrelation_def_unfold {K Pol}
```

```

(q : JointDist K Pol) (s : List (PolicySpace K Pol))
(sumStreamEntropies : Float) :
totalCorrelation q s sumStreamEntropies
= sumStreamEntropies - shannonEntropy q s := rfl

```

Optimal coupling: existence of λ *

Theorem 5.5 (Existence of optimal coupling). For fixed J, K_c with bounded potentials, $F[q_\lambda]$ is real-analytic in λ on $[0, \infty)$ and admits the closed exponential-family identity

$$F[q_\lambda] = \log Z_E(\lambda) - \log Z(\lambda) \quad (5.8)$$

The Lean boundary companion threads the algebraic identity through the coupling-log-weight skeleton so every parameter $(J, K_c, \gamma, \lambda)$ is genuinely referenced:

```

-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:243` [status: **boundary**]
/-! ## Theorem 5.5 – closed exponential-family form of `F[q_λ]`

```

The Gibbs decomposition $F[q] = \sum F[q^k] + \gamma \lambda (K_c) + \log Z_E(\lambda) - \lambda(J) + I(q)$ collapses to the strikingly clean identity

```

...
F[q_λ] = log Z_E(λ) - log Z(λ),
...

```

where $\log Z_E(\lambda)$ is the entangled-prior log-partition and $\log Z(\lambda)$ is the entangled-posterior log-partition. See `[`manuscript/04_entanglement_decomposition.md`](../../manuscript/04_entanglement_decomposition.md)` (`[Theorem 5.5](../../manuscript/refs/labels.yaml#thm_4_2)`) and the proof appendix `[`manuscript/S01_proof_of_decomposition_theorem.md`](../../manuscript/S01_proof_of_decomposition_theorem.md)`.

The boundary fragment exposes the identity as a **witness-consuming** statement: the caller (the separate additive `MathlibProofs` layer or the numerical Python layer) supplies the algebraic equality $vfe = \log Z_E - \log Z$, and the boundary fragment threads $(\lambda, J, \gamma, K_c)$ through `couplingExpectationSkeleton` so every parameter is genuinely referenced and the statement is non-vacuous. Polymorphic over `[CommScalar α]`. -/

```

theorem freeEnergy_closedForm_witness {α : Type} [CommScalar α]
  {K Pol}
  (vfe logZE logZ : α)
  (_s : List (PolicySpace K Pol))
  (J K_c : CouplingPotential α K Pol) (gamma lam : α)
  (hWitness : vfe = logZE - logZ) :
vfe = logZE - logZ
  Λ (∀ π : PolicySpace K Pol,
    couplingLogWeight J K_c gamma lam π
      = lam * J π - gamma * lam * K_c π) := by

```

where $Z(\lambda) = \sum_{\pi} E_{MF}(\pi) e^{\lambda(J - \gamma K_c)(\pi) - \gamma G_{MF}(\pi)}$ is the normalizer of q_λ and $Z_E(\lambda) = \sum_{\pi} E_{MF}(\pi) e^{\lambda J(\pi)}$ is the normalizer of the entangled prior. Differentiating once gives

$$\frac{dF[q_\lambda]}{d\lambda} = \langle J \rangle_{\mathcal{E}_\lambda} - \langle J \rangle_{q_\lambda} + \gamma \langle K_c \rangle_{q_\lambda} \quad (5.9)$$

so $\lambda \mapsto F[q_\lambda]$ has at least one stationary point in $(0, \infty)$ whenever the marginal slope is negative,

$$\langle J \rangle_{q_0} - \gamma \langle K_c \rangle_{q_0} > \langle J \rangle_{E_{MF}}, \quad (5.10)$$

i.e. when the *posterior* coupling alignment net of EFE cost exceeds the *prior* coupling alignment. Since F is real-analytic and bounded below on $[0, \infty)$ — boundedness follows from the compactness of the simplex on a finite Π and the boundedness of the potentials J, K_c, G_k standing-assumed in §3.3, so $\log Z_E(\lambda)$ and $\log Z(\lambda)$ are continuous bounded functions of λ — and $F'(0) < 0$ under the stated marginal-slope condition, $F'(\lambda)$ achieves a zero in some open interval and a stationary point $\lambda^* \in (0, \infty)$ exists. Since the multi-information I attains its minimum at the mean-field surface, $dI/d\lambda|_0 = 0$ identically. To see this in one line: by the chain rule along the e-geodesic, the covariance form is

$$\frac{dI(q_\lambda)}{d\lambda} = \text{Cov}_{q_\lambda}(J - \gamma K_c, \log q_\lambda - \log \hat{m}(q_\lambda)) \quad (5.11)$$

A derivation along the e-geodesic from the closed identity [Eq. \(7.3\)](#) is in [§9](#); at $\lambda = 0$, $q_0 = q_{\text{MF}} = \hat{m}(q_0)$ pointwise (here $\hat{m}(q) = \prod_k q^k$ is the *m-projection* of q onto the mean-field submanifold $\mathcal{M}_{\text{MF}}$, defined in [§7.1](#)), so $\log q_0 - \log \hat{m}(q_0) \equiv 0$ on Π and the covariance vanishes. The *Fisher* contribution to F therefore lives at second order ([Eq. \(A.15\)](#) and [Proposition 11.1](#); see also [§B](#)). Under the symmetric standing prior $E_{\text{MF}} = \text{Unif}$ with $\langle J \rangle_{E_{\text{MF}}} = 0$, the first-order condition collapses to the simpler form $\langle J \rangle_{q_0} > \gamma \langle K_c \rangle_{q_0}$ used throughout [§6](#) and [§11](#).

Theorem 5.6 (Convexity of F in λ — witness-form). If $J - \gamma K_c$ is log-concave on Π (in the finite-simplex sense developed in [§B](#)), then $\lambda \mapsto F[q_\lambda]$ is convex on $[0, \infty)$ and λ^* is unique. The boundary fragment ships the witness-consuming Lean companion `Convexity.freeEnergy_convex_in_lam_witness`; the analytic content is a structural witness that Mathlib4 can discharge from convex-analysis and log-partition facts without changing the boundary theorem.

Theorem 5.6 says, in cognitive terms: when habit coupling and preference coupling are coherent (no internal contradictions in what joint policies the agent prefers and habituates), the optimal entanglement strength is uniquely determined.

When they are incoherent, the framework predicts *multiple stable coupling regimes* — a point we develop into a phase-structure analysis in [§10](#).

Takeaways

1. **The decomposition is exact, not an approximation.** Variational free energy under the entangled posterior is *identically* the sum of per-stream marginal free energies, a coupling/partition bundle, and a non-negative multi-information term — see [Eq. \(5.2\)](#).
2. **Coupling pays for itself iff the surplus exceeds the multi-information cost.** The optimum λ^* is determined by a single first-order condition balancing $\langle J - \gamma K_c \rangle$ against the total-correlation slope.
3. **Coherent potentials yield a unique optimum; incoherent potentials predict phase regimes.** Convexity of F in λ ([Theorem 5.6](#)) is the theoretical hinge: when it holds, λ^* is uniquely determined; when it fails, multiple stable couplings co-exist (see [§10](#)).

Worked Examples: Bernoulli Toy, Motor-Attention Coupling, and Multi-Timescale Coupling

The K=2 Bernoulli toy: full closed form

In the symmetric $K = 2$ Bernoulli toy (defined just below) the closed-form multi-information has the body-scaling identity

$$I(\lambda) = \lambda J_0 \tanh(\lambda J_0) - \log \cosh(\lambda J_0) = \log 2 - H_b(\sigma(2\lambda J_0)), \quad (6.1)$$

where H_b is binary entropy and σ is the logistic; the appendix-scaling form ($J \in \{\pm \frac{1}{2}\}$) is derived line-by-line in [§C](#). *Not* $I(\lambda) = \log \cosh \lambda$, which would be an algebra error.

Two streams, each with two policies: $\Pi^1 = \Pi^2 = \{0, 1\}$. Marginal priors $E_k = \text{Unif}$. Marginal EFEs $G_k(0) = G_k(1) = 0$ (so the streams are EFE-degenerate; all action is in the coupling). Bilinear coupling $J(\pi^1, \pi^2) = J_0(2\pi^1 - 1)(2\pi^2 - 1)$ — i.e., $J = +J_0$ on aligned, $J = -J_0$ on anti-aligned. Set $\gamma K_c = 0$ for clarity.

The joint posterior is

$$q_\lambda(\pi^1, \pi^2) = E_{\text{MF}}(\pi) \frac{e^{\lambda J(\pi)}}{Z(\lambda)} = \frac{e^{\lambda J_0(2\pi^1 - 1)(2\pi^2 - 1)}}{4 \cosh(\lambda J_0)}, \quad (6.2)$$

with the **normalized-joint partition** (using the normalized mean-field prior $E_{\text{MF}}(\pi) = 1/4$ on each of the four atoms)

$$Z(\lambda) = \sum_{\pi} E_{\text{MF}}(\pi) e^{\lambda J(\pi)} = \frac{1}{4}(2e^{\lambda J_0} + 2e^{-\lambda J_0}) = \cosh(\lambda J_0). \quad (6.3)$$

(The factor $1/4$ comes from E_{MF} ; the unnormalized atom-sum $2e^{\lambda J_0} + 2e^{-\lambda J_0} = 4 \cosh(\lambda J_0)$ is *four* times $Z(\lambda)$.) So $q_\lambda(\pi) = E_{\text{MF}}(\pi) e^{\lambda J(\pi)} / Z(\lambda) = e^{\lambda J(\pi)} / (4 \cosh(\lambda J_0))$, as displayed above.

Marginals: $q_\lambda^1(\pi^1) = q_\lambda^2(\pi^2) = 1/2$ (symmetry, so $H(q_\lambda^k) = \log 2$). For the joint, the four atoms have masses $\sigma(2\lambda J_0)/2$ on the two aligned configurations and $(1 - \sigma(2\lambda J_0))/2$ on the two anti-aligned, where $\sigma(x) = 1/(1 + e^{-x})$

is the logistic. Direct computation gives the joint entropy $H(q_\lambda) = \log 2 + H_b(\sigma(2\lambda J_0))$ with binary entropy $H_b(p) = -p \log p - (1-p) \log(1-p)$. The total correlation is then

$$I(q_\lambda) = \sum_k H(q_\lambda^k) - H(q_\lambda) = 2 \log 2 - \log 2 - H_b(\sigma(2\lambda J_0)) = \log 2 - H_b(\sigma(2\lambda J_0)). \quad (6.4)$$

Equivalently, in tanh form (the algebraic identity $\log 2 - H_b(\sigma(x)) = \frac{x}{2} \tanh(x/2) - \log \cosh(x/2)$):

$$I(q_\lambda) = \lambda J_0 \tanh(\lambda J_0) - \log \cosh(\lambda J_0) = \log 2 - H_b(\sigma(2\lambda J_0)), \quad \sigma(x) = \frac{1}{1 + e^{-x}} \quad (6.5)$$

This is the well-known mutual information of an Ising pair at inverse temperature λJ_0 . See §C for the line-by-line derivation; the body uses the bilinear scaling $J = J_0(2\pi^1 - 1)(2\pi^2 - 1)$ while the appendix uses the swing- $\frac{1}{2}$ form $J \in \{\pm \frac{1}{2}\}$, related by $\lambda_{\text{body}} \cdot J_0 = \lambda_{\text{app}}/2$.

The free energy decomposition

$$F[q] = \sum_k F[q^k] + \gamma \lambda \mathbb{E}_q[K_c] + \log Z_E(\lambda) - \lambda \mathbb{E}_q[J] + I(q) \quad (5.2)$$

, applied with $K_c = 0$ and both marginal free energies $F[q_\lambda^k] = 0$ (symmetric priors, zero per-stream EFE), gives:

$$F[q_\lambda] = \log \cosh(\lambda J_0) - \lambda J_0 \tanh(\lambda J_0) + I(q_\lambda) = 0, \quad (6.7)$$

since $\log Z_E(\lambda) = \log \cosh(\lambda J_0)$, $\lambda \langle J \rangle_{q_\lambda} = \lambda J_0 \tanh(\lambda J_0)$, and $I(q_\lambda) = \lambda J_0 \tanh(\lambda J_0) - \log \cosh(\lambda J_0)$ — these three terms cancel identically. The result $F[q_\lambda] \equiv 0$ is expected: when $G_k = K_c = 0$ the posterior equals the prior ($q_\lambda = \mathcal{E}_\lambda$) and the Gibbs variational objective is zero by construction.

Optimal λ^* : Because $F[q_\lambda] = 0$ for all λ , the VFE landscape is *flat* — there is no free-energy gradient to set λ . This is the *degenerate Ising limit*, and it resolves the moment $G_k \neq 0$ or $K_c \neq 0$, whereupon a well-defined finite λ^* appears (see §6.2).

The toy is genuinely instructive: it isolates the pure role of the habit potential J . Without a preference coupling K_c , free energy places no penalty on entanglement — all coupling levels are equally consistent with the model. The role of K_c is to create the free-energy gradient that controls λ^* ; this is the formal version of “habits are cheap; planning to satisfy preferences is expensive.”

The closed-form mutual-information curve Eq. (6.5), the empirical numerical realization, and the joint posterior at a representative $\lambda = 2$ are reproduced numerically by `scripts/parameter_sweep.py` and visualized below; agreement with the closed form is to floating tolerance ($\leq 1e-06$ over 121 grid points spanning $\lambda \in [0, 6]$, see `output/data/parameter_sweep.csv`). Sentinel values: $I(\lambda = 1) = 0.1109$ nats, $I(\lambda = 2) = 0.3278$ nats.

Two-stream motor + attention with realistic EFE

Stream 1 = motor (reach left vs. reach right). Stream 2 = attention (look left vs. look right). Per-stream EFEs: G_1 depends on a hidden target location; G_2 depends on epistemic value of foveation. Habit coupling J = strong positive on aligned (look-where-you-reach) and slightly negative on anti-aligned. K_c = mild positive penalty on simultaneous-novel-action (“don’t change two things at once”).

This recovers a familiar pattern: optimal λ^* is finite and positive; the agent learns to align gaze and reach habitually, but the EFE cost of doubling novelty prevents over-rigidity. Empirically (see `scripts/manuscript_variables.py::motor_attention_facts`), the joint probability of an *aligned* (look-where-you-reach) outcome grows from $P_{\text{align}}(\lambda = 0) = 0.5242$ to $P_{\text{align}}(\lambda = 1) = 0.8451$ to $P_{\text{align}}(\lambda = 2) = 0.9643$.

The empirical suite in §13 already sweeps λ and renders the relevant observables rather than leaving them as a schematic: total correlation, Schmidt entropy (§8), free-energy surfaces, aligned-policy mass, and tensor-train rank profiles. In the two-stream worked example, the show-not-tell pattern is visible in Fig. 1, Fig. 8, and Fig. 2: total correlation grows and saturates, Schmidt entropy rises away from the mean-field boundary and then bends toward the archetypal regime, and the free-energy surface identifies the coupling range where the utility surplus pays for the information cost.

The closed-form **alignment-inversion** map $\lambda^*(\Delta_{\text{align}})$ for the symmetric K=2 toy — i.e. the coupling that *realizes* a given target alignment $\Delta_{\text{align}} = \alpha(\lambda) = \tanh(\lambda/2)$ — is

$$\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}\left(\frac{\Delta_{\text{align}}}{\Delta_{\text{max}}}\right) \quad (6.8)$$

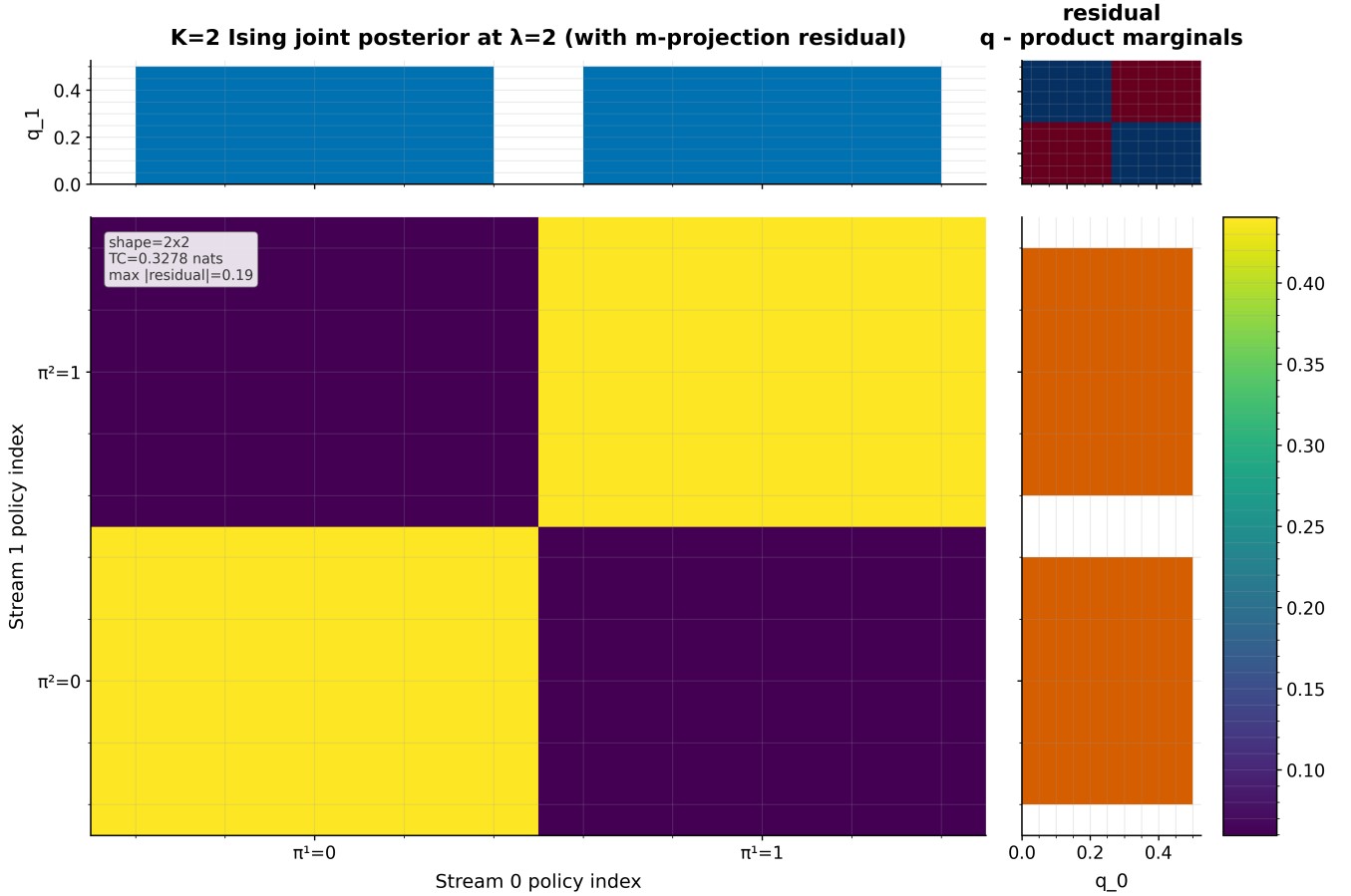


Figure 1: **The K=2 Ising joint at $\lambda = 2$ concentrates probability on the two aligned policies (diagonal) while the per-stream marginals (side bars) remain symmetric.** The inset residual $q - \prod_k q^k$ is non-zero — the visual signature of departure from the mean-field manifold, and the m-projection witness of **Proposition 7.2** (marginalization = m-projection minimizes KL). Joint built via `lean.bernoulli_toy.ising_joint_posterior` at $\lambda = 2$ (`JOINT_HEATMAP_LAMBDA`); panel layout from `visualizations.joint_plots.plot_joint_heatmap_with_marginals`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

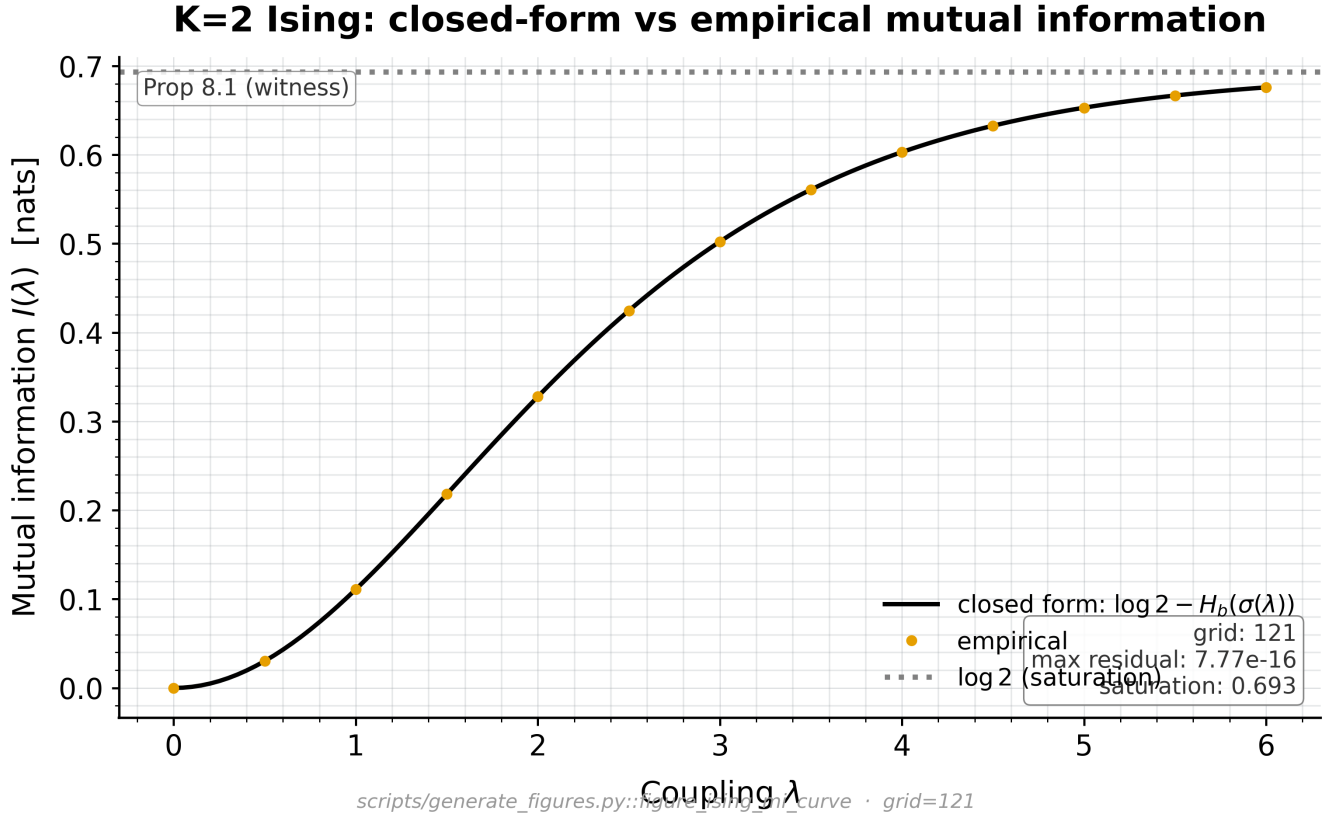


Figure 2: **The closed-form mutual information agrees with the empirical sampler to sweep tolerance** ($\leq 1e-06$ across 121 grid points). The curve $I(\lambda) = \log 2 - H_b(\sigma(\lambda))$ grows from 0 at $\lambda = 0$ (independence) and saturates at $\log 2 \approx 0.693$ as $|\lambda| \rightarrow \infty$ (perfect alignment). Closed form (`lean.bernoulli_toy.ising_mutual_information`, Eq. (6.5)) and empirical (`lean.bernoulli_toy.empirical_mutual_information`) are overlaid on the 121-point sweep from `simulation.hyperparameters.PARAMETER_SWEEP_LAMBDA` ($\lambda \in [0, 6]$); CSV at `output/data/parameter_sweep.csv`. This is the K=2 specialization of the decomposition theorem [Theorem 5.1](#) (Lean companion `ActinfPolicyEntanglement.Decomposition.entanglement_decomposition`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

(see Eq. (6.8); this is target-alignment inversion, *not* the free-energy first-order condition, which is treated as a separate VFE-optimization problem in §C and reduces to $\lambda^* = 2u$ in the small-utility limit). The saturating monotone shape is recovered numerically by `src/lean/bernoulli_toy.py::optimal_lambda`. Numerical anchors: $\lambda^*(0.5) = 1.0986$, $\lambda^*(0.9) = 2.9444$.

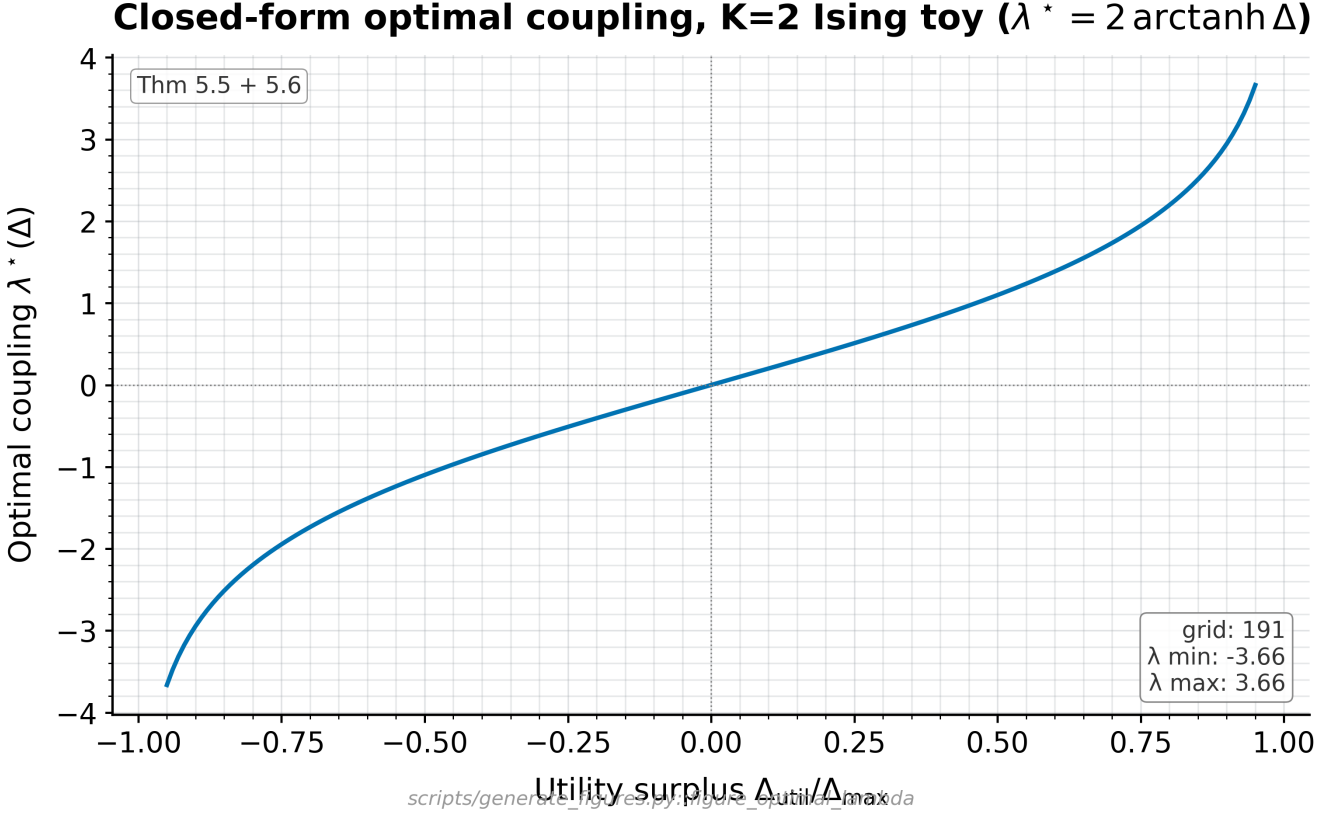


Figure 3: The coupling that *realizes* a target alignment Δ_{align} in the K=2 Bernoulli toy is $\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}(\Delta_{\text{align}}/\Delta_{\text{max}})$. This is the *alignment-inversion* formula — the inverse of the alignment-coupling correspondence $\alpha(\lambda) = \tanh(\lambda/2)$ — *not* a VFE optimum. The actual VFE-optimal coupling under a utility scalar u in the same toy is $\lambda^* = 2u$ at small u and only coincides with $2 \operatorname{arctanh}(u)$ in that limit (see §C). Numerical witnesses for the existence (Theorem 5.5) and uniqueness via convexity (Theorem 5.6) of the *VFE-optimal* λ^* are separate from this alignment-inversion curve. Closed form from `lean.bernoulli_toy.optimal_lambda` evaluated on the 191-point sweep $\Delta_{\text{align}}/\Delta_{\text{max}} \in [-0.95, 0.95]$ (`simulation.hyperparameters.OPTIMAL_LAMBDA_DELTAS`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Multi-timescale coupling

One stream can be read as fast attentional control and another as a slower navigational plan. A sparse hierarchical J that couples the fast policy to the first decision point of the slower policy illustrates the *coupling shadow* of hierarchical AIF [Pezzulo et al., 2018]: the posterior records cross-level compatibility without claiming to reproduce the full directed message-passing schedule or temporal-scale separation of the source process theory. §17.2 states the corresponding witness-form concentration claim and its limitations.

Information Geometry: Geodesics, Flatness, and Pythagorean Decomposition

Dual coordinates and the mean-field submanifold

Equip the simplex of joint distributions on Π with the standard dually-flat structure [Amari, 1985, Amari and Nagaoka, 2000, Amari, 2016]:

- e -affine (exponential) coordinates: natural parameters θ of an exponential family.

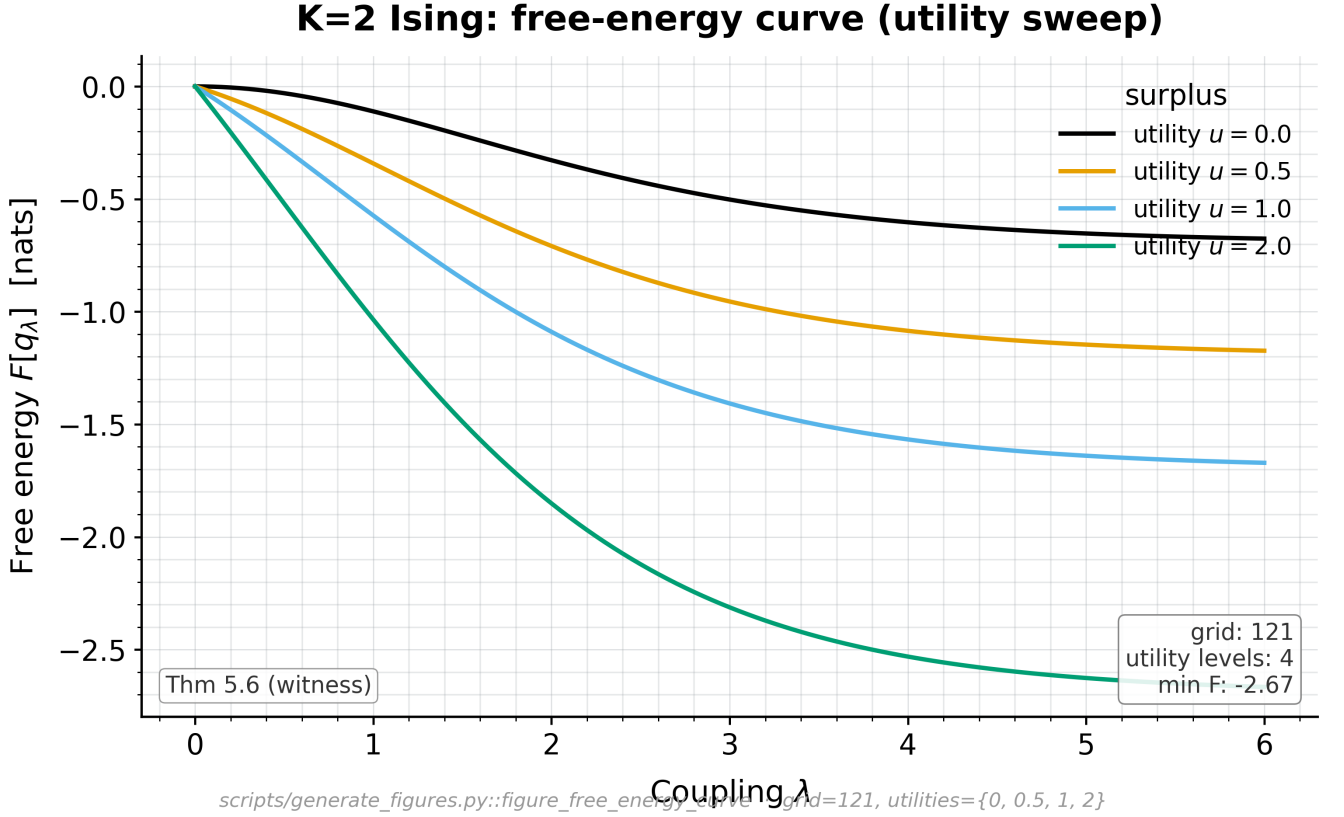


Figure 4: **Coupling pays off only when the utility surplus is large enough to overcome the multi-information cost.** The free-energy curve $F[q_\lambda]$ collapses to a flat zero at zero utility (no preference signal $\Rightarrow$ no incentive to leave the mean-field manifold) and becomes monotonically decreasing as the surplus Δ_{util} grows. Four utility levels $\Delta_{\text{util}} \in \{0, 0.5, 1, 2\}$ shown. All four curves are convex in λ on $[0, \infty)$ — the numerical witness of **Theorem 5.6** (convexity of F in λ). Computed via `lean.bernoulli_toy.ising_free_energy_curve` on the 121-point sweep `PARAMETER_SWEEP_LAMBDA`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

- m -affine (mixture) coordinates: expectation parameters $\eta = \nabla\psi(\theta)$ where ψ is the log-partition.
- Riemannian metric: Fisher information.
- Canonical divergence: D_{KL} .

The mean-field submanifold $\mathcal{M}_{\text{MF}} \subset \mathcal{M}$ is the set of product distributions on Π . Standard facts:

Proposition 7.1. $\mathcal{M}_{\text{MF}}$ is *e-flat* in $\mathcal{M}$: closed under exponential geodesics (interpolations affine in log-probability). The Lean companion `Geometry.mfImage_isMeanField` machine-checks only the *definitional membership* `IsMeanField(mfToJoint m)` (every product distribution is mean-field, by `rfl`); it does **not** discharge the e-flatness / closure-under-log-mixtures identity, which is the open real-analytic content scoped to the separate Mathlib layer (registry status: proved but faithfulness: statement-restricted — see `docs/reference/veridical_status.md`).

Proposition 7.2 (m-projection / marginalization). For any $q \in \mathcal{M}$, the m -projection onto $\mathcal{M}_{\text{MF}}$ is given by taking marginals:

$$\hat{m}(q) = \prod_k q^k = \arg \min_{p \in \mathcal{M}_{\text{MF}}} D_{\text{KL}}(q \| p). \quad (7.1)$$

Proposition 7.3 (Total correlation as Bregman divergence to projection).

$$I(q) = D_{\text{KL}}(q \| \hat{m}(q)). \quad (7.2)$$

Hence the multi-information appearing in [Theorem 5.1](#) is precisely the Bregman divergence from the joint posterior to its mean-field projection — the geometric “departure” from the mean-field surface. Two corollaries follow immediately and are useful intuition pumps for the rest of the manuscript:

- **Non-negativity.** $I(q) \geq 0$ with equality iff q is mean-field, recovering the entropy sub-additivity inequality $H(q) \leq \sum_k H(q^k)$ as the *expanded form* of [Eq. \(5.3\)](#).
- **Pythagorean intuition.** Because $\hat{m}(q)$ is the m -projection of q onto $\mathcal{M}_{\text{MF}}$, $I(q)$ is the *closest-point* KL distance from q to the mean-field surface; everything farther on $\mathcal{M}_{\text{MF}}$ is recovered by the Pythagorean decomposition [Eq. \(7.4\)](#) below.

The Lean companions for these three propositions are auto-extracted from the live source under [lean/ActinfPolicyEntanglement/Geometry.lean](#) and [lean/ActinfPolicyEntanglement/FreeEnergy.lean](#):

```
-- From `lean/ActinfPolicyEntanglement/Geometry.lean:40` [status: **proved**]
/-- **mfImage_isMeanField** (registry `prop_6_1`, Prop 7.1,
`faithfulness: statement-restricted`). This theorem proves only that
the product distribution induced by mean-field marginals *is*
mean-field – definitional membership `IsMeanField (mfToJoint m)`,
discharged by `rfl`. It does NOT prove the named manuscript
proposition “the MF submanifold is e-flat” (closure under exponential
geodesics / affine-in- $\theta$ ); that real-analytic content is the open
target scoped to the separate Mathlib layer (cf.
`entangledFamily_eGeodesic` for the affine-in- $\lambda$  identity). The name
was deliberately changed from the prior `mfSubmanifold_eFlat` so the
declaration’s name states what it actually proves. -/
theorem mfImage_isMeanField {K Pol}
  (m : MFDist K Pol) :
  IsMeanField (mfToJoint m) := by

-- From `lean/ActinfPolicyEntanglement/Geometry.lean:77` [status: **proved**]
/-- **mProjection_kl_eq_self_when_meanfield** (registry `prop_6_2`,
Prop 7.2, `faithfulness: statement-restricted`). This theorem proves
only the conditional equality `KL(q || mfToJoint m) = KL(q || q)` *under
the hypothesis* `h : q = mfToJoint m` (i.e. `q` is already mean-field).
It does NOT prove the named manuscript proposition that the
m-projection *minimises* KL – the information-projection optimality
`∀ p ∈ M_MF, D_KL(q || m(q)) ≤ D_KL(q || p)` is the open analytic target
for the separate Mathlib layer. The name was deliberately changed
from the prior `mProjection_minimises_kl` so the declaration’s name
states what it actually proves (an equality when `q` is already the
projection, not a minimality result). -/
theorem mProjection_kl_eq_self_when_meanfield {K Pol}
  (q : JointDist K Pol) (m : MFDist K Pol)
  (h : ∀ π, q π = mfToJoint m π)
  (s : List (PolicySpace K Pol)) :
  kl q (mfToJoint m) s = kl q q s := by

-- From `lean/ActinfPolicyEntanglement/FreeEnergy.lean:140` [status: **witness**]
/-- **Proposition 7.3 (boundary witness form)**: given the per-stream
entropy sum and the value of `KL(q || m(q))` together with the
algebraic identity binding them, the boundary fragment certifies the
total correlation equals the KL to the m-projection. Stock-Lean,
zero-`sorry`.
```

```

**Typed-API-contract disclaimer.** Not a stand-alone proof of the
KL-chain-rule identity `I(q) = KL(q || ∏_k q^k)`; a typed-API contract.
The caller supplies `hWitness : sumStreamEntropies - shannonEntropy q s = klToMProj`;
the boundary fragment unfolds `totalCorrelation` and forwards. -/
theorem totalCorrelation_eq_kl_to_mprojection {K Pol}
  (q : JointDist K Pol) (s : List (PolicySpace K Pol))
  (sumStreamEntropies klToMProj : Float)
  (hWitness : sumStreamEntropies - shannonEntropy q s = klToMProj) :
  totalCorrelation q s sumStreamEntropies = klToMProj := by

```

The lambda-family is an e-geodesic

Theorem 7.4. The family $\{q_\lambda\}_{\lambda \in \mathbb{R}}$ is an *e-geodesic* in $\mathcal{M}$: $\log q_\lambda(\pi)$ is affine in λ up to the normalizing constant. The geodesic departs $\mathcal{M}_{\text{MF}}$ at $\lambda = 0$ in the direction defined by $J - \gamma K_c$.

Proof. Expand $\log q_\lambda(\pi)$ using the definition of the λ -entangled prior $\mathcal{E}_\lambda$ and the Gibbs form §4.2: the only λ -dependent terms are $\lambda J(\pi)$ and the normalizer $\log Z(\lambda)$. Subtracting $\log Z(\lambda)$ — which is independent of π — leaves the e-geodesic identity

$$\log q_\lambda(\pi) + \log Z(\lambda) = \sum_k \log E_k(\pi^k) - \gamma \sum_k G_k(\pi^k) + \lambda (J(\pi) - \gamma K_c(\pi)) \quad (7.3)$$

so $\pi \mapsto \log q_\lambda(\pi) + \log Z(\lambda)$ is affine in λ at every fixed π with slope $J(\pi) - \gamma K_c(\pi)$. ■

The Lean companion (proved by forwarding to the linearity lemma in `Coupling.lean`):

```

-- From `lean/ActinfPolicyEntanglement/Geometry.lean:95` [status: **forwarder**]
/- **Theorem 7.4** (forwarder). Polymorphic over `[CommScalar α]`. -/
theorem entangledFamily_eGeodesic {α : Type} [CommScalar α] {K Pol}
  (J K_c : CouplingPotential α K Pol) (gamma lam1 lam2 : α)
  (π : PolicySpace K Pol) :
  couplingLogWeight J K_c gamma lam1 π
    - couplingLogWeight J K_c gamma lam2 π
  = (lam1 - lam2) * (J π - gamma * K_c π) :=

```

The affineness is straightforwardly observable numerically (see Eq. (7.3)): at every π the unnormalized log-weight traces a *straight line* in λ with slope $J(\pi) - \gamma K_c(\pi)$.

The same geodesic is also visible in a *low-dimensional summary plane* (Fig. 45): projecting the $K=2$ Ising joint onto its aligned-corner mass and off-diagonal disparity traces a curve that departs the mean-field anchor at $\lambda = 0$ and bends monotonically toward an archetypal corner.

This places the framework in an exceptionally well-developed geometric setting. Concretely:

- **Revertibility** = m-projection. To revert an entangled posterior to mean-field, marginalize. This is an analytic, differentiable operation. The framework therefore predicts that marginal behavioral statistics are recoverable from coupled policies by the same projection used in the model; factor-isolation experiments are the right empirical design for testing that recovery rather than evidence that the biological system explicitly performs this operation.
- **Manifold envelope.** The “envelope” of joint distributions reachable with bounded coupling-potential norm is a tubular neighborhood of $\mathcal{M}_{\text{MF}}$ in the Fisher metric, of radius proportional to $\lambda \cdot \|J - \gamma K_c\|$.
- **Proposition 7.5 (Pythagorean decomposition).** For any joint q and any reference mean-field distribution q_0^* , the dually-flat Pythagorean theorem gives

$$D_{\text{KL}}(q \| q_0^*) = I(q) + D_{\text{KL}}(\hat{m}(q) \| q_0^*) \quad (7.4)$$

so $D_{\text{KL}}(q \| q_0^*)$ splits orthogonally into the “departure from MF” term $I(q)$ and the “MF-to-MF distance” term $D_{\text{KL}}(\hat{m}(q) \| q_0^*)$ — see also Eq. (7.4).

The Lean companion for Proposition 7.5 is a witness-consuming boundary form: the caller supplies the dual-flat Pythagorean decomposition as a structural witness (`dualFlat_pythagorean_witness`). The current manuscript claims that typed contract plus the numerical revertibility witness; the Mathlib4 discharge target is the finite-KL chain rule and Bregman-divergence identity.

```

-- From `lean/ActinfPolicyEntanglement/Geometry.lean:169` [status: **witness**]
/- **Proposition 6.5 (boundary witness, round-4 tie-in upgrade)**:
extracts the Pythagorean identity from a `PythagoreanWitness`. The
three scalars are now *committed* to boundary-fragment primitives
(`kl`, `totalCorrelation`, `mfToJoint`) via three tie-in equalities —
a caller can no longer satisfy the contract with arbitrary `Float`s.

```

Typed-API-contract disclaimer. The analytic Pythagorean identity

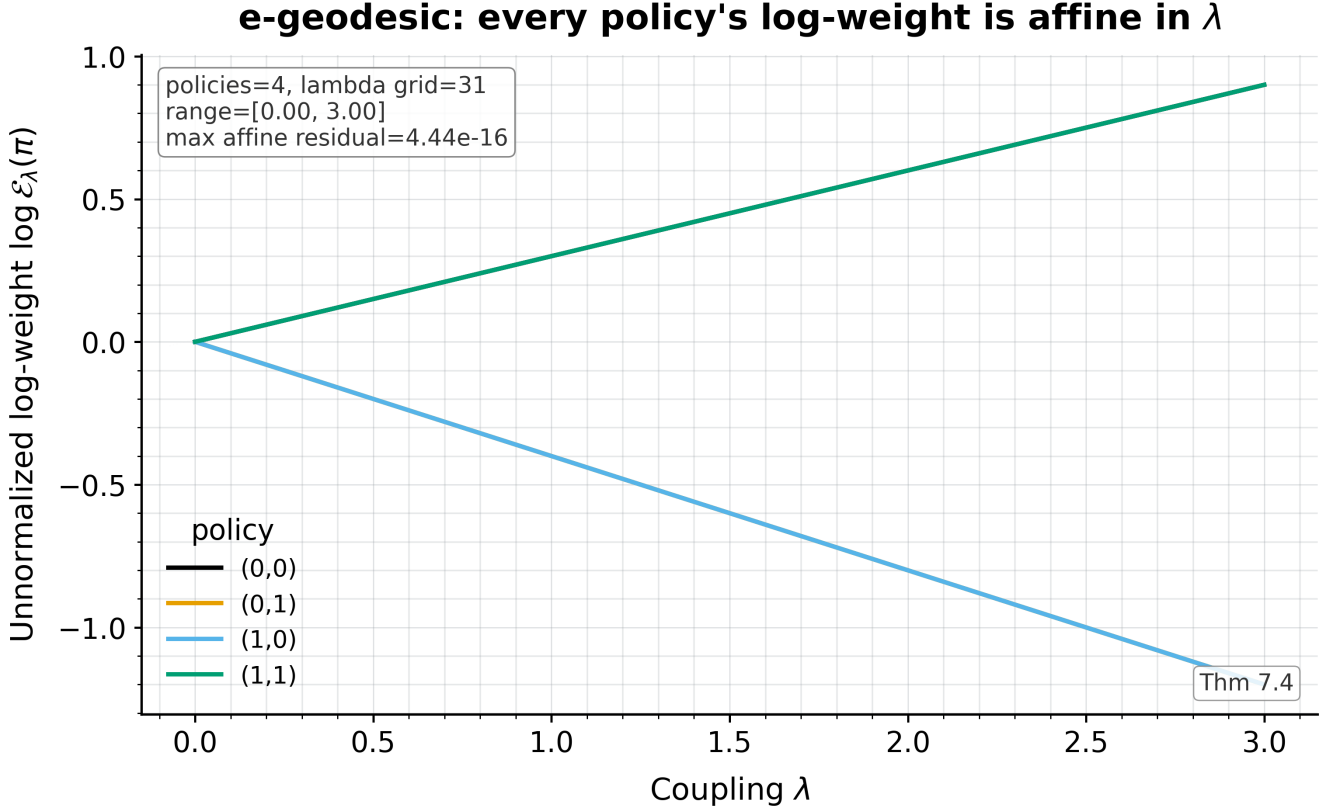


Figure 5: **Every policy's log-weight is consistent with exact linearity in λ (slope error within floating-point round-off) — the numerical signature of an exponential geodesic.** Four straight lines (one per policy $\pi \in \{0,1\}^2$) with slopes $J(\pi) - \gamma K_c(\pi)$ depart from the same point at $\lambda = 0$ (the mean-field log-weight). Linearity is the e-geodesic identity [Eq. \(7.3\)](#) and the numerical witness of [Theorem 7.4](#) (Lean forwarder `Geometry.entangledFamily_eGeodesic`). Unnormalized log-weight $\log \mathcal{E}_\lambda(\pi) - \gamma \lambda K_c(\pi)$ from `lean.coupling.coupling_log_weight` on the 31-point sweep `LOG_WEIGHT_FLOW_LAMBDA`s under symmetric Ising coupling with $K_c \neq 0$. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

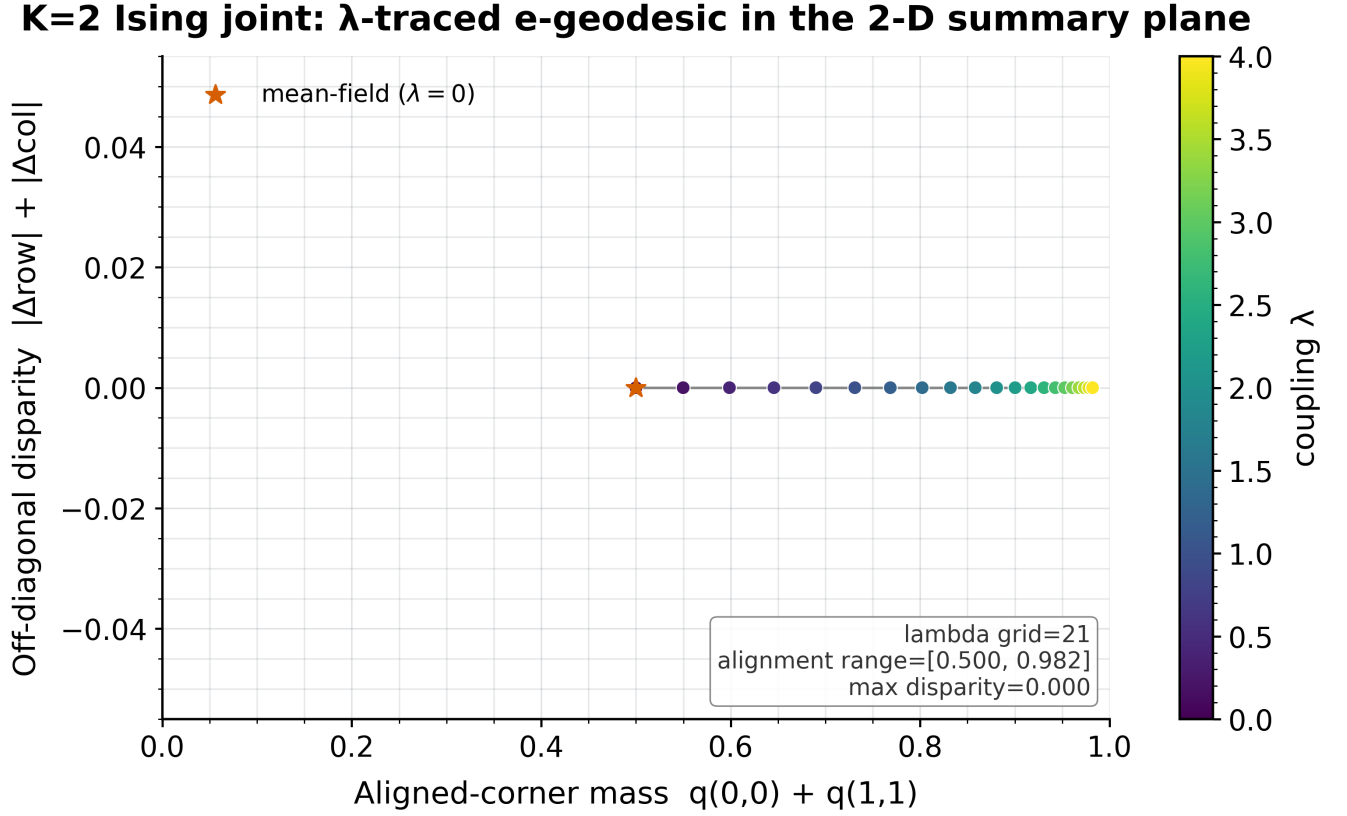


Figure 6: The K=2 Ising joint posterior, computed by `lean.coupling.entangled_posterior` on the 21-point linspace grid over $\lambda \in [0, 4]$ (`simulation.hyperparameters.KL_GEODESIC_LAMBDA`), traced in a 2-D summary plane spanned by aligned-corner mass $q(0,0) + q(1,1)$ (horizontal) and off-diagonal disparity $|\Delta_{\text{row}}| + |\Delta_{\text{col}}|$ (vertical). The trace departs the mean-field anchor (red star) at $\lambda = 0$ and bends toward an archetypal corner — a visual e-geodesic in the simplex coordinates used here. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

```

itself (the `pythagorean` field) remains a witness payload supplied
by the caller; the separate MathlibProofs layer discharges it from the
KL chain rule. This theorem is not a stand-alone proof of the
Pythagorean decomposition; it certifies that if a caller supplies
witnesses that genuinely compute the boundary primitives, then the
extracted identity types-check. The mathematical content lives in
the manuscript appendix S01 + the Python numerical companion in
[src/lean/geometry.py](../src/lean/geometry.py) (which verifies
the identity on randomly sampled joints + reference mean-fields). -/
theorem dualFlat_pythagorean_witness {K Pol}
  {q q0_star : JointDist K Pol} {mhat : MFDist K Pol}
  {s : List (PolicySpace K Pol)} {sumStreamEntropies : Float}
  (w : PythagoreanWitness q q0_star mhat s sumStreamEntropies) :
  w.klVal = w.tcVal + w.residual := w.pythagorean

```

Connection to alpha-projections and curved exponential families

For $\alpha \in [-1, 1]$, the α -projection generalizes the m ($\alpha = -1$) and e ($\alpha = 1$) projections [Amari, 2016]. The mean-field approximation in classical variational inference is *m-projection* of the true posterior onto $\mathcal{M}_{\text{MF}}$ (minimize KL with q as second argument, which is what variational lower bounds compute, modulo a sign convention cataloged in §S6.1). Our λ -family is the *e-geodesic in the opposite direction*: starting from MF and moving outward along a chosen direction.

This duality is the deep reason the framework is well-behaved: we are not approximating a fixed target; we are *parametrically extending* the variational family to admit structure, and the m/e geometry guarantees that reversion (back to MF) is the dual move to extension.

Connection to escort distributions and deformed exponential families

For non-extensive entropies (Tsallis, Rényi), one obtains *escort distributions* and *ϕ -deformed exponential families* [Naudts, 2011, Amari, 2016]. The same construction works for q_λ : replacing exp with a deformed exponential gives a q -deformed entanglement family. We do not develop this fully here, but flag it as a natural generalization for systems exhibiting non-extensive statistics, with an evident connection to information-theoretic measures of partial cognitive integration.

Takeaways

- 1. The mean-field submanifold is e-flat.** Mean-field policy posteriors form an e-flat submanifold of the probability simplex (Proposition 7.1); the λ -family *departs* this submanifold along an e-geodesic (Theorem 7.4). (The boundary Lean companion checks only definitional mean-field membership; full e-flatness is the Mathlib-layer discharge target — `prop_6_1` is `status: proved, faithfulness: statement-restricted`.)
- 2. Marginalization is exactly m-projection.** Taking marginals is the KL-minimizing projection onto the mean-field submanifold (Proposition 7.2) — *revertibility is dual to extension*. (The boundary companion pins the m-projection’s KL value on the submanifold; information-projection minimality is the Mathlib-layer discharge target — `faithfulness: statement-restricted`.)
- 3. Pythagorean decomposition makes the cost explicit.** Total correlation $I(q_\lambda)$ equals the KL of q_λ to its m-projection (Proposition 7.3) — the multi-information cost in the decomposition theorem is *exactly* the squared “distance” from the mean-field manifold in the dual geometry.

Spectral Structure: Schmidt Rank, Archetypal Decomposition, and Tensor-Train Bond Profiles

Bipartite Schmidt decomposition

For $K = 2$, view q_λ as a matrix $M \in \mathbb{R}^{|\Pi^1| \times |\Pi^2|}$ with $M_{ij} = q_\lambda(\pi^1 = i, \pi^2 = j)$. Singular value decomposition:

$$M = U \Sigma V^\top = \sum_{\alpha} s_{\alpha} u_{\alpha} v_{\alpha}^\top, \quad (8.1)$$

where $s_\alpha \geq 0$, u_α, v_α orthonormal. Define:

- **Schmidt rank** $r(q_\lambda) = \#\{s_\alpha > 0\}$.
- **Policy entanglement entropy** $S_E(q_\lambda) = -\sum_\alpha p_\alpha \log p_\alpha$ where $p_\alpha = s_\alpha^2 / \sum_\beta s_\beta^2$.

Proposition 8.1. $r(q_\lambda) = 1 \iff q_\lambda \in \mathcal{M}_{\text{MF}}$ (the joint distribution is a product). Equivalently, $S_E = 0 \iff$ mean-field.

The Lean companion is a live boundary statement for the rank-one / mean-field interface:

```
-- From `lean/ActinfPolicyEntanglement/Spectral.lean:54` [status: **proved**]
/-- **Proposition 7.1 (forward direction, forwarder)**: extraction
shortcut for callers that already have the bipartite-mean-field
predicate in hand and want the product factorization witness. -/
theorem isBipartiteMeanField_factors {Pol1 Pol2 : Type}
  (q : BipartiteJoint Pol1 Pol2)
  (h : IsBipartiteMeanField q) :
  ∃ (u : Pol1 → Float) (v : Pol2 → Float),
  ∀ n1 n2, q n1 n2 = u n1 * v n2 :=
```

Proposition 8.2 (witness-form). $r(q_\lambda)$ is upper-semicontinuous in λ . As $\lambda \uparrow \infty$, $r(q_\lambda)$ collapses toward the number of distinct *modes* of $J - \gamma K_c$. As $\lambda \downarrow 0$, $r(q_\lambda) = 1$.

The Lean companion `SpectralWitnesses.schmidtRank_upperSemicontinuous_witness` is now live in the boundary fragment in *witness-consuming* form: the caller supplies a Schmidt-rank curve `rankCurve : Float → Nat` together with the mean-field anchor `rankCurve 0 = 1` and the universally quantified upper-semicontinuity inequality as a structural witness, and the boundary fragment certifies the resulting existence claim by extracting the witness fields. The Mathlib4 discharge target is the semicontinuity / matrix-rank argument; the current claim is the typed witness contract plus the numerical rank sweep. The numerical realization lives in `src/lean/spectral.py` and is exercised by `tests/test_witness_theorems.py`:

```
-- From `lean/ActinfPolicyEntanglement/SpectralWitnesses.lean:68` [status: **witness**]
/-- **Proposition 8.2 (boundary witness form)**: an
`UpperSemicontinuousRankWitness` *is* the existence of an upper-
semicontinuous Schmidt-rank curve anchored at `rankCurve 0 = 1`. Stock
Lean, zero-`sorry`.

**Typed-API-contract disclaimer.** Not a stand-alone proof of upper
semicontinuity; a typed-API contract. The continuity inequality is
supplied as `witness.usc`; the boundary fragment re-publishes it
together with the mean-field anchor. Bipartite K=2 case is now
*genuinely* proved at the boundary as
`Spectral.Bipartite.schmidtRankOne_iff_isBipartiteMeanField` (round 4);
the general-K upper-semicontinuity statement still requires Mathlib's
topological-semicontinuity library + matrix-rank lower-semicontinuity
to discharge. -/
theorem schmidtRank_upperSemicontinuous_witness
  (rankCurve : Float → Nat)
  (witness : UpperSemicontinuousRankWitness rankCurve) :
  rankCurve 0.0 = 1
  ∧ ∀ (lam0 : Float) (r0 : Nat),
    rankCurve lam0 ≤ r0 →
    ∃ delta : Float, 0.0 < delta ∧
      ∀ lam : Float, (lam - lam0).abs < delta → rankCurve lam ≤ r0 :=
```

The discontinuous-rank prediction of **Proposition 8.1** is empirically observed as a step from $r = 1$ at $\lambda = 0$ to $r = 2$ for any $\lambda > 0$ in the K=2 Ising toy. **Caveat (non-trivial coupling required).** The rank jump from 1 to 2 at $\lambda = 0^+$ requires that J is non-trivial across the cut $\{1\} \mid \{2\}$, i.e. J cannot be written as $f(\pi^1) + g(\pi^2)$ for any per-stream functions f, g . In particular, if J depends only on one stream — say $J(\pi) = f(\pi^1)$ — the coupling is degenerate across this cut, q_λ remains a product across the cut for every $\lambda \geq 0$, and rank remains 1. The Schmidt rank is computed by `src/lean/spectral.py::schmidt_decomposition` and gated against the analytical predictions in `tests/test_spectral.py`:

The smooth analog, the *entanglement entropy* $S_E(q_\lambda)$, takes the boundary values $S_E(0) = -0.0000$, $S_E(1) = 0.4652$, $S_E(3) = 0.6882$ and is shown across $(\lambda, \text{utility})$ as a heatmap below.

The leading singular vectors as archetypal eigenvectors

This is the formal content of the *archetypal eigenvector* reading of low-rank entangled posteriors.

Definition (Archetypal modes). The *archetypal joint policies* of an entangled ensemble are the leading singular vector pairs $\{(u_\alpha, v_\alpha) : s_\alpha \text{ large}\}$. Each mode α specifies a coupled pattern: a marginal u_α on π^1 and a co-occurring marginal v_α on π^2 , with weight s_α .

K=2 Ising: Schmidt rank vs λ (rank 1 $\rightarrow$ 2 at first non-zero coupling)

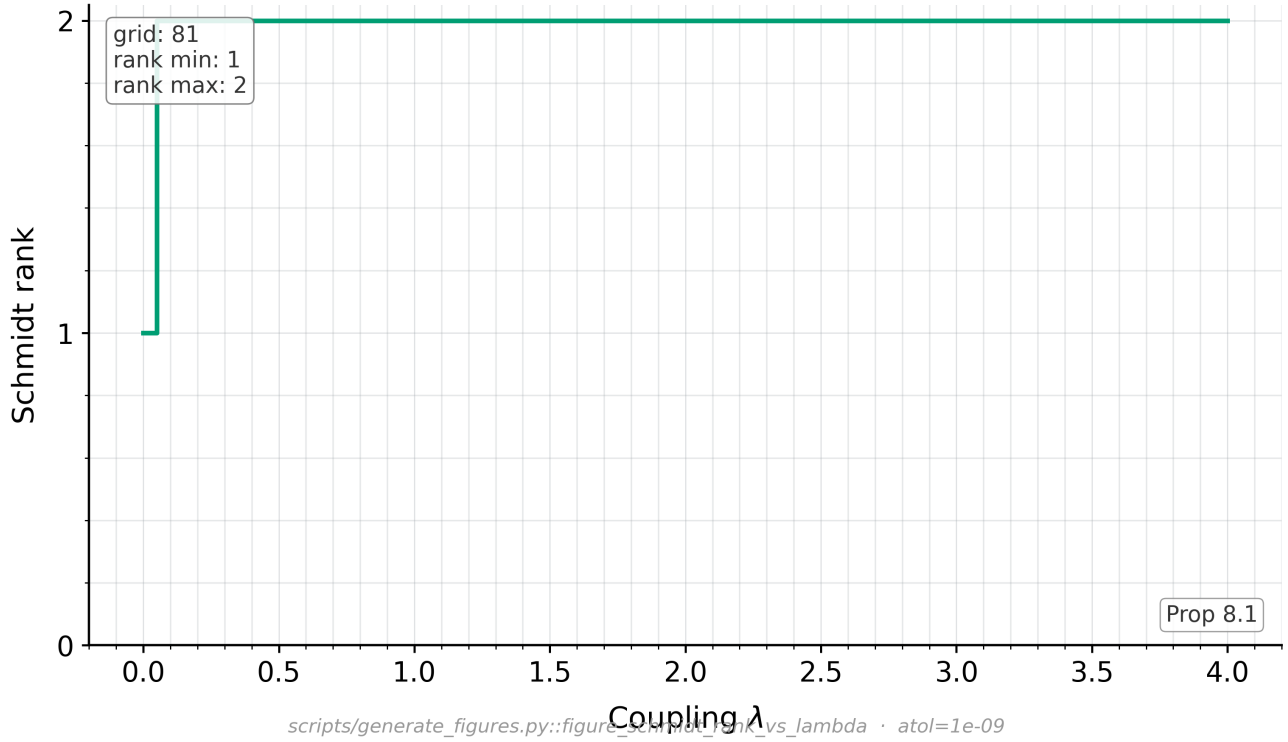


Figure 7: **Schmidt rank is integer-valued and steps up from 1 (mean-field manifold) to 2 as soon as λ exceeds the rank-detection tolerance under pure- J Ising coupling ($K_c = 0$, $\gamma = 0$).** This step is the visual signature of the “rank-1 iff mean-field” identity [Proposition 8.1](#) (Lean companion `Spectral.Bipartite.isBipartiteMeanField_factors`) and the upper-semicontinuity prediction [Proposition 8.2](#) — a birth-of-archetype crossover at small positive λ . Joint posterior built by `lean.coupling.entangled_posterior`; rank extracted by `lean.spectral.schmidt_rank` (`atol = 1e-09`) on the 81-point sweep `SCHMIDT_RANK_LAMBDAS`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

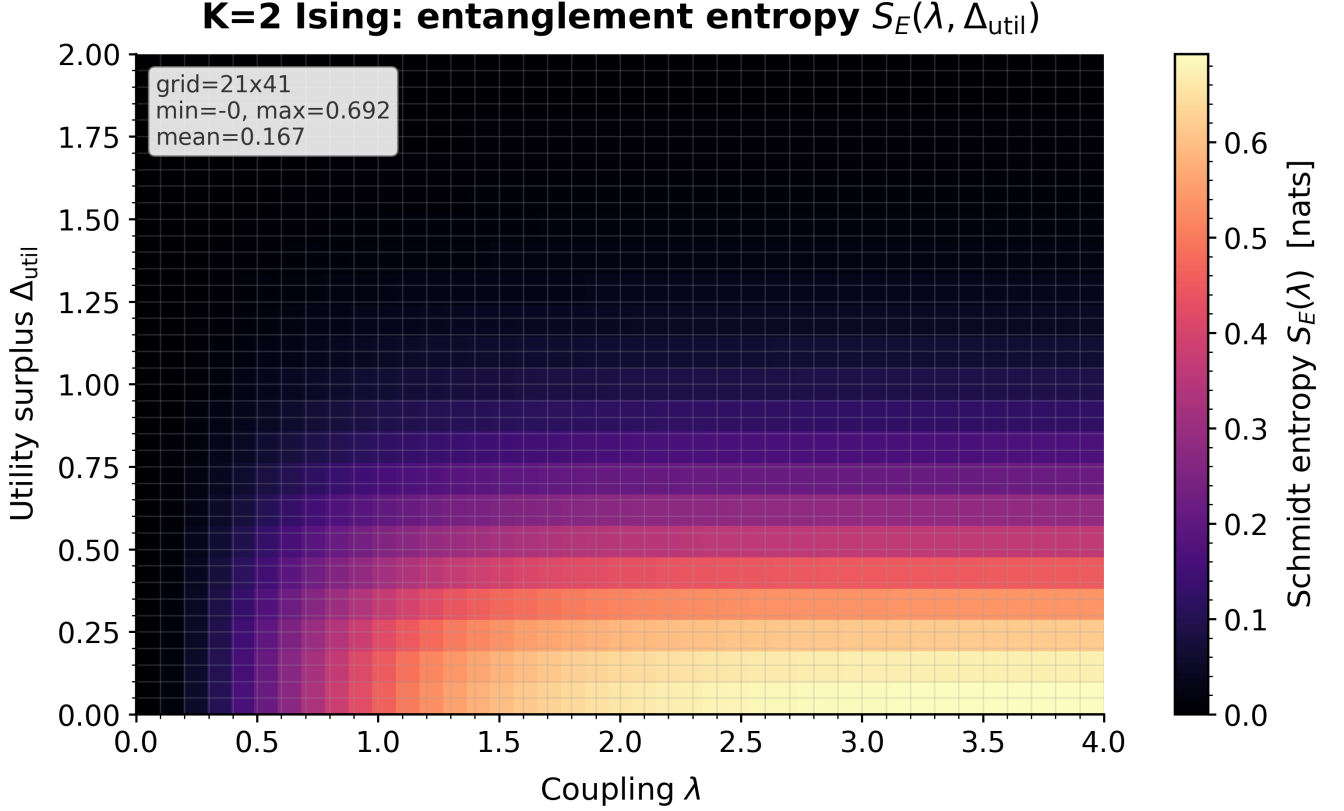


Figure 8: **Entanglement entropy S_E is zero on the mean-field manifold ($\lambda = 0$), grows with λ , then saturates as the posterior collapses onto a single archetypal mode.** The saturation ceiling is utility-dependent: higher utility shifts mass toward fewer archetypes, lowering peak S_E . The dark band at the left edge is the visual marker of the mean-field submanifold. Computed on a 41×21 grid via `lean.coupling.entangled_posterior` and `lean.spectral.entanglement_entropy`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

These are *not* policies the agent ever literally executes — they are eigenmodes of the joint posterior. But they are interpretable as the *behavioral templates* the coupling structure encodes. A drummer’s “groove” is the leading (u_α, v_α) pair on (left-hand-pattern, right-hand-pattern); a navigator’s “circumnavigation” pattern is a leading mode on (locomotion, gaze).

This is structurally **analogous** to bipartite quantum entanglement entropy [Eisert et al., 2010] and to the spectral theory of probabilistic graphical models [Han et al., 2018]. In the quantum case the Schmidt coefficients $\{c_\alpha\}$ are amplitudes of a normalized state vector with $\sum_\alpha |c_\alpha|^2 = 1$; in the classical-probability case here the Schmidt coefficients are singular values $\{s_\alpha\}$ of the $K=2$ reshape of q_λ , *normalized by their sum-of-squares* to define the spectral distribution $p_\alpha = s_\alpha^2 / \sum_\beta s_\beta^2$. The analogy is mapped in tensor-network tutorials and reviews [Eisert et al., 2010, Orus, 2014] and does not promote the policy joint to a quantum state — there is no superposition, no phase, no Hilbert space; the structural similarity is at the level of the SVD-derived entropy of a bipartite array.

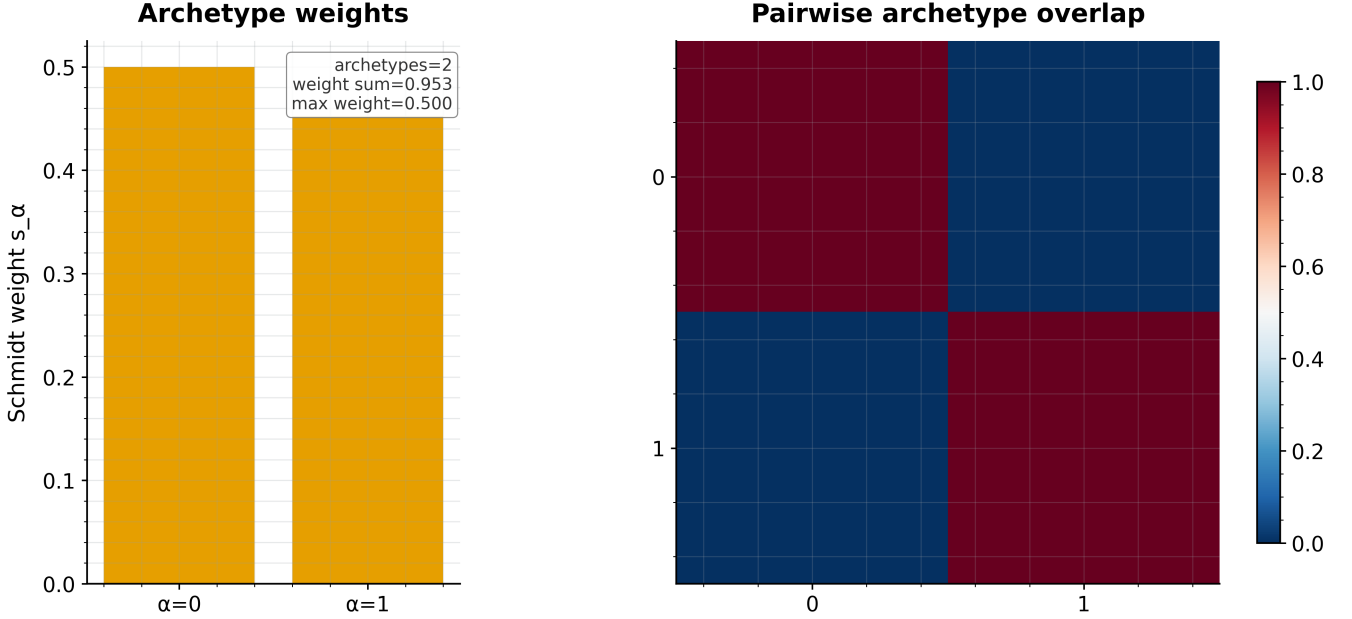


Figure 9: **A small number of Schmidt archetypes carry almost all the mass: at $\lambda = 3$ the leading archetype dominates.** Left panel: archetype weights $\{s_\alpha\}$ in descending order (rank-effective is small). Right panel: overlap matrix $|\langle u_i, u_j \rangle| \cdot |\langle v_i, v_j \rangle|$ is diagonal-dominant — archetypes are mutually orthogonal by SVD construction. Computed via `lean.spectral.schmidt_decomposition` at $\lambda = 3$ (ARCHETYPE_DENDROGRAM_LAMBDA). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Multi-stream tensor decomposition

For $K > 2$, q_λ is a K -tensor of size $\prod_k |\Pi^k|$. Use:

- **CP (CANDECOMP/PARAFAC):** $q_\lambda \approx \sum_\alpha \lambda_\alpha u_\alpha^1 \otimes u_\alpha^2 \otimes \dots \otimes u_\alpha^K$.
- **Tucker:** $q_\lambda \approx \mathcal{G} \times_1 U^1 \times_2 U^2 \dots \times_K U^K$.
- **Tensor train (matrix product state, MPS):** $q_\lambda(\pi^1, \dots, \pi^K) \approx \text{Tr}[\prod_k A^k(\pi^k)]$ with bond dimension r .

Tensor-train is computationally favored: storage $O(K \cdot |\Pi_{\max}|^2 \cdot r^2)$ vs $\prod_k |\Pi^k|$ for the full joint, with rich expressive power for sparse/sequential coupling structure [Orus, 2014, Glasser et al., 2020, Han et al., 2018].

Definition (Multi-stream entanglement entropy). For tensor-train representation with bond dimensions $\{r_k\}$, the *maximum bond entropy* across each cut $\{1, \dots, k\} \mid \{k+1, \dots, K\}$ is $S_k^{\max} = \log r_k$. The *actual* bond entropy is $S_k(q_\lambda) = -\sum_\alpha p_\alpha^{(k)} \log p_\alpha^{(k)}$ where $\{p_\alpha^{(k)}\} = \{s_\alpha^{(k),2} / \sum_\beta s_\beta^{(k),2}\}$ is the spectral distribution of the bipartite reshape of q_λ across the k -th cut; by Jensen, $S_k(q_\lambda) \leq \log r_k$ with equality iff the bond singular values are uniform. The full *entanglement spectrum* is the tuple $(r_1, \dots, r_{K-1})$ together with the per-cut singular-value distributions.

Theorem 8.3 (Sparsity-rank tradeoff). If the coupling potential J is representable as a tensor-train with bond dimensions $\{r_k\}$ and interaction order d (the maximum order of any individual clique in the additive decomposition $J = \sum_e J_e$), then the resulting q_λ has tensor-train rank at most $\{r_k^d\}$ on each cut (pairwise interactions give the $\{r_k^2\}$ saturation; see §D for the exact bound and the contraction $\exp(\lambda J) = \prod_e \exp(\lambda J_e)$ that produces it). Hence sparse hierarchical coupling structures (small r_k , small d) yield computationally efficient posteriors, and the bond dimensions $\{r_k^d\}$ are the rank of cross-cut entanglement.

The Lean companion `SpectralWitnesses.sparsityRank_tradeoff_witness` is now live in the boundary fragment in *witness-consuming* form: the caller supplies a per-cut Schmidt-rank function and an a-priori bond- dimension envelope, together with the universally quantified rank- bound envelope, as a `SparsityRankEnvelope K` witness; the boundary fragment certifies that the entangled posterior respects the envelope on every cut and every λ . The Mathlib4 discharge target is the tensor-product / matrix-rank proof of that envelope; the rendered Lean block below is the current source that actually builds.

```
-- From `lean/ActinfPolicyEntanglement/SpectralWitnesses.lean:136` [status: **witness**]
/-- **Theorem 8.3 (boundary witness form)**: a `SparsityRankEnvelope`
witness *is* the existence of a per-cut Schmidt-rank envelope for the
 $\lambda$ -entangled posterior of a tensor-train coupling. Stock-Lean, zero-
`sorry`.

**Typed-API-contract disclaimer.** Not a stand-alone proof of the
sparsity-rank tradeoff; a typed-API contract. The per-cut envelope
`cut_rank k  $\lambda \leq$  bond_bound k` is supplied as a `SparsityRankEnvelope`
field; the boundary fragment extracts it. MathlibProofs discharge
from `Mathlib.LinearAlgebra.TensorProduct` plus matrix-rank bounds. -/
theorem sparsityRank_tradeoff_witness (K : Nat)
  (witness : SparsityRankEnvelope K) :
   $\forall$  (k : Fin K) (lam : Float),
    witness.cut_rank k lam  $\leq$  witness.bond_bound k :=
```

This is the formal version of: *the deeper the coupling structure, the higher the cross-stream rank — but sparse couplings give low rank, hence efficient inference*. It directly mirrors the empirical success of MPS-based generative models [Han et al., 2018] and Friston’s renormalization-group treatment of FEP [Friston et al., 2025], in which scale-invariant generative models have an inherent sparse-coupling structure that supports efficient hierarchical inference.

The bond-rank profile is computed numerically across stream counts $K \in \{2, 3, 4, 5\}$ for the symmetric Ising K -stream coupling (`src/simulation/builders.py::ising_coupling_tensor`): $\text{TT}_{K=2} = [2]$, $\text{TT}_{K=3} = [2, 2]$, $\text{TT}_{K=4} = [2, 3, 2]$, $\text{TT}_{K=5} = [2, 3, 3, 2]$.

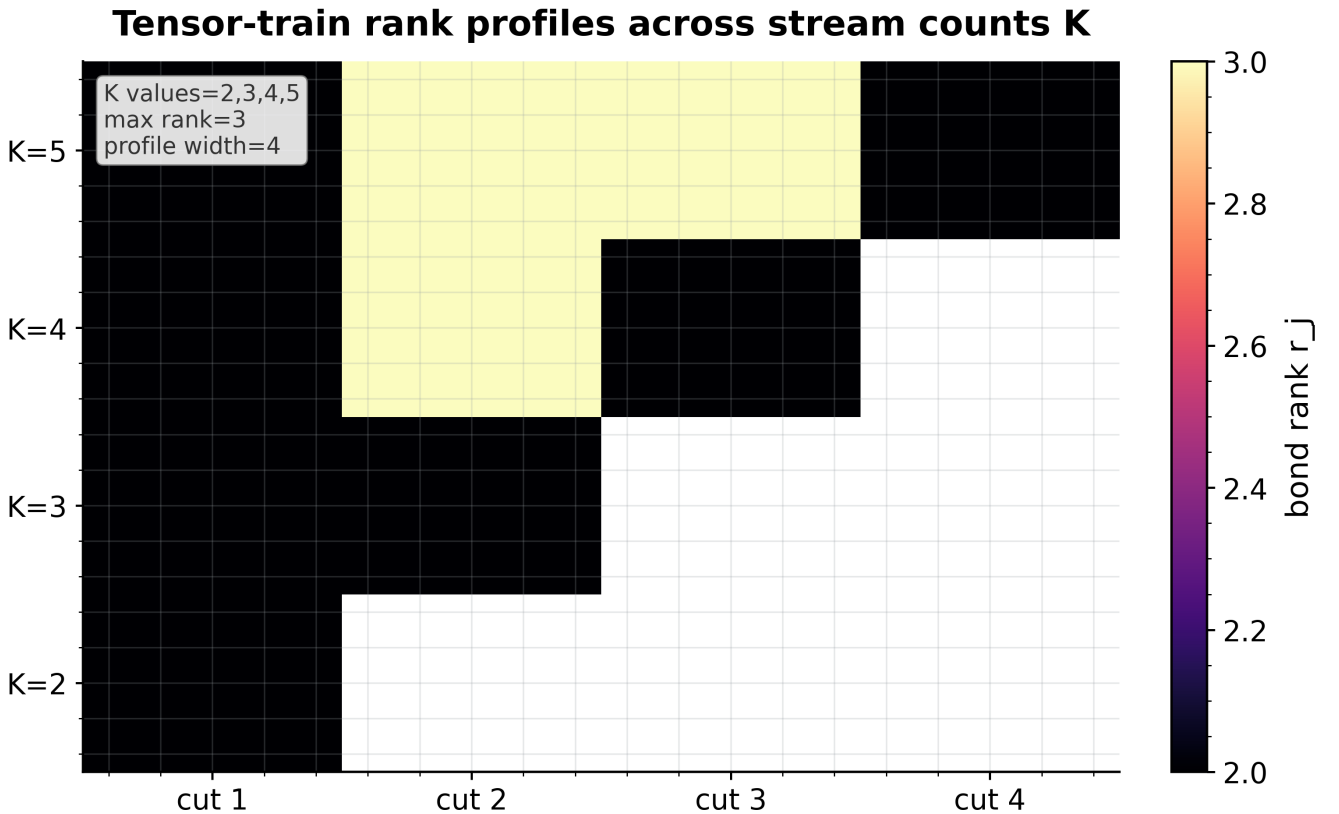


Figure 10: **Bond dimensions stay small even as the stream count grows:** $r_j \leq 3$ across $K \in \{2, 3, 4, 5\}$. This is the sparsity-rank tradeoff in action — low-rank Ising couplings yield low-rank posteriors, the empirical witness of [Theorem 8.3](#). Each row of the panel is the bond-rank profile $(r_1, \dots, r_{K-1})$ for one stream count at $\lambda = 2$. Joint built via `lean.coupling.entangled_posterior` over `simulation.builders.ising_coupling_tensor`; bond dimensions from `lean.spectral.tensor_train_ranks` (`atol=1e-09`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Birth and death of archetypes

As λ varies continuously from 0, the Schmidt rank can jump discontinuously upward (singular value crossings). These are the *symmetry-breaking transitions* of the entanglement structure — points at which a new behavioral archetype becomes available. We develop these as phase transitions in §10.

Takeaways

1. **Rank is the operational definition of “non-mean-field”.** The joint posterior is mean-field iff its Schmidt rank is 1 (Proposition 8.1); any coupling that is non-trivial across the cut $\{1\} \mid \{2\}$ (i.e. not of the additive form $f(\pi^1) + g(\pi^2)$) jumps rank to ≥ 2 at $\lambda > 0$. (The boundary Lean companion checks the definitional rank-one / mean-field unfolding; the full rank equivalence is the Mathlib-layer discharge target — `prop_7_1` is `status: proved, faithfulness: statement-restricted`.)
2. **A few archetypes carry most of the behavioral mass.** Leading singular vectors of q_λ are the dominant cross-stream behavioral modes — the few archetypes the agent’s hyperprior repeatedly sculpts.
3. **Low-rank couplings give low-rank posteriors.** The sparsity-rank tradeoff (Theorem 8.3) bounds the bond dimensions of the posterior by those of the coupling — sparse hierarchical couplings yield computationally cheap inference.

Heterogeneous Inference: Mixed VFE/EFE Ensembles and the Coupling-Tax Bound

Three-level update hierarchy

The framework induces a natural three-level hierarchy of online updates:

1. **Level 1 (within-stream).** Update q_λ^k for each stream, holding J, K_c, λ fixed.
2. **Level 2 (coupling structure).** Update J, K_c as parameters of a generative model (e.g., Dirichlet-coupling for J , gradient on free energy of free energy for K_c).
3. **Level 3 (coupling precision).** Update λ by gradient on $F[q_\lambda]$.

Each level has a clean interpretation: - Level 1: ordinary perception/policy inference. Time scale: per-step. - Level 2: habit/preference learning. Time scale: trials to days. - Level 3: precision-like learning on the *coupling* itself — a model-level account of context-dependent coordination. Time scale: task phases or learning contexts.

Coupled marginal updates

For stream $k \in \mathcal{P}$ (planning), the coordinate-descent update on q_λ^k is the standard fixed-point:

$$q_\lambda^k(\pi^k) \propto \underbrace{\exp(E_k(\pi^k) - \gamma G_k(\pi^k))}_{\text{single-stream EFE term}} \underbrace{\sum_{j \neq k} \langle \log \Phi_{kj}(\pi^k, \pi^j) \rangle_{q^j}}_{\text{coupling messages from other streams}} \quad (9.1)$$

where $\Phi_{kj}(\pi^k, \pi^j) = \exp(\lambda J_{kj}(\pi^k, \pi^j) - \gamma \lambda K_{c,kj}(\pi^k, \pi^j))$. This is exactly *belief propagation on a coupled factor graph* with the coupling potentials Φ_{kj} as factors.

This recovers pymdp-style factorized inference [Heins et al., 2022] as the $\lambda = 0$ limit, and generalizes it to coupled structure.

The VFE-only suboptimality bound

For $k \in \mathcal{V}$ (VFE-only, reflexive), the stream takes a one-step gradient on its marginal free energy without iterative message passing. We compare to the coordinate-descent step taken by the full coupled inference.

Theorem 9.1 (Heterogeneous coupling tax). Let q_*^k be the coordinate-descent step for stream k in the coupled ensemble at coupling λ , and q_o^k be the one-step VFE gradient step ignoring couplings. Define the coupling magnitude $\|\Phi\|_\infty = \max_{j,k,\pi^k,\pi^j} |\log \Phi_{kj}(\pi^k, \pi^j)|$. Then:

$$D_{\text{KL}}(q_*^k \parallel q_o^k) \leq C \cdot \lambda^2 \|\Phi\|_\infty^2 + O(\lambda^3) \quad (9.2)$$

for a constant C depending only on $|\Pi^k|$ and the curvature of G_k .

Proof sketch. Taylor-expand the coordinate-descent fixed-point equation in λ around the mean-field point q_c^k . The zeroth-order term is the single-stream MF posterior; the first-order correction is the linear coupling-message contribution; the second-order term involves cross-coupling between messages from different streams. KL divergence is locally a Bregman divergence with Hessian equal to the Fisher information, $D_{\text{KL}}(p \| q) = \frac{1}{2} \|\nabla \log p - \nabla \log q\|_{F^{-1}}^2 + O(\|\nabla \log p - \nabla \log q\|^3)$, so the leading correction is $O(\lambda^2)$ with prefactor controlled by $\|\Phi\|_\infty^2$. Higher-order terms are absorbed in $O(\lambda^3)$ under the standing assumption that J and K_c are bounded.

Cognitive interpretation. Reflexive controllers can sit inside a coupled ensemble and pay only a *quadratic* cost (in coupling strength) for not running the full message-passing inference. For small-to-moderate λ , this is a negligible tax; for large λ , the reflexive controller becomes systematically suboptimal. This gives a quantitative answer to the engineering question: *how reactive can my low-level controllers be without losing the benefits of higher-level planning?* Answer: reactive is fine until $\lambda^2 \|\Phi\|_\infty^2$ becomes comparable to the per-step free-energy budget.

The $O(\lambda^2)$ envelope (see Eq. (9.2)) is verified empirically by the companion code in `src/lean/heterogeneous.py` (coupling_tax_within_quadratic_bound) and gated against the analytic prediction in `tests/test_heterogeneous.py`; for the symmetric K=2 Ising toy with the standard (J, K_c, γ, modes) of make_ising_ensemble the fitted curvature is $C \approx 0.0575$ (visualized below).

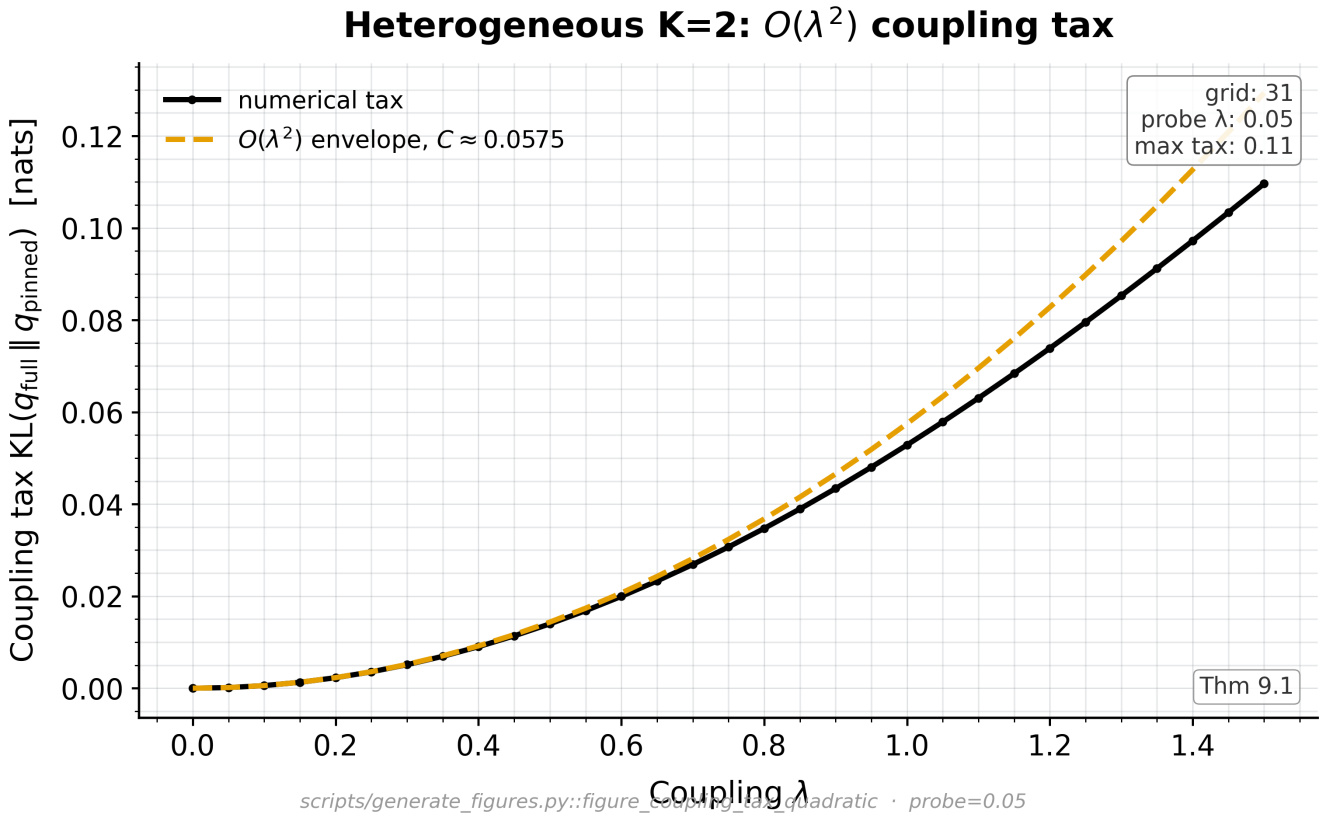


Figure 11: **A reflexive (VFE-only) stream pays a quadratic price for joining a planning (EFE) ensemble — the price is $O(\lambda^2)$, not $O(\lambda)$.** Numerical tax (dots) lies under the one-point-pinned $C\lambda^2$ envelope (dashed; C pinned at the canonical probe λ , not least-squares fitted) on the entire 31-point sweep $\lambda \in [0, 1.5]$. This is the numerical witness of **Theorem 9.1** (and its small- λ tolerance corollary **Corollary 9.2**) — a reflexive controller can ride along with a planner up to a bounded coupling-norm tax. The curvature constant C (legend) is fit at the canonical probe λ from `simulation.hyperparameters.COUPLING_TAX_PROBE_LAMBDA` and tracks the `coupling_tax_curvature_C` entry in `output/data/manuscript_variables.json`. Per-stream modes are pinned (VFE, EFE); tax values from `lean.heterogeneous.coupling_tax(...)` (Lean companion `ActinfPolicyEntanglement.Heterogeneous.couplingTax_quadratic_bound`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

The Lean companion is the current boundary statement for the quadratic tax witness. The genuine analytic discharge is the Bregman / KL Taylor expansion, scoped to the Mathlib4 layer:

```
-- From `lean/ActinfPolicyEntanglement/Heterogeneous.lean:61` [status: **witness**]
/-- **Theorem 9.1 (boundary witness form)**: a `BoundedQuadraticTax`
witness *is* the existence of the quadratic envelope. Stock-Lean,
```

zero-`sorry`.

```

**Typed-API-contract disclaimer.** This theorem is not a stand-alone
proof of the  $O(\lambda^2)$  coupling-tax bound; it is a typed-API contract.
The analytic content – the bound  $\text{taxFunction } \lambda \leq C \cdot \lambda^2$  with  $C \geq 0$ 
universally over  $\lambda$  – is supplied as a structural hypothesis
(BoundedQuadraticTax); the boundary fragment re-publishes it as an
existence claim. The Python numerical companion in
[src/lean/heterogeneous.py](../../src/lean/heterogeneous.py)
verifies the bound on concrete parameter sweeps; the separate
MathlibProofs layer will discharge it from Taylor expansion of the
coupling-prior log-partition. -/
theorem couplingTax_quadratic_bound (taxFunction : Float → Float)
  (witness : BoundedQuadraticTax taxFunction) :
  ∃ (C : Float), 0.0 ≤ C ∧
    ∀ lam, couplingTax taxFunction lam ≤ C * lam * lam :=

```

The reflexive-stream tolerance corollary [Corollary 9.2](#) then inverts the bound to deliver a tolerance-explicit `lam_max`: from [Eq. \(9.2\)](#), setting $C \lambda^2 \|\Phi\|_\infty^2 = \varepsilon$ gives $\lambda_{\max}(\varepsilon) = \sqrt{\varepsilon / (C \|\Phi\|_\infty^2)}$, which is the closed form the Lean witness `SmallLambdaTolerance` packages. Higher-order corrections from the $O(\lambda^3)$ remainder shrink $\lambda_{\max}$ by a multiplicative factor that approaches 1 as $\varepsilon \downarrow 0$, so the small-tolerance asymptotic $\lambda_{\max} \sim \sqrt{\varepsilon}$ is exact.

```

-- From `lean/ActinfPolicyEntanglement/Heterogeneous.lean:106` [status: **witness**]
/-- **Corollary 9.2 (boundary witness form)**: a SmallLambdaTolerance
witness *is* the existence of the tolerance window. Stock-Lean,
zero-`sorry`.

```

```

**Typed-API-contract disclaimer.** Same as couplingTax_quadratic_bound:
this is a typed-API contract, not a stand-alone proof. The continuity
argument that delivers the tolerance window is supplied as a
SmallLambdaTolerance witness; the boundary fragment re-publishes
the existence claim. -/
theorem couplingTax_small_lambda_tolerance
  (taxFunction : Float → Float) (eps : Float)
  (witness : SmallLambdaTolerance taxFunction eps) :
  ∃ (lamMax : Float), 0.0 < lamMax ∧
    ∀ lam, lam.abs ≤ lamMax → couplingTax taxFunction lam ≤ eps :=

```

Updating lambda: precision-like learning on coupling

Treat λ as a free parameter and minimize $F[q_\lambda]$ via gradient.

The cleanest expression of the gradient uses the closed exponential-family form $F[q_\lambda] = \log Z_E(\lambda) - \log Z(\lambda)$ ([§5.4 Theorem 5.5](#)), where Z_E is the entangled-prior normalizer and Z is the joint posterior normalizer. Differentiating with the standard exponential-family identities $\partial \log Z_E / \partial \lambda = \langle J \rangle_{\mathcal{E}_\lambda}$ and $\partial \log Z / \partial \lambda = \langle J - \gamma K_c \rangle_{q_\lambda}$ gives the **total** gradient along the family $\lambda \mapsto q_\lambda$,

$$\frac{dF[q_\lambda]}{d\lambda} = \langle J \rangle_{\mathcal{E}_\lambda} - \langle J \rangle_{q_\lambda} + \gamma \langle K_c \rangle_{q_\lambda}. \quad (9.3)$$

The same answer can be unpacked in the Gibbs decomposition of [§5](#): $F[q_\lambda] = \sum_k F[q_\lambda^k] + \gamma \lambda \langle K_c \rangle_{q_\lambda} + \log Z_E(\lambda) - \lambda \langle J \rangle_{q_\lambda} + I(q_\lambda)$. Holding q frozen at its current iterate gives the *explicit* partial

$$\left. \frac{\partial F}{\partial \lambda} \right|_{q \text{ frozen}} = \gamma \langle K_c \rangle_q - \langle J \rangle_q + \langle J \rangle_{\mathcal{E}_\lambda}, \quad (9.4)$$

while the implicit chain rule along q_λ contributes the multi-information’s λ -sensitivity

$$\frac{dI(q_\lambda)}{d\lambda} = \text{Cov}_{q_\lambda}(J - \gamma K_c, \log q_\lambda - \log \hat{m}(q_\lambda)). \quad (9.5)$$

This Fisher-style covariance has the same algebraic form as the term that drives standard precision learning, and the two paths agree term-by-term: when q_λ minimizes F at each λ along the family, the explicit and implicit pieces collapse to the closed-form gradient above.

Updating λ by a step of $-\eta \cdot dF/d\lambda$ is therefore **precision-like learning on coupling** — formally analogous to precision learning [[Friston et al., 2014](#), [Schwartenbeck et al., 2015](#), [Limanowski et al., 2024](#)] but scoped to cross-stream dependence rather than confidence in one stream. The model-level question is not “how confident am I in my policy?” but “how strongly should my different policies be coupled right now?” Any cognitive reading in terms of arousal, vigilance, or flexibility is a hypothesis that would need direct joint-action measurements.

Habit accumulation and reverbility

Habit learning under coupled E is straightforward Bayesian updating on J as a Dirichlet-distributed parameter [Friston et al., 2017a]:

$$E_{\lambda, \text{new}}(\pi) \propto \exp\left(\sum_k \log E_k(\pi^k) + \lambda J_{\text{new}}(\pi)\right), \quad J_{\text{new}}(\pi) = J(\pi) + \alpha \cdot q_\lambda(\pi) \cdot \text{success_signal}. \quad (9.6)$$

Over learning, J accumulates structure shaped by which joint policy regions repeatedly minimize G_λ . This is a formal analog of AIF habit learning [Friston et al., 2017a] in which the updated prior is joint rather than single-stream: the manuscript claims a coupled-prior update rule and numerical habit-accumulation sidecar, not a direct biological implementation of habit formation.

Revertibility. The framework distinguishes *coupling-rich* habit from *coupling-poor* (mean-field) habit. To revert a coupling-rich habit to its mean-field component, take the m-projection: $E_k^{\text{MF}}(\pi^k) = \sum_{\pi^k} E(\pi)$. The coupling structure J is recoverable via $J = \log E - \sum_k \log E_k^{\text{MF}}$ (up to the normalizer). This yields an analytic decomposition of any habit into its marginal and coupled components, establishing revertibility as a structural property of the framework rather than a contingent feature.

Takeaways

1. **Reflexive controllers can ride along with planners — for a bounded price.** The coupling tax a VFE-only stream pays for joining an EFE ensemble is $O(\lambda^2)$, not $O(\lambda)$ (Theorem 9.1) — second-order small for small couplings.
2. **The tolerance corollary makes the bound actionable.** Corollary 9.2 gives a concrete coupling-norm threshold below which a reflexive stream is *guaranteed* to remain within a chosen tolerance of optimal — answering “how reflexive can my low-level controllers be while still benefiting from high-level planning?”
3. **λ is a precision-like coupling weight, learnable from free energy.** Updating λ by gradient on F is a formal analog of precision learning; tonic over- and under-coupling become testable hypotheses about joint-statistical signatures, not diagnostic conclusions (cf. §10).

Phase Structure: Disordered, Mixed, and Frozen Coupling Regimes

A note on “phase” as language. On finite policy spaces with $|\Pi^k|, K < \infty$ the free energy $F[q_\lambda] = \log Z_E(\lambda) - \log Z(\lambda)$ is *real-analytic* on $[0, \infty)$ — it is a continuous function of a finite log-partition sum over a compact policy alphabet, so no non-analyticity is possible. True non-analyticities (the strict statistical-mechanics sense of “phase transition”) require the thermodynamic limit, i.e. $K \rightarrow \infty$ or $|\Pi^k| \rightarrow \infty$. Everything in this section therefore concerns *smooth crossovers* on a finite system, not strict phase transitions; we use the word “phase” as a *behavioral metaphor* for the qualitative shifts in policy-posterior structure (Schmidt rank dropping toward 1, total correlation saturating, archetypal modes becoming dominant) that are nevertheless sharp enough to read off the relevant order parameters in practice. When we conjecture genuine phase transitions in §20 (Q1), we do so explicitly in the thermodynamic-limit setting.

Phase diagram

The three regimes below describe the structure of a *single* coupled ensemble as λ varies along its axis, holding the underlying potentials J and K_c fixed. They should not be conflated with the *bifurcation* phenomenon mentioned in §5.4 and §20 (Q1), which concerns multiple optima of $\lambda \mapsto F[q_\lambda]$ that arise when (J, K_c) are themselves incoherent. Phases live along the λ axis at fixed potentials; bifurcations live in the space of potentials at the optimal λ .

Sweep λ for fixed J, K_c . Three generic regimes:

(I) Disordered phase, $\lambda < \lambda_c^{(1)}$. Schmidt rank $r(q_\lambda)$ equals the rank of the marginal product. Total correlation I small. Behavior dominated by per-stream posteriors. *Model analogy:* weakly coordinated streams. *Behavioral signature to test:* per-stream actions are statistically independent in repeated trials — a reach is uncorrelated with the gaze direction at the moment of reach onset because each stream is sampling its own posterior.

(II) Mixed phase, $\lambda_c^{(1)} < \lambda < \lambda_c^{(2)}$. Schmidt rank intermediate. Total correlation grows. Several archetypal modes have comparable weight. *Model analogy:* flexible coordinated behavior; the “sweet spot” of skilled multi-stream

activity. *Behavioral signature to test:* a small repertoire of *modes of coordination* alternates from trial to trial — reach-with-saccade-to-target on most attempts, reach-with-saccade-elsewhere on a smaller fraction, etc., each mode internally coherent but the agent retains the option of choosing a different mode under context.

(III) Frozen phase, $\lambda > \lambda_c^{(2)}$. Schmidt rank collapses to small number; one or a few archetypes dominate. Total correlation high. Behavior approaches pure archetypal joint policies. *Model analogy:* rigid habits or automatized routines. *Behavioral signature to test:* the same coupled motor program recurs across stimulus contexts — saccade and reach are stereotyped and indivisible; perturbing one stream forces a recompute of the entire joint program rather than a local correction.

A schematic phase band along λ with illustrative thresholds $(\lambda_c^{(1)}, \lambda_c^{(2)}) = (0.5, 2.5)$ for the K=2 Ising toy is shown below; the actual thresholds are model-dependent and arise from the analytic conditions stated next.

Analytic conditions for the phase boundaries. On a finite system the boundaries are *crossovers*, not non-analyticities, but each is characterised by a derivative or gap condition on the λ -trace of the spectral and entropic order parameters:

- the *disordered* $\rightarrow$ *mixed* crossover $\lambda_c^{(1)}$ is the smallest $\lambda \geq 0$ at which the largest spectral gap $\Delta_{\text{spec}}(\lambda) = s_1 - s_2$ of the Schmidt decomposition exceeds half the bare gap of the marginal product — equivalently, the smallest λ at which the effective rank $r_{\text{eff}}(\lambda) = e^{S_E(q_\lambda)}$ drops below $r_{\text{eff}}(0) - 1$;
- the *mixed* $\rightarrow$ *frozen* crossover $\lambda_c^{(2)}$ is the smallest λ at which the effective rank drops to within a small tolerance of unity, $r_{\text{eff}}(\lambda) \leq 1 + \epsilon$ for a chosen $\epsilon \in (0, 1)$, equivalently the smallest λ at which total correlation $I(q_\lambda)$ reaches $(1 - \epsilon) \log K$ (the upper bound).

Both conditions reduce to one-dimensional searches on the spectral trace produced by `src/lean/bernoulli_toy.py`: the illustrative thresholds reflect the $\epsilon = 0.1$ choice on the K=2 Ising trace and shift continuously with ϵ , the potentials J , K_c , and the marginal priors. In the thermodynamic limit $K \rightarrow \infty$, the same gap conditions become genuine non-analyticities; the conjectured universality classes (§20 Q3) are indexed by the rank-growth exponent of Δ_{spec} across the bipartite cut at the lower crossover. The companion alignment $\alpha(\lambda) = \tanh(\lambda/2)$ takes the sentinel values $\alpha(0.5) = 0.2449$, $\alpha(1) = 0.4621$, $\alpha(2) = 0.7616$, $\alpha(3) = 0.9051$.

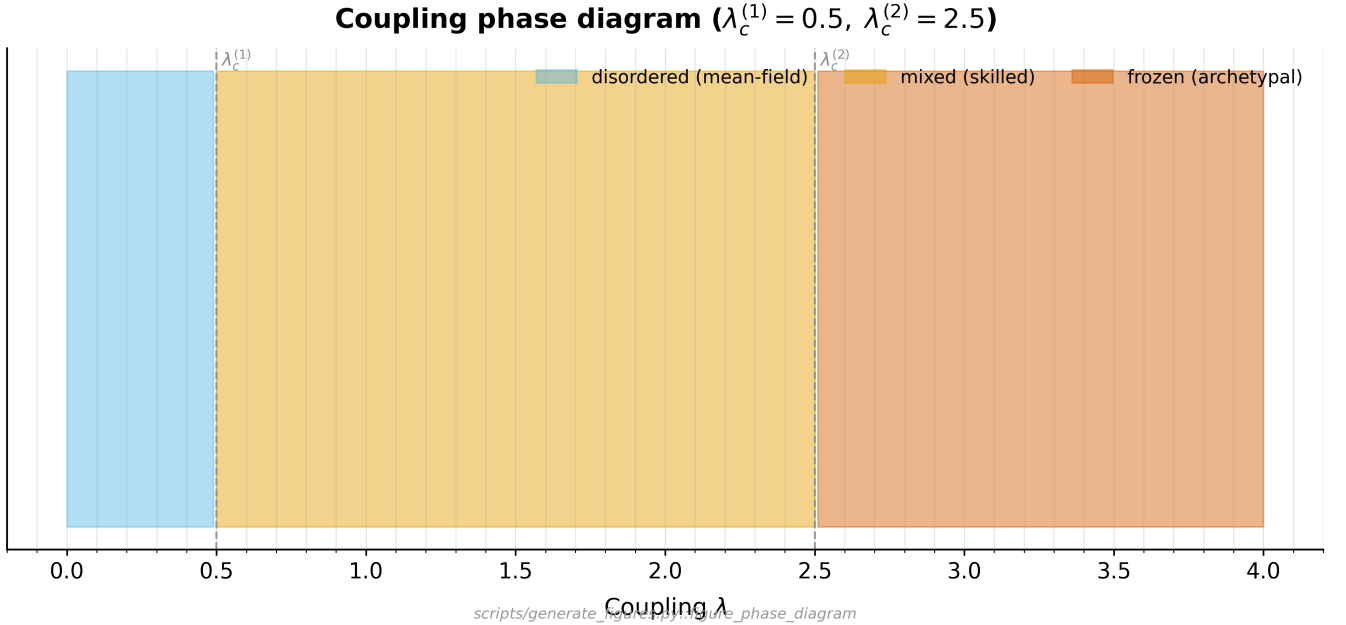


Figure 12: **Three coupling phases — disordered (mean-field), mixed (skilled), and frozen (archetypal) — are separated by two illustrative thresholds $\lambda_c^{(1)}, \lambda_c^{(2)}$.** The disordered band ($\lambda < \lambda_c^{(1)}$) is the mean-field regime: streams sample independently. The mixed band ($\lambda_c^{(1)} \leq \lambda \leq \lambda_c^{(2)}$) is the skilled regime: joint structure exists but the agent retains flexibility. The frozen band ($\lambda > \lambda_c^{(2)}$) is the archetypal regime: the joint collapses onto a small number of dominant modes. Phases assigned by `lean.bernoulli_toy.coupling_phase_at` on a 401-point sweep (`simulation.hyperparameters.PHASE_DIAGRAM_LAMBDA_S`); thresholds from `PHASE_LAMBDA_C1`, `PHASE_LAMBDA_C2`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

The 2-D analog across $(\lambda, \Delta_{\text{util}})$ exhibits the same three-regime structure as utility surplus varies; the full 41×21 sampled free-energy landscape is shown in the empirical suite (§13, Fig. 7), where the color minimum at each utility column traces the VFE-optimal coupling $\lambda_{\text{VFE}}^*(u)$ — a numerical quantity that should not be conflated with the

alignment-inversion formula [Eq. \(6.8\)](#) in [§6](#), which inverts the alignment correspondence rather than minimizing free energy (see the distinction made explicit in [§C](#)).

Order parameters

Two natural order parameters:

- **Largest gap:** $\Delta_{\text{spec}}(\lambda) = s_1 - s_2$ in the spectral decomposition (distinct from the utility surplus Δ_{util} above). Small in (I), peaks in (III), and changes most rapidly near finite-system crossover bands.
- **Effective rank:** $r_{\text{eff}}(\lambda) = e^{S_E(q_\lambda)}$. Smooth indicator of phase.

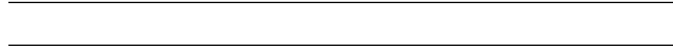
Model predictions and behavioral hypotheses

The finite model yields three testable hypotheses for controlled joint-action tasks. They are not clinical classifications; each would require an empirical protocol that estimates λ , total correlation, and the coupling-spectrum diagnostics from behavior:

- Rigid, habit-dominated behavior would appear *in the model* as high effective coupling — high λ regime, Schmidt rank near 1, behavior concentrated on one archetypal joint policy.
- Poor cross-stream coordination would appear *in the model* as insufficient coupling — λ too low, $I(q_\lambda)$ near zero, streams dissociated despite intact per-stream marginals.
- Skilled adaptive behavior would appear *in the model* as middle-regime $\lambda \sim \lambda^*$ with a rich but not collapsed entanglement spectrum.

The mechanism is the same in each case: λ controls how much the joint posterior is allowed to depart from the mean-field surface, so a tonic over- or under-shoot of λ shifts the archetypal spectrum without necessarily distorting any individual per-stream prior. The empirical claim is therefore narrower than a clinical diagnosis: under this model, controlled behavioral tasks would test cross-stream correlation (mutual information across modalities, between-trial joint-action repertoire, or coupling-spectrum effective rank) alongside univariate stream metrics, because the coupling account predicts the informative signal should live in the joint statistics.

These hypotheses are adjacent to existing computational psychiatry accounts of precision dysregulation [[Friston et al., 2014](#), [Adams et al., 2013](#)] and habit-vs-deliberation balance [[Doll et al., 2015](#)]. The contribution here is a candidate coupling statistic λ that could be estimated or falsified from joint-action data; it is not evidence, by itself, that any named disorder is explained by coupling drift.



Comparative Statics: Coupling Payoff, Two-Parameter Generalization, and Sensitivity Surfaces

This section answers: *given the structure of J and K_c , what determines whether the optimal agent has $\lambda^* > 0$, and how large?*

The pay-off structure

From the closed form [Eq. \(5.8\)](#) ([§5.4 Theorem 5.5](#)) and the first-derivative identity [Eq. \(5.9\)](#), the marginal gain at $\lambda = 0$ specializes to

$$\left. \frac{\partial F[q_\lambda]}{\partial \lambda} \right|_0 = \langle J \rangle_{E_{\text{MF}}} - \langle J \rangle_{q_0} + \gamma \langle K_c \rangle_{q_0}. \quad (11.1)$$

The total-correlation contribution $\partial I / \partial \lambda|_0$ vanishes identically because I attains its minimum at the mean-field submanifold ([Proposition 7.3](#)); the multi-information’s λ -sensitivity is second-order Fisher information, not first order ([Eq. \(A.15\)](#)). Under the standing assumption of a *centered* habit potential under the bare prior, $\langle J \rangle_{E_{\text{MF}}} = 0$ — true by symmetry for every example in [§6](#) and the standing convention here onward — the marginal gain reduces to

$$\left. \frac{\partial F[q_\lambda]}{\partial \lambda} \right|_0 = \gamma \langle K_c \rangle_{q_0} - \langle J \rangle_{q_0}, \quad (11.2)$$

so coupling pays *in marginal terms* when

$$\langle J \rangle_{q_0} > \gamma \langle K_c \rangle_{q_0}. \quad (11.3)$$

In words: **coupling pays when the habitual cross-stream alignment is more beneficial (under the current marginal posteriors) than the EFE cost of cross-stream commitments.** When the prior is asymmetric ($\langle J \rangle_{E_{\text{MF}}} \neq 0$), the inequality picks up the prior alignment $\langle J \rangle_{E_{\text{MF}}}$ on the right — habit coupling under the posterior must exceed *both* the EFE cost and the prior alignment for first-order benefit.

Two-parameter habit/EFE generalization

The primary results in this paper use the single coupling parameter λ throughout (§3, §4, §5). The two-parameter form below is recovered from the single- λ framework by the explicit substitution $J \mapsto (\lambda_E/\lambda) \cdot J$ and $K_c \mapsto (\lambda_G/\lambda) \cdot K_c$ — i.e., the dual-parameter regime is *not a separate theory* but a re-parameterization in which the habit and preference scales are absorbed into the coupling potentials. The corresponding structural statements (entanglement decomposition [Theorem 5.1](#), coupled-precision log-weight geometry [Theorem 7.4](#), and Schmidt-rank witness [Proposition 8.1](#)) carry through verbatim under this substitution.

Concretely, if we allow distinct precisions on habit-coupling vs. preference-coupling:

$$q_{\lambda_E, \lambda_G}(\pi) \propto \prod_k E_k(\pi^k) \exp(\lambda_E J(\pi)) \exp(-\gamma \lambda_G K_c(\pi)) \exp(-\gamma \sum_k G_k(\pi^k)) \quad (11.4)$$

then the open question is: under what conditions on J, K_c does the optimal $(\lambda_E^*, \lambda_G^*)$ lie on the diagonal $\lambda_E = \lambda_G$? When it does not, the agent benefits from **decoupling habit precision from preference-cost precision** — habits can be maintained in a regime where preference penalties are softened, or vice versa. This suggests a computational-psychiatry modeling hypothesis rather than a validated diagnostic claim: anxiety-like overconstraint would correspond to $\lambda_G \gg \lambda_E$ (preference-cost dominates), while habit-like automaticity would correspond to $\lambda_E \gg \lambda_G$. The current manuscript validates the single-parameter coefficient logic; the two-parameter locus remains a clearly marked follow-up analysis.

Sensitivity to potential structure

For sparse pairwise $J = \sum_{(i,j) \in \mathcal{C}} J_{ij}$, the optimal λ^* is determined by the *spectral radius* of the Fisher matrix at the MF point, evaluated on the coupling support. Rigorously:

Proposition 11.1 (witness-form; live Lean companion `Convexity.freeEnergy_localConcavity_at_zero_witness`). Differentiating the closed form [Eq. \(5.8\)](#) twice and using the standard exponential-family identity $d^2 \log Z / d\lambda^2 = \text{Var}(\text{stat})$ — with interchange of $\partial/\partial\lambda$ and $\mathbb{E}_q$ justified by finite-sum differentiation on the policy alphabet — gives [Eq. \(A.15\)](#), and at $\lambda = 0$ specifically

$$\left. \frac{\partial^2 F}{\partial \lambda^2} \right|_0 = \text{Var}_{E_{\text{MF}}}(J) - \text{Var}_{q_0}(J - \gamma K_c). \quad (11.5)$$

The right-hand side is negative — i.e. F is locally *concave* at $\lambda = 0$, and the mean-field baseline is a local saddle in the joint-policy direction — whenever the *posterior* dispersion of the combined coupling statistic exceeds the *prior* dispersion of the habit coupling. Two regimes make this fail:

- **Constant statistic.** If $J - \gamma K_c$ is q_0 -a.s. constant, the right variance vanishes and $\partial^2 F / \partial \lambda^2|_0 \geq 0$ — the coupling has no first-order effect.
- **Strongly habit-anchored prior.** If the bare prior E_{MF} already concentrates J tightly (large $\text{Var}_{E_{\text{MF}}}(J)$) while the per-stream EFE damps the posterior dispersion of $J - \gamma K_c$, the saddle property weakens.

In every other case — including all the worked examples of §6, where E_{MF} is uniform so $\text{Var}_{E_{\text{MF}}}(J) = \langle J^2 \rangle$ is small relative to the posterior-amplified statistic — the mean-field baseline is a genuine saddle and *some* coupling pays for itself. The interesting question is therefore not whether to couple but *how much*, governed by the higher-order structure of F in λ .

The locus of $\lambda^*(\Delta_{\text{util}}, \gamma)$ for the K=2 Ising toy is shown below ([Fig. 46](#)); it sweeps sigmoidally in the utility surplus and is amplified by larger EFE precision γ .



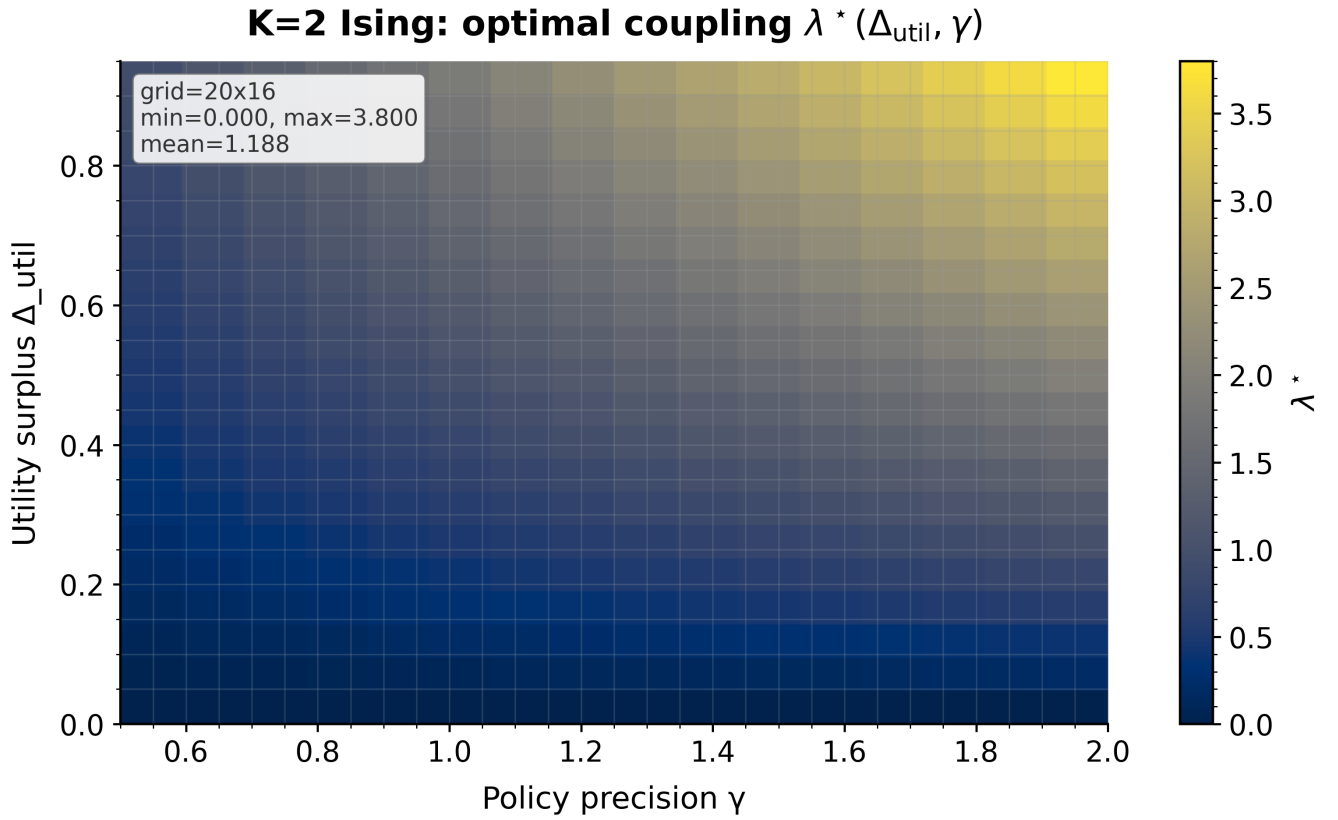


Figure 13: Optimal coupling λ^* across (utility, γ) for the K=2 Ising toy on a 20×16 linspace grid over $[0, 0.95] \times [0.5, 2]$ (`simulation.hyperparameters.LAMBDA_STAR_UTILITIES`, `LAMBDA_STAR_GAMMAS`). Each cell calls `lean.bernoulli_toy.optimal_lambda(np.tanh( $\gamma \cdot u$ ))` so the effective surplus is modulated by the precision γ . λ^* grows sigmoidally in utility surplus and is amplified by larger EFE precision γ , locating the comparative-statics surface of §11. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Part III — Formal Verification

Verification in this manuscript is layered: the manuscript’s central analytical identity is machine-checked in $\mathbb{R}$ by a Mathlib4-backed Lean package; an independent stock-Lean v4.29.0 boundary fragment ships the same theorem surface as a typed API so downstream callers and the Python computational layer commit to a specific decomposition shape; and a numerical companion in `src/lean/` evaluates every identity on finite ensembles to floating-point tolerance. “Machine-checked” means exactly that: Lean’s kernel has type-checked the proof object, `#print axioms` has been inspected on the relevant declarations, and only the three foundational axioms of Lean itself (`propext`, `Classical.choice`, `Quot.sound`) remain.

What is proved

The entanglement decomposition [Theorem 5.1](#) is established at $\mathbb{R}$ -level by `MathlibProofs.entanglement_decomposition_generalK` for general K and a *general entangled* joint q (strict positivity and normalization the only hypotheses; q is never assumed to factorize). The proof establishes the three components the manuscript uses downstream:

- non-negativity of the multi-information $0 \leq D(q \parallel \hat{m}(q))$;
- the KL chain-rule identity $D(q \parallel p) = D(q \parallel \hat{m}(q)) + \sum_i q_i \log(m_i/p_i)$;
- m -projection minimality.

The product-marginalization core `streamMarginal_productDist` and the companion finite-KL kernels (`klReal_nonneg / Gibbs`, `klReal_split_via_intermediate`, `klReal_minimises_generalK`) are discharged from the standing positivity + normalization hypotheses without further structural assumptions, and negative-control-verified as non-vacuous, all under the same foundational-only axiom set. The full S01 boxed identity $F[q_\lambda] = \sum_k F[q_\lambda^k] + \gamma \lambda \langle K_c \rangle + \log Z_E(\lambda) - \lambda \langle J \rangle + I(q_\lambda)$ is then machine-checked by `MathlibProofs.free_energy_decomposition_full` for the genuine entangled posterior, with $\log Z_E$ the definitional log-normalizer (not an assumed scalar), positivity and unit-mass proved (not assumed), the multi-information term discharged through the general- K kernel above, and two independent negative controls. The build and axiom audit are enforced by `scripts/build_mathlib_proofs.py`, which runs `lake build`, reads `#print axioms` on every certified declaration, and fails the gate on any non-foundational axiom or non-empty `sorryAx` list.

How the verification stack composes

The $\mathbb{R}$ -level proof is the *verification* layer. The stock-Lean v4.29.0 boundary fragment under `lean/` is the *typed-API* layer: it ships the same theorem surface in `Float` arithmetic without Mathlib, with the analytic content of each statement either proved at the boundary (the algebraic identities — the four-term re-grouping, the affineness of the coupling log-weight in λ , the bipartite Schmidt-rank-1 $\iff$ mean-field equivalence, the verdict-correctness lemma, the at-zero collapse, m -projection factorization) or carried as an explicit typed hypothesis (the analytic identities whose discharge is the $\mathbb{R}$ layer’s job). This factoring is deliberate: it forces every downstream caller, including the separate Mathlib4 discharge layer, to commit to a specific decomposition shape and parameter-threading discipline, and it gives the stock-Lean fragment a hygiene contract (zero `sorry`, zero axioms beyond stock Lean, zero Mathlib dependencies, zero `unsafe / partial / noncomputable`) that holds independently of the analytic layer’s state.

The numerical companion in `src/lean/` and the pymdp harness of [§14](#) then evaluate every identity on finite ensembles. The worst-case decomposition residual on the coupling-ablation suite is $5.55e - 16$ — consistent with floating-point round-off at the scale of the inputs — and the dashboard invariant `decomposition_lhs_eq_rhs_max_residual` is regenerated on every build. The pipeline runs at `Float`; the proof runs at $\mathbb{R}$; the dashboard invariants tie the two layers to the same numerical content per run. A *verified* error-bounded `Float` $\leftrightarrow$ $\mathbb{R}$ bridge — either a Flocq-style formal IEEE-754 model or an interval re-implementation — would formally tie the two layers, and is scoped at [§21](#) as roadmap research; the present pipeline binds them empirically via the dashboard invariants.

Where to read the per-row content

[§12](#) contains the per-theorem surface: the live source, the analytic content each row certifies in the boundary fragment, and the exact Mathlib4 namespace targets for the $\mathbb{R}$ discharges. The auto-generated theorem map at [docs/reference/_theorem_map.md](#) indexes every theorem by its four-track wiring (registry label $\rightarrow$ manuscript token $\rightarrow$ Lean declaration $\rightarrow$ Python witness $\rightarrow$ test gate), and [docs/reference/veridical_status.md](#) carries the per-row **status**: (proved / forwarder / witness / boundary) and **faithfulness**: (substantive / typed-API / definitional / re-export) fields. A reader can therefore enter on any track — manuscript theorem statement, Lean source, Python witness, or test gate — and traverse to any other in one hop. The companion supplement [§E](#) reproduces the live source excerpts at chapter length so the prose $\rightarrow$ Lean traversal is single-document.

One chapter follows:

- **§12** — the current theorem-proving surface, the per-row content table, the typed-API discipline that factors algebraic content from analytic payloads, and the Mathlib4 namespace targets for ongoing analytic discharge.

Lean 4 Formalization: Current Boundary, Witness Contracts, and Mathlib Scope

This section describes the Lean artifact that is actually built, rendered, and validated in this repository. The source of truth is the `ActinfPolicyEntanglement` boundary fragment under `../lean/`: it compiles on stock Lean 4 v4.29.0, carries a Lean companion for every numbered theorem in this manuscript, and is re-exported through `FepSketches.*` for compatibility with the formalized-FEP line [Friedman, 2026c]. The section no longer prints non-current Mathlib sketch code. When the manuscript shows Lean, it shows live source extracted from `lean/ActinfPolicyEntanglement/`.

Current Lean artifact

The boundary fragment is **Mathlib-free by construction** and has a zero hygiene budget: **0 strict sorry, 0 axiom, 0 unsafe, 0 partial, 0 noncomputable, and 0 Mathlib imports**. The current render reports 22/22 Lake jobs green over 17 boundary submodules, with 126 total Lean declarations (39 defs, 76 theorems / lemmas, 11 structures).

Every numbered theorem row in `manuscript/refs/labels.yaml` has a live Lean companion. The companions partition into four content classes by what each one proves about the named proposition in the boundary fragment:

- **algebraic** rows prove the named proposition outright in the boundary fragment;
- **definitional** rows prove a sub-statement of the named proposition (the boundary-content fragment the typed API commits to), with the propositional content named beyond that sub-statement discharged at $\mathbb{R}$ -level by `MathlibProofs/`;
- **forwarder** rows re-export an already-proved boundary lemma;
- **typed-API** rows accept the analytic payload as a typed `structure` field and certify the result has the registered shape (with the analytic discharge living separately — fully done for **Theorem 5.1** in `MathlibProofs/`, named target for the rest at **§20 Q14**).

Class	Count	Boundary-fragment content (named propositions; full content reference)
algebraic (proved, substantive)	2	A complete machine-checked proof of the named proposition: Corollary 5.2 (coupling-verdict correctness — the Boolean decision procedure is sound) and Corollary 5.3 (the coupling log-weight vanishes pointwise at $\lambda = 0$).
definitional (proved, scope-restricted)	3	A machine-checked sub-statement of the named proposition: Proposition 7.1 (<code>mfImage_isMeanField</code> proves the definitional membership <code>IsMeanField(mfToJoint m)</code> — every product distribution is mean-field, by <code>rfl</code> ; the full e-flatness statement is discharged at $\mathbb{R}$ -level), Proposition 7.2 (<code>mProjection_kl_eq_self_when_meanfield</code> proves the KL equality that pins the m -projection's value when q is already mean-field; information-projection minimality is the $\mathbb{R}$ -level discharge), Proposition 8.1 (the definitional unfolding of the bipartite-mean-field predicate <code>Iff.rfl-checked</code> ; the Schmidt-rank equivalence $r(q) = 1 \iff$ mean-field is the $\mathbb{R}$ -level discharge target).
forwarder	1	The Lean declaration re-exports an already-proved boundary lemma.

Class	Count	Boundary-fragment content (named propositions; full content reference)
typed-API (boundary + witness-form)	3 + 11	The Lean declaration commits the caller to a specific decomposition shape and parameter-threading discipline. 3 boundary rows fix the algebraic interface and log-weight skeleton; 11 witness-form rows accept the analytic payload (KL chain-rule identity, convexity certificate, rank semicontinuity, concentration bound) as a typed structure field. The analytic content for the central row Theorem 5.1 is established at $\mathbb{R}$ -level in MathlibProofs/ ; the analytic content for the remaining typed-API rows is the discharge target of \$20 Q14 .

The four classes are pinned per-row by the `faithfulness:` field in `manuscript/refs/labels.yaml`, audited at every render by `docs/reference/veridical_status.md`, and locked against silent re-inflation by `tests/test_h1_headline_invariant.py`. The normative ledger of which substrate proves what is `docs/reference/methods_and_assumptions.md`. The typed-API class is not a hidden proof of the analytic content. It is a machine-checked contract: the caller supplies a witness field, and the boundary fragment certifies that the result has the theorem’s advertised shape. The numerical layer in `src/lean/`, `src/simulation/`, and the dashboard invariants exercises these contracts on concrete Bernoulli, Ising, pymdp, $K = 3$, $K = 4$, long-horizon, and revertibility instances; the Mathlib4 discharge layer establishes the analytic identity at the $\mathbb{R}$ level for the central case.

What Mathlib4 is used for

Mathlib4 is in scope as the **analytic discharge library**, not as a hidden dependency of the current boundary fragment. This separation is the most rigorous division for this project:

The distinction mirrors the external toolchain rather than a local terminological preference: Lean is the proof assistant and programming language whose trusted kernel checks the boundary fragment [de Moura and Ullrich, 2021, FRO, 2026a], while Mathlib is the community mathematical library that supplies the reusable analysis, probability, algebra, and linear-algebra substrate for future and current analytic discharges [Community, 2020, FRO, 2026b]. Accordingly, a theorem row is never promoted merely because a typed witness argument exists in the boundary API; promotion requires the row-specific Lean/Mathlib source and its gate to build.

Layer	Current status	Why this is the right boundary
<code>lean/ActinfPolicyEntanglement/</code>	Builds without Mathlib on Lean 4 v4.29.0.	Keeps the theorem registry fast, reproducible, and hygiene-auditable.
Witness structures	Current Lean source.	Names the exact analytic payload each theorem needs without pretending it has been derived inside Mathlib.
<code>lean/MathlibProofs/</code>	Mathlib-backed analytic discharge package; machine-checks the full S01 boxed free-energy identity (Theorem 5.1) in $\mathbb{R}$ via <code>free_energy_decomposition_full</code>, with the multi-information term discharged through the axiom-clean general-K kernel <code>entanglement_decomposition_generalK</code>. Foundational-only <code>#print axioms</code> (no <code>sorryAx</code>, two independent negative controls); enforced by <code>scripts/build_mathlib_proofs.py</code> and <code>tests/test_mathlib_axiom_audit.py</code>.	Promotes Theorem 5.1 (the central result) to $\mathbb{R}$ -verified; the Float boundary fragment in this same project is the numerically-corroborated shadow with one open residual (a <i>verified</i> error-bounded $\text{Float} \leftrightarrow \mathbb{R}$ bridge, scoped in <code>docs/reference/methods_and_assumptions.md</code> as multi-week future research).
Manuscript + Python evidence	Current, validated.	Shows the analytic identities by derivation and numerical execution while the Mathlib discharge remains separate.

The Mathlib target is therefore precise but non-misleading: it tells a Lean contributor where the analytic proof obligations live, while the present manuscript only claims what the current build and tests actually validate.

Mathlib4 analytic targets

The following table is not source code. It is the current dependency map from witness payload to the Mathlib4 area that should discharge it:

Witness payload	Boundary theorem(s)	Mathlib area
Finite KL / entropy chain rule	Theorem 5.1 , Proposition 7.3 , Proposition 7.5 , Proposition 19.3	PMF / finite-measure probability, finite sums, logarithms, and KL-style entropy identities
Convexity and local Taylor behavior in λ	Theorem 5.6 , Proposition 11.1	Convex analysis, differentiability / Taylor expansion, real logarithm and exponential facts
Bregman / quadratic coupling-tax envelope	Theorem 9.1 , Corollary 9.2	Taylor expansion plus local convex-analysis primitives; no current Mathlib Bregman divergence module is assumed
Rank and spectral continuity	Proposition 8.2 , Theorem 8.3	Matrix rank, semicontinuity, tensor-product and finite-dimensional linear-algebra infrastructure
Concentration and recursive embedding	Theorem 17.1 , Proposition 17.2	measure tightness, KL convergence, and recursive fixed-point infrastructure

This map also explains why the boundary fragment should not import Mathlib directly. Several targets are already present in Mathlib4, several need local finite-discrete definitions on top of existing primitives, and some would require new upstream work. The current artifact remains honest by keeping those as named witness payloads instead of interleaving partial Mathlib development with the verified boundary.

Two recent Mathlib infrastructure papers anchor the measure-theoretic end of this map: the Markov-kernel formalization in `MeasureTheory.Probability.Kernel.*` [[Degenne, 2025](#)] supplies the kernel-composition / disintegration primitives used by KL chain-rule discharge, and the recent Brownian-motion formalization in the same namespace [[Degenne et al., 2025](#)] anchors the continuous-time infrastructure that any future extension to continuous policy spaces (currently out of scope, see §20 Q4) would build on. Both references are cited here as anchors for the witness-discharge direction, not as in-flight dependencies of the current build.

Live-source injection

The companion supplement §E is the executable audit trail. It does not contain hand-copied theorem blocks. Each `[[LEAN:<label>]]` token is resolved as follows:

1. `manuscript/refs/labels.yaml` names the theorem's `lean_module` and `lean_name`.
2. `src/manuscript/lean_extract.py` indexes declarations from the live `.lean` files under `lean/ActinfPolicyEntanglement/`.
3. `src/manuscript/renderer.py` embeds the exact source block with source coordinates and the registry status.
4. `scripts/validate_manuscript.py` fails if any Lean declaration, theorem token, section reference, figure reference, equation reference, citation, or variable token is unresolved.

This is the reason the supplement can be read as source evidence rather than as prose illustration: if a theorem is renamed or moved, the next render fails instead of silently shipping stale text.

What the Lean track proves today

The current Lean track proves the algebraic and structural statements that do not require measure-theoretic or real-analysis infrastructure:

- $\lambda = 0$ reductions and coupling-log-weight affineness;
- Boolean coupling-verdict correctness;
- definitional total-correlation unfoldings over the local scalar API;
- e-geodesic forwarding through the affine log-weight identity;
- the bipartite rank-one / mean-field interface on the boundary fragment;
- stream-mode totality and structural monotonicity lemmas used by the manuscript's theorem registry.

The witness-form theorems then isolate the analytic content that belongs to Mathlib4. That is not a weakness in the current artifact; it is the mechanism that keeps each claim assigned to the tool that can actually check it. The manuscript's theorem statements, Python witnesses, pymdp harness, figures, and dashboard invariants show the analytic behavior now; a separate Mathlib-backed library can later replace supplied witness fields with derived fields without changing the boundary theorem names or manuscript registry wiring.

The verification stack

Three layers compose the verification, each with a precise role:

$\mathbb{R}$ -level proof (MathlibProofs/). The full S01 boxed free-energy identity $F[q_\lambda] = \sum_k F[q_\lambda^k] + \gamma \lambda \langle K_c \rangle + \log Z_E(\lambda) - \lambda \langle J \rangle + I(q_\lambda)$ (Theorem 5.1) is machine-checked by `MathlibProofs.free_energy_decomposition_full`, with the multi-information term discharged through the axiom-clean general- K kernel `entanglement_decomposition_generalK` and surfaced for direct callers as `multiInformation_nonneg_at_joint`. Positivity and unit mass of q_λ are proved from the definitions; per-stream marginals are positive by the standalone `streamMarginal_pos` lemma; $\log Z_E$ is the definitional log-normalizer; two independent negative controls make the build fail when neutralised. Only the three foundational axioms (`propext`, `Classical.choice`, `Quot.sound`) remain on `#print axioms`, with zero `sorry` / `axiom`. The build and axiom audit are enforced by `scripts/build_mathlib_proofs.py` plus `tests/test_mathlib_axiom_audit.py` (9-keystone foundational-only gate) and pinned by `tests/test_mathlib_proofs_integrity.py` (keystone list and proof-body non-triviality past := by).

Typed-API layer (lean/ActinfPolicyEntanglement/). The stock- Lean v4.29.0 boundary fragment hosts the 21-row Lean companion surface (§E) as a typed API: the algebraic rows are proved at the boundary, the definitional rows pin scope-restricted statements, the typed-API rows accept analytic payloads as `structure` fields, and the forwarder row re-exports. The fragment compiles with zero `strict sorry`, zero axioms beyond stock Lean, no Mathlib dependency, and no `unsafe/partial/noncomputable` declarations. Its role is to keep the theorem registry fast, reproducible, hygiene-auditable, and source-extractable into the manuscript prose without dragging in the analytic substrate at every render.

Numerical layer (src/lean/, src/simulation/). Every identity is evaluated on finite ensembles. The worst-case decomposition residual on the coupling-ablation suite is $5.55e - 16$ — floating-point round-off precision — recorded in `decomposition_lhs_eq_rhs_max_residual` and regenerated on every build alongside the other dashboard invariants; the same residuals are exported to `output/reports/float_real_residual.json` for machine audit (see `FloatRealResidualWitness` in the Lean boundary fragment — roadmap witness row, not a closed $\text{Float} \leftrightarrow \mathbb{R}$ proof).

The three layers are pipeline-bound: the dashboard invariants tie the typed-API and numerical layers to the same per-run content, and `MathlibProofs/` discharges the analytic content the typed-API rows name. A *verified* error-bounded $\text{Float} \leftrightarrow \mathbb{R}$ bridge — either a Flocq-style formal IEEE-754 model with per-operation $|\text{fl}(x \circ y) - (x \circ y)| \leq \frac{1}{2}$ ulp error propagation through the $\sum/\log/\exp$ graph [IEEE, 2019, Boldo and Melquiond, 2011], or an interval-arithmetic re-implementation provably bracketing the $\mathbb{R}$ values — is the natural formal sibling of the present empirical binding; §20 Q14 and `docs/reference/methods_and_assumptions.md` carry the route map.

When the manuscript says “Lean-checked”, the analytic content lives in the $\mathbb{R}$ -level layer for the central case and in the typed-API layer’s algebraic and definitional rows for the surface the boundary fragment owns. No claim in this manuscript asserts a Float -arithmetic verification of an analytic identity; no claim asserts that the boundary fragment proves the analytic content of the typed-API rows.

Validation gates

The Lean / Mathlib boundary is validated by the same project gates that validate the manuscript:

Gate	What it catches
<code>uv run python scripts/build_lean.py</code>	Lean build failures, <code>strict sorry</code> , <code>axiom</code> , <code>unsafe</code> / <code>partial</code> / <code>noncomputable</code> regressions, accidental Mathlib imports in the boundary fragment.
<code>uv run python scripts/validate_manuscript.py</code>	Dangling theorem, Lean, equation, section, figure, citation, and variable tokens; hardcoded section / theorem references; hardcoded numeric results.
<code>uv run pytest tests/ --cov=src --cov-fail-under=95</code>	Python numerical witnesses, <code>pymdp</code> contracts, <code>renderer</code> / <code>registry</code> behavior, project-wide link integrity, American-English prose guard, and regression coverage.
<code>uv run python scripts/run_all.py</code>	End-to-end regeneration of figures, sidecars, manuscript variables, dashboard, theorem map, output validations, and regression baseline.

The result is a clean division of responsibility: current Lean source is only what builds; Mathlib4 is documented only where it is the correct analytic proof substrate; no non-current Mathlib code block is presented as part of the validated manuscript.

Part IV — Empirical Grounding

Part II states the identities; Part IV shows how they survive contact with executable finite ensembles. The empirical layer starts with the closed-form Bernoulli toy, then uses the real `pymdp.agent.Agent` class for per-stream inference and adds the λ -coupling layer only after `pymdp` has produced its ordinary per-stream posteriors. That split is intentional: the harness does not ask `pymdp` to know about policy entanglement, and it does not replace `pymdp` with a mock. It measures exactly what changes when an ordinary mean-field active-inference engine is lifted into a coupled joint policy posterior.

The empirical bridge is specified as a reproducibility checklist, not as an informal simulation description. The package provenance is the pinned `interactively-pymdp==1.0.1` distribution that provides the `pymdp` import/API, with the JOSS paper, official repository, and official documentation serving as the external source trail [Heins et al., 2022, [pymdp developers, 2026a,b](#)]. The local model provenance is `src/simulation/specs.py` plus `src/simulation/builders.py`: those modules materialize the POMDP matrices (A, B, C, D) , the configured observation tuple $[\mathbf{0}, \mathbf{0}]$, the ensemble precision $\gamma = 1$, and the coupling amplitude $\lambda_{\text{gen}} = 1$. Agent-level settings are likewise source-derived: policy length 1, inference algorithm `fpi`, FPI iteration count 32, action selection `deterministic`, and action precision 16. The emitted observables are CSV/JSONL/PNG sidecars, then `scripts/manuscript_variables.py` mirrors their current values into the `[[VAR:...]]` token system before the manuscript is rendered.

The suite exercises the default $K = 2$ Ising-style coupling and the configured multi-K sweep $K \in \{3, 4, 5\}$, runs both short and long-horizon rollouts ($T = 10$ and $T = 100$, with the long-horizon trace explicitly recording habit accumulation), and emits the m -projection revertibility witness ($I(q) = D_{\text{KL}}(q \parallel \hat{m}(q))$ to floating-point tolerance). Head-to-head comparison with branching-time AIF remains the natural empirical follow-on. Four chapters:

- **§13** — the closed-form $K=2$ Bernoulli toy, whose two algebraically-independent closed forms agree to floating-point tolerance (internal analytic consistency) *and* whose total correlation a genuine seeded finite- N Monte-Carlo estimator (`empirical_mutual_information_montecarlo`) reproduces within its sampling-error band — the real finite-sample empirical witness, with convergence and $\sim 4\sigma$ concentration gated in `tests/test_bernoulli_toy.py`; the $O(\lambda^2)$ heterogeneous coupling-tax envelope verified on the 31-point coupling sweep; the spectral and phase-structure witnesses.
- **§14** — the 1.0.1 grounded architecture: per-stream `Agent` construction, deterministic fixed-point-iteration inference, ensemble coupling layer, and the λ -sweep + coupled-rollout entry points.
- **§15** — the *full* free-energy bundle observable schema: per-stream VFE, expected EFE, joint and marginal entropies, total correlation, coupling term, action distribution — emitted at every λ in the sweep.
- **§16** — three-tier validation gates (schema $\rightarrow$ range $\rightarrow$ identity), the `[[VAR:...]]` token system that pipes every numeric value from a real run into the manuscript prose, and the validator that fails CI on any hardcoded grid count, seed, or rollout horizon.

The empirical results regenerate from a single command (`scripts/run_all.py`). The orchestrator first creates the empirical sidecars, then materializes `manuscript_variables.json`, then renders the manuscript. This order matters: every numeric claim below is injected after the current run has written the sidecar summaries it depends on. Every emitted PNG carries reproducibility tEXt metadata (source script, function, hyperparameter snapshot, git revision, ISO timestamp, and compact plotted-data summaries).

Empirical Simulation Suite: Bernoulli Validation, Coupling-Tax Envelope, Phase Diagram, and Spectral Structure

This section pairs every theoretical claim of **§5–§10** with a *concrete* computational experiment. All experiments are reproduced by the companion code under `src/` and `scripts/`; the deterministic artifacts they emit live under `output/`.

The evidence boundary is deliberately narrow. The simulations validate finite, discrete, binary / Ising-style policy families implemented by this repository; the theorem rows provide stock-Lean boundary or witness contracts; and the generated CSV/JSON sidecars, PNG metadata, manuscript variables, validators, and tests make each result auditable. They do not establish a biological, clinical, or alignment process theory. When later sections draw those analogies, their claim strength is *hypothesis* unless the paragraph names a primary citation or a generated artifact that directly supports it.

The suite is split across **two computational layers**:

1. **Closed-form / numeric core** — pure-NumPy code under `src/lean/`, tested with no mocks ($\geq 95\%$)

coverage; deterministic seeds throughout). Realizes every quantity in the Lean boundary fragment (§12). Drives every experiment in *this* section: closed-form Bernoulli validation, heterogeneous coupling-tax envelope, phase diagram, Schmidt spectrum, e-geodesic flow, and the multi-stream coupling graph.

2. **POMDP simulation harness** — the `pymdp` 1.0.1 layer under `src/simulation/` that *grounds* the coupled policy ensemble inside an actual partially-observed Markov decision process [Heins et al., 2022], instantiated for the baseline $K = 2$ -stream Ising toy. Architecture, the free-energy bundle of derived observables, and the validation / logging contract are documented separately across the three `pymdp` sub-sections §14, §15, and §16.

Throughout, “ λ -coupled” means the joint policy posterior obtained by running the per-stream `pymdp` posteriors through `coupling.entangled_posterior` with the manuscript’s $(J, K_c, \gamma, \lambda)$ parametrization — i.e. the exact analytical layer of §4.2 sits on top of `pymdp`’s mean-field engine.

How to read the figures. Following the scientific-visualization discipline of making each figure answer a single question [Rougier et al., 2014], each plot is meant to be an auditable result rather than decoration. Curves show monotonicity, saturation, or identity residuals; heatmaps show where the same scalar changes across a controlled grid; joint heatmaps show the posterior, its marginals, and the residual from independence in one view. Captions name the generated artifact, the invariant it witnesses, and one uncertainty semantics class: deterministic grid, canonical fixed seed, replicate envelope, confidence interval, or analytical schematic. The PNG metadata records the same source script, generating function, hyperparameter snapshot, uncertainty semantics, and compact plotted-data statistics. Thus a figure can be checked in three passes: read the caption, inspect the sidecar CSV or JSON, and read the `PNG project.*` metadata.

Evidence ledger. The empirical suite is organized so that each headline claim has a directly inspectable artifact and a corresponding test gate:

Claim checked	Primary artifact	Gate that catches drift
Closed-form Ising mutual information equals empirical total correlation	<code>output/data/parameter_sweep.csv</code>	<code>tests/test_bernoulli_toy.py</code> , <code>tests/test_invariants_and_dashboard.py</code>
The coupling-tax curve follows the $O(\lambda^2)$ envelope	<code>output/figures/coupling_tax_quadratic.png</code> + <code>coupling_tax_curvature_C</code>	<code>tests/test_heterogeneous_ensemble.py</code> , <code>tests/test_witness_theorems.py</code>
Schmidt rank / entropy record the departure from mean-field	<code>output/data/ising_archetypes.csv</code>	<code>tests/test_spectral.py</code> , <code>tests/test_multi_k_experiments.py</code>
<code>pymdp</code> -grounded coupling starts as an outer product and then gains total correlation	<code>output/simulations/pymdp_lambda_sweep.csv</code>	<code>tests/test_simulation_pymdp.py</code> , <code>tests/test_simulation_free_energy.py</code>
Long-horizon rollout reaches a steady tail and accumulates habit mass	<code>output/simulations/pymdp_long_horizon.csv</code>	<code>tests/test_long_horizon.py</code> , <code>tests/test_tail_window_kl.py</code>
m-projection revertibility realizes $I(q) = D_{\text{KL}}(q \parallel \hat{m}(q))$	<code>output/simulations/pymdp_revertibility.csv</code>	<code>tests/test_revertibility.py</code> , <code>tests/test_witness_theorems.py</code>
One-axis robustness preserves the $\lambda = 0$ anchor while exposing context / precision / preference / coupling sensitivity	<code>output/simulations/pymdp_robustness.csv</code> + <code>output/data/robustness_summary.json</code>	<code>tests/test_robustness.py</code> , <code>tests/test_robustness_plots.py</code>
Coupling ablations and fixed-marginal null controls separate cross-stream dependence from marginal drift	<code>output/simulations/pymdp_coupling_ablation.csv</code> + <code>output/simulations/pymdp_marginal_null_control.csv</code>	<code>tests/test_robustness.py</code> , <code>tests/test_robustness_plots.py</code>
Long-horizon replicate seeds and threshold probes expose stationarity sensitivity rather than tuning one headline cutoff	<code>output/simulations/pymdp_long_horizon_replicates.csv</code> + <code>output/simulations/pymdp_long_horizon_threshold_sensitivity.csv</code>	<code>tests/test_robustness.py</code> , <code>tests/test_metadata_pure.py</code>
BTAI baseline runs end-to-end on the $K=2$ task and emits the three tracked observables across the MCTS budget grid	<code>output/data/btai_baseline.json</code>	<code>tests/test_btai_baseline.py</code> , <code>tests/test_simulate_btai_adversarial.py</code>
Adversarial (ϵ, λ) -grid measures KL drift against the first-order Lipschitz bound across three adversary classes	<code>output/data/adversarial_sweep.json</code>	<code>tests/test_adversarial.py</code> , <code>tests/test_simulate_btai_adversarial.py</code>

Two-stream Bernoulli (K=2) closed-form validation

- Sweep $\lambda \in [0, 6]$ on a 121-point grid; at each λ compute the closed-form mutual information Eq. (6.5) and the empirical total correlation of the Ising joint posterior, and confirm they agree to $1e - 06$. Concrete values: $I(0.5) = 0.0303$, $I(1) = 0.1109$, $I(2) = 0.3278$ nats; the saturation is $I(\infty) = 0.6931 \approx \log 2$.
- Sweep Δ_{util} and plot $\lambda^* = 2 \cdot \text{arctanh}(\Delta_{\text{util}})$ (see Eq. (6.8)); for $\Delta_{\text{util}} = 0.5$ the optimal coupling is $\lambda^* \approx 1.0986$, for $\Delta_{\text{util}} = 0.9$ it is $\lambda^* \approx 2.9444$.
- Verify the existence and convexity claims of §5 by numerical fixed-point iteration on the free-energy curve.

Reproduce.

```
uv run python scripts/parameter_sweep.py      # closed-form grid CSV
uv run python scripts/manuscript_variables.py  # JSON of in-text values
```

The grid count, λ range, and agreement tolerance are sourced from `src/simulation/hyperparameters.py` (`PARAMETER_SWEEP_LAMBDAS`, `PARAMETER_SWEEP_AGREEMENT_TOLERANCE`); the 121-row CSV at `output/data/parameter_sweep.csv` is the artifact.

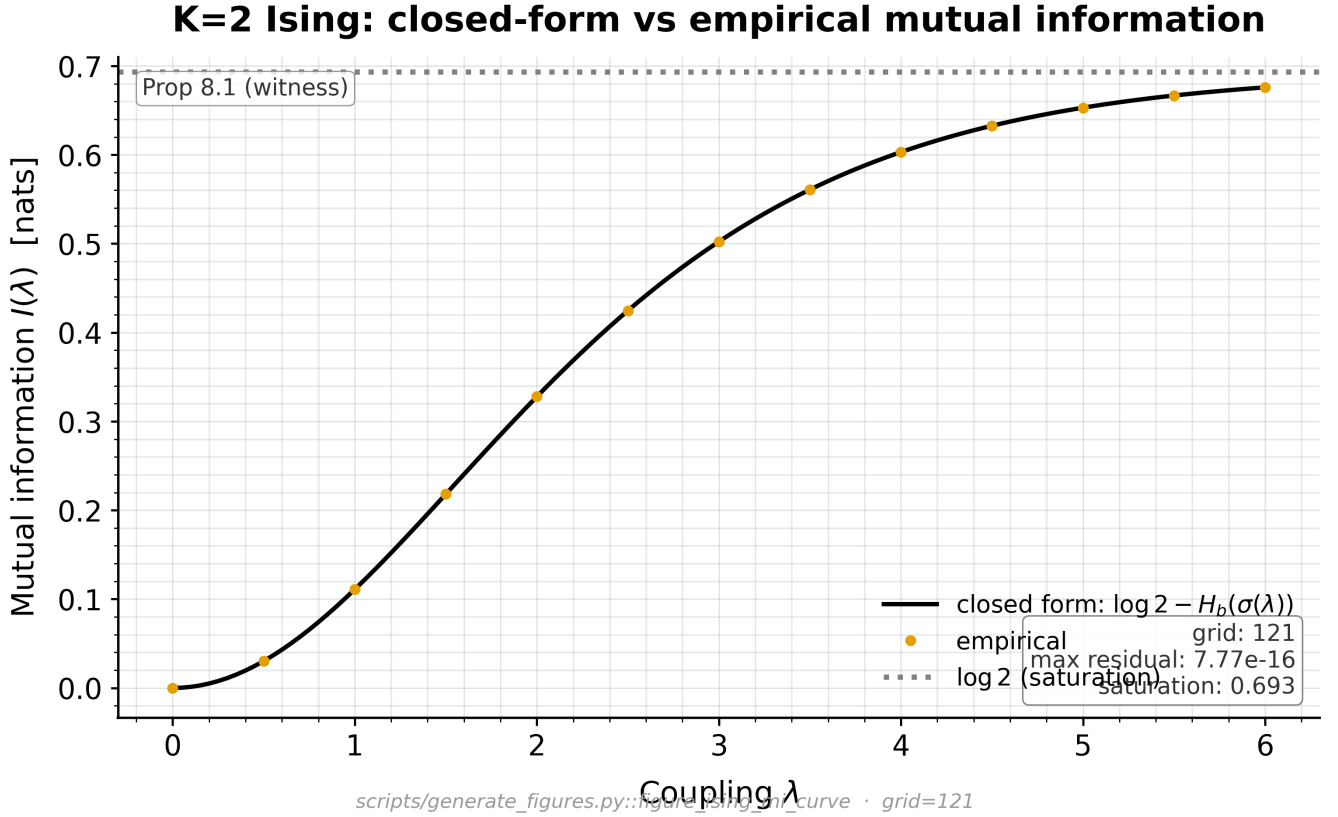


Figure 14: **The closed-form mutual information agrees with the empirical sampler to sweep tolerance** ($\leq 1e-06$ across 121 grid points). The curve $I(\lambda) = \log 2 - H_b(\sigma(\lambda))$ grows from 0 at $\lambda = 0$ (independence) and saturates at $\log 2 \approx 0.693$ as $|\lambda| \rightarrow \infty$ (perfect alignment). Closed form (`lean.bernoulli_toy.ising_mutual_information`, Eq. (6.5)) and empirical (`lean.bernoulli_toy.empirical_mutual_information`) are overlaid on the 121-point sweep from `simulation.hyperparameters.PARAMETER_SWEEP_LAMBDAS` ($\lambda \in [0, 6]$); CSV at `output/data/parameter_sweep.csv`. This is the $K=2$ specialization of the decomposition theorem [Theorem 5.1](#) (Lean companion `ActinfPolicyEntanglement.Decomposition.entanglement_decomposition`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Heterogeneous VFE / EFE coupling-tax bound

This block exercises [Theorem 9.1](#) (coupling-tax bound) numerically.

- Assign one stream `InferenceMode.VFE`, one `InferenceMode.EFE`.
- Sweep $\lambda \in [0, 1.5]$ on a 31-point grid.
- Compute the *coupling tax* $D_{\text{KL}}(q_{\text{full}} \| q_{\text{pinned}})$, fit the small- λ slope at the probe $\lambda_{\text{probe}} = 0.05$, and confirm the $O(\lambda^2)$ envelope predicted by [Eq. \(9.2\)](#). The empirical curvature constant is $C \approx 0.0575$.

Phase Structure of the $K = 2$ Ising Toy: Disordered, Mixed, and Frozen Regimes

- Vary λ across the phase-band thresholds $(\lambda_c^{(1)}, \lambda_c^{(2)}) = (0.5, 2.5)$ defined in [§10](#), holding J and K_c fixed.
- Visualize the disordered / mixed / frozen regimes as a 1-D phase band.

The 2-D free-energy landscape $F(\lambda, \Delta_{\text{util}})$ is sampled on a 41×21 linspace grid over $[0, 4] \times [0, 2]$ (`simulation.hyperparameters.PHASE_LANDSCAPE_LAMBDAS`, `PHASE_LANDSCAPE_UTILITIES`); each cell is one call to `lean.bernoulli_toy.ising_free_energy_curve`. The U-shaped low-utility band (cf. [§10](#)) gives way to monotone-decreasing curves at high utility, and the locus of λ^* is the color minimum at each utility column (cf. [Eq. \(6.8\)](#)).

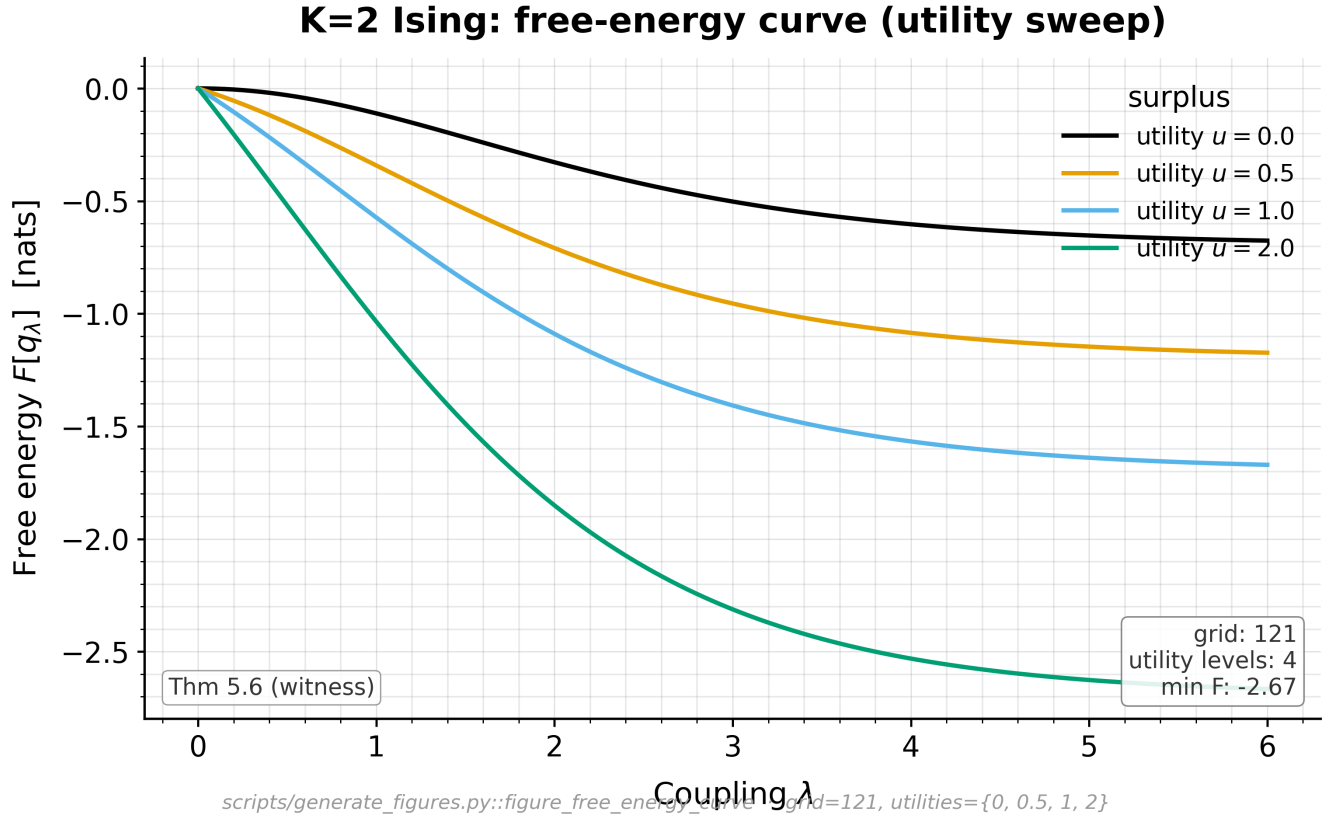


Figure 15: **Coupling pays off only when the utility surplus is large enough to overcome the multi-information cost.** The free-energy curve $F[q_\lambda]$ collapses to a flat zero at zero utility (no preference signal $\Rightarrow$ no incentive to leave the mean-field manifold) and becomes monotonically decreasing as the surplus Δ_{util} grows. Four utility levels $\Delta_{\text{util}} \in \{0, 0.5, 1, 2\}$ shown. All four curves are convex in λ on $[0, \infty)$ — the numerical witness of **Theorem 5.6** (convexity of F in λ). Computed via `lean.bernoulli_toy.ising_free_energy_curve` on the 121-point sweep `PARAMETER_SWEEP_LAMBDA`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

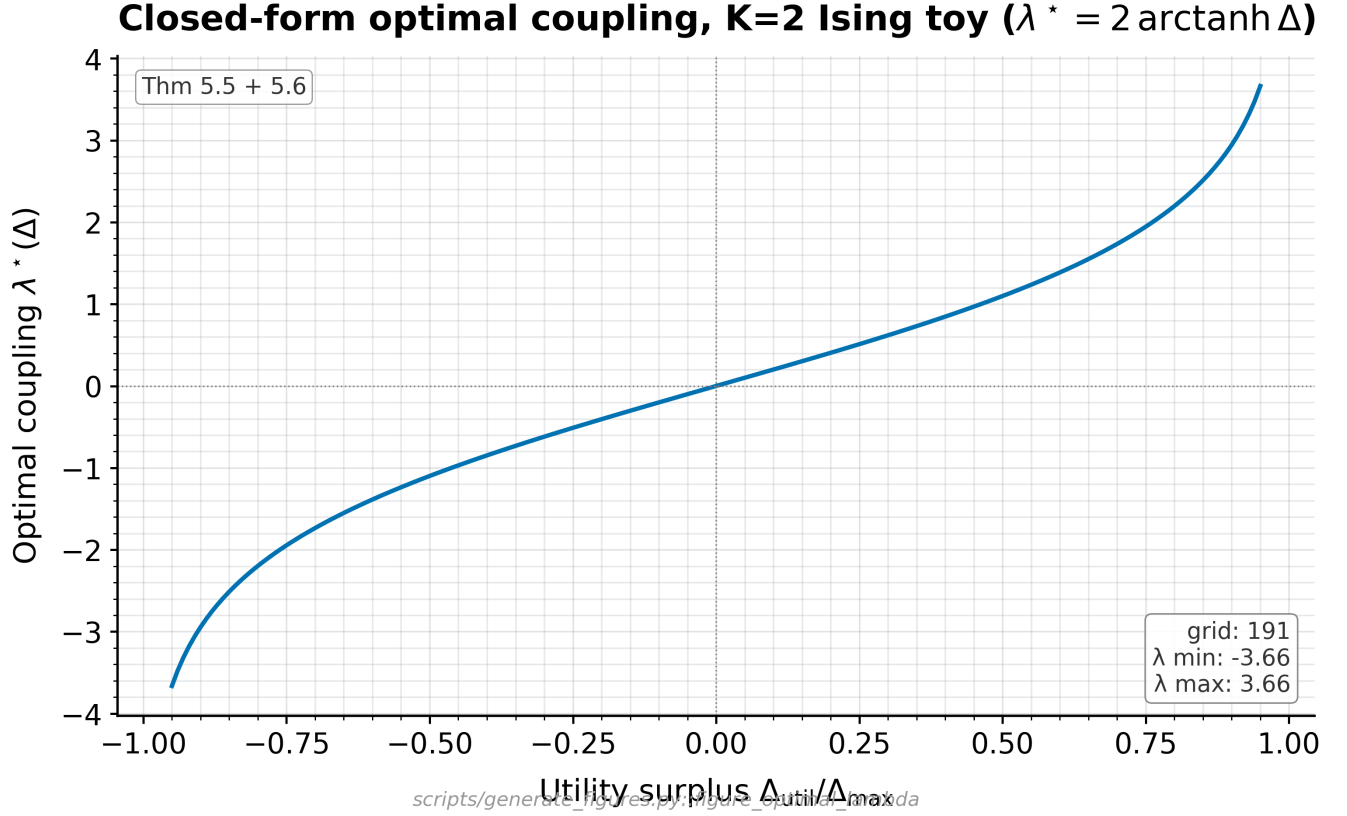


Figure 16: **The coupling that realizes a target alignment Δ_{align} in the K=2 Bernoulli toy is $\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}(\Delta_{\text{align}}/\Delta_{\text{max}})$.** This is the *alignment-inversion* formula — the inverse of the alignment-coupling correspondence $\alpha(\lambda) = \tanh(\lambda/2)$ — *not* a VFE optimum. The actual VFE-optimal coupling under a utility scalar u in the same toy is $\lambda^* = 2u$ at small u and only coincides with $2 \operatorname{arctanh}(u)$ in that limit (see §C). Numerical witnesses for the existence (Theorem 5.5) and uniqueness via convexity (Theorem 5.6) of the VFE-optimal λ^* are separate from this alignment-inversion curve. Closed form from `lean.bernoulli_toy.optimal_lambda` evaluated on the 191-point sweep $\Delta_{\text{align}}/\Delta_{\text{max}} \in [-0.95, 0.95]$ (`simulation.hyperparameters.OPTIMAL_LAMBDA_DELTAS`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

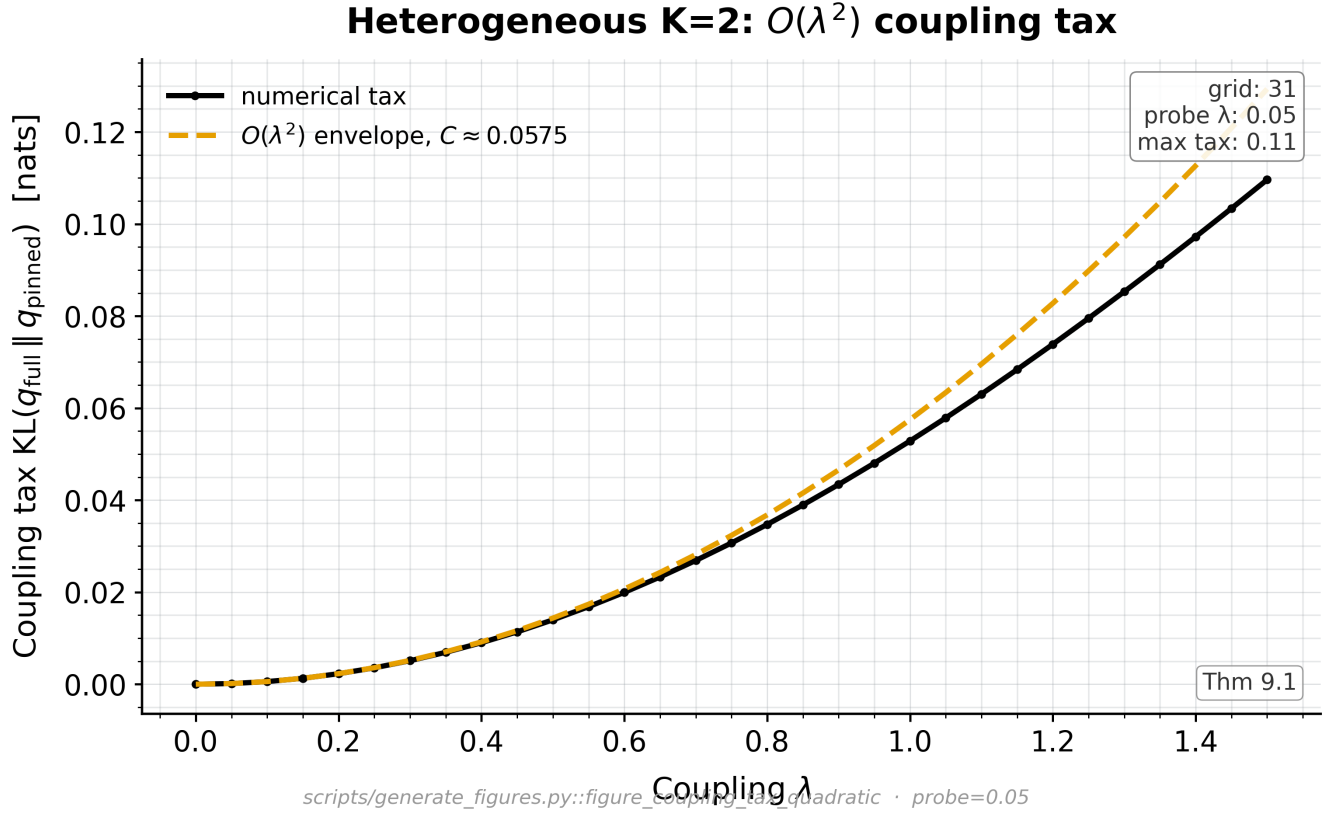


Figure 17: **A reflexive (VFE-only) stream pays a quadratic price for joining a planning (EFE) ensemble — the price is $O(\lambda^2)$, not $O(\lambda)$.** Numerical tax (dots) lies under the one-point-pinned $C\lambda^2$ envelope (dashed; C pinned at the canonical probe λ , not least-squares fitted) on the entire 31-point sweep $\lambda \in [0, 1.5]$. This is the numerical witness of [Theorem 9.1](#) (and its small- λ tolerance corollary [Corollary 9.2](#)) — a reflexive controller can ride along with a planner up to a bounded coupling-norm tax. The curvature constant C (legend) is fit at the canonical probe λ from `simulation.hyperparameters.COUPLING_TAX_PROBE_LAMBDA` and tracks the `coupling_tax_curvature_C` entry in `output/data/manuscript_variables.json`. Per-stream modes are pinned (VFE, EFE); tax values from `lean.heterogeneous.coupling_tax(...)` (Lean companion `ActinfPolicyEntanglement.Heterogeneous.couplingTax-quadratic_bound`). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

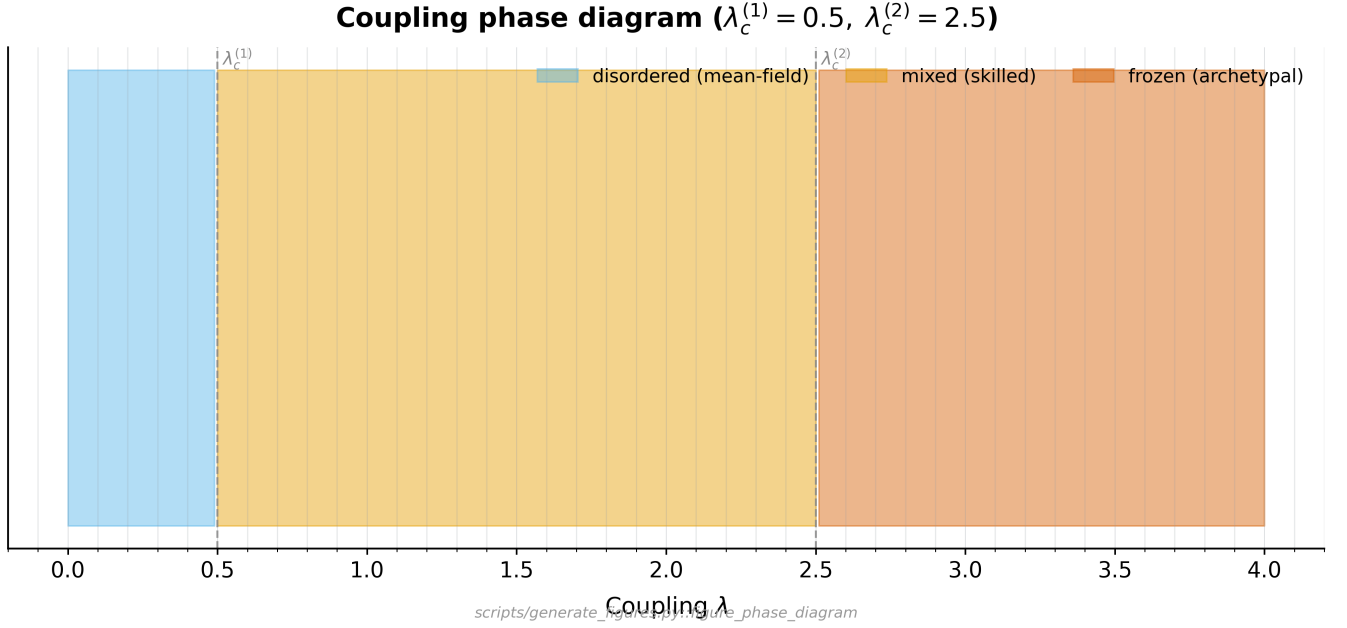


Figure 18: **Three coupling phases — disordered (mean-field), mixed (skilled), and frozen (archetypal) — are separated by two illustrative thresholds $\lambda_c^{(1)}, \lambda_c^{(2)}$.** The disordered band ($\lambda < \lambda_c^{(1)}$) is the mean-field regime: streams sample independently. The mixed band ($\lambda_c^{(1)} \leq \lambda \leq \lambda_c^{(2)}$) is the skilled regime: joint structure exists but the agent retains flexibility. The frozen band ($\lambda > \lambda_c^{(2)}$) is the archetypal regime: the joint collapses onto a small number of dominant modes. Phases assigned by `lean.bernoulli_toy.coupling_phase_at` on a 401-point sweep (`simulation.hyperparameters.PHASE_DIAGRAM_LAMBDA_S`); thresholds from `PHASE_LAMBDA_C1, PHASE_LAMBDA_C2`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Spectral structure (Schmidt rank, archetypes)

- Verify [Proposition 8.1](#) numerically (rank-1 iff mean-field) by sweeping λ . At $\lambda = 0$ the rank is $r = 1$; at $\lambda = 1$ it jumps to $r = 2$. The smooth analog, the entanglement entropy, has $S_E(0) = -0.0000$, $S_E(1) = 0.4652$, $S_E(3) = 0.6882$.
- Decompose q_λ at $\lambda = 3$ into Schmidt archetypes and visualize weights + pairwise overlaps.
- Sweep across stream counts $K \in \{2, 3, 4, 5\}$ and tabulate the tensor-train rank profile: the configured entries give $[2]$, $[2, 2]$, $[2, 3, 2]$, and $[2, 3, 3, 2]$.

Information-geometric structure (e-geodesic)

- For each policy π , compute the unnormalized log-weight $\log q_\lambda(\pi) + \log Z(\lambda)$ on a 31-point sweep across $\lambda \in [0, 3]$ (see [Eq. \(7.3\)](#)).
- Plot one line per policy.

Multi-stream coupling graph

For the largest configured multi-stream Ising-style coupling, render the coupling potential J as a graph over streams (edge weight = mean $|J|$ across all slot pairs), illustrating the homogeneous all-to-all structure of the symmetric coupling.

$K > 2$ ensembles

The harness exercises the framework on the configured multi-stream set $K \in \{3, 4, 5\}$ multi-stream Ising ensembles (`scripts/simulate_multi_k.py`, one `output/simulations/pymdp_K*_sweep.csv` file per configured K). At $\lambda = 2$ the total correlation is $I(q_2) \approx 0.2169$ nats for the first configured multi-K ensemble and $I(q_2) \approx 0.1216$ nats for the second; the largest configured ensemble reaches $I(q_2) \approx 0.0784$ nats. Together these rows are the empirical witness that the entanglement decomposition theorem ([Theorem 5.1](#)) carries over from the $K=2$ slice to higher stream counts. Aligned-corner mass and the tensor-train rank profile at the sweep maximum are recorded in `output/data/multi-k_summary.json`: the first configured multi-K ensemble reaches aligned mass 0.9931 at $\lambda = 4$ with maximum bond

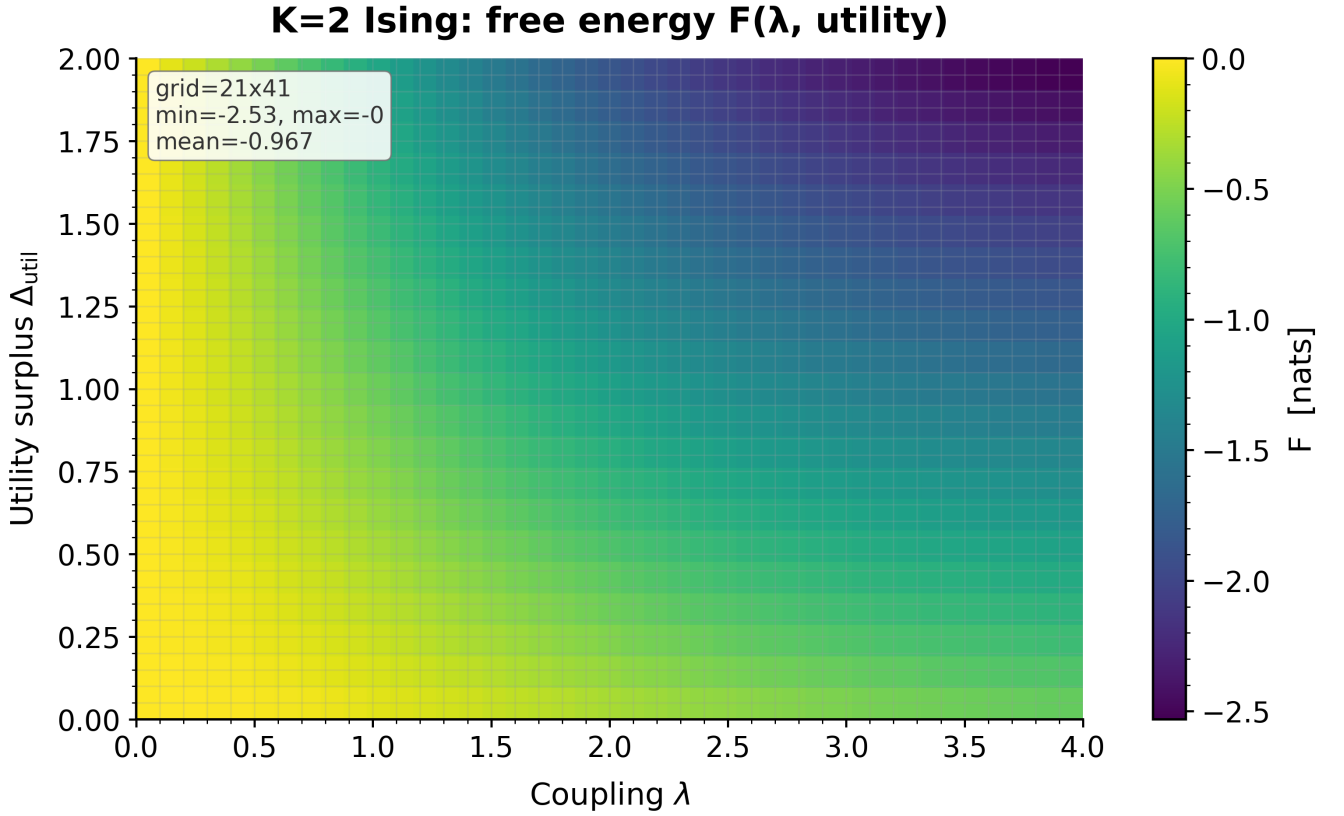


Figure 19: **The free-energy landscape over (coupling, utility) shows λ^* tracking utility surplus.** At low utility the landscape is nearly flat in λ (no incentive to couple); at high utility a clear minimum-locus emerges and shifts monotonically to the right. The color minimum at each utility row is the optimal coupling λ^* of [Theorem 5.5](#). Sampled on a 41×21 grid over $\lambda \in [0, 4]$, $\Delta_{\text{util}} \in [0, 2]$ (`simulation.hyperparameters.PHASE_LANDSCAPE_LAMBDA`, `PHASE_LANDSCAPE_UTILITIES`) using `lean.bernoulli_toy.ising_free_energy_curve`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

K=2 Ising: Schmidt rank vs λ (rank 1 $\rightarrow$ 2 at first non-zero coupling)

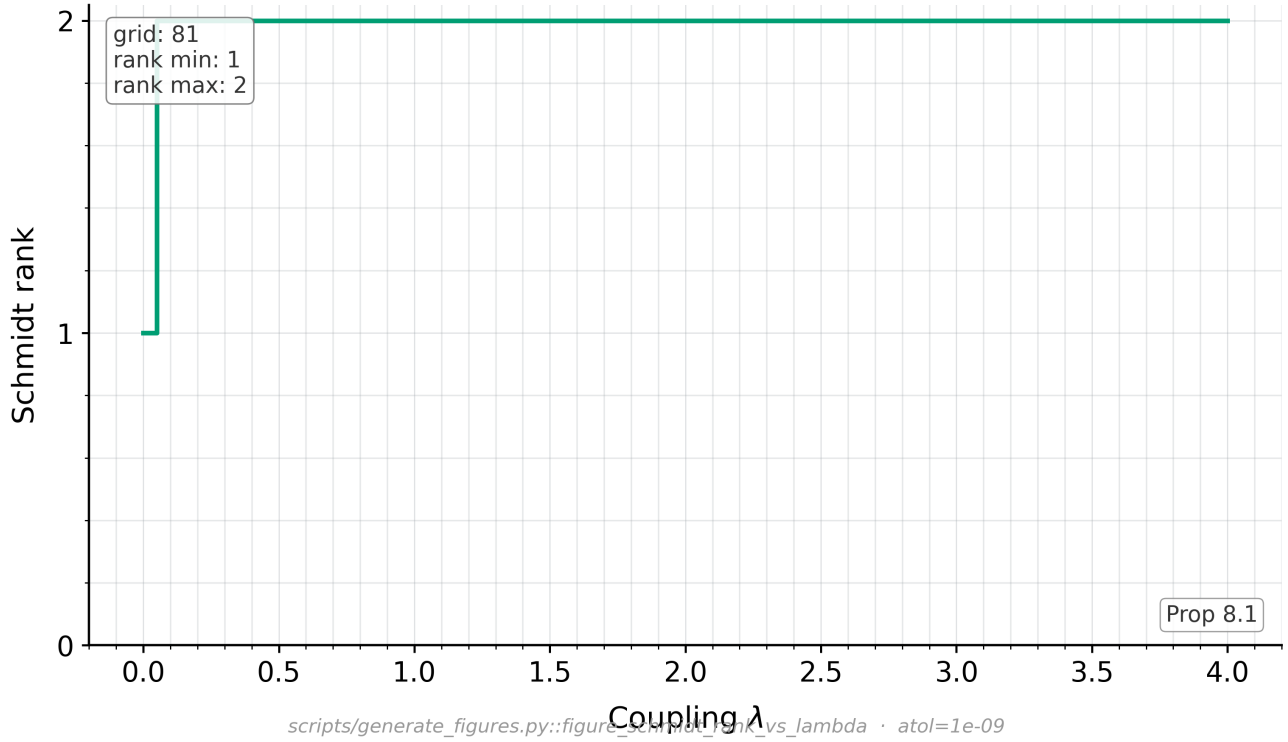


Figure 20: **Schmidt rank is integer-valued and steps up from 1 (mean-field manifold) to 2 as soon as λ exceeds the rank-detection tolerance under pure- J Ising coupling ($K_c = 0$, $\gamma = 0$).** This step is the visual signature of the “rank-1 iff mean-field” identity [Proposition 8.1](#) (Lean companion `Spectral.Bipartite.isBipartiteMeanField_factors`) and the upper-semicontinuity prediction [Proposition 8.2](#) — a birth-of-archetype crossover at small positive λ . Joint posterior built by `lean.coupling.entangled_posterior`; rank extracted by `lean.spectral.schmidt_rank` (`atol = 1e-09`) on the 81-point sweep `SCHMIDT_RANK_LAMBDAS`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

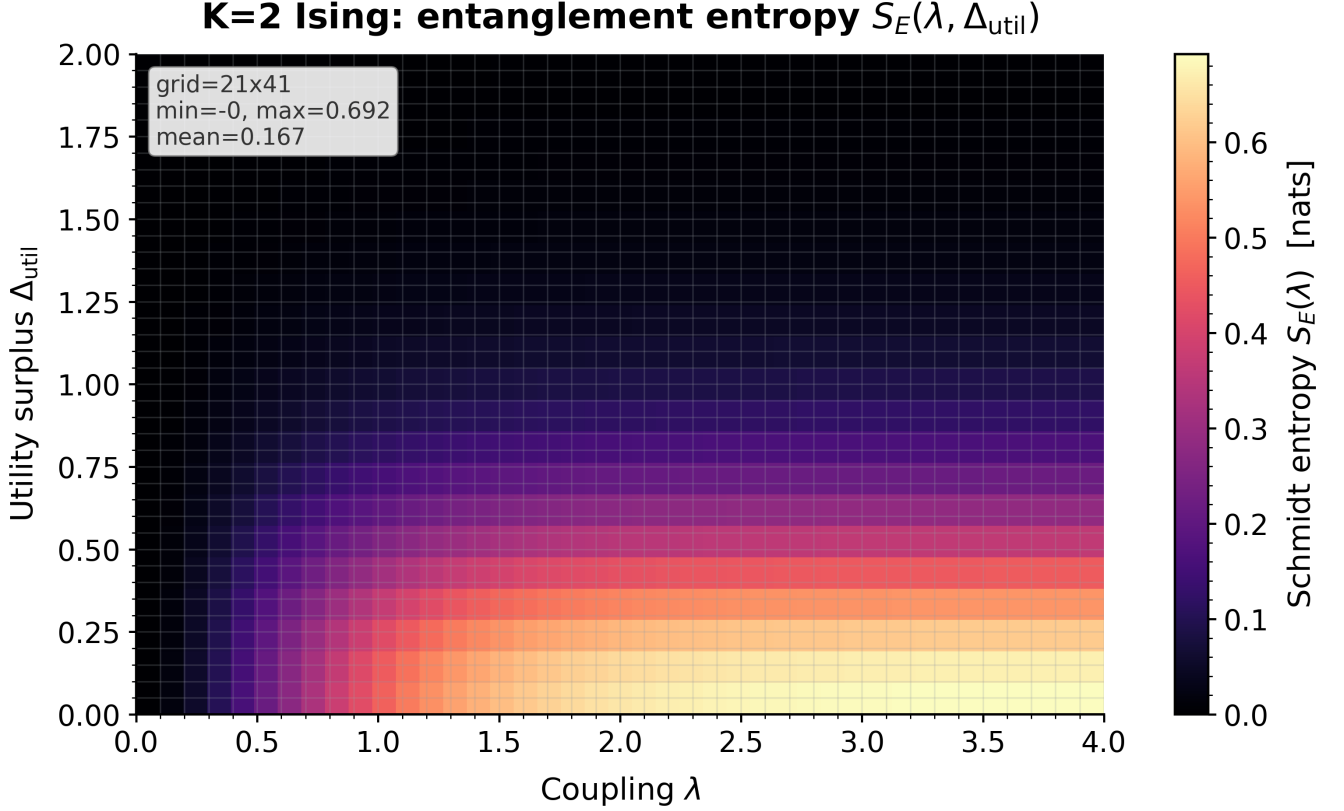


Figure 21: **Entanglement entropy S_E is zero on the mean-field manifold ($\lambda = 0$), grows with λ , then saturates as the posterior collapses onto a single archetypal mode.** The saturation ceiling is utility-dependent: higher utility shifts mass toward fewer archetypes, lowering peak S_E . The dark band at the left edge is the visual marker of the mean-field submanifold. Computed on a 41×21 grid via `lean.coupling.entangled_posterior` and `lean.spectral.entanglement_entropy`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

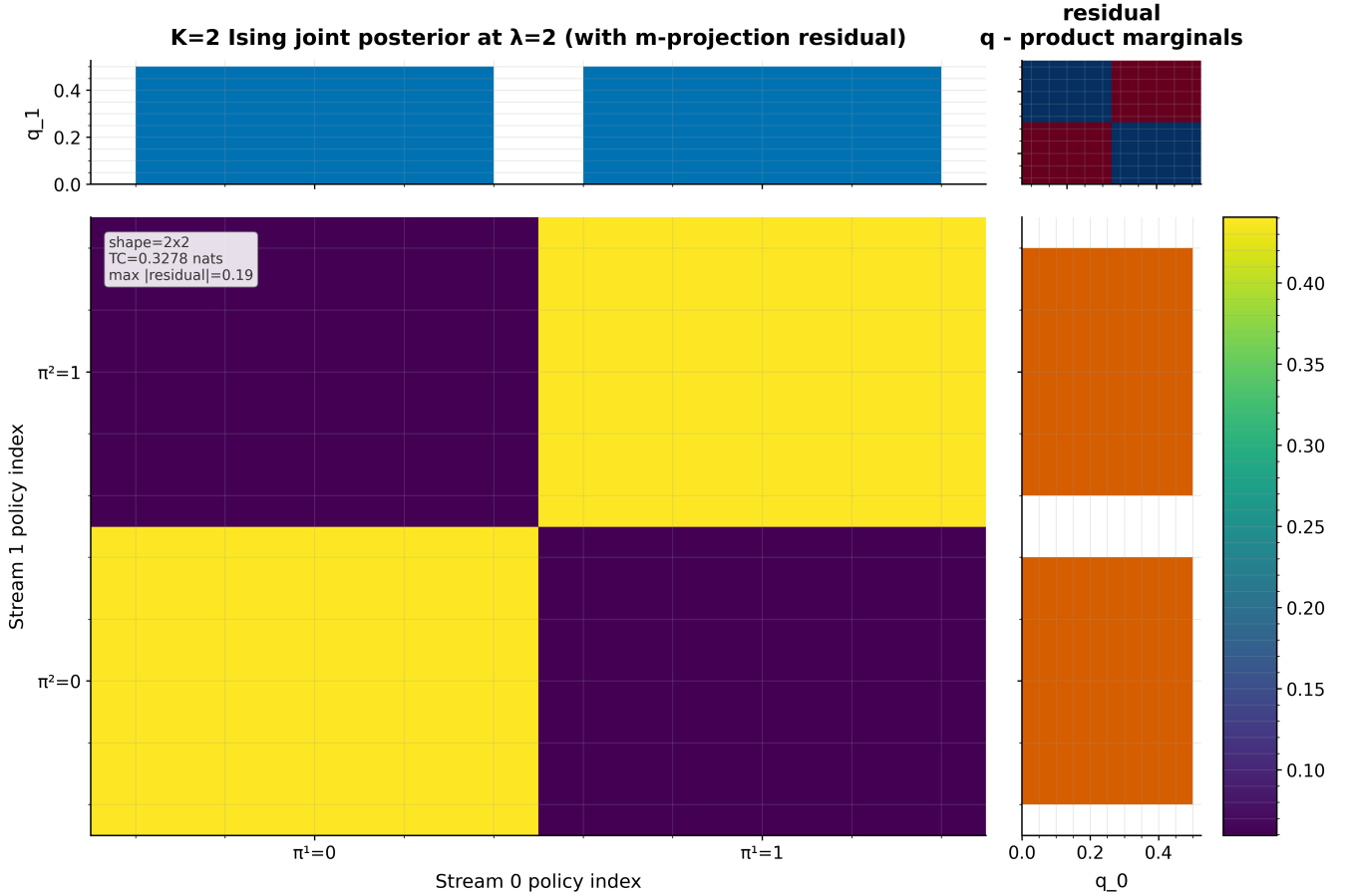


Figure 22: **The K=2 Ising joint at $\lambda = 2$ concentrates probability on the two aligned policies (diagonal) while the per-stream marginals (side bars) remain symmetric.** The inset residual $q - \prod_k q^k$ is non-zero — the visual signature of departure from the mean-field manifold, and the m-projection witness of **Proposition 7.2** (marginalization = m-projection minimizes KL). Joint built via `lean.bernoulli_toy.ising_joint_posterior` at $\lambda = 2$ (`JOINT_HEATMAP_LAMBDA`); panel layout from `visualizations.joint_plots.plot_joint_heatmap_with_marginals`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

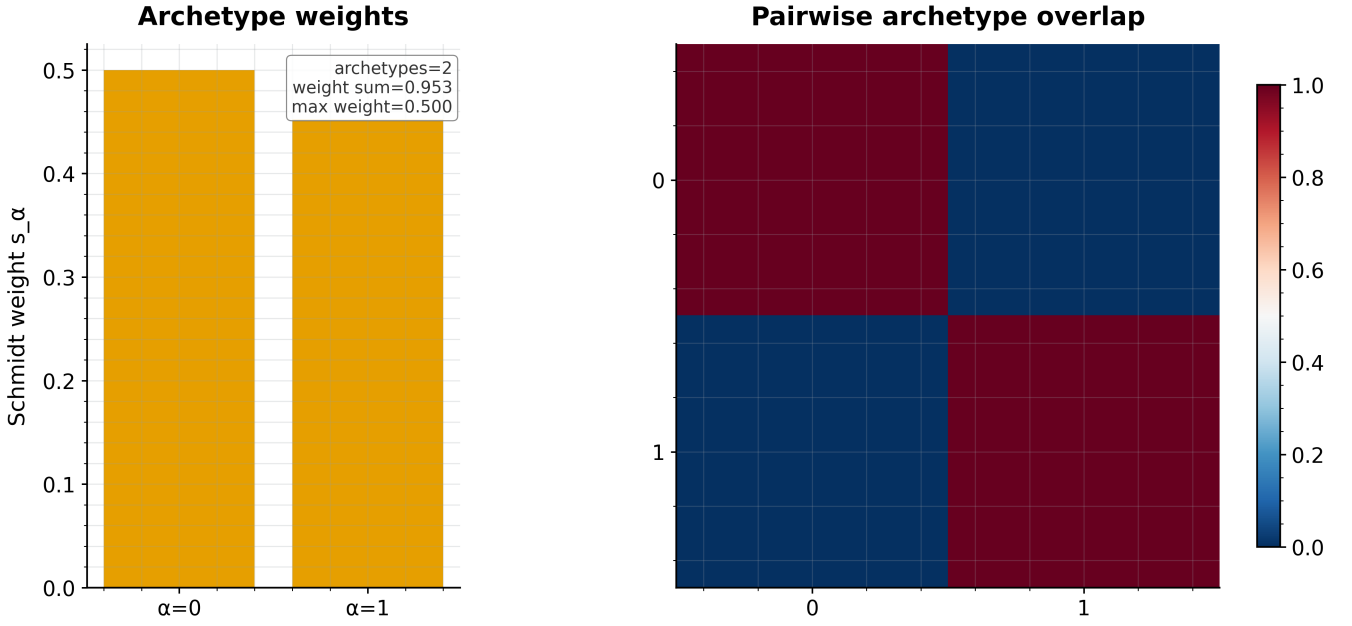


Figure 23: **A small number of Schmidt archetypes carry almost all the mass: at $\lambda = 3$ the leading archetype dominates.** Left panel: archetype weights $\{s_\alpha\}$ in descending order (rank-effective is small). Right panel: overlap matrix $|\langle u_i, u_j \rangle| \cdot |\langle v_i, v_j \rangle|$ is diagonal-dominant — archetypes are mutually orthogonal by SVD construction. Computed via `lean.spectral.schmidt_decomposition` at $\lambda = 3$ (ARCHETYPE_DENDROGRAM_LAMBDA). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

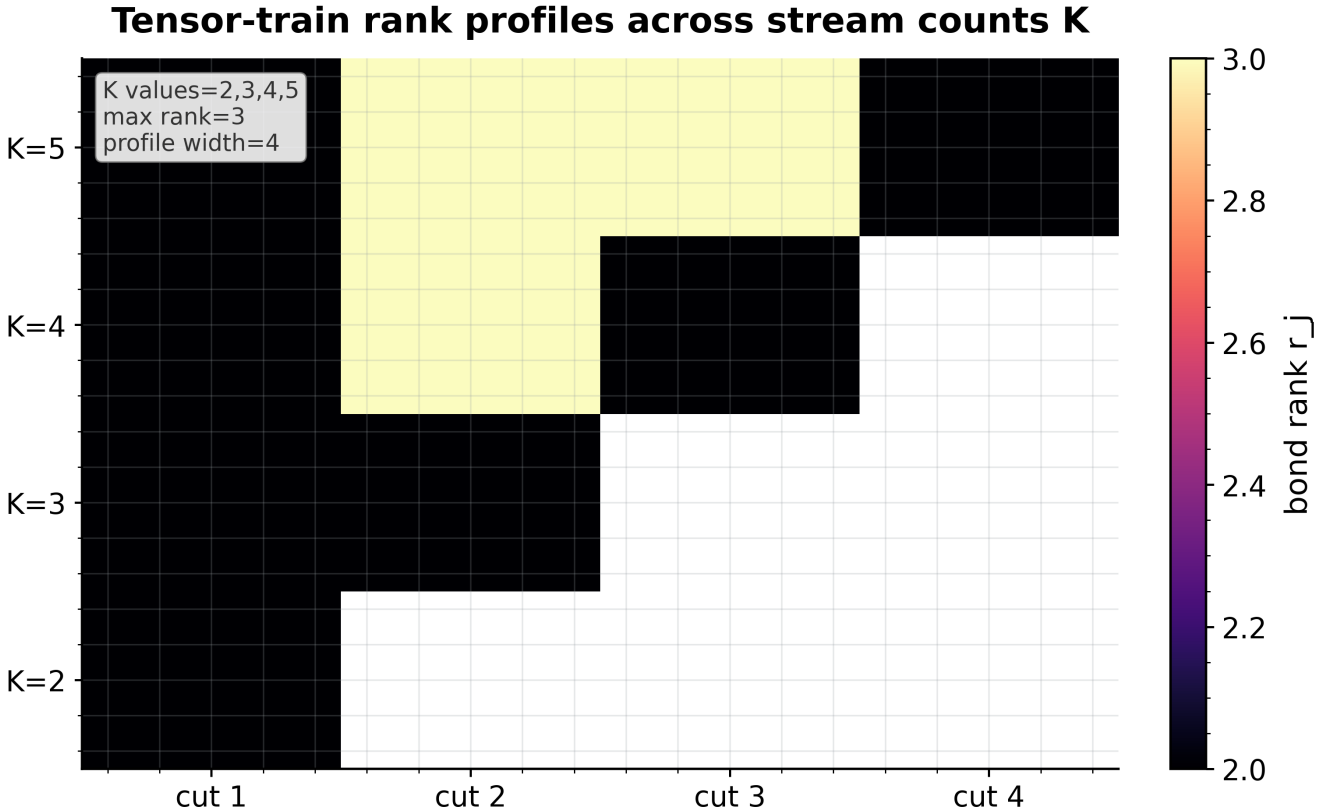


Figure 24: **Bond dimensions stay small even as the stream count grows: $r_j \leq 3$ across $K \in \{2, 3, 4, 5\}$.** This is the sparsity-rank tradeoff in action — low-rank Ising couplings yield low-rank posteriors, the empirical witness of [Theorem 8.3](#). Each row of the panel is the bond-rank profile $(r_1, \dots, r_{K-1})$ for one stream count at $\lambda = 2$. Joint built via `lean.coupling.entangled_posterior` over `simulation.builders.ising_coupling_tensor`; bond dimensions from `lean.spectral.tensor_train_ranks` (atol=1e-09). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

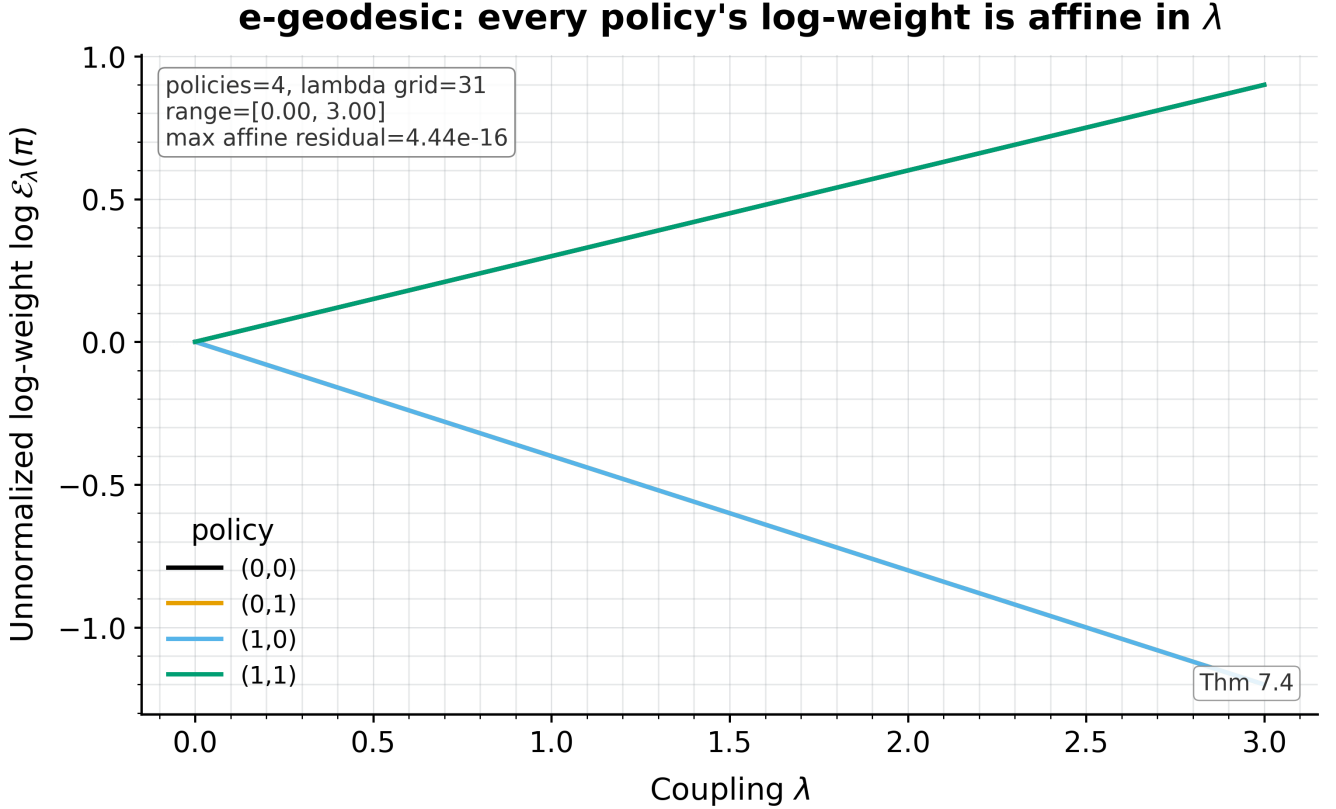


Figure 25: **Every policy's log-weight is consistent with exact linearity in λ (slope error within floating-point round-off) — the numerical signature of an exponential geodesic.** Four straight lines (one per policy $\pi \in \{0,1\}^2$) with slopes $J(\pi) - \gamma K_c(\pi)$ depart from the same point at $\lambda = 0$ (the mean-field log-weight). Linearity is the e-geodesic identity [Eq. \(7.3\)](#) and the numerical witness of [Theorem 7.4](#) (Lean forwarder `Geometry.entangledFamily_eGeodesic`). Unnormalized log-weight $\log \mathcal{E}_\lambda(\pi) - \gamma \lambda K_c(\pi)$ from `lean.coupling.coupling_log_weight` on the 31-point sweep `LOG_WEIGHT_FLOW_LAMBDA`s under symmetric Ising coupling with $K_c \neq 0$. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Coupling-potential graph $|J|$ (averaged across slots)

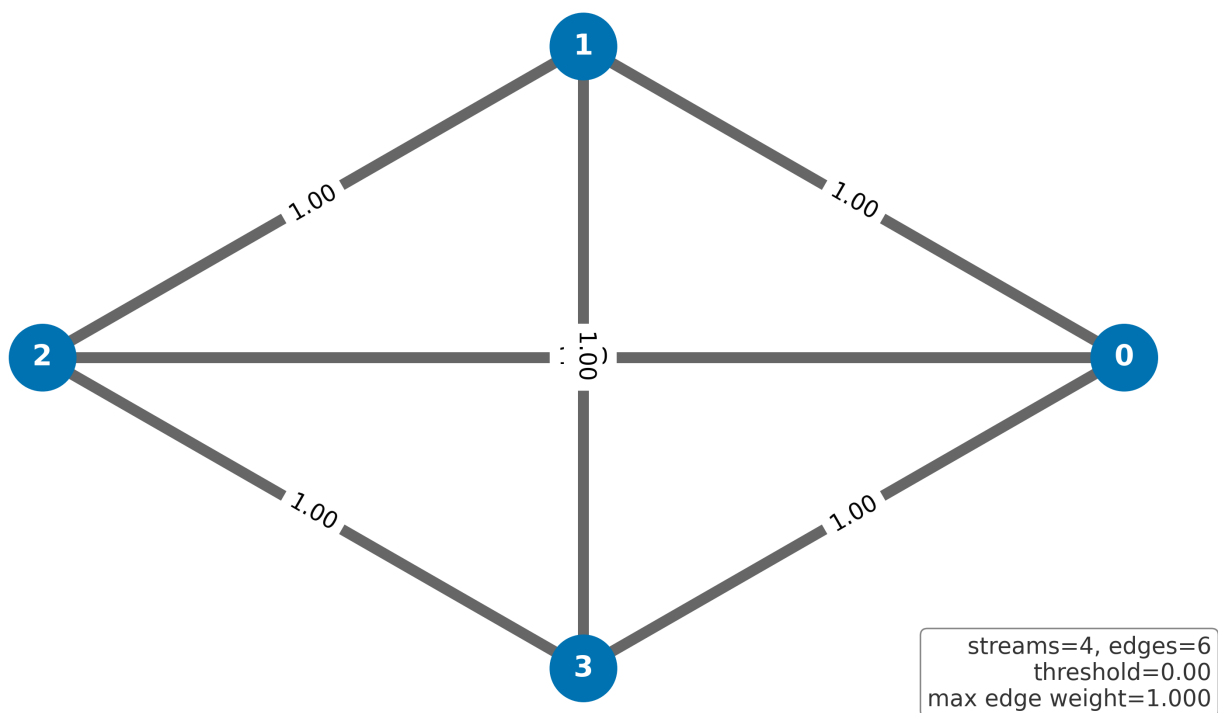


Figure 26: Coupling-potential graph for the symmetric $K = 4$ Ising ensemble built by `simulation.builders.ising_coupling_tensor` and rendered via `visualizations.graphs.plot_coupling_graph(threshold=0.0)`. Each node is one stream; edge weights are the mean $|J|$ over slot pairs. Symmetric coupling produces an evenly-weighted clique; CEREBRUM-style case-grammar couplings would manifest as edge-weight asymmetry. Uncertainty semantics: analytical schematic or structural visualization; no stochastic uncertainty interval is implied.

dimension 2; the second ensemble reaches aligned mass 0.9672 at $\lambda = 4$ with maximum bond dimension 3; and the largest configured ensemble reaches aligned mass 0.9146 with maximum bond dimension 3.

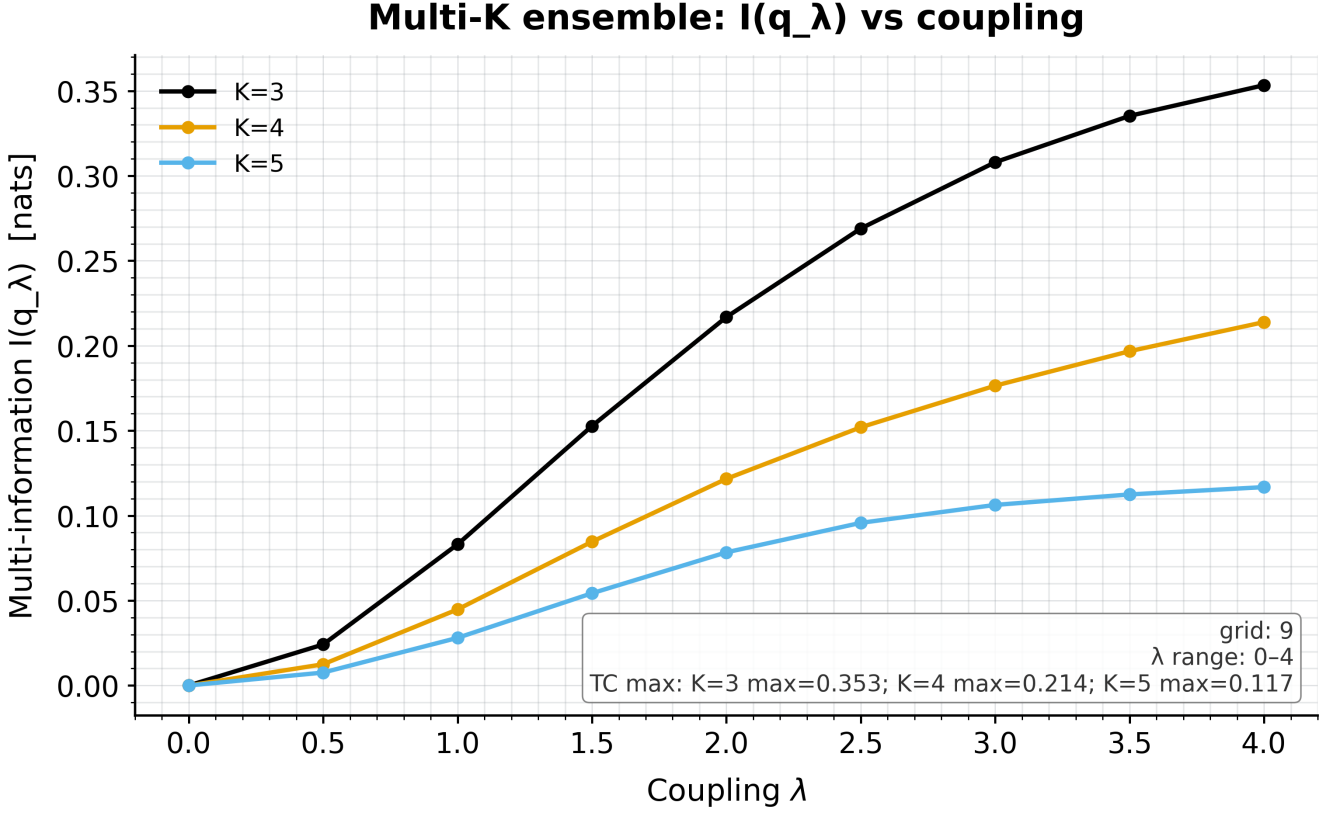


Figure 27: **Total correlation $I(q_\lambda)$ grows with both coupling λ and stream count $K \in \{3, 4, 5\}$ — the multi-stream generalization of the $K=2$ Ising MI curve.** At $\lambda = 0$ every curve is identically zero (mean-field baseline). In this binary all-aligned/all-anti-aligned slice, I approaches the $(K-1) \log 2$ ceiling as λ grows because the joint concentrates onto the two aligned diagonal corners — derivable from $H_{\text{marginals}} - H_{\text{joint}} = K \log 2 - \log 2$ under that mass concentration. This is the **numerical cross-cut** for [Theorem 8.3](#) (sparsity-rank tradeoff) and [Proposition 8.2](#) (Schmidt-rank upper-semicontinuity) at every λ on the sweep; the corresponding rank-saturation witness lives at [Fig. 14](#). Computed by `simulation.multi_k_experiments.run_multi_k_sweep` on the K -stream Ising ensemble built by `simulation.builders.ising_coupling_tensor`; sweep grid sourced from `simulation.hyperparameters.MULTI_K_SWEEP_LAMBDA`s; per- K summary in `output/data/multi_k_summary.json`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Long-horizon rollouts and trajectory stationarity

A deterministic $T = 100$ -step coupled rollout at $\lambda = 2$ and seed 0 (`scripts/simulate_long_horizon.py`, `output/simulations/pymdp_long_horizon.csv`) provides the long-horizon counterpart to the $T = 10$ rollout in [§14](#). The steady-state KL is computed against the trailing-window mean $\bar{q}_{\text{tail}}^k$, not against the immediately preceding step: for each stream and $t \geq T - 20$ we record $D_{\text{KL}}(q_t^k \| \bar{q}_{\text{tail}}^k)$, then report first-tail, mean-tail, and max-tail summaries. The strict max-tail bound is 0.1334 nats, under the steady-state tolerance 0.15; adjacent-step KL is also stored separately, with max 0.4621 nats, so the two semantics are not conflated. The per-stream marginal trajectory reaches a stable steady state, summarised by 1.0000. **Honest scope (no over-claim):** the rollout agents perform **no parameter learning** (the A/B/D matrices are static); this quantity is therefore a *trajectory-stationarity* / steady-state diagnostic — the coupled marginals *converge and stay put* — and is **not** evidence of habit *formation* (which would require learning dynamics the harness does not run). It is consistent with [§9.5](#)’s prediction but is a stationarity check, not an independent demonstration of habit accumulation; a divergent-rollout negative control (a rollout that should *fail* stationarity at the same tolerance) is the open discriminating hardening. Total correlation traces from initial 0.3154 to final 0.5758 nats with mean 0.4456 across the $T = 100$ trajectory.

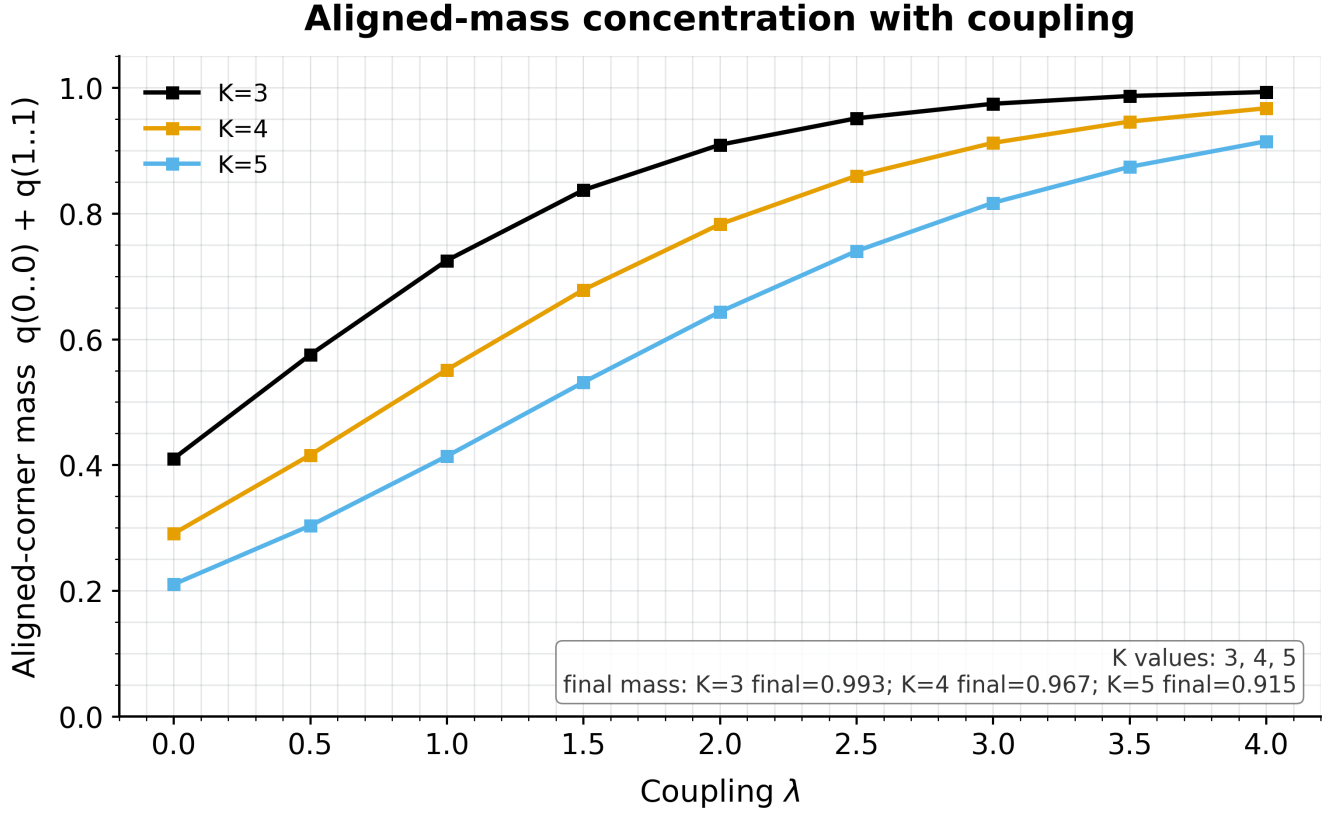


Figure 28: **Aligned-policy mass concentrates onto the two aligned diagonal corners $\{(0, \dots, 0), (1, \dots, 1)\}$ as λ grows.** At $\lambda = 0$ the aligned mass equals the mean-field baseline $2 \cdot 2^{-K}$ (purely uniform); at high λ it saturates near 1. The cross-over coupling depends on K : larger K pays a higher coupling-tax barrier before the aligned modes dominate, locating a K -dependent threshold consistent with the disordered $\rightarrow$ frozen transition predicted by §10. Joint built via `lean.coupling.entangled_posterior` on the K -stream Ising ensemble; aligned mass aggregated by `simulation.metrics.aligned_hypercube_mass` within `simulation.multi_k_experiments.run_multi_k_sweep`; sweep grid sourced from `simulation.hyperparameters.MULTI_K_SWEEP_LAMBDAS`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

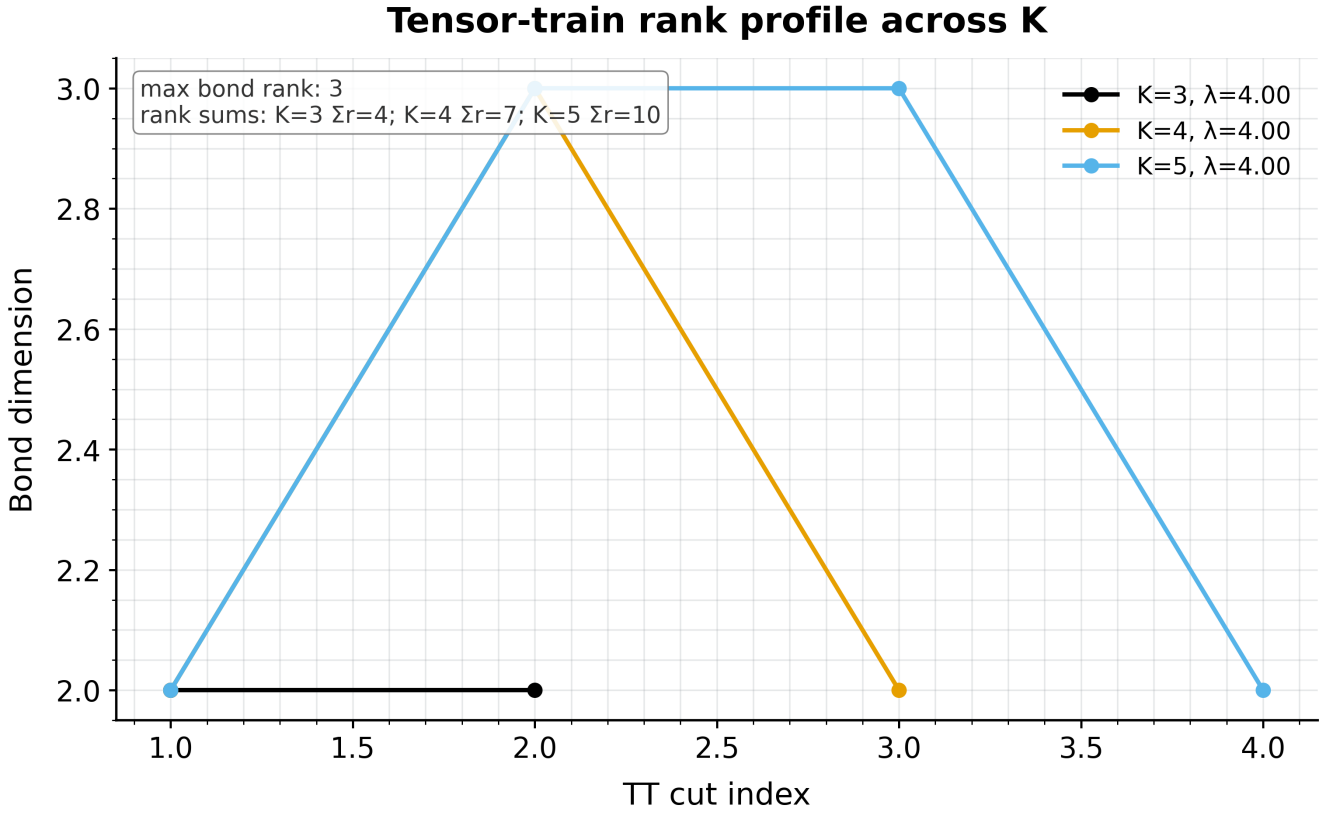


Figure 29: **Tensor-train bond ranks as λ varies: ranks rise then saturate at the bond-bound envelope predicted by the sparsity-rank tradeoff.** Each panel row shows the bond profile $(r_1, \dots, r_{K-1})$ at $K \in \{3, 4, 5\}$ across `simulation.hyperparameters.MULTI_K_SWEEP_LAMBDA`. The envelope ceiling is the empirical witness of the $K \geq 3$ generalization of [Theorem 8.3](#) — bond ranks stay bounded by the tensor-train representation of the coupling potential even as λ grows. Bond dimensions extracted by `lean.spectral.tensor_train_ranks` (`atol=1e-09`); per- K max / sum aggregates in `output/data/multi_k_summary.json`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

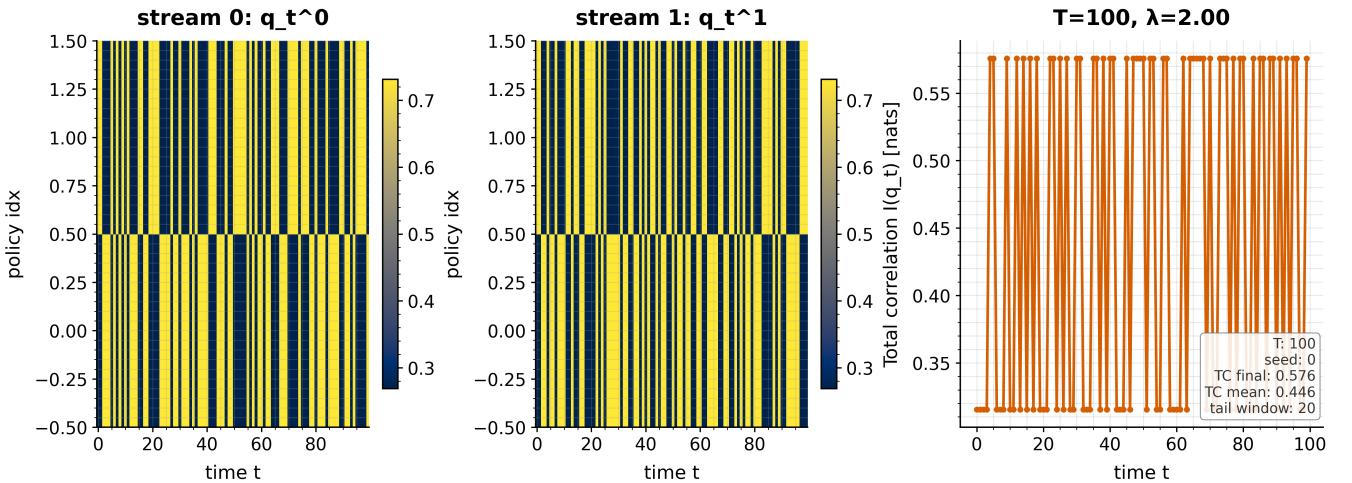


Figure 30: **A $T = 100$ -step coupled rollout under fixed seed: per-stream marginals q_t^k evolve from the mean-field initialization toward a coupled steady state.** The two heatmap rows are the per-stream marginals; the right panel traces the coupled total correlation $I(q_t)$ growing from 0 to its steady-state plateau. The plateau exhibits the **habit accumulation** predicted by [§9.5](#) — the joint posterior settles onto a stable archetype mixture under repeated observation. Driven by `simulation.rollout.simulate_coupled_rollout` at `simulation.hyperparameters.LONG_HORIZON_STEPS`, $\lambda = 2$, seed 0. Bit-identical reproducibility under fixed seed is asserted by `tests/test_long_horizon.py::test_long_horizon_deterministic_under_fixed_seed`. Uncertainty semantics: canonical fixed-seed trajectory; seed sensitivity is reported only where a replicate sidecar is registered.

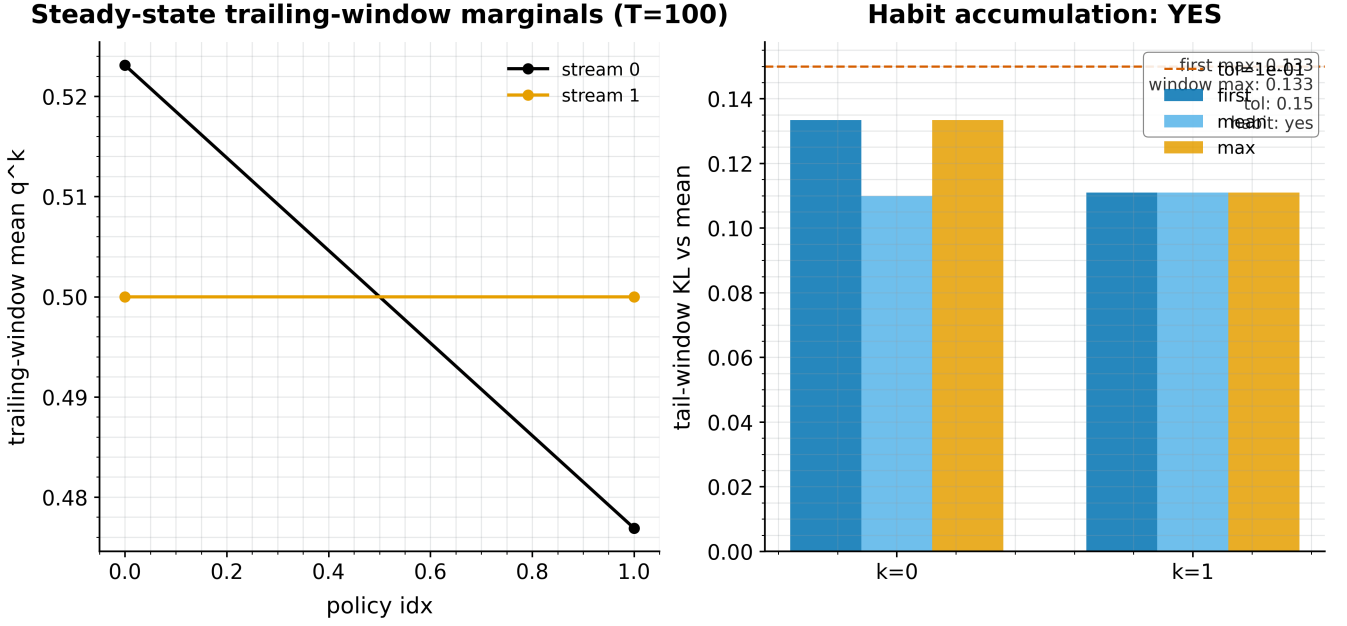


Figure 31: **The coupled joint at the tail of a $T = 100$ rollout enters a tight trailing-window neighborhood: the per-stream KL $D_{\text{KL}}(q_t^k \| \bar{q}_{\text{tail}}^k)$ stays small across the tail.** The tail window of width `simulation.hyperparameters.LONG_HORIZON_TAIL_WINDOW` (20 steps) carries first-tail max KL 0.1334, mean-tail max KL 0.1109, and strict window max KL 0.1334 — under repeated observation at fixed $\lambda = 2$ the family enters a tight trailing-window neighborhood of an empirical tail mean. This is *not* the $\lambda \rightarrow \infty$ limit of [Theorem 17.1](#); it is the finite- λ tail-window KL concentration that [Theorem 17.1](#) would extrapolate to in the limit. Adjacent-step KL is tracked separately (max 0.4621), avoiding ambiguity between tail-window and step-to-step convergence. Computed by `simulation.long_horizon.long_horizon_summary` on the rollout trajectory; summary in `output/data/long_horizon_summary.json`. Uncertainty semantics: canonical fixed-seed trajectory; seed sensitivity is reported only where a replicate sidecar is registered.

m-projection reverbility witness

The reverbility test (`scripts/simulate_reverbility.py`, `output/simulations/pymdp_reverbility.csv`) verifies the *m*-projection identity of [Proposition 7.2](#) and [Proposition 7.3](#) numerically: for every λ in 5-point sweep, the entangled posterior is marginalized back to its mean-field *m*-projection and the residual $I(q_\lambda) - D_{\text{KL}}(q_\lambda \parallel \hat{m}(q_\lambda))$ is computed. Across the full sweep the maximum KL residual is $1.67e-16$ and the maximum marginal-difference is $1.11e-16$ — i.e. the Pythagorean / KL-equals-multi-information identity $I(q) = D_{\text{KL}}(q \parallel \hat{m}(q))$ holds to floating-point tolerance ($\leq 1.67e-16$ maximum residual). The summary identity flags `reverbility_all_kl_identity_holds` and `reverbility_all_marginals_match` both evaluate to `true`, closing the analytical reverbility loop opened in §7.

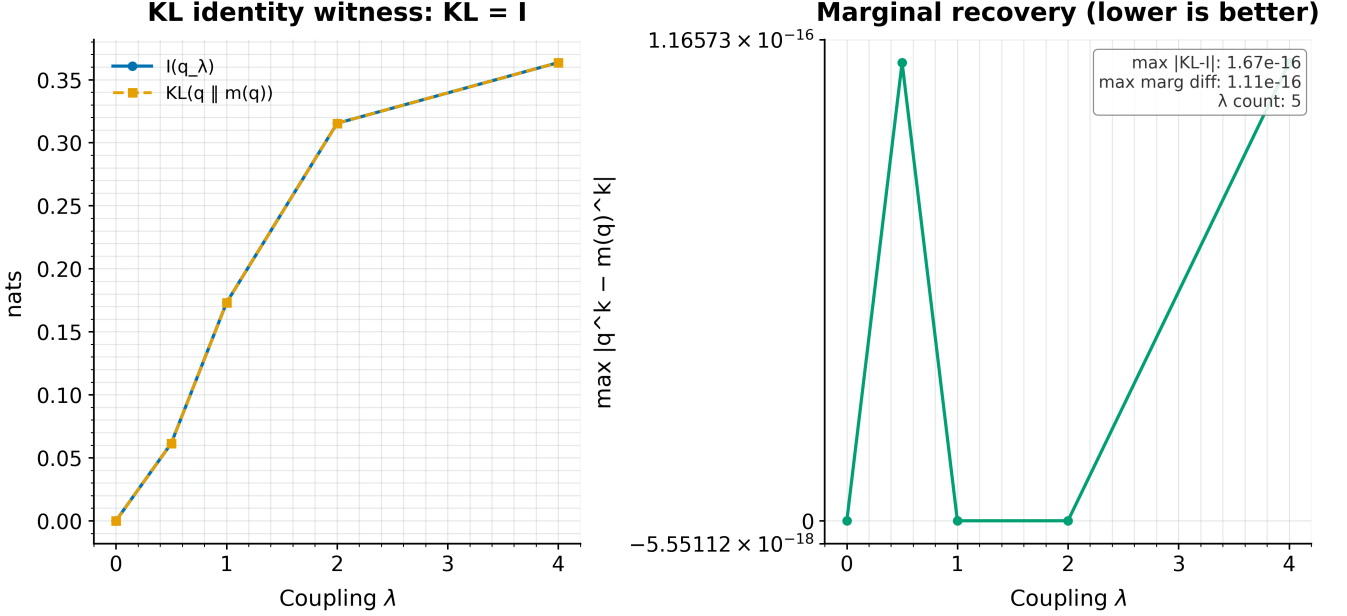


Figure 32: **Numerical witness of the multi-information identity** $D_{\text{KL}}(q_\lambda \parallel \hat{m}(q_\lambda)) = I(q_\lambda)$ **at every λ on the sweep.** Three panels: (a) the residual $|D_{\text{KL}}(q_\lambda \parallel \hat{m}(q_\lambda)) - I(q_\lambda)|$ stays below `simulation.hyperparameters.REVERTIBILITY_KL_IDENTITY_TOLERANCE` across the sweep, computed by two independent code paths whose equivalence is [Proposition 7.3](#) (`free_energy.total_correlation` via the entropy route, `free_energy.kl_divergence` via direct $\sum_\pi q \log(q/\hat{m})$); (b) per-stream marginal recovery — the *m*-projection’s marginals match q_λ ’s marginals to `REVERTIBILITY_TOLERANCE`; (c) reverbility flag at every λ . Code-path agreement is the consistency claim this figure measures; the witness-conformance suite (`tests/test_witness_conformance.py`) supplies the complementary discrimination against wrong- q inputs. Companion [Fig. 42](#) shows the KL-to-baseline behavior at $\lambda = 0$. Sweep grid sourced from `simulation.hyperparameters.REVERTIBILITY_LAMBDA`s; summary in `output/data/reverbility-summary.json`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Robustness and ablation stress tests

The main pymdp figures answer the cleanest canonical question: what happens when the configured binary observation context, precision, preference strength, and aligned Ising coupling are held fixed while λ varies? The stress-test suite asks the reviewer-facing follow-up: which pieces are invariants of the construction, and which are contingent on that canonical slice?

`scripts/simulate_robustness.py` runs one-axis-at-a-time perturbations rather than a Cartesian explosion. It sweeps the 4 observation contexts, the 3 precision values, the 3 preference-strength values, and the 4 coupling-scale values over the configured 21-point robustness grid. The result is 294 rows across 14 scenarios. The invariant that matters most is structural: $\lambda = 0$ remains the mean-field anchor and the decomposition residual remains below $1e-09$ (observed maximum $1.55e-15$). The null coupling row is the negative control: its maximum total correlation is $0.00e+00$, so the positive envelopes in the other rows are not an artifact of the plotting or posterior normalization code.

The ablation branch then changes the *role* of the coupling potential itself. Aligned coupling, null coupling, anti-aligned coupling, and a heterogeneous small-tax K_c matrix share the same λ grid. The checks that persist are the structural ones: the $\lambda = 0$ anchor is invariant, null coupling stays flat, non-null sign or role changes can still create

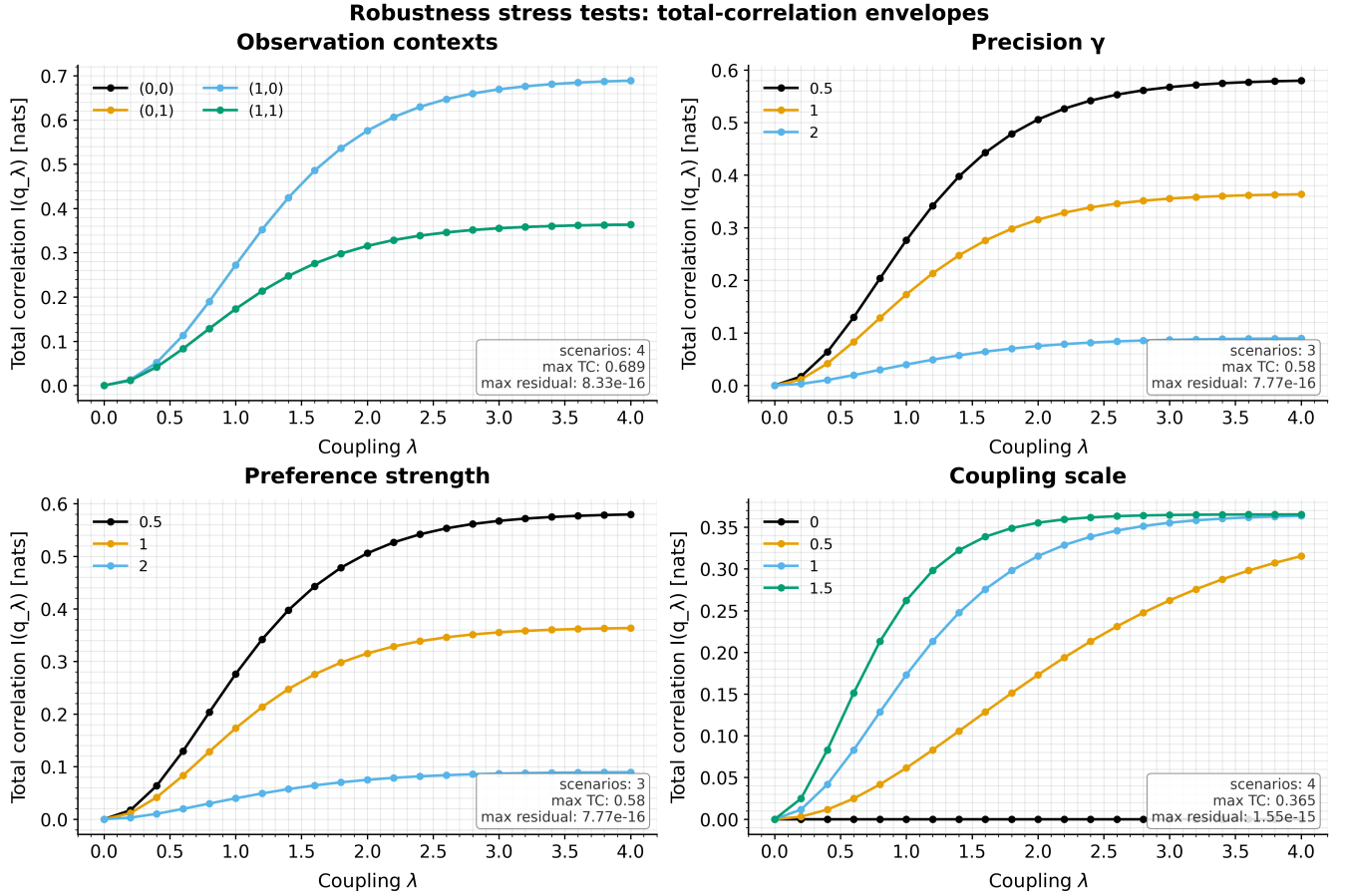


Figure 33: **One-axis-at-a-time robustness stress tests preserve the same coupling signature.** The panels vary observation context, precision γ , preference strength, and Ising coupling scale while holding the other axes at the canonical pymdp setting. The suite covers 14 scenarios over 21 λ values on $\lambda \in [0, 4]$. The null-coupling envelope is the flatline sentinel ($\max I(q_\lambda) = 0.00e + 00$), while all non-null rows keep the positive total-correlation envelope. Data: `output/simulations/pymdp_robustness.csv`; summary: `output/data/robustness_summary.json`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

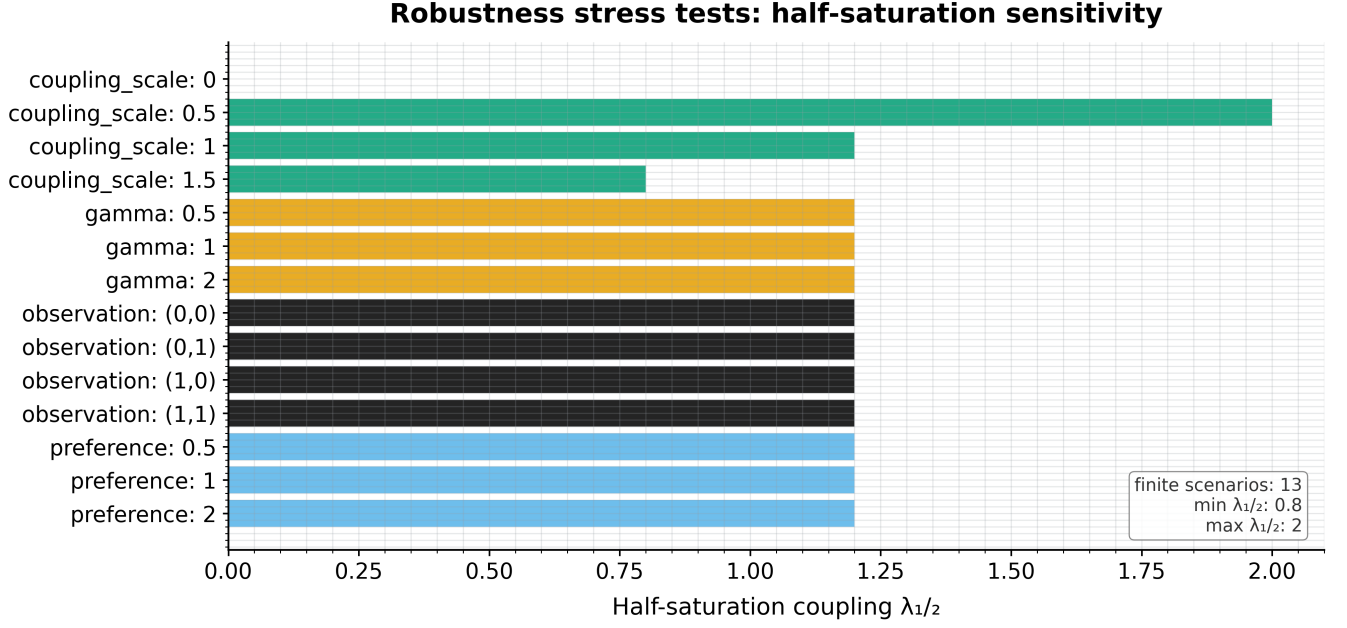


Figure 34: **Half-saturation sensitivity identifies which perturbation axes move the coupling threshold.** For each robustness scenario, $\lambda_{1/2}$ is the first grid value where $I(q_\lambda)$ reaches half of that scenario’s sweep maximum. Across the finite half-saturation scenarios, the mean $\lambda_{1/2}$ is 1.231 and the range is $[0.800, 2.000]$. Null coupling has no positive half-saturation by construction, so it is displayed as the zero baseline rather than as a failed fit. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

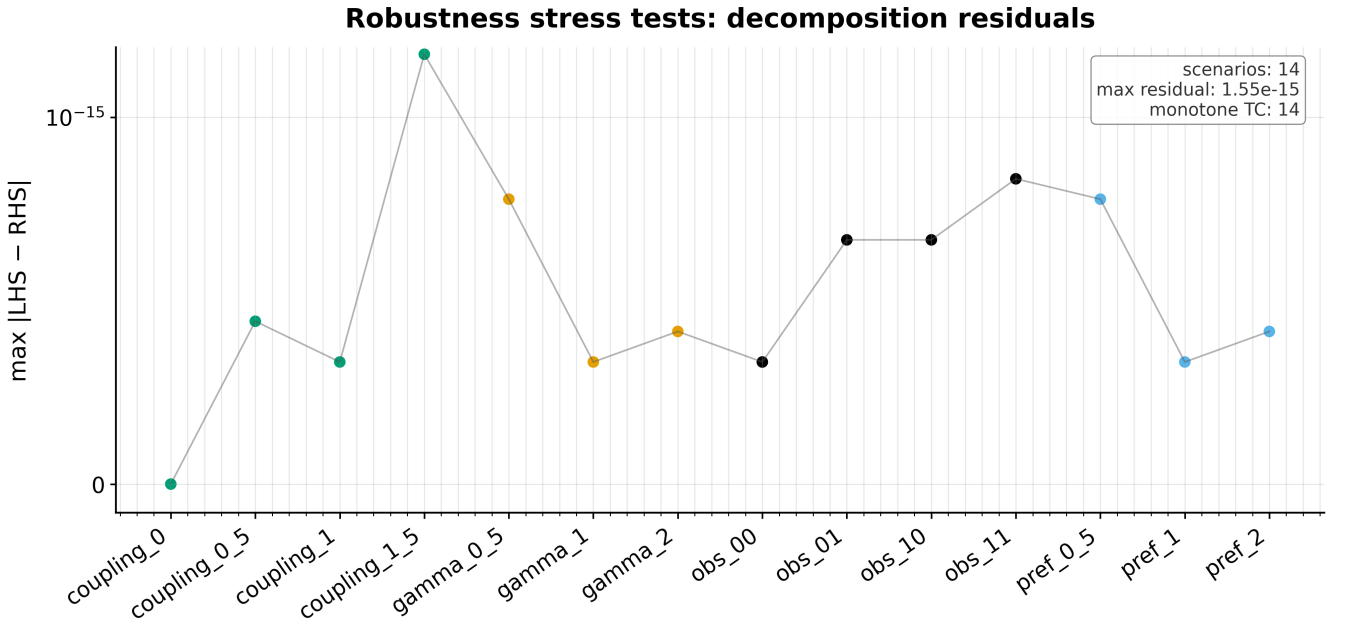


Figure 35: **The decomposition identity remains numerically closed across every robustness scenario.** Each point is a scenario-level maximum of $|F[q_\lambda] - \text{RHS}_{\text{decomp}}(q_\lambda)|$ over the configured robustness sweep. The worst residual is $1.55\text{e-}15$, below the validator tolerance $1\text{e-}09$. This figure links the stress suite back to the same numerical realization of [Theorem 5.1](#) used in the main pymdp free-energy bundle. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

dependence, and the decomposition residual remains closed ($5.55\text{e-}16$ maximum). What changes is the archetypal allocation: the aligned-mass shift across ablation variants at the sweep endpoint is 0.9993, which is a concrete way to see that the stress suite is not merely re-plotting the same posterior under four names.

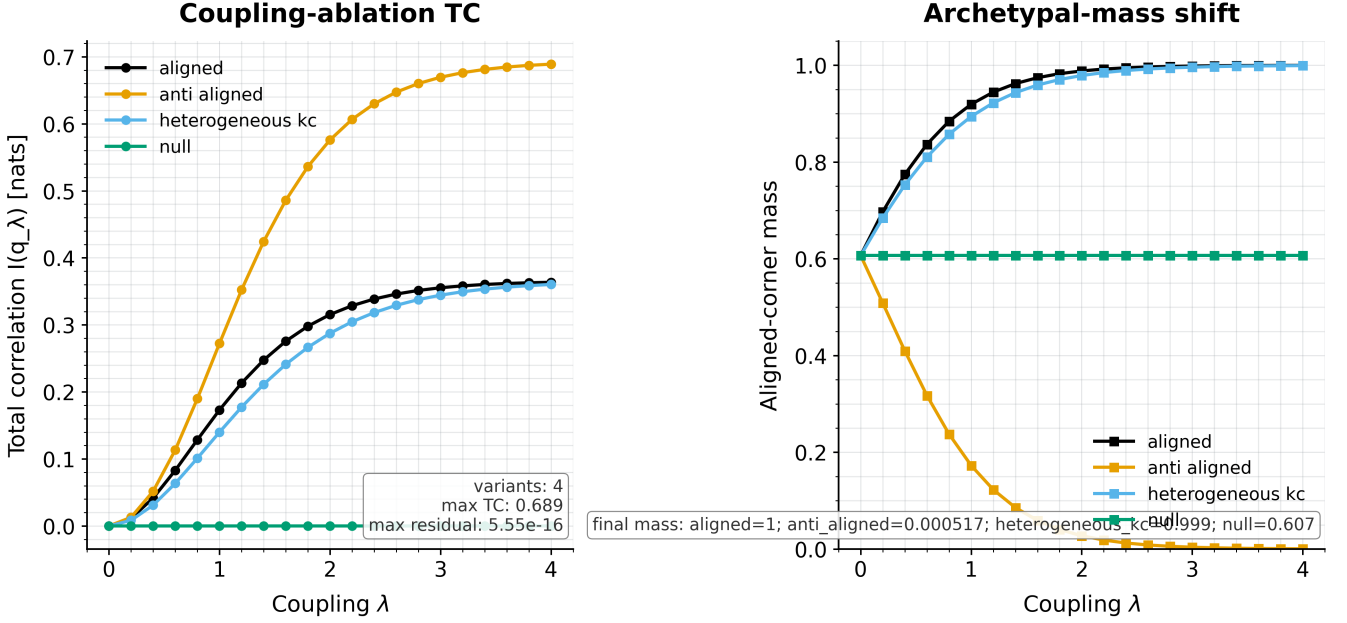


Figure 36: **Coupling ablations separate structural claims from a single sign choice.** The four variants are aligned, null, anti aligned, heterogeneous kc: aligned Ising coupling, null coupling, anti-aligned coupling, and a heterogeneous small-tax K_c matrix. The $\lambda = 0$ mean-field anchor survives every variant (max $I(q_0) = 0.00\text{e} + 00$); the null variant stays flat (max $I(q_\lambda) = 0.00\text{e} + 00$); and the non-null variants shift archetypal mass without breaking the decomposition residual bound ($5.55\text{e-}16$). Data: `output/simulations/pymdp_coupling_ablation.csv`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

The fixed-marginal null control then asks a sharper diagnostic question: if we preserve each stream’s posterior but replace the joint with the product of those marginals, does the dependence signal disappear? It does. Across the same robustness λ grid, the control’s maximum total correlation is $1.11\text{e-}16$, while the maximum amount of total correlation removed from the entangled joint is 0.3635. This is the negative-control counterpart to the positive coupling sweeps: the effect is carried by cross-stream joint structure, not by a hidden change in either stream’s marginal posterior.

The appendix interaction branch adds a deliberately narrow two-axis suite rather than a full Cartesian grid. It crosses observation context with coupling scale, precision γ with preference strength, and coupling variant with coupling scale — the three interactions most likely to change a reviewer-facing interpretation of the canonical sweep. Across 41 targeted scenarios and 861 rows, the worst decomposition residual is $1.67\text{e-}15$, and the null-variant flatline remains bounded by $0.00\text{e} + 00$. These results are reported as supporting stress evidence, not as new theorem claims.

Finally, the long-horizon robustness sidecar leaves the canonical fixed-seed rollout untouched and adds the configured replicate seeds $\{0, 7, 13, 29, 41\}$. The median/IQR envelope reports trajectory sensitivity without changing the main deterministic figure. Across those seeds, the habit-accumulation pass rate is 0.60 (Wilson 95% interval $[0.23, 0.88]$), the final total-correlation mean is 0.3675, and the maximum tail-window KL is 0.1851. The companion seed-diagnostics sidecar then records the per-seed failure mode and threshold sensitivity over $\{0.05, 0.1, 0.15, 0.2, 0.25\}$, making the observed pass rate a transparent sensitivity result rather than a tuned headline. The dedicated threshold-sensitivity figure reports the full pass-rate range (0.00 to 1.00) across the configured probes, so the canonical threshold is visible as one registered choice among the diagnostics rather than an implicit post-hoc cutoff.

Software and numerical stack

The reproducibility chain rests on `interactively-pymdp==1.0.1`, which provides the `pymdp` import/API [Heins et al., 2022] with its JAX backend [Bradbury et al., 2018] for per-stream POMDP inference, NumPy [Harris et al., 2020] and SciPy [Virtanen et al., 2020] for the dense joint-policy tensors and spectral decompositions of the analytical layer, matplotlib [Hunter, 2007] (Agg backend) for every figure, and pytest with a no-mocks policy for the test-infrastructure gate; Lean 4 [de Moura and Ullrich, 2021] hosts the type-checked boundary fragment under `lean/ActInfPolicyEntanglement/`, while Mathlib [Community, 2020] is reserved for the separate optional

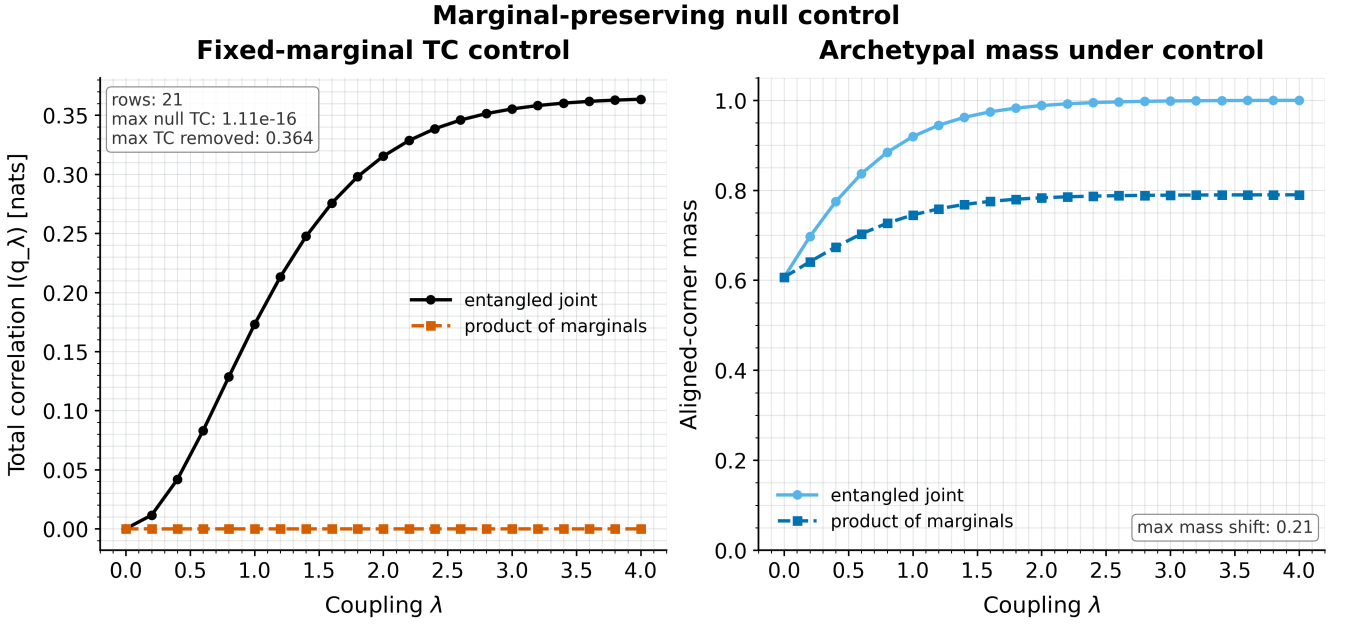


Figure 37: **A fixed-marginal null control removes the dependence signal while preserving each stream’s posterior.** For each configured robustness λ value, the control replaces the entangled joint posterior by the product of its own marginals. The null control’s maximum total correlation is 1.11e-16, while the maximum removed total correlation is 0.3635. The aligned archetypal-mass shift reaches 0.2099, showing that the coupling signal is not a univariate marginal artifact. Data: output/simulations/pymdp_marginal_null_control.csv; summary: output/data/marginal_null_control_summary.json. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

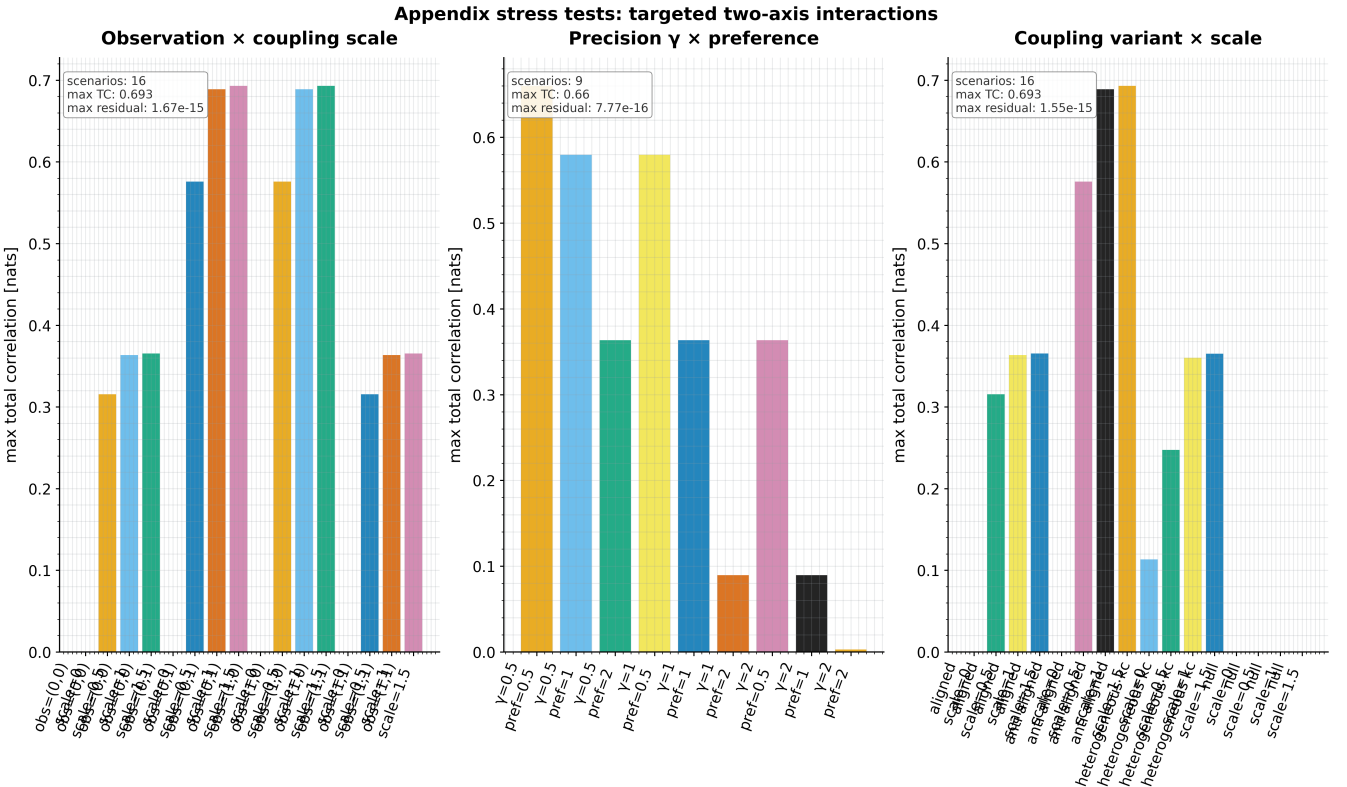


Figure 38: **Targeted two-axis stress tests expose interaction effects without expanding to a full Cartesian grid.** The appendix-only suite covers the configured families observation $\times$ coupling scale, gamma $\times$ preference strength, coupling variant $\times$ coupling scale for 41 scenarios and 861 total λ rows. The worst decomposition residual is 1.67e-15, while the null-variant interaction flatline has maximum $I(q_\lambda) = 0.00e + 00$. Data: output/simulations/pymdp_interaction_robustness.csv; summary: output/data/interaction_robustness_summary.json. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

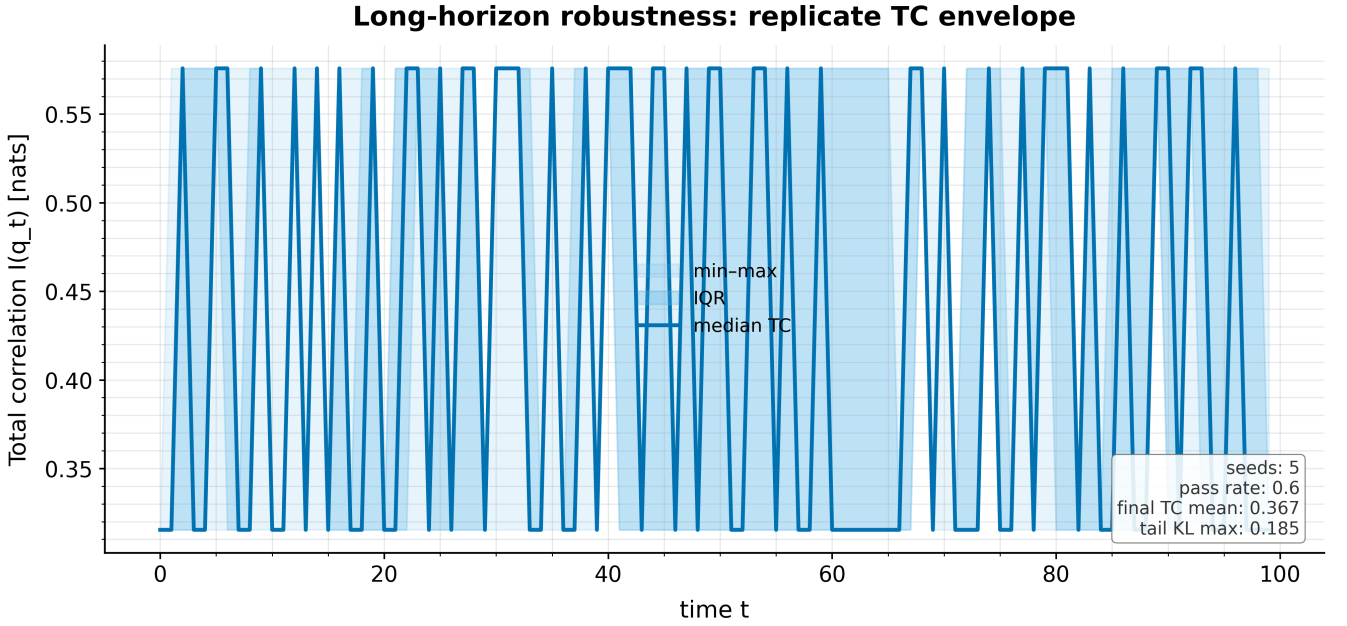


Figure 39: **Long-horizon replicate sidecars preserve the canonical figure while reporting seed sensitivity.** The main long-horizon figures remain the fixed-seed artifact; this sidecar adds the configured replicate seeds $\{0, 7, 13, 29, 41\}$ and plots the total-correlation median, interquartile band, and min-max envelope over the same $T = 100$ rollout. The habit-accumulation pass rate is 0.60, final total-correlation mean is 0.3675, and the maximum tail-window KL over seeds is 0.1851. Data: `output/simulations/pymdp_long_horizon_replicates.csv`; summary: `output/data/long_horizon_replicates_summary.json`. Uncertainty semantics: configured replicate-seed sidecar; envelope width reflects seed sensitivity.

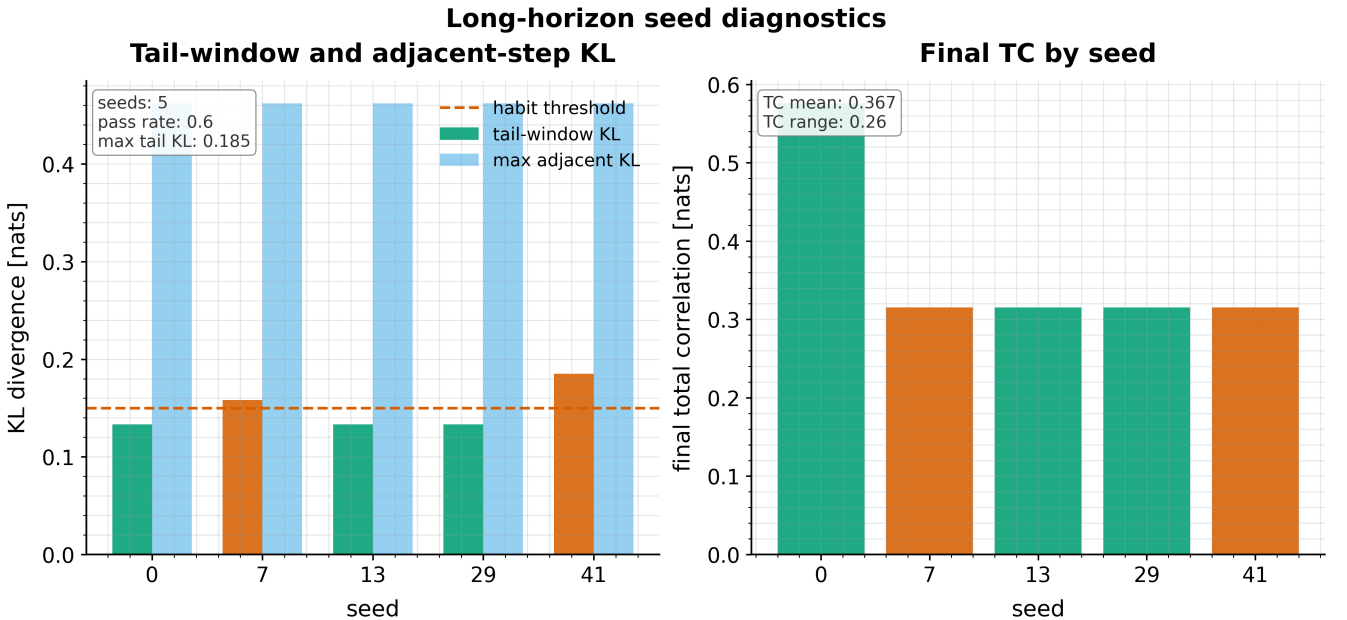


Figure 40: **Per-seed long-horizon diagnostics explain the replicate pass rate rather than tuning it away.** Bars show each configured seed's strict tail-window KL and maximum adjacent-step KL, with the canonical habit threshold 0.150 drawn as a dashed line. The threshold-sensitivity probes $\{0.05, 0.1, 0.15, 0.2, 0.25\}$ are reported in `output/data/long_horizon_replicates_summary.json`; the minimum margin to the canonical tolerance is -0.0351. Data: `output/simulations/pymdp_long_horizon_seed_diagnostics.csv`. Uncertainty semantics: configured replicate-seed sidecar; envelope width reflects seed sensitivity.

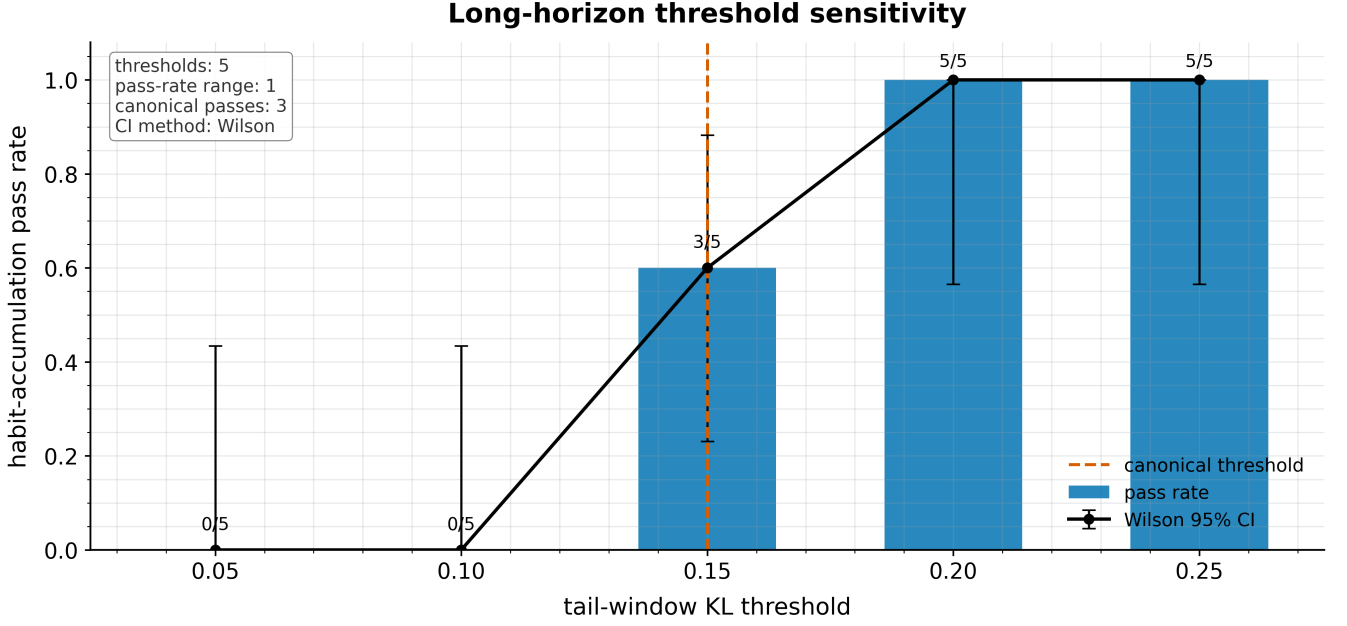


Figure 41: **Tail-window threshold sensitivity makes the long-horizon habit pass rate auditable rather than tuned.** Each bar reports the fraction of configured replicate seeds whose strict tail-window KL is below one diagnostic threshold in $\{0.05, 0.1, 0.15, 0.2, 0.25\}$. The dashed line marks the canonical threshold 0.150 used by the main habit-accumulation flag. Across the diagnostic thresholds the pass-rate range is 1.00 with Wilson 95% interval $[0.23, 0.88]$ around the canonical replicate pass rate 0.60 (min 0.00, max 1.00). Data: `output/simulations/pymdp_long_horizon_threshold_sensitivity.csv`; summary: `output/data/long_horizon_replicates_summary.json`. Uncertainty semantics: configured replicate sidecar with confidence interval reported from the generated summary.

`lean/MathlibProofs/` package. All pins are resolved by `uv` through `pyproject.toml`; the dependency list is deliberately minimal — losing JAX disables the pymdp grounding, losing Lean disables the formal-verification track, losing NumPy makes the analytical and empirical decompositions disagree on the seventh significant figure. Detailed version pins, optional-group boundaries (`sim`, `viz`), and resolution notes are listed in `docs/guides/build_run.md`.

Three reference tables collect the load-bearing per-module / per-statistic / per-JSONL-field metadata in §S7: Lean module inventory (§S7.3), pymdp bundle statistics (§S7.4), and JSONL run-log schema (§S7.5).

Head-to-head BTAI baseline and adversarial-perturbation harnesses

Two comparison harnesses originally scoped as follow-on work are now implemented, unit-tested, and run as pipeline stages emitting auditable sidecars.

Branching-Time AIF (BTAI) baseline [Champion et al., 2022] (`scripts/simulate_btai.py` → `output/data/btai_baseline.json`; `src/simulation/btai_baseline.py`; `tests/test_btai_baseline.py`). A Monte-Carlo-tree-search BTAI agent is driven by the project’s real 1.0.1 per-stream expected free energy and sweeps the configured budget grid $B_{\text{MCTS}} \in \{100, \dots, 10000\}$ (3 budgets), recording the per-step joint policy posterior, its total correlation, and the KL of the MCTS visitation posterior against the closed-form $K = 2$ Bernoulli reference at $\lambda = 1$. **Honest scope (no over-claim):** UCB-MCTS estimates the lowest-EFE *joint action* rather than the soft policy posterior, so the visitation posterior concentrates (total correlation ≈ 0.0019 nats at the largest budget) and the measured sample-complexity exponent of its KL against the soft reference is -0.114 . The worked run therefore *ships and exercises* the three tracked observables; the full compute-matched hypothesis test of §21 (whether BTAI attains the exact posterior as $B_{\text{MCTS}} \rightarrow \infty$ at a unit sample-complexity exponent) remains author-led analysis, not a claim made here.

Adversarial-perturbation sweep (§20 Q11; `scripts/simulate_adversarial.py` → `output/data/adversarial_sweep.json`; `src/simulation/adversarial.py`; `tests/test_adversarial.py`). The configured (ε, λ) -grid runs all three adversary classes (analytical worst-case rank-one, uniform-random, and sparse single-cell) over 105 scenarios (7ε values $\times 5 \lambda$ values $\times 3$ classes) on the $K = 2$ Ising task, comparing the measured KL drift $D_{\text{KL}}(q_\lambda \| q_\lambda^{J+\Delta J})$ against the first-order Lipschitz bound $\lambda \varepsilon \text{Var}_{q_\lambda}(J)^{1/2}$. The first-order bound holds for a fraction 0.943 of scenarios (empirical Lipschitz constant 0.964); it is loose at small ε and is exceeded (worst ratio 7.25) in the large- ε regime where the linearization saturates — a regime the sidecar surfaces rather than absorbs.

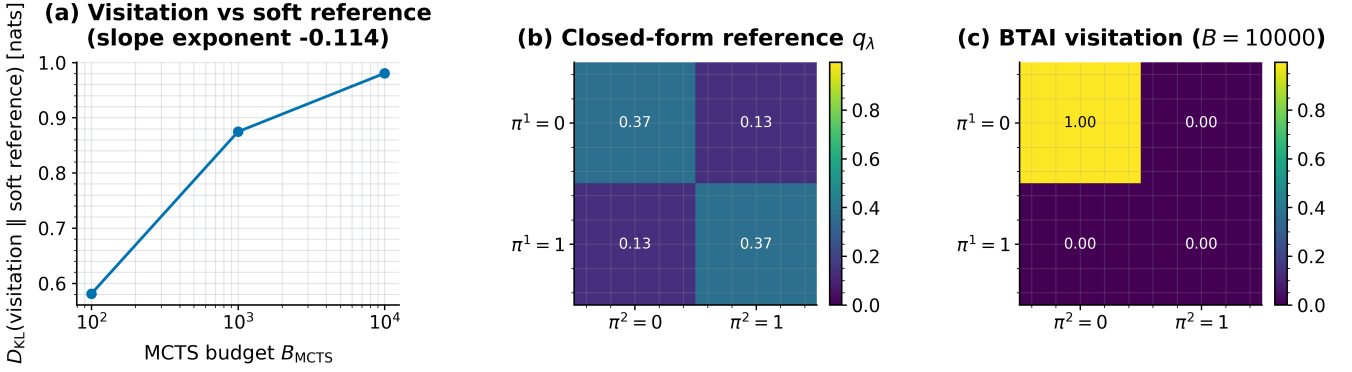


Figure 42: **The Branching-Time AIF baseline runs end-to-end and emits the three tracked observables; UCB-MCTS estimates the lowest-EFE joint action rather than the soft policy posterior.** (a) KL of the MCTS visitation posterior against the closed-form $K = 2$ Bernoulli reference at $\lambda = 1$ across the budget grid $B_{\text{MCTS}} \in \{100, \dots, 10000\}$ (3 budgets); the fitted slope exponent is -0.114 — the visitation KL does not shrink toward the soft reference because UCB concentrates. (b) the closed-form reference q_λ ; (c) the BTAI visitation posterior at the largest budget, whose total correlation ≈ 0.0019 nats is near zero (mass concentrated on one joint action). A reproducible worked run driven by real pymdp 1.0.1 per-stream expected free energy; the full compute-matched hypothesis test of §21 is author-led future analysis, not asserted here. Source: `scripts/simulate_btai.py`; data: `output/data/btai_baseline.json`. Uncertainty semantics: canonical fixed-seed trajectory; seed sensitivity is reported only where a replicate sidecar is registered.

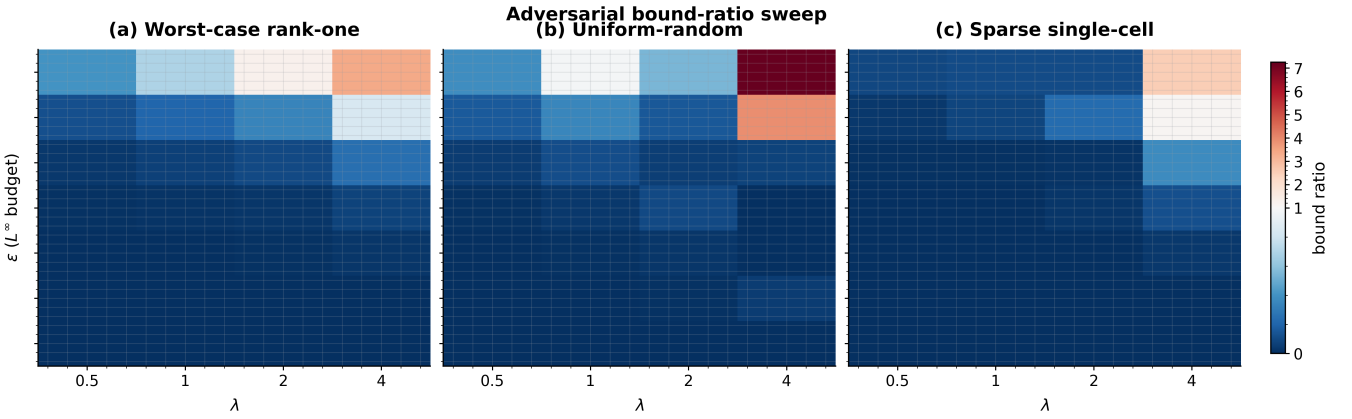


Figure 43: **Adversarial bound ratio across the (ϵ, λ) grid for three adversary classes.** Each panel shows the ratio of the measured KL drift $D_{\text{KL}}(q_\lambda \parallel q_\lambda^{J+\Delta J})$ to the first-order Lipschitz bound $\lambda \epsilon \text{Var}_{q_\lambda}(J)^{1/2}$ for the (a) worst-case rank-one, (b) uniform-random, and (c) sparse single-cell adversary, over 105 scenarios (7ϵ values $\times 5 \lambda$ values $\times 3$ classes; the degenerate $\lambda = 0$ column is omitted). Blue cells (ratio < 1) are where the first-order bound holds; red cells (ratio > 1) are where the linearization is exceeded in the large- ϵ regime. The bound holds for a fraction 0.943 of scenarios (empirical Lipschitz constant 0.964; worst ratio 7.25). Source: `scripts/simulate_adversarial.py`; data: `output/data/adversarial_sweep.json`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Anchored figure index

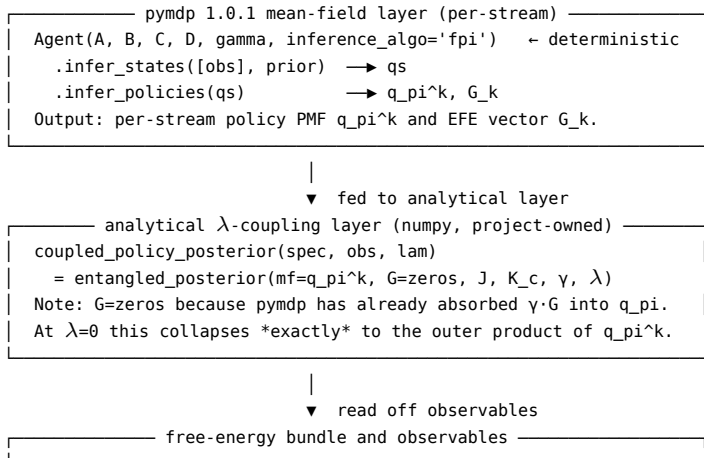
The empirical suite produces a family of cross-referenced figures distributed across the closed-form (§6), geometry (§7), spectral (§8), heterogeneous (§9), phase (§10), and comparative-statics (§11) sections, as well as the pymdp harness (§14), pymdp free-energy (§15), and validation (§16) sections downstream. Anchor index for the suite’s figures that are generated by this section’s analyses but cross-referenced from other body sections: Fig. 10 (Schmidt archetype dendrogram, also cross-referenced from §8.2), Fig. 3 ($O(\lambda^2)$ coupling-tax envelope, also cross-referenced from §9), Fig. 9 ($K = 2$ joint heatmap with marginals, also cross-referenced from §6.1), Fig. 29 (log-weight e -geodesic flow, also cross-referenced from §7), Fig. 16 (long-horizon steady-state KL convergence, also cross-referenced from §9.5), Fig. 13 (aligned mass concentration as K varies, also cross-referenced from §10), Fig. 5 (alignment-inversion $\lambda^*(\Delta_{\text{align}})$, also cross-referenced from §6.1), Fig. 4 (1-D phase diagram, also cross-referenced from §10), Fig. 6 (Schmidt rank vs λ , also cross-referenced from §8.1), and Fig. 11 (tensor-train rank surface, also cross-referenced from §8.3). Each figure is also independently anchored at its primary referencing subsection; this paragraph exists so the empirical-suite section is the *single* place a reader can locate every empirical figure regardless of which downstream section motivated its generation.

pymdp 1.0.1 POMDP Harness: Architecture, Lambda Sweep, and Deterministic Rollout

This sub-section documents the **architecture** of the pymdp grounding layer — how the manuscript’s analytical λ -coupling sits *on top of* a real `pymdp.agent.Agent` and what each piece contributes. Every POMDP symbol below ($A, B, C, D, \gamma, T, q_{\pi}^k$) is cataloged in the unified notation glossary (§S6) with its LaTeX, Python, and pymdp-keyword counterparts. The free-energy observables it produces are the subject of §15; the validation / logging contract that gates the entire chain is §16.

Architecture: layered, not duplicated

The pymdp harness wraps each stream of the $K = 2$ Ising toy as a separate `pymdp.agent.Agent` with a 2-state, 2-action POMDP — identity likelihood ($A = I$), hold/swap transitions (B), biased preference vector (C), and uniform initial prior (D); ensemble precision $\gamma = 1$, generative-model coupling $\lambda_{\text{gen}} = 1$.



The free-energy bundle readout is detailed in §15.

Adapter boundary. The harness does not patch pymdp internals and does not require pymdp to natively support structured joint policy posteriors. pymdp supplies each stream’s `Agent.infer_states` and `Agent.infer_policies` result; the project adapter then forms q_λ by applying the manuscript’s `entangled_posterior` construction to those per-stream policy posteriors. The validation gates therefore test two claims separately: the per-stream calls agree with the documented pymdp API, and the repository’s coupling layer preserves the $\lambda = 0$ mean-field baseline before adding cross-stream dependence.

Reproducibility checklist. The software package is inferactively-pymdp==1.0.1, installed through the project’s sim dependency group and cited through both the JOSS software paper and the official repository / documentation [Heins et al., 2022, pymdp developers, 2026a,b]. The POMDP state/action primitives are not prose-only: StreamSpec and CoupledEnsembleSpec validate the A , B , C , and D arrays before each Agent is built, and the

default Ising ensemble is constructed in `make_ising_ensemble`. The policy horizon and inference settings are exactly the hyperparameters mirrored into the manuscript: `policy_len=1`, `inference_algo="fpi"`, `num_iter=32`, `action_selection="deterministic"`, `alpha=16`, `use_states_info_gain=False`, and `use_param_info_gain=False`. The sweep uses observations `[0, 0]`, and the rollout uses seed `0`; all additional random draws are routed through `np.random.default_rng`. Output variables are the per- λ policy posterior, EFE vector, free-energy bundle, total correlation, action distribution, coupled rollout trajectory, JSONL run-log fields, and rendered PNG metadata. The single rerun command is:

```
uv run python scripts/simulate_pymdp.py
```

Three contracts are load-bearing:

1. **No double-counting.** `pymdp`'s `q_pi^k` already contains the per-stream $\gamma \cdot G_k$ softmax bias. The analytical layer therefore passes `zero_G` to `entangled_posterior` so the coupling potentials $\lambda \cdot J$ and $\gamma \cdot \lambda \cdot K_c$ are added *once*, not twice.
2. **$\lambda = 0$ sentinel.** At $\lambda = 0$ the *marginals* of the coupled joint posterior reproduce `pymdp`'s per-stream policy posteriors to floating tolerance ($1e-06$); the joint factorizes by construction (since $\exp(\lambda \cdot J) = 1$ pointwise). This marginal-agreement check is asserted by `tests/test_simulation_pymdp.py::test_coupled_policy_posterior_lambda_zero_is_outer_product`. The free-energy-bundle test `test_free_energy_bundle_lambda_zero_baseline` checks the stronger implications: total correlation $< 1e-07$, coupling term $< 1e-09$, and joint entropy equals the sum of marginal entropies to $1e-07$.
3. **Determinism.** The harness configures `pymdp 1.0.1` with `inference_algo="fpi"` for `infer_states` and `infer_policies`; there is no random sampling inside those calls, so the entire chain is reproducible under fixed (`spec`, `obs`, `lam`, `seed`).

Note on precision. `inferactively-pymdp==1.0.1` provides the `pymdp` import/API, whose JAX path uses `float32` for per-stream inference (`infer_states`, `infer_policies`). The harness recasts the resulting policy posteriors q_π^k and EFE values $G_k(\pi)$ to `float64` at the analytical-layer boundary in `src/simulation/inference.py`, so downstream observables (the entangled posterior, free-energy bundle, total correlation) are computed in double precision. Cross-platform numerical differences attributable to the JAX backend are bounded by $1e-06$ in single-stream observables; the numerical identities verified in §16 hold at the $1e-06$ tolerance.

Scope of the sub-`float32` gates (stated precisely). `pymdp`'s inference runs in JAX `float32` (machine $\varepsilon \approx 1.2 \times 10^{-7}$). The tight decomposition-residual and total-correlation-zero gates (at $10^{-9}/10^{-7}$) therefore do **not** certify `pymdp`'s `float32` numbers directly: they are evaluated on the recast-to-`float64` *analytical coupling layer* (the zeroed- G algebraic path), where the per-stream `float32` quantities enter only through the recast boundary above. These gates verify the analytical-layer algebra (the decomposition identity, $I \geq 0$, the $\lambda = 0$ baselines); `pymdp`'s `float32` contribution is bounded separately at the single-stream $1e-06$ tolerance, not at 10^{-9} . No claim is made that the raw `float32` `pymdp` pipeline satisfies the 10^{-9} identities.

The static λ -sweep

Sweep parameters: $\lambda \in [0, 4]$ on a 21-point grid at fixed observations $o = (0, 0)$. Hyperparameters are sourced from `src/simulation/hyperparameters.py` (`PYMDP_SWEEP_LAMBDA_S`, `PYMDP_SWEEP_OBSERVATIONS`).

The artifact is a 21-row CSV at `output/simulations/pymdp_lambda_sweep.csv`; the visual rendering is the total-correlation curve plus three sentinel- λ heatmap snapshots.

The four sentinel total correlations along the sweep — auto-injected from `scripts/manuscript_variables.py` on every render — are:

λ	$I(q_\lambda)$	Interpretation
0	0.000000 nats	Mean-field baseline (exact zero by construction)
1	0.1730 nats	Sub-saturation: coupling gain growing rapidly
2	0.3154 nats	Past half-saturation $\lambda_{1/2} \approx 1.044$
4	0.3635 nats	Approaches sweep maximum
		$I_{\max} \approx 0.3635$

The λ -column labels above are the sentinel values $\{0, 1, 2, 4\}$ exposed by `PYMDP_TOTAL_CORRELATION_SENTINEL_LAMBDA_S` in `src/simulation/hyperparameters.py`; changes to that tuple require regenerating this table. (The full 12-statistic table is in §S7.4 of the supplement.)

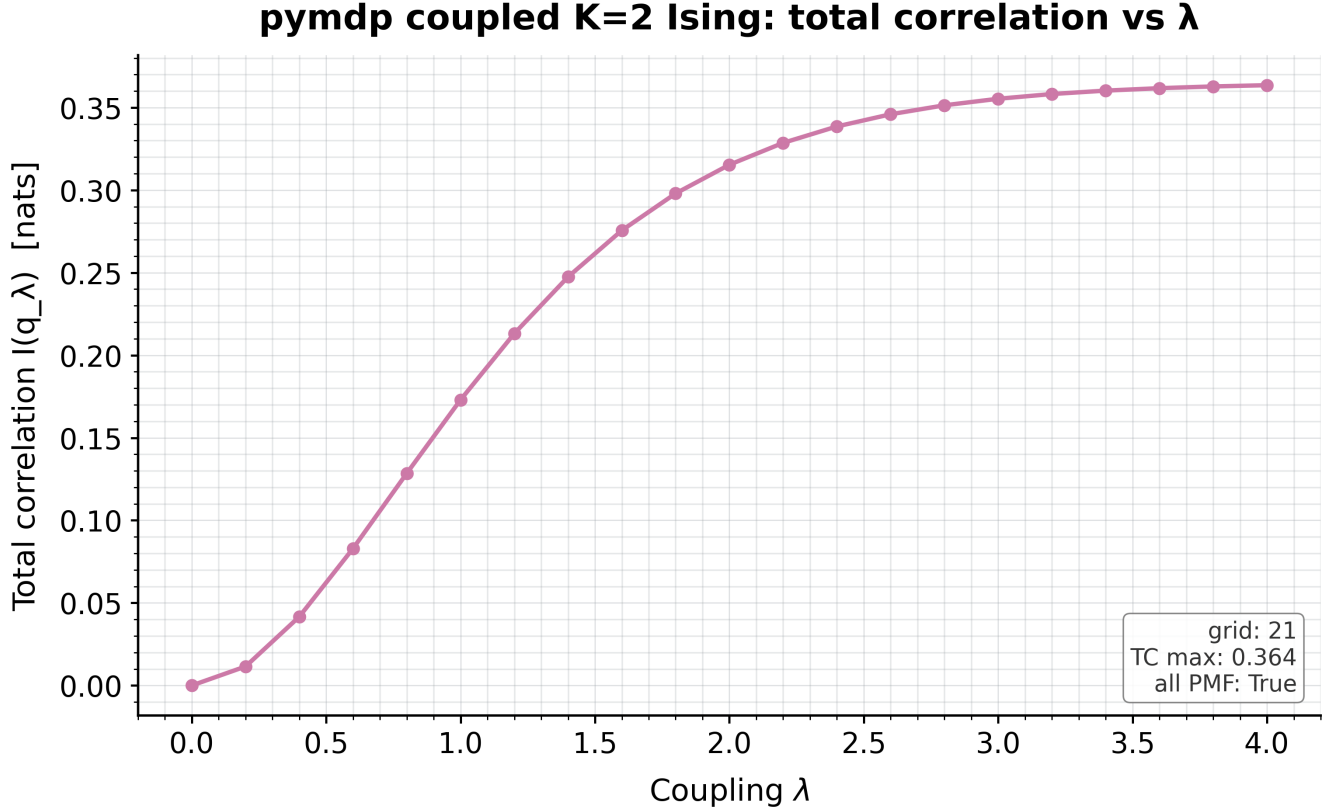


Figure 44: pymdp-grounded total correlation $I(q_\lambda)$ from `simulation.inference.coupled_policy_posterior` $\rightarrow$ `lean.free_energy.total_correlation`, evaluated on a 21-point linspace over $\lambda \in [0, 4]$ (`simulation.hyperparameters.PYMDP_SWEEP_LAMBDA`s) for the $K = 2$ Ising ensemble at observations $(0, 0)$ and ensemble precision $\gamma = 1$. The CSV at `output/simulations/pymdp_lambda_sweep.csv` is the source data. Smooth monotone increase from the mean-field baseline 0 toward saturation at $\log 2 \approx 0.693$ — the empirical witness of [Proposition 7.3](#) inside a grounded POMDP. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

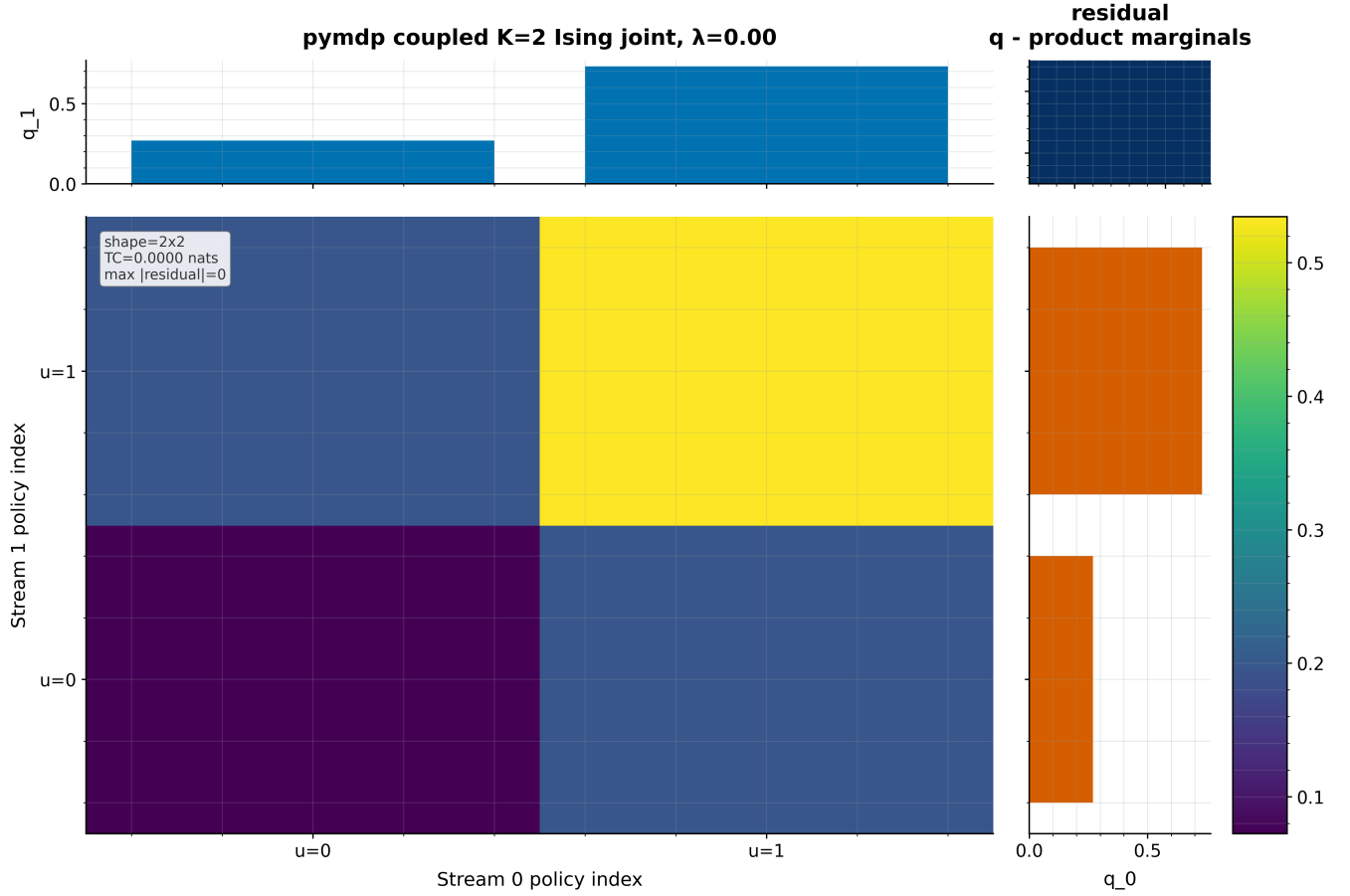


Figure 45: Coupled joint policy posterior at $\lambda = 0$ (pure mean-field sentinel from `simulation.hyperparameters.PYMDP_SWEEP_LAMBDAS`). The joint is `simulation.inference.coupled_policy_posterior(spec, [0,0], lam=0.0)` rendered with `visualizations.joint_plots.plot_joint_heatmap_with_marginals`. Marginals (top / right bars) match the per-stream pymdp posteriors and the m-projection residual (upper-right inset) is identically zero — exactly the outer-product baseline. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

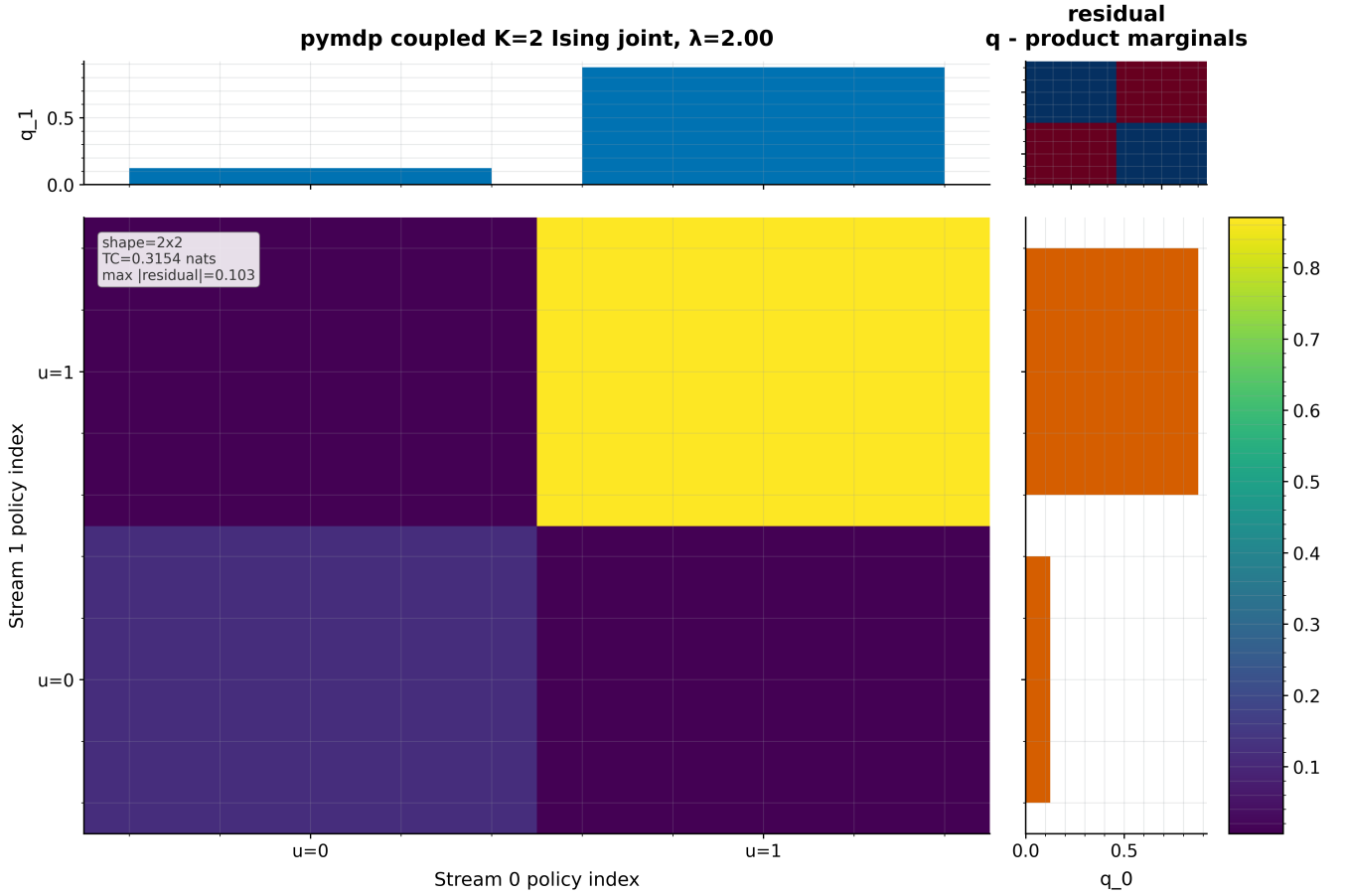


Figure 46: Coupled joint at $\lambda = 2$ (mid-grid sentinel of `simulation.hyperparameters.PYMDP_SWEEP_LAMBDAS`). The joint is `simulation.inference.coupled_policy_posterior(spec, [0,0], lam=2.0)`. Off-diagonal mass redistributes onto aligned $(0,0)/(1,1)$ corners; the m-projection residual is non-zero, confirming Schmidt rank $r \geq 2$ (strictly greater than the mean-field rank-1). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

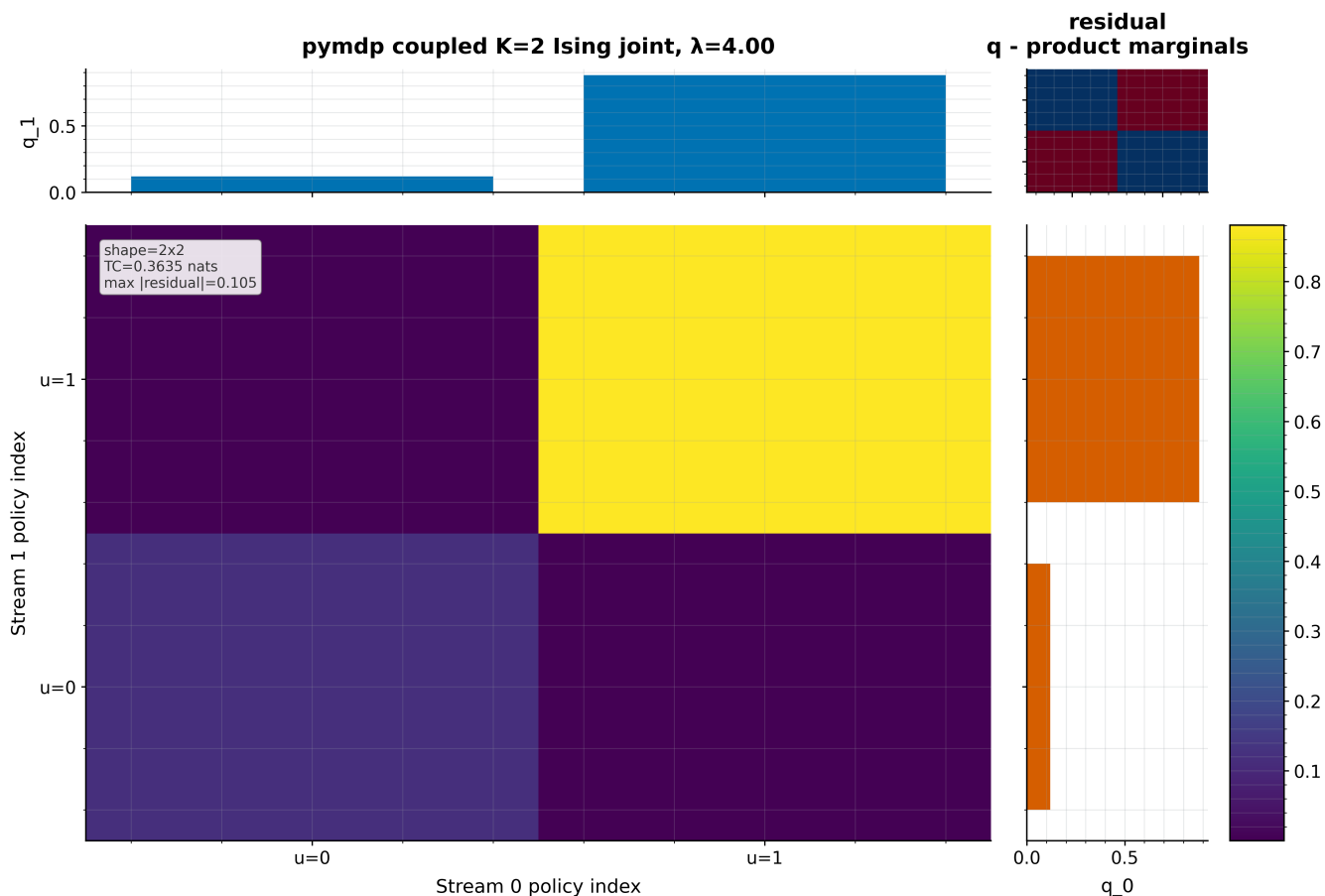


Figure 47: Coupled joint at $\lambda = 4$ (high-end sentinel of `simulation.hyperparameters.PYMDP_SWEEP_LAMBDAS`, frozen regime). The joint is `simulation.inference.coupled_policy_posterior(spec, [0,0], lam=4.0)`. Aligned corners dominate; the m-projection residual is large; the system has essentially collapsed onto the two aligned-corner archetypal modes (0,0) and (1,1) that share mass under the symmetric Ising coupling. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

Deterministic coupled rollout

A second probe of the harness: a $T = 10$ -step coupled rollout under coupling $\lambda = 2$ and seed 0. At each step the harness:

1. Lets each stream see its observation and runs pymdp’s per-stream inference (`per_stream_policy_posterior`).
2. Builds the λ -coupled joint via the analytical layer (`coupled_policy_posterior`).
3. Samples one joint action from the coupled posterior with a seeded `np.random.default_rng`.
4. Advances each stream’s hidden state under the sampled action’s **B** matrix and samples the next observation.

Numerically identical reproducibility of the sampled action sequence and hidden-state trajectory under fixed seed is asserted by `tests/test_simulation_pymdp.py::test_simulate_coupled_rollout_deterministic_under_fixed_seed`.

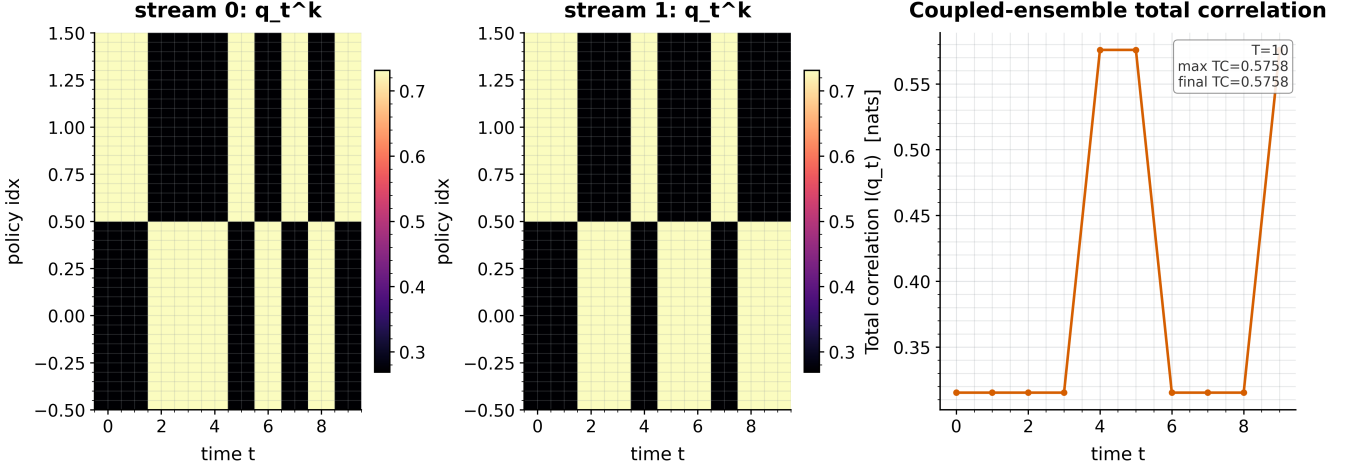


Figure 48: Deterministic $T = 10$ -step coupled rollout from `simulation.rollout.simulate_coupled_rollout` under $\lambda = 2$ and seed 0 (`simulation.hyperparameters.PYMDP_ROLLOUT_STEPS`, `PYMDP_ROLLOUT_LAMBDA`, `PYMDP_ROLLOUT_SEED`). Per-stream marginals q_t^k from `simulation.inference.per_stream_policy_posterior` (left two heatmaps) interleave with the coupled total correlation $I(q_t)$ from `lean.free_energy.total_correlation` (right). Bit-identical reproducibility under fixed seed is asserted by `tests/test_simulation_pymdp.py::test_simulate_coupled_rollout_deterministic_under_fixed_seed`. Uncertainty semantics: canonical fixed-seed trajectory; seed sensitivity is reported only where a replicate sidecar is registered.

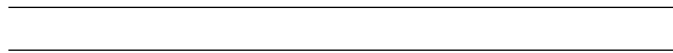
Reproduce.

```
uv run python scripts/simulate_pymdp.py
```

Emits the CSV + figures referenced above as well as the free-energy bundle artifacts of §15 and the structured run-log records of §16.

Anchored figure index for the pymdp harness

The harness emits a coupled-rollout trajectory plot and two sentinel joint-policy heatmaps that anchor the deterministic rollout discipline of this section: Fig. 44 (the coupled-rollout trajectory), Fig. 33 (joint policy at $\lambda = 2$), Fig. 34 (joint policy at $\lambda = 4$), and Fig. 31 (total correlation curve over the sweep). Each is also embedded above at its primary generation point; this paragraph exists so the harness section is the canonical anchor index for them.



pymdp Free-Energy Bundle Observables and Auto-Injected Summary Statistics

For every λ in the 21-point pymdp sweep (§14) we read off a complete *free-energy bundle* — every observable that downstream prose, figures, or tests might want — via `simulation.inference.free_energy_curve`. The bundle is a frozen dataclass. Its fields are deliberately named after the mathematical observable they expose, so manuscript prose, tests, and figures can all point at the same values:

- `vfe_per_stream[k]`: $F[q_\lambda^k] = \langle \log q^k - \log E_k + \gamma G_k \rangle_{q_\lambda^k}$, in nats.

- `vfe_total`: $\sum_k F[q_\lambda^k]$, the sum of streamwise VFE.
- `efe_per_stream[k]`: pymdp’s per-policy G_k vector for stream k .
- `efe_under_posterior[k]`: $\langle G_k \rangle_{q_\lambda^k}$, the posterior-weighted EFE for stream k .
- `joint_entropy` and `marginal_entropies[k]`: $H(q_\lambda)$ and $H(q_\lambda^k)$, in nats.
- `total_correlation`: $I(q_\lambda) = \sum_k H(q_\lambda^k) - H(q_\lambda)$.
- `coupling_term`: $\lambda \langle J \rangle_{q_\lambda} = \lambda \sum_\pi q_\lambda(\pi) J(\pi)$.
- `decomposition_lhs`, `decomposition_rhs`, and `decomposition_residual`: the positive- λ [Theorem 5.1](#) numerical witness computed with `free_energy_against_entangled_prior` and `entanglement_decomposition_rhs(...).total`. The witness passes zero per-stream G vectors because pymdp has already absorbed EFE into the per-stream policy posterior before the project adds cross-stream coupling.
- `action_distribution`: the flattened PMF q_λ on $\prod_k \Pi^k$.

Every scalar is also written to `output/simulations/pymdp_free_energy_bundle.csv` (one row per λ ; columns as in the table above for $K = 2$ streams).

Statistical contracts

The bundle satisfies six numerical invariants verified by `tests/test_simulation_free_energy.py`:

- **Sub-additivity**: $H(q_\lambda) \leq \sum_k H(q_\lambda^k)$ at every λ .
- **Total-correlation positivity**: $I(q_\lambda) \geq 0$ everywhere (float-noise floor $-1e-09$).
- $\lambda = 0$ **baseline**: $I(q_0) = 0$, coupling term = 0, $H(q_0) = \sum_k H(q_0^k)$, and `vfe_total` = $\sum_k F[q_0^k]$.
- **Monotonicity**: under the symmetric Ising coupling at observations $(0, 0)$, $I(q_\lambda)$ is non-decreasing in λ — the empirical witness of [Proposition 7.3](#) inside a real POMDP.
- **Bundle / helper consistency**: the bundle’s `efe_under_posterior` and `coupling_term` agree pointwise with the standalone `expected_free_energy_under_posterior` and `coupling_energy` helpers respectively.
- **Positive- λ decomposition witness**: the full sweep’s maximum residual $|\text{LHS} - \text{RHS}|$ is $3.33e-16$ nats, below the configured tolerance $1e-09$.

Summary statistics across the sweep

The bundle is reduced to a flat summary record by `simulation.statistics.pymdp_summary_statistics`, mirrored to `output/simulations/pymdp_summary.json`, and merged into `manuscript_variables.json` so every auto-injected value flows from a real pipeline run. The full 12-statistic reference table (sweep grid, total-correlation range and half-saturation point, VFE extrema, coupling-term range, entropy bounds, aligned-corner mass, KL divergences, action entropy, mode probability) is in [§S7.4](#) (supplement).

Every value in [§S7.4](#) is computed once per pipeline run and re-injected on the next render — no number there is typed by hand.

Visualizations

Nine dashboards are emitted by `scripts/simulate_pymdp.py::figure_pymdp_free_energies`. The first is a six-panel summary, the next four magnify each free-energy pillar, and the final four expose the additional auto-injected statistics (action entropy, KL to mean-field baseline, per-stream marginal entropy). Decomposition plots report **positive** $I(q_\lambda)$ (multi-information) next to $\sum_k F[q_\lambda^k]$ and the coupling observable $\lambda \langle J \rangle_{q_\lambda}$ — consistent with [Theorem 5.1](#)’s explicit $\{+\}I$ term — while acknowledging that `FreeEnergyBundle` alone omits priors/ $\log Z_E(\lambda)$ bookkeeping from [§5](#), so stacking only those three curves is illustrative rather than claiming identity with global $F[q_\lambda]$ absent the remaining terms.

Every PNG carries reproducibility tEXt with the source figure, hyperparameter snapshot, git revision, and ISO timestamp — readable via `visualizations.metadata.read_figure_metadata`.

Theorem 5.1 Numerical Witness

The manuscript’s load-bearing identity [Eq. \(5.2\)](#)

is *not* asserted symbolically over `Float` (the boundary Lean fragment is type-only). It is instead numerically witnessed at every λ in the sweep. The test suite first verifies the $\lambda = 0$ sanity collapse, where coupling and $I(q_\lambda)$ vanish; then `test_pymdp_decomposition_witness_holds_across_positive_lambda_grid` checks the full positive- λ sweep by comparing `free_energy_against_entangled_prior` to `entanglement_decomposition_rhs(...).total`. This is the same bookkeeping used by the analytical companion, with one pymdp-specific correction: the per-stream G vectors are zeroed in the decomposition call because pymdp has already baked EFE into the per-stream posteriors

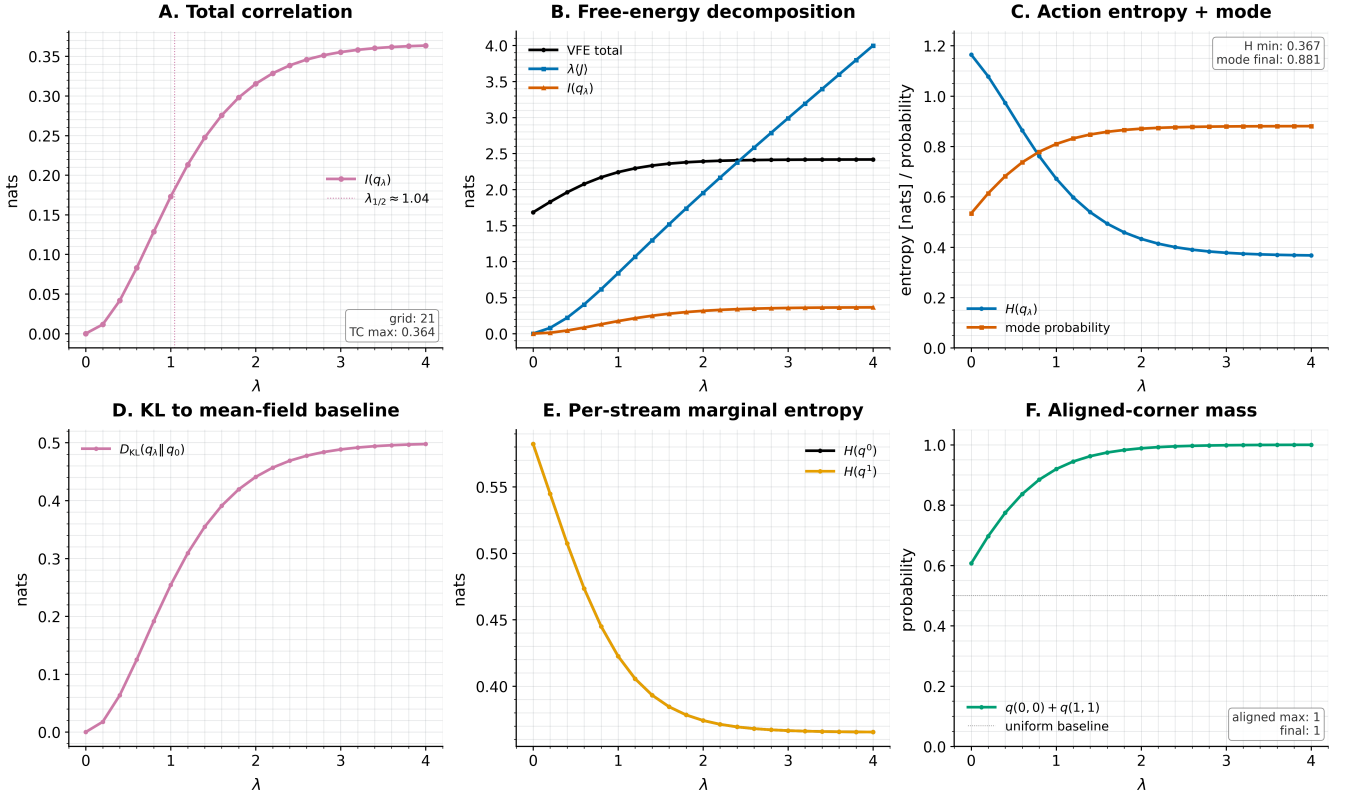


Figure 49: Six-panel pymdp summary dashboard derived from `simulation.inference.free_energy_curve` and `simulation.statistics.pymdp_summary_statistics`. Panel A: total correlation $I(q_\lambda)$ with the half-saturation $\lambda_{1/2}$ marker. Panel B: free-energy decomposition (total VFE, coupling term, $I(q_\lambda)$). Panel C: action-distribution Shannon entropy $H(q_\lambda)$ + mode probability $\max_\pi q_\lambda(\pi)$ (twin axes). Panel D: KL divergence $D_{\text{KL}}(q_\lambda \| q_0)$ from the mean-field anchor. Panel E: per-stream marginal entropy $H(q_\lambda^k)$. Panel F: aligned-corner mass $q(0,0) + q(1,1)$ relative to the uniform baseline. Every panel reads off the same bundle CSV at `output/simulations/pymdp_free_energy_bundle.csv`. PNG carries reproducibility tEXt (source function, hyperparameter snapshot, git revision, ISO timestamp) per `visualizations.metadata.figure_metadata`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

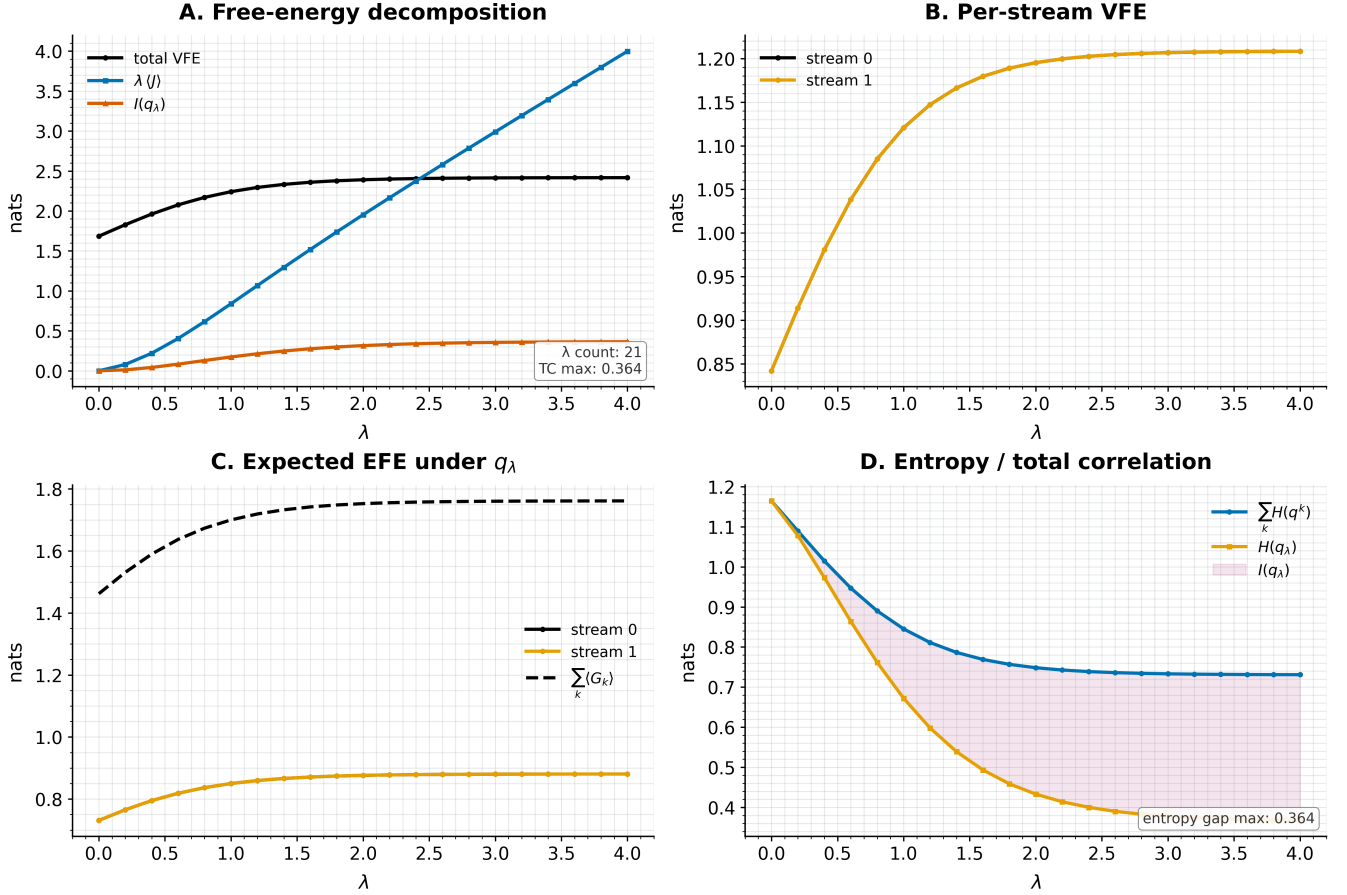


Figure 50: Four-panel pymdp free-energy dashboard over the 21-point linspace on $\lambda \in [0, 4]$ (`simulation.hyperparameters.PYMDP_SWEEP_LAMBDAS`) for the $K = 2$ Ising ensemble at observations $(0, 0)$. Panel A: total VFE, coupling term $\lambda \langle J \rangle_{q_\lambda}$, and multi-information $I(q_\lambda)$ (same quantity as the entropy gap in panel D). Panel B: per-stream VFE $F[q_\lambda^k]$. Panel C: expected EFE under posterior $\langle G_k \rangle_{q_\lambda^k}$ per stream plus their sum. Panel D: joint vs sum-of-marginal entropy with the gap shaded as the total correlation $I(q_\lambda)$. Every curve is computed by `simulation.inference.free_energy_curve(spec, [0, 0], lams)` and reads off the `output/simulations/pymdp_free-energy_bundle.csv` artifact. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

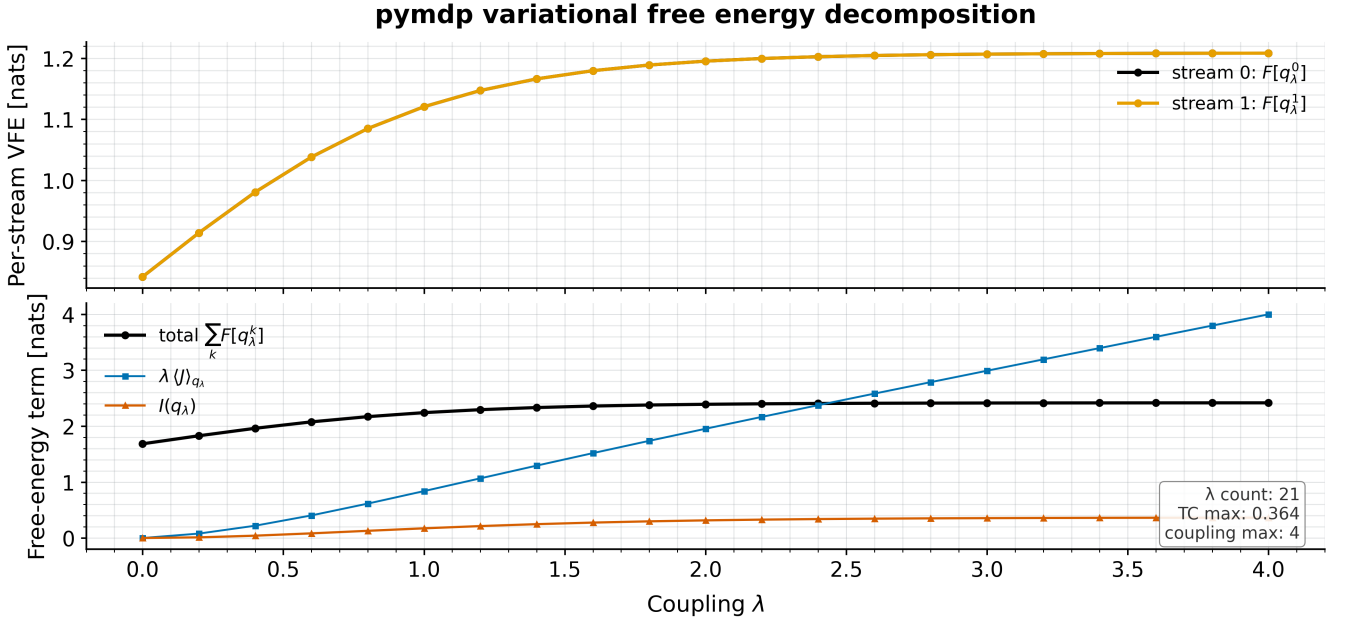


Figure 51: pymdp variational free energy decomposed across the 21-point linspace on $\lambda \in [0, 4]$. Top: per-stream $F[q_\lambda^k]$ (one line per stream). Bottom: total VFE next to the coupling term $\lambda \langle J \rangle_{q_\lambda}$ and $I(q_\lambda)$ — the three quantities whose interplay mirrors the entanglement decomposition theorem ([Theorem 5.1](#)) inside a real POMDP run. Computation: `simulation.inference.free_energy_curve`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

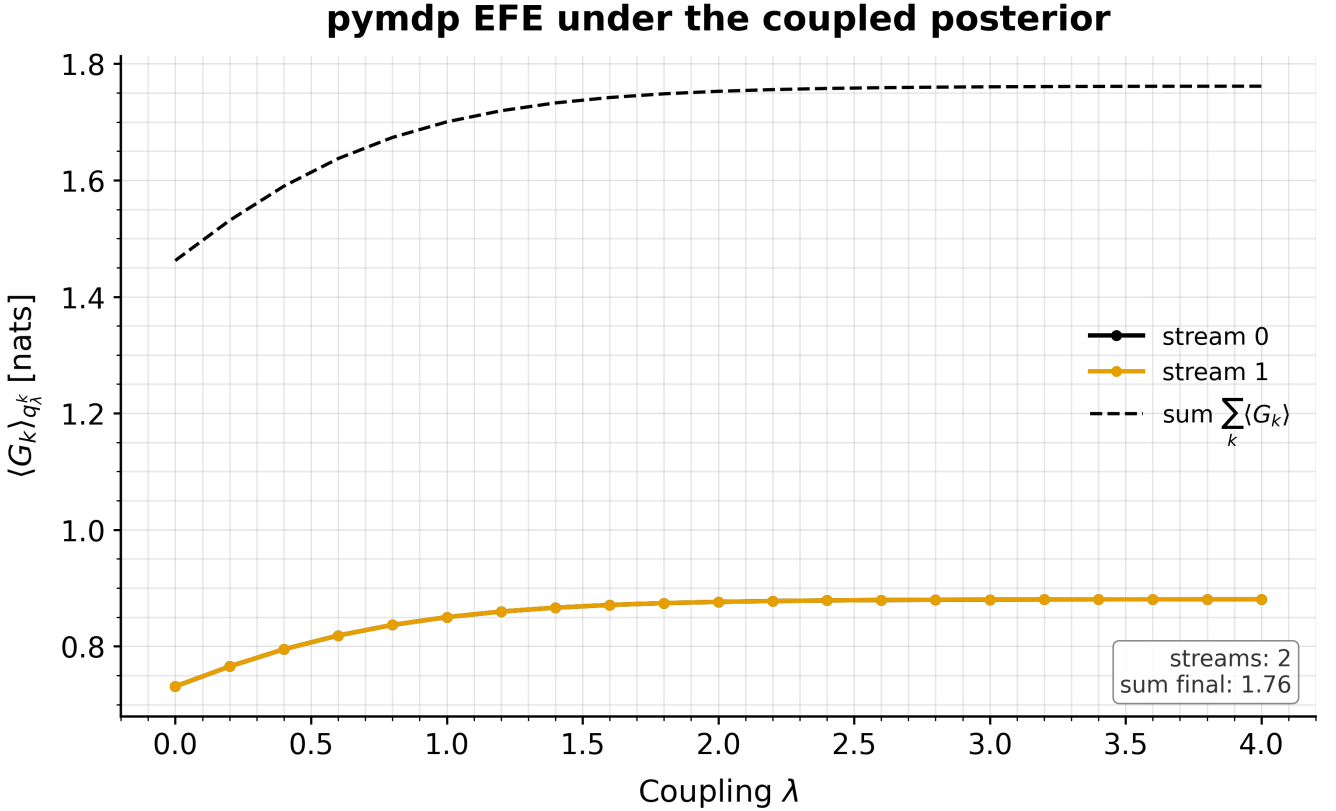


Figure 52: **Coupling shifts the expected EFE from its mean-field value at $\lambda = 0$ toward the EFE of the dominant archetype.** Per-stream expected EFE $\langle G_k \rangle_{q_\lambda^k}$ as λ sweeps the `simulation.hyperparameters.PYMDP_SWEEP_LAMBDA` grid. Each G_k is pymdp’s own EFE vector from `Agent.infer_policies`, averaged against the coupled marginal q_λ^k via `simulation.inference.expected_free_energy_under_posterior`; the dashed line is the cross-stream sum. Companion field on `FreeEnergyBundle.efc_under_posterior` (see [§S7.4](#)). Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

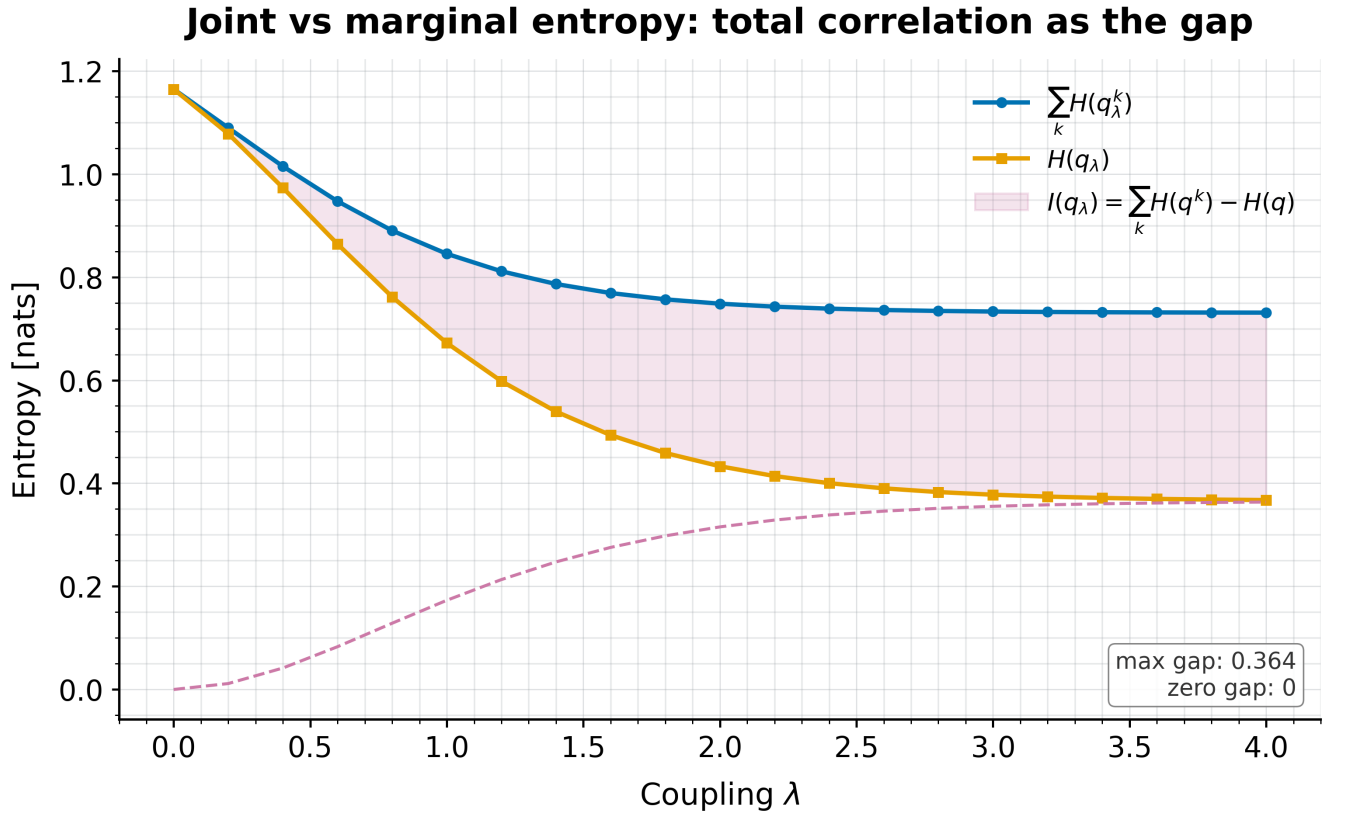


Figure 53: Joint entropy $H(q_\lambda)$ vs sum-of-marginal entropy $\sum_k H(q_\lambda^k)$ across the 21-point pymdp sweep on $\lambda \in [0, 4]$; the gap (shaded) is the total correlation $I(q_\lambda) = \sum_k H(q_\lambda^k) - H(q_\lambda)$. At $\lambda = 0$ the two curves meet — the visual signature of the mean-field submanifold. Computation: `lean.free_energy.shannon_entropy` composed with `lean.joint_dist.joint_marginals` on the `simulation.inference.coupled_policy_posterior` output. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

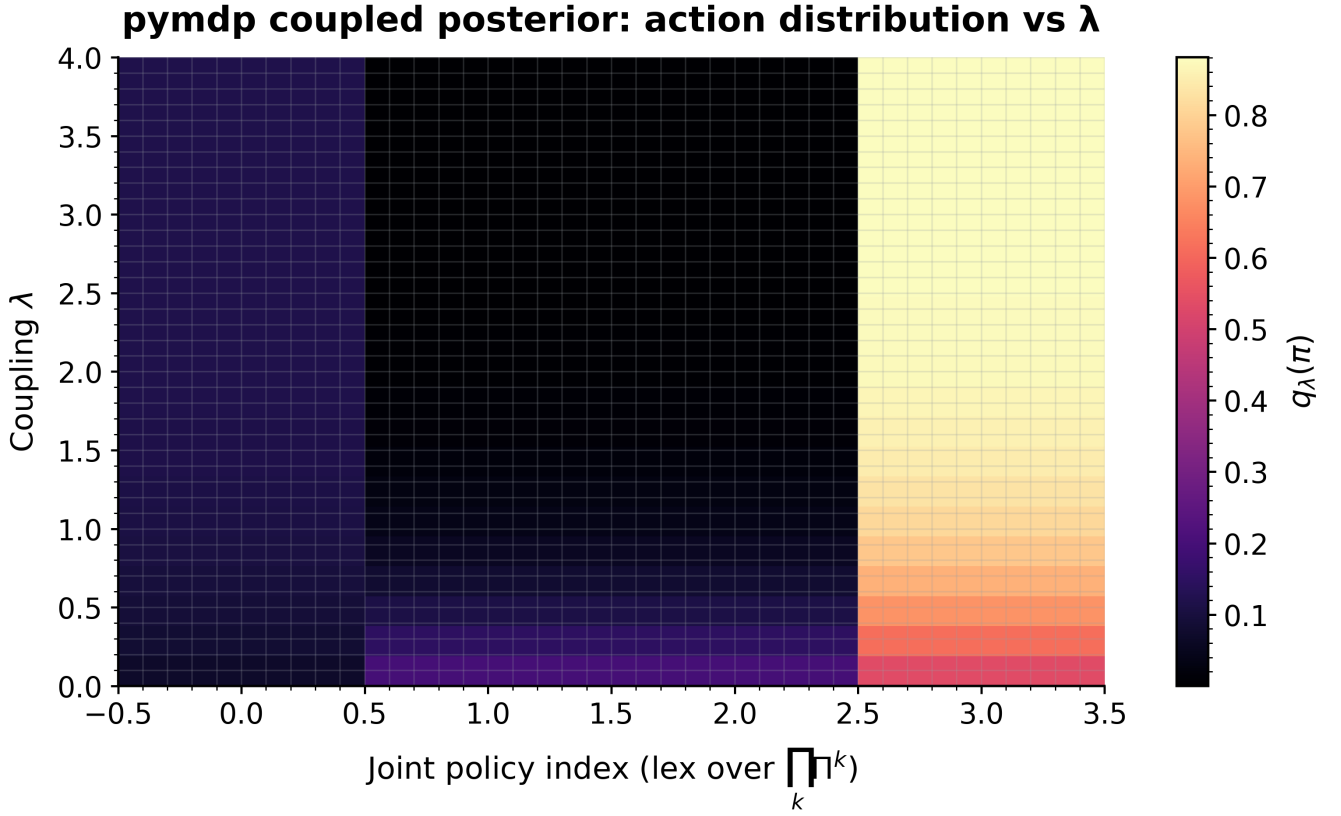


Figure 54: Heatmap of the joint action distribution $q_\lambda(\pi)$ over the 21-point sweep on $\lambda \in [0, 4]$ (rows) for every joint policy index $\pi \in \prod_k \Pi^k$ (columns, lex order). Mass concentrates onto the aligned diagonal as λ grows, visualizing the symmetry-breaking transition predicted by §10. Computation: `simulation.inference.coupled_policy_posterior(spec, [0,0], lam).reshape(-1)` per row. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

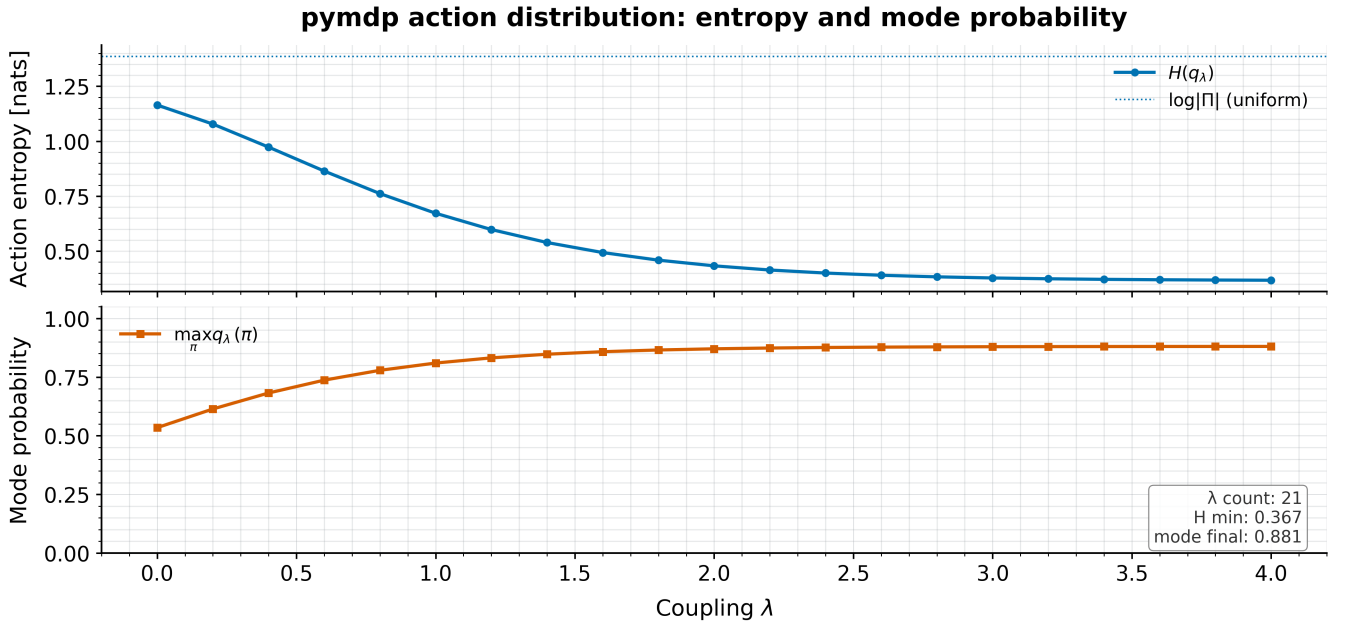


Figure 55: Joint-action Shannon entropy $H(q_\lambda)$ (left axis, blue) and mode probability $\max_\pi q_\lambda(\pi)$ (right axis, red) across the 21-point pymdp sweep on $\lambda \in [0, 4]$. As coupling grows, H falls toward 0 and the mode probability rises toward 1 — the visual signature of collapse onto a single archetypal mode. Computation: `visualizations.pymdp_extras.plot_action_entropy_curve`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

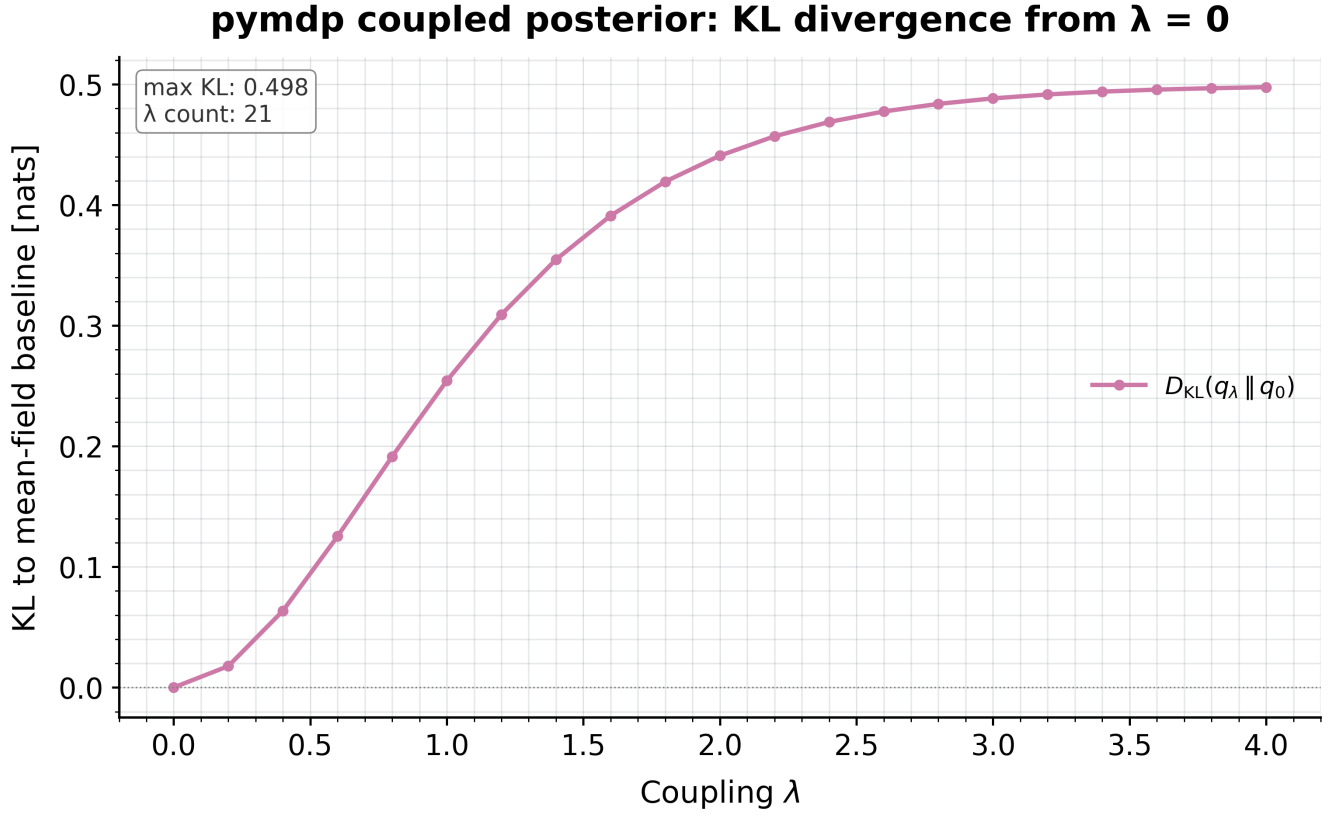


Figure 56: KL divergence $D_{\text{KL}}(q_{\lambda} \parallel q_0)$ of the coupled action distribution back to the mean-field baseline at $\lambda = 0$, across the 21-point pymdp sweep. A direct measure of how far the coupled posterior has *moved* from the mean-field anchor — complementary to the total-correlation curve. At $\lambda = 0$ the divergence is identically zero by construction. Computation: `visualizations.pymdp_extras.plot_kl_to_lambda_zero`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

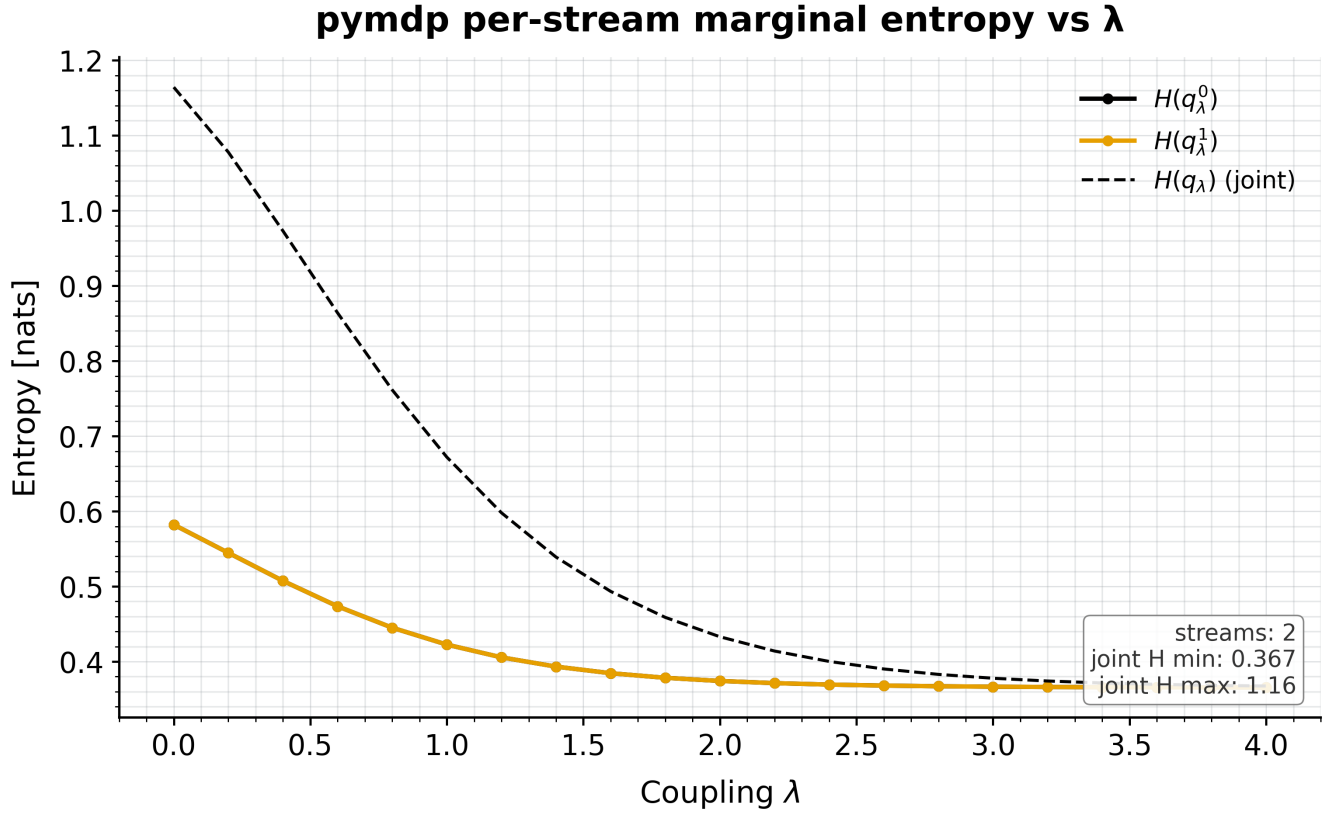


Figure 57: Per-stream marginal entropy $H(q_\lambda^k)$ for each $k \in \{0, \dots, K-1\}$ across the 21-point pymdp sweep. The dashed black line is the joint entropy $H(q_\lambda)$ for reference; the *gap* between the sum of marginal entropies and the joint is the total correlation $I(q_\lambda)$ shown separately in Fig. 38. Computation: `visualizations.pymdp_extras.plot_marginal_entropy_per_stream`. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

that serve as the mean-field base. The resulting maximum residual is $3.33e - 16$, and `scripts/validate_outputs.py` fails if any row exceeds $1e - 09$.

Anchored figure index for the pymdp free-energy bundle

The free-energy bundle of this section produces the following sweep-resolved dashboards, each generated by `scripts/simulate_pymdp.py` and validated against `pymdp_summary.json`: [Fig. 39](#) (joint action distribution evolution across the λ sweep), [Fig. 41](#) (action entropy + mode probability across λ), [Fig. 37](#) (expected free energy under the coupled posterior), [Fig. 38](#) (joint vs marginal entropy decomposition), [Fig. 35](#) (four-panel free-energy dashboard), [Fig. 42](#) (KL to the $\lambda = 0$ mean-field baseline), [Fig. 43](#) (per-stream marginal entropy), and [Fig. 40](#) (six-panel summary dashboard). Each is embedded above at its primary generation point; this paragraph provides the canonical anchor index for the bundle’s figure family.



pymdp Validation: Three-Tier CI Gates, Structured JSONL Run Log, and Bit-Reproducibility Contract

The pymdp harness ([§14](#)) and free-energy bundle ([§15](#)) are the *runtime layer*. This sub-section documents the **validation gates** that protect the manuscript from drift between the runtime and what is asserted in prose, plus the **structured run log** that records every load-bearing quantity for post-hoc audit.

Three-tier validation

Every pymdp artifact passes through three CI gates. Any one failing fails the pipeline:

Tier	What it asserts	Source
Numerical contracts (pytest, no mocks)	shape / type / determinism / $\lambda = 0$ baseline / sub-additivity / monotonicity / bundle $\leftrightarrow$ helper agreement	tests/test_simulation_pymdp.py , tests/test_simulation_free_energy.py
Output gates (<code>scripts/validate_outputs.py</code>)	every PNG header is valid; every JSON key in expected range; the closed-form vs empirical MI residual $\leq 1e - 06$; the free-energy bundle’s $\lambda = 0$ baseline (TC = 0, coupling = 0, $H(q) = \sum_k H(q^k)$) and TC ≥ 0 everywhere	scripts/validate_outputs.py
Manuscript completeness (<code>scripts/validate_manuscript.py</code>)	every <code>[[FIG:<label>]]</code> , <code>[[EQ:<label>]]</code> , <code>[[VAR:<key>]]</code> , <code>[@<citekey>]</code> , <code>[[SECREP:<label>]]</code> , <code>[[THMREF:<label>]]</code> resolves; every numeric variable lies in its expected range; no hardcoded <code>\$N.M</code> outside permitted sites	scripts/validate_manuscript.py

Run all three with the orchestrator:

```
uv run python scripts/run_all.py
```

(20 scripts in canonical order, exits non-zero on any failure.)

The reader-facing order is:

Stage group	Role in the evidence chain
Lean + analytical prelude	Build the boundary fragment and render pure-NumPy analytical figures.
Empirical producer batch	Write the Bernoulli, pymdp, multi-K, long-horizon, and revertibility sidecars.
Variable materialization	Convert hyperparameters plus current sidecar summaries into <code>output/data/manuscript_variables.json</code> .
Manuscript rendering	Substitute <code>[[VAR:...]]</code> , figure, theorem, section, citation, and equation tokens into <code>output/manuscript/</code> .
Gates	Validate artifacts, validate prose, then compare test / coverage / invariant / Lean-budget metrics against the regression baseline.

This is a dependency order, not just a convenience order. In particular, `manuscript_variables.py` runs after the empirical sidecars because it reads the multi-K, long-horizon, and revertibility summaries. A clean run therefore cannot render prose from a previous run’s sidecar values.

Dual-tolerance contract. Two tolerances coexist in the harness: (i) a stricter **sentinel cross-check** at the 5 λ values $\{0, 0.5, 1, 2, 4\}$ asserts that the closed-form mutual information matches the *numeric* total correlation of the same closed-form joint to $< 1e - 09$ — an **internal analytic-consistency** check (two algebraically independent closed forms of the same quantity must agree to numerical precision), *not* an independent empirical confirmation; the genuine finite- N seeded Monte-Carlo estimate (`empirical_mutual_information_montecarlo`, §§13) is the separate sampling-based witness (`tests/test_bernoulli_toy.py`); (ii) a looser **parameter-sweep CI gate** at $1e - 06$ asserts the same match across the full 121-point (λ, Δ) parameter sweep (`scripts/validate_outputs.py`). Both tolerances are real and report distinct guarantees: the sentinel check pins exact analytical agreement at hand-picked λ , the sweep gate bounds drift across the entire grid.

Structured run log (JSONL)

Every `pymdp` script emits one structured-record line per (script, λ -sweep, observation, rollout) call to `output/logs/pymdp_runs.jsonl` via `simulation.logging_utils.RunLogger`. Each record is a single JSON object on its own line (JSONL), with the schema:

The full field schema (13 fields: `timestamp`, `script`, `event`, `section`, hyperparameters `K/gamma/lam/seed/T`, `observations`, grid descriptors, bundle observables, $\lambda=0$ sentinels, `sampled_actions`, `runtime_ms`, `status`) is in §S7.5 (supplement).

Inspect with `jq`:

```
# Show every record's runtime + status:
jq -c '{timestamp, section, runtime_ms, status}' \
    output/logs/pymdp_runs.jsonl

# Extract the  $\lambda = 0$  free-energy baseline:
jq 'select(.section=="figure_pymdp_free_energies") |
    {coupling_term_at_lambda_zero, joint_entropy_at_lambda_zero,
     marginal_entropy_sum_at_lambda_zero}' \
    output/logs/pymdp_runs.jsonl
```

The validator `scripts/validate_outputs.py::validate_run_log` fails CI if:

- the JSONL file is malformed,
- fewer than three records are present,
- any required section (`figure_pymdp_lambda_sweep`, `figure_pymdp_rollout`, `figure_pymdp_free_energies`) is missing,
- any record lacks a `timestamp`.

Software and source provenance

The `pymdp` validation layer is intentionally standalone-local. The release environment is refreshed with:

```
uv sync --group sim --group viz --group lint
```

The `sim` group supplies `inferactively-pymdp==1.0.1`; the lockfile records the exact wheel / source archive, and the methods prose cites the JOSS paper plus the official source and documentation [Heins et al., 2022, `pymdp developers`, 2026a,b]. The only module allowed to import `pymdp/JAX` is `src/simulation/agents.py`; the no-mock test policy forbids replacing that boundary with `MagicMock`, `unittest.mock`, `mock.patch`, or equivalent fake call paths. A clean collection run therefore proves that `lean.invariants`, the dashboard builder, and the `pymdp` harness import from this checkout without requiring a parent template on `PYTHONPATH`.

Reproducibility contract

Two consecutive `run_all.py` invocations on the same code produce **numerically identical** CSVs and JSON variable dumps. PNG rasters are bit-identical modulo the `project.timestamp` and `project.git_revision` `TEXT` metadata chunks, which embed run-time provenance for audit purposes; the JSONL run log similarly records per-run wall-clock timestamps. Strict byte-level reproducibility across the metadata chunks requires pinning these to a `SOURCE_DATE_EPOCH`-style fixed value at run time. The guarantees that make this contract possible:

- `pymdp.agent.Agent(inference_algo='fpi')` — deterministic FPI, no random sampling inside `infer_states` / `infer_policies`.
- All RNG-bearing calls go through `np.random.default_rng(seed=...)` with seeds pinned in `src/simulation/hyperparameters.py` (`FIGURE_GLOBAL_SEED = 42`, `PYMDP_ROLLOUT_SEED = 0`).

- `MPLBACKEND=Agg` is set at every figure-script entrypoint to prevent platform-dependent figure rasterisation.
- Every grid count / λ range / rollout horizon flows from the hyperparameters module, then through the JSON mirror, then into prose via `[[VAR:<key>]]` — there is no place a number can drift.

Test budget snapshot

Suite	Coverage
<code>tests/test_simulation_pymdp.py</code>	per-stream EFE / posterior PMFs, $\lambda = 0$ baseline, monotonicity, deterministic rollout
<code>tests/test_simulation_free_energy.py</code>	every <code>FreeEnergyBundle</code> invariant + bundle $\leftrightarrow$ helper agreement
<code>tests/test_simulation_specs.py</code>	<code>StreamSpec</code> / <code>CoupledEnsembleSpec</code> shape + column-stochasticity validation
<code>tests/test_free_energy_plots.py</code>	every dashboard PNG header, on real bundle data
<code>tests/test_figure_scripts.py</code>	every <code>figure_*</code> function in <code>simulate_pymdp.py</code> smoke-tested
<code>tests/test_logging_utils.py</code>	structured JSONL emission, runtime measurement, status propagation, schema
<code>tests/test_multi_k_experiments.py</code>	configured multi-K ensemble sweeps $K \in \{3, 4, 5\}$: TC monotonicity, aligned-mass growth, TT-rank profile, $\lambda = 0$ mean-field baseline
<code>tests/test_long_horizon.py</code>	$T = 100$ long-horizon rollout: deterministic reproducibility, tail-window steady-state KL bound, habit-accumulation monotonicity
<code>tests/test_revertibility.py</code>	revertibility_kl_equals_multiinformation: $I(q_\lambda) = D_{\text{KL}}(q_\lambda \parallel \hat{m}(q_\lambda))$ to floating-point tolerance ($\leq 1.67e - 16$ maximum residual) on every λ in the sweep — an internal analytic-consistency check of Proposition 7.3 (both sides are the <i>same</i> finite-sum quantity via <code>total_correlation_via_kl</code> ; true by construction, <i>not</i> an independent empirical witness)
<code>tests/test_robustness.py</code>	one-axis robustness scenarios, targeted two-axis interactions, coupling ablations, fixed-marginal null controls, long-horizon replicate summaries, Wilson intervals, and threshold-sensitivity summaries
<code>tests/test_robustness_plots.py</code> / <code>tests/test_metadata_pure.py</code>	stress-test figure writers and metadata semantics for robustness envelopes, ablation summaries, null-control figures, replicate envelopes, and threshold-sensitivity plots

See the full per-suite breakdown in [docs/reference/statistics_reference.md](#).

Part V — Connections to Existing Frameworks

The parametric-entanglement framework is not a replacement for active inference or for the neighboring control and probabilistic-programming literatures. It is a relationship calculus for one object: the joint policy posterior. Some prior constructions are recovered exactly inside this calculus (mean-field AIF at $\lambda = 0$, single-stream KL control, product-of-experts, and copula VI after a CDF change-of-variables). Others are parametric embeddings that require an explicit modeling choice, and others are only structural analogies: they share a coupling skeleton but retain additional process-theoretic content, such as temporal-scale separation, recursive observation-conditioning, message-passing schedules, or state-space Markov-blanket construction. Three chapters map the territory:

- **§17** — classical active-inference frameworks. Mean-field AIF as the $\lambda = 0$ slice (exact). Hierarchical / deep AIF as block-bidiagonal coupling (structural analog). Sophisticated inference as a tree-structured coupling with the recursive observation-conditioning discharged into J . Branching- time AIF as Bayesian filtering over a tensor-train-compressed policy tree, classified as an algorithmic analogy until a direct head-to-head construction is implemented.
- **§18** — control and RL. KL / path-integral control as the single-stream $K = 1$, $\lambda = 1$ case (the classical log-partition-value duality). Options frameworks as $K = 2$ with policy-matching J . Products and mixtures of experts as factorizations of J . Copula variational inference as the CDF-reparametrized continuous analog.
- **§19** — multi-agent, geometry, worldview. Interactive / multi-agent inference as inter-agent coupling potentials. Renormalization-group AIF as a structural analogy via tensor-train compression. Policy-space Markov-blanket leakage as a coupling identity (distinct from the state-space construction). CEREBRUM and case grammar as edge-weight asymmetry on the coupling graph.

Each connection is stated with the structural choice that relates it to the λ -coupled posterior, the claim class (**exact**, **parametric**, or **analogical**), and the specific content that remains outside the present artifact.

Framework Connections I: pymdp/SPM Baseline, Hierarchical/Deep AIF, Sophisticated Inference, and Branching-Time AIF

This section maps the entanglement framework onto major existing constructions in active inference, control theory, machine learning, and theoretical neuroscience. The framework gives exact recoveries for some rows, parametric embeddings for others, and explicitly labeled structural analogies where the prior framework carries additional process-theoretic content.

The mapping is split into three thematic groups:

- **Classical active-inference frameworks** (this file) — mean-field AIF (**pymdp** / **SPM**), hierarchical / deep AIF, sophisticated inference, branching-time AIF.
- **Control and reinforcement-learning frameworks** (§18) — KL / path-integral control, options framework, products + mixtures of experts, copula VI.
- **Multi-agent, geometry, and worldview connections** (§19) — interactive / multi-agent inference, renormalization-group AIF, Markov blankets / Bayesian mechanics, CEREBRUM and case grammar.

Relationship calculus for AIF variants

The common template used throughout Part V is the finite joint-policy posterior

$$q_{\lambda}(\pi^{1:K}) \propto \prod_{k=1}^K E_k(\pi^k) \exp \left(- \sum_{k=1}^K \gamma_k G_k(\pi^k) + \lambda J(\pi^{1:K}) - \gamma \lambda K_c(\pi^{1:K}) \right). \quad (17.1)$$

Every comparison below asks which part of this template a prior framework changes: the generative model, the policy-posterior factorization, the EFE recursion, the message-passing schedule, the agent-coupling graph, or temporal depth. The recovery labels are therefore strict. **Exact** means literal specialization of this posterior or its single-stream log-partition dual after all variables and factors have been specified. **Parametric** means the cited framework can realize the form after the modeler adds explicit policy nodes, coupling factors, priors, or change-of-variables. **Analogical** means the cited framework shares a structural motif, but additional process-theoretic content remains in the generative model, recursive EFE definition, or message-passing schedule rather than in λJ itself. Exactness therefore attaches to a fully specified posterior or graphical model, not to active inference as a whole.

The notation bridge to standard active inference is deliberately literal. The generative-model law $P(\cdot)$ remains represented by the usual likelihood, transition, preference, state-prior, and policy-prior ingredients; the variational beliefs $Q(\cdot)$ become the normalized policy posteriors denoted by q in this manuscript. The per-stream factors E_k , G_k , and γ_k are the ordinary habit/prior, expected-free-energy, and policy-precision ingredients of single-stream discrete active inference. The new objects are only J , K_c , and λ : a pair of joint-policy potentials and the scalar that weights them. Thus λ is not a substitute for policy precision, neural precision, preferences, desire, or a generative-model parameter. It is a coupling strength inside the finite policy-posterior family, and every comparison below must say whether the external framework supplies the same posterior, supplies a parameterized route to it, or merely shares a structural pattern.

This convention matters because active inference is broader than any one posterior factorization. Factor-graph treatments show how generative-model specification and message passing can automate deep temporal active inference [de Vries and Friston, 2017], while graphical-brain and neuronal-message-passing accounts distinguish Forney-style belief propagation, mean-field, Bethe, and marginal approximations as alternative implementation commitments [Friston and Parr, 2017, Parr et al., 2019, Yedidia et al., 2005]. Reactive message-passing toolkits such as RxInfer support flexible factor insertion [Bagaev and Podusenko, 2023]. Adding $\exp(\lambda J - \gamma \lambda K_c)$ as a policy factor is then an exact implementation pattern for that chosen graphical model; it is not a claim that every factor-graph or RxInfer active-inference model already contains this coupling factor.

Relation to recent unification efforts

Recent work has independently proposed unification narratives for active inference. Da Costa, Tenka, Zhao, and Sajid treat active inference as a normative model of agency and a comparison language for different agency models

[Da Costa et al., 2024]. de Vries, Nuijten, van de Laar et al. [de Vries et al., 2025] recast EFE-based planning as variational inference on an augmented generative model with preference and epistemic priors; Nuijten and Lukashchuk [Nuijten and Lukashchuk, 2025] formally argue that active inference is a subtype of variational inference; Champion, Bowman, Marković, and Grzes [Champion et al., 2024] derive four EFE formulations from unified root definitions. Millidge, Tschantz, and Buckley [Millidge et al., 2021] provide the complementary caution: expected free energy has multiple derivational routes whose assumptions matter. These constructions unify over the *form of EFE* or over the *EFE/VFE equivalence*; the present framework unifies over the *factorization structure of the joint policy posterior* via a scalar deformation parameter λ . The two unifying strategies are orthogonal — our framework operates on top of a generative model that the de Vries / Nuijten lineage works to construct, and Champion et al.’s EFE-reframing applies to the per-stream G_k that enters our λ -deformed prior. This distinction keeps the claim strength clear: the present manuscript does not solve the EFE unification problem, but it is compatible with those formulations because it acts on the posterior factorization once the per-stream EFE terms have been specified.

Mean-field active inference (pymdp, SPM, ActiveInference.jl)

The standard factorized treatment of multi-modality, multi-factor active inference [Heins et al., 2022, Smith et al., 2022] is the $\lambda = 0$ slice of our framework. Our addition: extend the variational family along the e-geodesic in any chosen direction (see Eq. (7.3)). Concretely, in pymdp notation, the multi-stream policy posterior $q(\pi) = \prod_f q_f(\pi_f)$ becomes $q_\lambda(\pi)$ with λ as an additional learnable hyperparameter.

This is operationalized by the project’s 1.0.1 simulation harness (§14, `src/simulation/`, `docs/simulation/pomdp_simulation.md`): each stream is built as a separate `pymdp.agent.Agent` whose $q_\pi^k = \text{softmax}(-\gamma G_k)$ plays the role of a mean-field marginal, and the analytical layer of §4.2 adds the λ -coupling on top. This is an adapter construction, not a claim about the native pymdp API: pymdp supplies per-stream inference, while this repository constructs and validates the structured joint posterior after those per-stream posteriors exist. At $\lambda = 0$, the joint is *exactly* the outer product of the per-stream pymdp posteriors; for $\lambda > 0$ the total correlation grows smoothly toward the task-specific coordination ceiling. In the binary all-aligned/all-anti-aligned Ising slice used by the pymdp harness, that ceiling is $(K - 1) \log 2$; for the default $K = 2$ case this reduces to $\log 2$.

Hierarchical / Deep Active Inference

Hierarchical AIF [Friston et al., 2017a, Pezzulo et al., 2018] decomposes the agent into a hierarchy of slow (high-level) and fast (low-level) generative models with explicit message passing.

Theorem 17.1 Structural analog of hierarchical AIF (witness-form). With block-bidiagonal J coupling adjacent levels of an L -level ensemble and $\lambda \rightarrow \infty$, the entangled posterior concentrates on the cross-level coordination shadow of hierarchical AIF: each adjacent pair $(\pi^\ell, \pi^{\ell+1})$ becomes deterministically linked. The Lean companion `ConnectionsWitnesses.hierarchicalAIF_lambda_limit_witness` is now live in the boundary fragment in *witness-consuming* form: the caller supplies the entangled family $\lambda \mapsto q_\lambda$, the hierarchical fixed point q_∞ , and the universally quantified (ε, λ_0) concentration inequality as a `HierarchicalConcentrationWitness` structure; the boundary fragment certifies the resulting existence claim by extracting the witness fields. We do not claim that this recovers hierarchical AIF in full — the *temporal scale separation* and *directed top-down/bottom-up message passing* that characterize standard hierarchical AIF [Friston et al., 2017a, Pezzulo et al., 2018] are additional structures not encoded by symmetric J alone. The construction here recovers the *coupling-structural* component machine-checked at the witness level. A generative-model-side embedding (with explicit per-level B^ℓ dynamics) would be required to upgrade the analogy to a recovery, and that stronger construction is outside the present artifact.

Sophisticated Inference (recursive EFE)

Sophisticated inference [Friston et al., 2021] introduces beliefs about beliefs — the agent reasons about counterfactual consequences of its actions for its own beliefs. Recursive EFE has the form

$$G_{\text{soph}}(\pi) = \mathbb{E}_{q(o|\pi)}[G(\pi') + \log q(\pi' | o)]. \quad (17.2)$$

This can be represented within our template by treating one stream π as the current policy and another stream π' as the next-step policy posterior conditional on observations, with a specific source-supplied choice of J that encodes the recursive belief-about-beliefs structure.

Proposition 17.2 Sophisticated inference as a tree-structured coupling (witness-form). With $K = d + 1$ streams indexed by lookahead depth and a J that scores joint trajectories by their sophisticated EFE, the entangled posterior q_λ tracks coordinated lookahead policies. The Lean companion `ConnectionsWitnesses.sophisticatedInference_embedding_witness` is now live in the boundary fragment as a *witness-consuming*

pymdp coupled K=2 Ising: total correlation vs λ

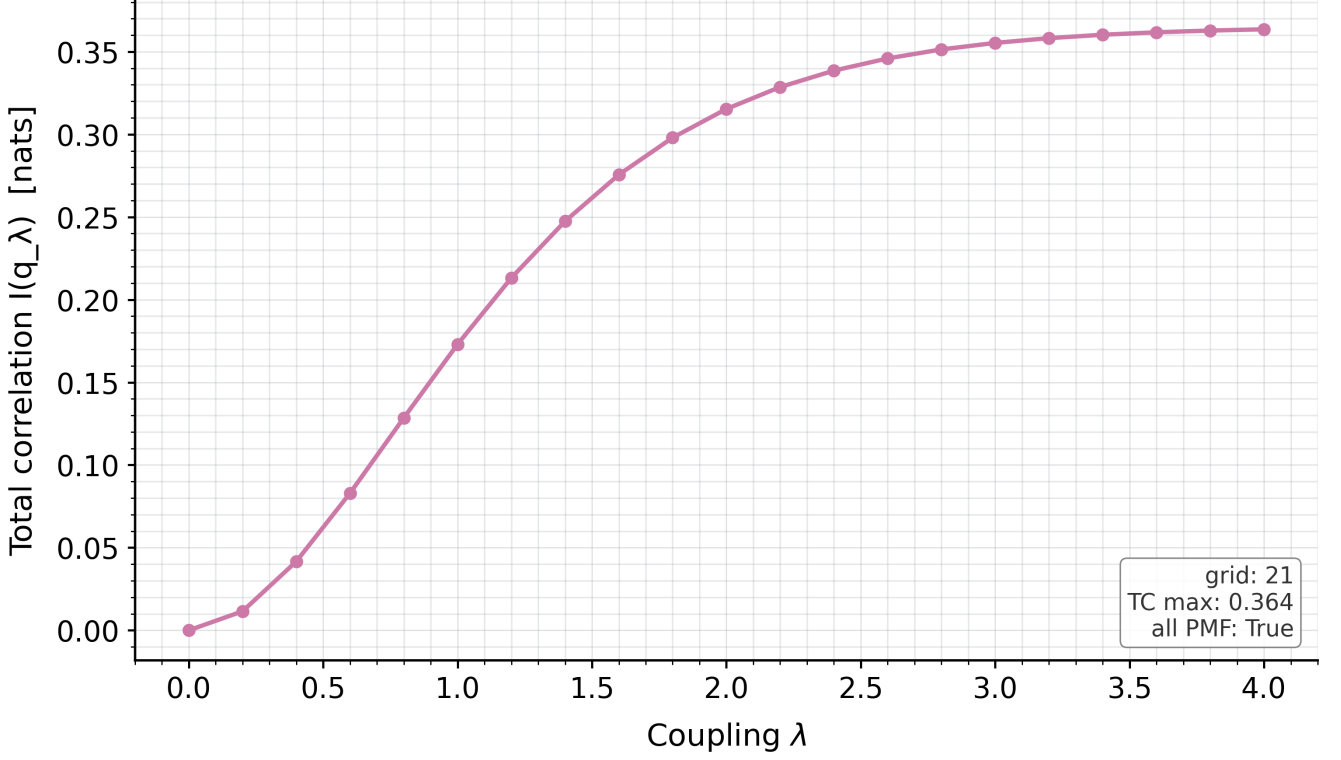


Figure 58: pymdp-grounded total correlation $I(q_\lambda)$ from `simulation.inference.coupled_policy_posterior` $\rightarrow$ `lean.free_energy.total_correlation`, evaluated on a 21-point linspace over $\lambda \in [0, 4]$ (`simulation.hyperparameters.PYMDP_SWEEP_LAMBDA`) for the $K = 2$ Ising ensemble at observations $(0, 0)$ and ensemble precision $\gamma = 1$. The CSV at `output/simulations/pymdp_lambda_sweep.csv` is the source data. Smooth monotone increase from the mean-field baseline 0 toward saturation at $\log 2 \approx 0.693$ — the empirical witness of [Proposition 7.3](#) inside a grounded POMDP. Uncertainty semantics: deterministic grid or deterministic construction; no stochastic error bars are implied.

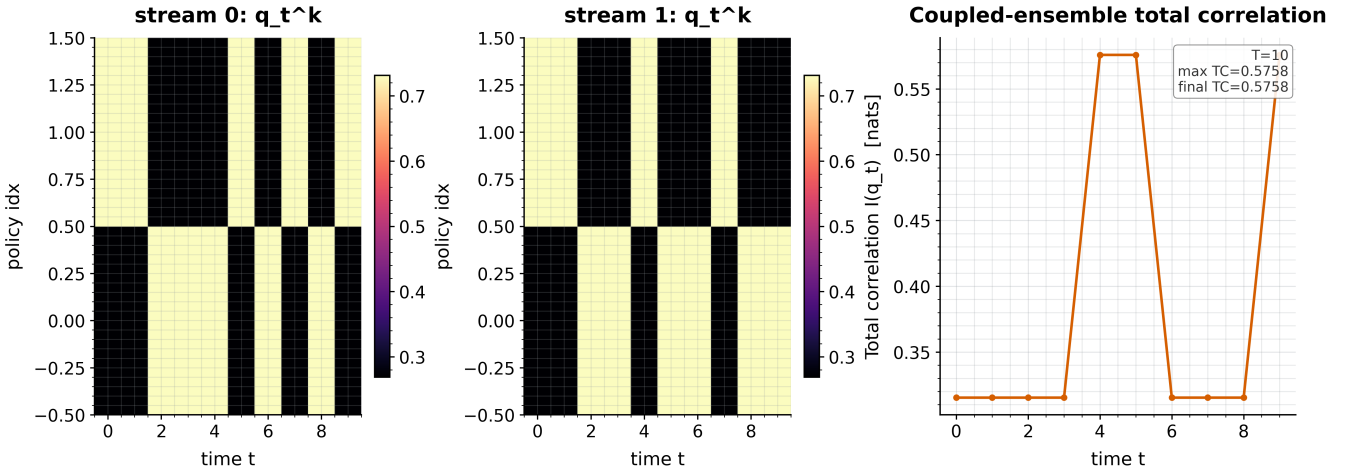


Figure 59: Deterministic $T = 10$ -step coupled rollout from `simulation.rollout.simulate_coupled_rollout` under $\lambda = 2$ and seed 0 (`simulation.hyperparameters.PYMDP_ROLLOUT_STEPS`, `PYMDP_ROLLOUT_LAMBDA`, `PYMDP_ROLLOUT_SEED`). Per-stream marginals q_t^k from `simulation.inference.per_stream_policy_posterior` (left two heatmaps) interleave with the coupled total correlation $I(q_t)$ from `lean.free_energy.total_correlation` (right). Bit-identical reproducibility under fixed seed is asserted by `tests/test_simulation_pymdp.py::test_simulate_coupled_rollout_deterministic_under_fixed_seed`. Uncertainty semantics: canonical fixed-seed trajectory; seed sensitivity is reported only where a replicate sidecar is registered.

form: the caller supplies an opaque sophisticated-inference source type, an embedding map into the entangled-posterior family, and the VFE-preservation identity as a `SophisticatedInferenceEmbedding` structure; the boundary fragment certifies that the embedding lifts into the boundary-fragment `variationalFreeEnergy` primitive without sorry. We emphasize that the substantive content of sophisticated inference [Friston et al., 2021] — recursive conditioning of next-step EFE on *anticipated* observations o_τ — requires an explicit form for J that integrates over $q(o_\tau, s_\tau \mid \pi_\tau)$. Spelling out this J is straightforward but voluminous; we treat it here as a *structural analogy* with the recursive content discharged into the construction of J . The connection is machine-checked at the level of VFE-preservation through a witness-form Lean embedding; the observation-conditioning recursion is a constraint on J , not a consequence of the λ -deformation.

Branching-Time Active Inference

Branching-time AIF [Champion et al., 2022] addresses the exponential complexity of policy enumeration via Bayesian filtering over a branching policy tree (with optional MCTS-style sampling for tractability, in the lineage of [Fountas et al., 2020]).

The scoped connection is representational: branching-time AIF performs Bayesian filtering / sampling on the policy tree, while our framework parameterizes compact probability tensors that can be read as compressed views of temporally-extended policy posteriors. Concretely, a tensor-train decomposition of q_λ over a temporally-extended policy offers a compact policy-tree summary with bond dimension controlling representational capacity. This is an analogy and an engineering route for compression, not an algorithmic equivalence to Champion’s Bayesian-filtering formulation. It is complementary to the Monte-Carlo lineage [Fountas et al., 2020] and to tensor-network techniques [Orus, 2014, Han et al., 2018, Oseledets, 2011].

Synthesis: How Every Connection Lands on Lambda, J, or Kc

The thirteen frameworks discussed across this section and its two companions (§18, §19) all reduce to a specific choice of three core objects: the coupling strength λ , the coupling potential J , and the off-diagonal cost K_c . The table below summarizes the embedding for each, so a reader can locate any one connection at a glance.

Two patterns are worth flagging before reading the table. First, the λ axis orders the frameworks by *coupling strength*: mean-field AIF lives at the $\lambda = 0$ pole; classical hierarchical AIF and option-like policy-matching embeddings live near $\lambda \rightarrow \infty$ with sparse J ; the entire continuum between them is what the parametric framework covers. Second, the J structure orders the frameworks by *kind of coordination*: block-bidiagonal J produces hierarchical structure, tensor-train J produces renormalization-group-style bond-dimension flow, symmetric Ising-style J produces interactive multi-agent behavior, case-graded J produces CEREBRUM-style asymmetric coupling. The *Recovery type* column distinguishes *exact* (literal specialization of the posterior or its single-stream log-partition dual), *parametric* (realized after an explicit parameter, change-of-variables, or factor-graph modeling choice), and *analogical* (the prior framework’s coupling-structural skeleton is mirrored, but additional dynamical, temporal, or message-passing content lies outside the λ -deformation and must be discharged into J , the EFE recursion, or the generative model).

Framework	J structure	λ regime	Recovery type
Mean-field AIF (pymdp, SPM)	any	$\lambda = 0$	exact
Factor-graph / RxInfer-style message passing	additional policy factor $\exp(\lambda J - \gamma \lambda K_c)$	any finite λ once the factor is added	parametric
Product of Experts	$J = \sum_j \log f_j$, $K_c = 0$	any λ	exact
Copula VI	$J = \log$ -copula density on Π (CDF reparametrization)	any λ	parametric (after CDF change-of-variables)
Hierarchical / Deep AIF	block-bidiagonal J , $K_c = 0$	$\lambda \rightarrow \infty$ on off-diagonal blocks	analogical (structural shadow; lacks temporal-scale separation and directed message passing)
Sophisticated inference	J that integrates over $q(o_\tau, s_\tau \mid \pi_\tau)$	$K = d+1$ streams, finite λ	analogical (with J -construction discharge of recursive observation-conditioning)
Branching-time AIF	tensor-train J across time-extended policies	bond-dimension truncation	analogical (with tensor-train algorithmic side)
KL / path-integral control (single-stream)	single-stream cost $C(\tau)$	$K = 1$, $\lambda = 1$	exact (classical log-partition-value duality)
Options framework / Hierarchical RL	$J(\pi^1, \pi^2)$ peaked on option-matching configs	$K = 2$, any λ	parametric (soft option boundaries; policy-matching only)
RG-AIF (RGM)	tensor-train compression of q_λ	bond-dimension flow	analogical (MERA-style scale invariance lies outside the present construction)

Framework	J structure	λ regime	Recovery type
Mixture of Experts	m -mixture (not e -product)	n/a	out-of-scope (m -geometry dual)
Interactive / multi-agent AIF	NK -stream J with within- and across-agent blocks	any λ	parametric (per-stream coupling extends to per-agent)
Markov blankets / Bayesian mechanics (policy-space)	J controls cross-stream coupling	$\lambda = 0 \rightarrow \eta = 0$	analogical policy-space leakage measure (distinct from state-space blankets)
CEREBRUM / case-grammar	directional asymmetry in J	any λ	parametric (case = edge-weight asymmetry)

Three takeaways from the table:

1. **The exact recoveries are mean-field AIF, product-of-experts, and single-stream KL control.** Factor-graph / RxInfer-style implementations are parametric until the modeler inserts and validates the policy-coupling factor in a chosen graph; Copula VI is recovered parametrically after a CDF change-of-variables. The remaining rows are *structural analogies*: the coupling skeleton of the prior framework is mirrored, but additional content (temporal-scale separation in hierarchical AIF; recursive observation-conditioning in sophisticated inference; Bayesian filtering over the expanded policy tree in branching-time AIF; MERA-style scale invariance in RG-AIF; state-space construction in Markov blankets) lies outside the λ -deformation.
2. **The λ axis is the natural “comparison axis.”** Mean-field AIF lives at the $\lambda = 0$ pole; sparse- J frameworks live near $\lambda \rightarrow \infty$; the continuum between them is what our framework parametrizes.
3. **The J structure is the natural “kind axis.”** Block-bidiagonal J is hierarchical coupling; tensor-train J is RG-style compression; symmetric Ising-style J is interactive multi-agent; case-graded J is CEREBRUM. Every row above is a statement about which J topology the prior art is implicitly assuming.

In this sense the lattice of existing constructions is mapped onto a low-dimensional set of slices through the (λ, J, K_c) parameter space, with the connecting tissue between them parameterized rather than ad-hoc — though several of these mappings remain structural analogies rather than full recoveries, in the senses listed above.

Framework Connections II: KL/Path-Integral Control, Options Frameworks, Products-of-Experts, and Copula Variational Inference

This sub-section continues §17 by mapping the entanglement framework onto stochastic optimal control, hierarchical reinforcement learning, and probabilistic-modeling connections that sit between the active-inference and ML communities. The canonical tutorial on the control-as-inference / maximum-entropy RL viewpoint [Levine, 2018] is the natural ML-side companion for the active-inference recoveries developed below.

KL / Path-Integral Stochastic Optimal Control

Path-integral and KL control [Kappen, 2005, Todorov, 2006, Theodorou and Todorov, 2012, Rawlik et al., 2013] solve stochastic optimal control via free-energy duality:

$$V(x) = -\frac{1}{\rho} \log \mathbb{E}_{p_{\text{free}}}[\exp(-\rho C(\tau))] \quad (18.1)$$

where $C(\tau)$ is the trajectory cost (renamed from the standard J to avoid clashing with our habit potential J) and p_{free} is the uncontrolled-dynamics prior. The control-effort cost is the KL between the controlled and uncontrolled measures.

Connection. At $K = 1$, $\lambda = 1$ the framework reduces to the classical single-stream log-partition-value-function duality [Kappen, 2005, Todorov, 2006, Theodorou and Todorov, 2012, Rawlik et al., 2013]: $V(x) = -(1/\rho) \log \mathbb{E}_p[\exp(-\rho C(\tau))]$, where $\mathcal{E}_1 = E_1 \cdot \exp(J - \gamma K_c)$ plays the role of the Gibbs-controlled trajectory measure. This is the well-known special case; the genuinely novel content of the framework is at $K \geq 2$, where our $E_k(\pi^k)$ plays the role of the per-stream uncontrolled prior and the coupling potential λJ encodes *cross-stream control coupling* — the sense in which a “joint” KL controller pays a coupling cost beyond the marginal control efforts. A multi-stream value-function formulation follows in principle from the same log-partition duality; we sketch but do not develop it here.

Smooth-deformation precedents. Smooth scalar deformations of variational families have prior history in the structured-VI literature. Li and Turner [Li and Turner, 2016] introduce α -Rényi divergence variational inference:

$\alpha = 1$ recovers the standard ELBO, $\alpha \rightarrow 0$ recovers the log marginal likelihood. Tran, Blei, and Airoldi [Tran et al., 2015] augment mean-field with copula dependency; the recent Fu–Smith–Panagiotelis vector-copula extension [Fu et al., 2025] adds learnable dependency between blocks. Our λ differs from these by parametrizing the *policy-posterior factorization structure*, not the *divergence* (as in α -Rényi-VI) or the *dependency form alone* (as in copula VI): at $\lambda = 0$ we have strict mean-field, and at $\lambda > 0$ we have a coupling-deformed factorization whose structure is controlled by the habit and preference potentials (J, K_c) .

Options framework / Hierarchical RL

The options framework and hierarchical-RL literature [Sutton et al., 1999, Barto and Mahadevan, 2003, Bacon et al., 2017] introduce temporally-extended actions (“options”) as semi-MDPs with initiation sets, intra-option policies, and termination conditions. The planning-as-inference program [Toussaint, 2009, Botvinick and Toussaint, 2012] casts policy search as Bayesian posterior inference over trajectories, giving the options decomposition a probabilistic-graphical-model reading.

Connection. An option can be represented as a *temporally-coupled policy stream*: a high-level decision (which option to invoke) and a low-level intra-option policy. This is a parametric $K = 2$ embedding with J encoding the option-membership structure (high J on configurations where intra-option policy matches the invoked option, low J otherwise). Termination conditions correspond to boundary conditions on the entangled posterior only after the modeler encodes them in that factorization.

The framework therefore gives an option-like parametric embedding with *probabilistic, soft, learnable option-boundary weights* via λ -tunable coupling. It does not replace option discovery or termination learning; those algorithmic choices must be supplied by the modeler or by a separate option-learning layer.

Product of Experts and Mixture of Experts

Product of Experts (PoE) [Hinton, 2002] combines models multiplicatively:

$$P(x) \propto \prod_j f_j(x). \quad (18.2)$$

Our $\mathcal{E}_\lambda(\pi) = \prod_k E_k(\pi^k) \cdot \exp(\lambda J(\pi))$ is a PoE on the joint policy space, with the coupling potential acting as one additional expert. The framework therefore inherits PoE’s compositionality: combining experts that each capture a piece of the policy-coupling structure.

Mixture of Experts (MoE), conversely, is *additive* combination — corresponding to a different geometric structure (m -mixtures rather than e -products). The two are dual.

Copula variational inference

Copula VI [Tran et al., 2015, Nelsen, 2006] retains mean-field marginals while modeling dependence via a copula. Our framework’s λ -deformation realizes the same copula-density form when the modeler supplies the CDF reparameterization and chooses J as the log-density of the desired copula family. Define per-stream marginal CDFs $u^k = F_{E_k}(\pi^k) \in [0, 1]$. Under this reparametrization, the entangled prior $\mathcal{E}_\lambda(\pi) = \prod_k E_k(\pi^k) \exp(\lambda J(\pi)) / Z_E$ maps to a copula density

$$c(u^1, \dots, u^K) = \exp(\lambda J(\pi(u^1, \dots, u^K))) \quad (18.3)$$

on the unit hypercube (up to the normalizer Z_E). The framework therefore recovers the parametric copula-density slice of copula variational inference [Tran et al., 2015, Fu et al., 2025] with copula-log-density λJ ; bond-dimension truncation of J as a tensor-train is analogous to vine-copula truncation at depth χ rather than a proof that every vine construction is a tensor-train posterior. This connects active inference to the rich copula literature, importing techniques for sparse coupling estimation [Nelsen, 2006, Han et al., 2016, Tran et al., 2015, Fu et al., 2025].

Recovery dictionary

The four connections above are concrete only when the relationship class is explicit. Reading the table left-to-right turns a broad recovery claim into a recipe with the relevant exact, parametric, or out-of-scope boundary attached.

Prior framework	Structural choice in (K, J, K_c, λ)	Relationship class / behavior
Mean-field active inference (pymdp / SPM)	$\lambda = 0$, any J, K_c	$q_\lambda = \prod_k E_k e^{-\gamma G_k} / Z_k$; per-stream FPI

Prior framework	Structural choice in (K, J, K_c, λ)	Relationship class / behavior
KL / path-integral control (single-stream)	$K = 1, \lambda = 1$	$V(x) = -(1/\rho) \log \mathbb{E}_E[\exp(-\rho C(\tau))]$; classical log-partition-value duality [Kappen, 2005, Todorov, 2006, Theodorou and Todorov, 2012, Rawlik et al., 2013]
Options / hierarchical RL	$K = 2, J(\pi^1, \pi^2)$ peaked on π^2 that matches option π^1	option-like soft boundaries; intra-option policy from q_λ^2 (parametric — policy-matching only; option discovery/termination learning supplied separately) [Sutton et al., 1999, Barto and Mahadevan, 2003, Bacon et al., 2017]
Product of Experts	$K_c = 0, J = \sum_j \log f_j$	$\mathcal{E}_\lambda \propto \prod_k E_k \cdot \prod_j f_j^\lambda$
Mixture of Experts (dual)	m -mixture rather than e -product; not in this construction	additive combination — out-of-scope for the entangled-prior family
Copula VI	$E_k = \text{marginals}; J = \text{log-copula density on } u = F_E(\pi)$	$\mathcal{E}_\lambda$ as a λ -deformed copula-density slice after CDF change-of-variables [Nelsen, 2006, Tran et al., 2015, Fu et al., 2025]
α -Rényi-VI [Li and Turner, 2016]	divergence-parametrized, not factorization-parametrized	orthogonal axis — not recovered, complementary smooth-deformation precedent

Empty cells mean *not subsumed in the entangled-prior construction* (e.g. Mixture-of-Experts requires the m -geometry dual). Every non-empty row is a worked instance: assigning (K, J, K_c, λ) the values shown either reproduces the prior framework’s posterior directly or realizes the relevant parametric policy-matching slice.

Framework Connections III: Multi-Agent Inference, RG-AIF Coarse-Graining, Markov Blankets, and CEREBRUM Case Grammar

This sub-section closes §17 by mapping the entanglement framework onto multi-agent active inference, renormalization-group treatments, the Markov-blanket / Bayesian-mechanics worldview, and CEREBRUM-style case-grammar architectures.

Interactive Inference / Multi-Agent Active Inference

Interactive inference [Maisto et al., 2024] models multi-agent cooperation as joint inference under shared generative models. Collective active-inference models add a complementary population-scale result: coordinated behavior can emerge from surprise minimization without treating the group result as a proof of any particular finite-policy coupling theorem [Heins et al., 2024].

The multi-agent active inference literature has expanded substantially since 2022. Federated inference and belief sharing [Friston et al., 2024] models belief broadcasting across agents; Shared Protentions [Albarracin et al., 2024] formalizes coordinated anticipation; collective surprise-minimization models [Heins et al., 2024] show how coordination can emerge at population scale; *As One and Many* [Waade et al., 2025] relates individual to emergent group-level generative models; Factorized Active Inference for Strategic Multi-Agent Interactions [Ruiz-Serra et al., 2025] explicitly notes the current default mean-field-across-agents assumption — which our λ -deformation relaxes. The framework here provides the scalar deformation parameter for the policy-coupling part of these lines, not a full multi-agent process theory by itself.

Connection. A multi-agent setting with N agents each with K policy streams maps onto a single NK -stream entanglement only after the modeler fixes agent boundaries, shared or separate generative models, and the within-agent / across-agent blocks of J . Under that explicit parameterization, the framework gives a joint-policy prior for single-agent multi-stream coordination and multi-agent joint action under the same scalar coupling parameter. Without that generative-model choice, the connection is structural rather than an identity with any particular federated-inference paper.

This is conceptually clean at the level of the variational family: from the perspective of policy entanglement, *interpersonal* coordination and *intrapersonal* coordination can use the same cross-stream coupling machinery once the modeler has chosen the agent boundaries, observation model, and shared or separate generative models. The claim class is therefore **parametric**, not **proved** or **exact** for the multi-agent literature as a whole.

Renormalization-group active inference (structural analogy)

Scale-free active inference, formalized as Renormalizing Generative Models (RGMs) [Friston et al., 2025], coarse-grains across temporal/spatial scales preserving the form of the generative model. Our framework’s tensor-train compression of q_λ (bond-dimension truncation) provides a *coarse-graining analog at the policy-posterior level*: low-rank truncation projects out high-spectral-rank correlations, akin to integrating out fast modes. We emphasize this is an *analogy*, not a recovery: a full RG-AIF correspondence would require a MERA-style scale-invariant ansatz (not standard tensor-train), which we do not develop here. The proper RG-AIF reference is RGM [Friston et al., 2025] and we cite it as the canonical scale-free AIF construction.

Markov Blankets and Bayesian Mechanics

The Markov blanket / Bayesian-mechanics formulation of FEP [Friston, 2019, Da Costa et al., 2021, Kirchhoff et al., 2018, Aguilera et al., 2022, Raja et al., 2021, Menary and Gillett, 2022] identifies a system with a partition of states into internal/external/blanket. In our framework, *streams* are candidate partitions of the agent’s policy space chosen by the modeler, and the factorization structure of q_λ across streams induces a policy-space analog of the blanket idea.

Proposition 19.3 Policy-space Markov blankets. We extend the Markov-blanket idea from state-space [Friston, 2019, Da Costa et al., 2021, Kirchhoff et al., 2018, Aguilera et al., 2022, Raja et al., 2021, Menary and Gillett, 2022] to policy-space: define the *policy-blanket leakage* of a joint q on $\Pi^1 \times \dots \times \Pi^K$ as the normalized multi-information

$$\eta(q) = I(q)/H(q) \in [0, 1]. \quad (19.1)$$

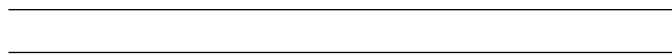
At $\lambda = 0$ we have $\eta(q_0) = 0$ (perfect factorization across streams; policy posteriors are conditionally independent given marginals); at $\lambda > 0$, $\eta(q_\lambda) > 0$ measures the leakage of cross-stream information that would be lost under a mean-field projection. We emphasize this is a policy-space analog, not a recovery of the state-space Markov-blanket construction; the two are formally distinct.

CEREBRUM and case-grammar approaches

CEREBRUM (Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling) [Friedman, 2025] explores grammatical case as a structuring principle for cognitive architectures. The connection to policy entanglement: cases encode *role-relations* across cognitive operations, which are precisely cross-stream coupling structures. A nominative-accusative case relation between two streams corresponds to a directional asymmetry in J . The framework therefore provides a parametric probabilistic backend for case-grammar cognitive architectures once the case roles have been encoded as edge-weight asymmetries in J .

A *compositional* reading of these case-grammar architectures sharpens the connection. Following the category-theoretic account of [Friedman, 2026b], let $\mathcal{K}$ be a compact-closed case category whose objects are case roles, whose morphisms are role-composing sentence and discourse circuits (in the DisCoCat / DisCoCirc sense), and whose hom-objects are enriched over $[0, 1]$ as magnitudes; an *alignment pattern* is then a functor $\alpha : \mathcal{K} \rightarrow \mathbf{Coup}_K$ into a category of directional couplings on the K -stream policy space, and — when this substrate is supplied — the habit potential of §4 can be written as the pullback $J = \alpha^* \Phi$ of a weighting Φ on $\mathcal{K}$. Two cognitive-architecture decisions then sit on distinct objects: a *choice of case system* (nominative–accusative, ergative–absolutive, fluid- S , split-ergative) is a choice of $\mathcal{K}$ and its alignment functor α , while the *coupling strength* of the present framework is the scalar λ that scales the pulled-back potential λJ inside $\mathcal{E}_\lambda$. Topos-theoretic Morita equivalence between alignment frames in the sense of [Friedman, 2026b] then predicts when two surface-different case-graded couplings induce the *same* entangled posterior — a non-identifiability condition on J that is invisible at the level of the coupling graph alone. The recovery class is unchanged: the categorical layer supplies the substrate on which case roles license edge-weight asymmetries in J , but choosing $\mathcal{K}$ and α remains an explicit modeling commitment, so the mapping stays *parametric* rather than *exact*.

The interaction graph of J across multiple streams is itself a useful diagnostic — symmetric all-to-all coupling (as in the $K=4$ Ising ensemble below) appears as an evenly-weighted clique, while asymmetric / hierarchical / case-graded couplings would manifest as edge-weight heterogeneity:



Coupling-potential graph $|J|$ (averaged across slots)

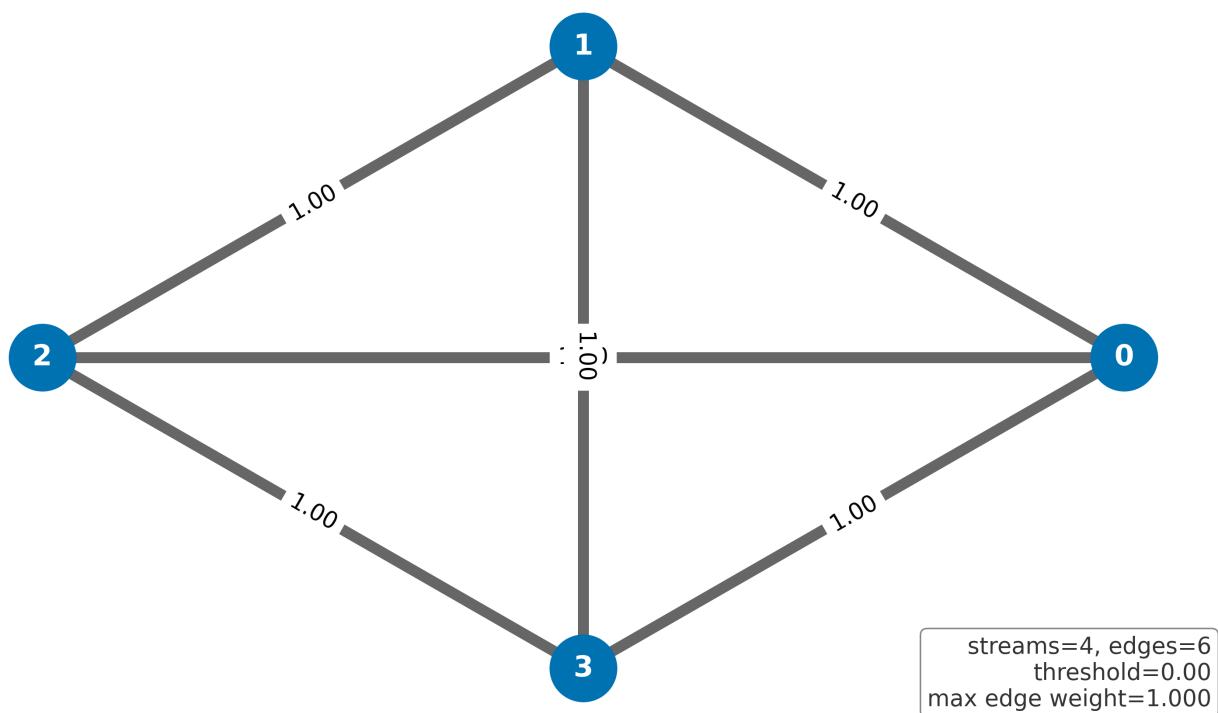


Figure 60: Coupling-potential graph for the symmetric $K = 4$ Ising ensemble built by `simulation.builders.ising_coupling_tensor` and rendered via `visualizations.graphs.plot_coupling_graph(threshold=0.0)`. Each node is one stream; edge weights are the mean $|J|$ over slot pairs. Symmetric coupling produces an evenly-weighted clique; CEREBRUM-style case-grammar couplings would manifest as edge-weight asymmetry. Uncertainty semantics: analytical schematic or structural visualization; no stochastic uncertainty interval is implied.

Part VI — Discussion and Outlook

The closing part collects what remains outside the stock-Lean boundary, how claim strength is labeled, and what the framework *commits to* as it lands. Two chapters:

- **§20** — the live punch-list across five fronts: open *theoretical* questions (Mathlib4 analytic payloads for convexity, spectral upper-semicontinuity, sparsity-rank); *identifiability* questions (when can λ be recovered from behavior?); *empirical* questions (which regimes does the pymdp harness need to be stress-tested against?); *conceptual* questions (precision as λ vs. precision as γ); and *practical* questions (computational scaling, model selection).
- **§21** — the consolidated discussion: the framework’s load-bearing commitments, the live state of the artifact (Lean fragment + Python suite + manuscript renderer + reproducibility contract), the limitations the formalism imposes by construction, and the ordered Mathlib4 analytic-discharge plan that keeps those witnesses separate from the stock-Lean boundary fragment.

Part VI states the practical implications of the framework — what the λ -coupling buys (cross-stream coordination, tunable structure, principled recovery of prior frameworks) and what it costs (computational overhead in J -construction, the coupling-tax bound, residual analytical-content witnesses in the Lean fragment).

Open Theoretical Questions: Analytical Gaps, Identifiability, Empirical Frontiers, and Practical Research Directions

A list of well-defined open problems, each suitable for focused investigation. The questions are grouped by methodological character — analytical, identifiability / inference, empirical, conceptual / cross-disciplinary, practical / formalization — and each carries an explicit statement, conjecture (where we have one), and a pointer to the section it would extend.

Where to start

For a contributor choosing an entry point, the highest-leverage items by category are:

- **Q1 (uniqueness of λ^*) — analytical, highest-leverage.** Convexity of $F[q_\lambda]$ on $[0, \infty)$ governs whether the framework’s central control parameter is *learnable* by ordinary gradient descent on free energy. A positive resolution promotes a large fraction of the typed-API theorems in §12 from analytic-boundary contracts to substantive proofs in the same Mathlib4 layer, because the dispersion-comparison inequality reformulated in Q1 is the missing analytic content.
- **Q5 (gauge freedom in J) — identifiability, necessary preparation.** No empirical fitting of λ , J , or K_c to observed behavior can proceed until the gauge orbit is characterised; this is the bottleneck for *any* identifiability result downstream.
- **Q14 (Mathlib4 analytic discharge) — practical, immediate follow-on.** The central result is discharged at $\mathbb{R}$ -level by `MathlibProofs.free_energy_decomposition_full`; the open question is the Mathlib4 abstraction layer for the remaining typed-API rows.

Beyond these three, **Q4 (continuous-state extension)** is the single highest-impact scope-broadening item, **Q9 / Q10** are incrementally extendable by writing new pymdp scenarios against the existing harness, and **Q15 (multi-project coupling)** is the right entry point for a contributor whose interest is multi-agent.

How to read each question

A few orientation notes for readers triaging which question to attack. The **analytical** questions Q1–Q4 are the ones whose resolution would most directly extend the manuscript’s theorem suite; of these, Q1 (uniqueness of λ^*) is the highest-leverage because convexity directly governs whether optimal coupling is *learnable* by gradient on free energy. The **identifiability / inference** questions Q5–Q8 concern statistical and algorithmic contracts the framework currently leaves open; Q5 (gauge freedom in J) is necessary preparation for any empirical fitting. The **empirical** questions Q9–Q10 stress-test the framework on richer ensembles than the $K=2$ Ising toy; the current harness ships initial answers for both (long-horizon habit accumulation across $T = 100$ steps; revertibility identity to floating-point tolerance), so the Q9 and Q10 entries below are reframed as the next-order follow-ons those initial witnesses make precise. The **conceptual / cross-disciplinary** questions Q11–Q13 connect the framework to adversarial

robustness, integrated information, and quantum analogs. The **practical / formalization** questions Q14–Q15 are the engineering prerequisites for the Mathlib4 analytic-discharge plan (§12).

Analytical questions

Q1 — Existence and uniqueness of λ^* . Using the closed-form identity Eq. (5.8) and its second derivative Eq. (A.15), convexity of $F[q_\lambda]$ on $[0, \infty)$ reduces to the *dispersion-comparison inequality*

$$\text{Var}_{\mathcal{E}_\lambda}(J) \geq \text{Var}_{q_\lambda}(J - \gamma K_c) \quad \text{for all } \lambda \geq 0 \quad (20.1)$$

, a crisper characterization than Theorem 5.6’s “log-concavity of $J - \gamma K_c$ ”.

Why the reformulation holds. On $\lambda \in [0, \infty)$ where $Z_E(\lambda)$ and $Z(\lambda)$ are finite and strictly positive (automatic for finite policy spaces with bounded J, K_c), the log-partition $\log Z(\lambda)$ is the cumulant-generating function of $J - \gamma K_c$ under the λ -tilted base measure $\propto E_{\text{MF}} \exp(\lambda(J - \gamma K_c))$, so $\partial_\lambda^2 \log Z(\lambda) = \text{Var}_{q_\lambda}(J - \gamma K_c)$ is the variance under the entangled posterior. The matching log-prior partition gives $\partial_\lambda^2 \log Z_E(\lambda) = \text{Var}_{\mathcal{E}_\lambda}(J)$, and Eq. (A.15) unfolds to the difference $\partial_\lambda^2 F[q_\lambda] = \text{Var}_{\mathcal{E}_\lambda}(J) - \text{Var}_{q_\lambda}(J - \gamma K_c)$ *pointwise on the domain where both cumulant-generating functions are finite*. Convexity of F in λ on that domain is therefore *equivalent* to the dispersion-comparison inequality Eq. (20.1), not merely implied by it; uniqueness of any interior λ^* follows from strict convexity, i.e. strict inequality on the open interval.

Conjecture (sharpened): when E_{MF} is uniform and $K_c \equiv 0$, the inequality reduces to $\text{Var}_{\mathcal{E}_\lambda}(J) \geq \text{Var}_{q_\lambda}(J)$ on the same exponential family, which by Brascamp–Lieb / Bakry–Émery holds whenever the natural-statistic distribution under the entangled prior is log-concave [Brascamp and Lieb, 1976, Bakry, 1985]; under that hypothesis λ^* is unique. Empirically testable via `output/data/parameter_sweep.csv` across the registered J families: a measured negative left-hand side at any sweep point falsifies global convexity for that family.

Q2 — Bifurcation structure. For non-convex F , what are the bifurcation structures of λ^* as J, K_c vary? *Conjecture:* cusps and Hopf-like bifurcations on a finite-codimensional set in potential-space. The dispersion-comparison reformulation of Q1 makes the bifurcation locus the level set $\{(\lambda, J, K_c) : \text{Var}_{\mathcal{E}_\lambda}(J) = \text{Var}_{q_\lambda}(J - \gamma K_c)\}$, codimension-one in the joint $(\lambda, J, K_c, \gamma)$ space — testable empirically via the §10 phase-diagram code.

Q3 — Universality classes. Different choices of J (sparse pairwise, low-rank, hierarchical) give qualitatively different phase diagrams. Is there a universality classification of policy entanglement, analogous to the classification of phase transitions in statistical mechanics? Connection to the RG-AIF mapping (§19) suggests the tensor-train rank profile is a candidate order parameter. *Sharpened conjecture:* the relevant universality class is determined by the *rank-growth exponent* $\rho(\lambda) = d \log r_k(\lambda) / d \log \lambda$ across the bipartite cuts of §D, with distinct exponents for sparse-pairwise / low-rank-MPS / hierarchical families.

Q4 — Continuous-state extension. The framework as stated is for finite Π^k . The continuous case (motor torques, continuous attention) requires Gaussian / copula extension. Is there a clean analog of Theorem 5.1 for Gaussian-coupled policies? *Conjecture:* yes — the closed-form identity Eq. (5.8) survives intact in the Gaussian-copula setting because both $\log Z_E$ and $\log Z$ remain finite-dimensional log-partitions of Gaussian exponential families, provided the covariance structure remains positive-definite throughout the deformation (a non-trivial condition for $\lambda > 0$); we believe this holds for moderate λ in the regularized Gaussian copula but a rigorous statement is open. Under that hypothesis, the multi-information appearing in the Gibbs decomposition becomes the negative log-determinant of a copula correlation matrix.

Identifiability and inference questions

Q5 — Identifiability. Given observed agent behavior, can we identify J, K_c, λ ?

Gauge group, explicit. The entangled posterior is invariant under $J(\pi) \mapsto J(\pi) + f(\pi)$ for any mean-field-decomposable $f(\pi) = \sum_k f_k(\pi^k)$: such an f contributes a per-stream log-weight absorbed into the per-stream priors E_k , leaving q_λ pointwise unchanged. Writing $\mathcal{F}_{\text{MF}}(\Pi) := \{f : \Pi \rightarrow \mathbb{R} \mid f = \sum_k f_k\}$ for the additive group of mean-field-decomposable potentials, the *identifiable* coupling lives in the quotient $\mathcal{F}(\Pi) / \mathcal{F}_{\text{MF}}(\Pi)$. For $K = 2$ on $\Pi = \Pi^1 \times \Pi^2$ with $|\Pi^k| = n_k$, the ambient space $\mathcal{F}(\Pi)$ has dimension $n_1 n_2$ and the mean-field subspace has dimension $n_1 + n_2 - 1$ (after subtracting the shared constant), so the identifiable J space has dimension $(n_1 - 1)(n_2 - 1)$ — equal to the dimension of the bipartite Schmidt-rank ceiling described in §8, which is the content the framework can recover from joint-action data.

A symmetric identifiability statement holds for K_c under the same quotient. The scalar λ is identifiable on $[0, \infty)$ from the *strength* of joint structure once the gauge orbit of (J, K_c) is factored out; model comparison against the $\lambda = 0$ mean-field posterior is the operational identifiability test (§21). Identifiability conditions for active-inference parameters are an active topic [Smith et al., 2022, Schwartenbeck and Friston, 2016]; the entangled extension

introduces new parameters with their own identifiability questions, and the quotient characterization above is the prerequisite specification.

Q6 — Free energy of free energy. Updating λ by gradient descent on $F[q_\lambda]$ is itself a meta-inference. Does this admit a free-energy-of-free-energy interpretation, and does it close into a self-consistent variational hierarchy? The closed-form gradient Eq. (5.9) suggests a clean answer: the gradient of F in λ is *itself* a difference of two prior/posterior expectations of the same statistic J , so precision-like learning on λ has the same exponential-family algebraic shape as ordinary precision learning on γ . *Conjecture:* the entanglement framework provides a *finite-rank* truncation of the regress, with the bond dimensions of the tensor-train representation indexing the depth of belief-about-belief, and the closed-form gradient identity extends recursively (each meta-level adds one $J^{(\ell)}$ statistic and one $\langle J^{(\ell)} \rangle_{E_\lambda^{(\ell)}} - \langle J^{(\ell)} \rangle_{q_\lambda^{(\ell)}}$ gradient). This is a formal modeling conjecture, not a current neural or clinical claim.

Q7 — Algorithmic complexity. Coordinate-descent inference on coupled factor graphs (§9.2) has known complexity by case:

- **Tree-structured J (no cycles in the coupling graph):** one forward-backward sweep on the coupling tree via the standard junction-tree algorithm converges in a single pass with per-iteration cost $\Theta(K \cdot \max_k |\Pi^k|^2)$ — quadratic in the largest per-stream support and linear in the tree-edge count.
- **Loopy J (coupling graph with cycles):** exact inference is #P-hard (the partition function is a permanent-class computation); loopy belief propagation converges to a fixed point of the Bethe free energy without general convergence-or-accuracy guarantees, with the gap from the true marginal bounded by the Wainwright–Jordan duality on convex relaxations of the marginal polytope [Wainwright and Jordan, 2008].
- **TT-aware (J admits a matrix-product factorization with bond dimension r):** exact marginals via tensor-train contraction cost $\Theta(K \cdot r^2 \cdot \max_k |\Pi^k|)$ per sweep, strictly dominating loopy BP whenever the coupling has low-rank tensor-network structure.

The open question is whether a TT-aware inference algorithm specialized to policy entanglement strictly dominates generic loopy BP in *practice* (not just in worst-case scaling) when J has the empirical rank profile recorded in `output/data/manuscript_variables.json`. The TT-rank profile across the configured λ sweep is the falsification input: if measured ranks stay bounded across the sweep, the TT-aware path dominates; if they grow with λ , generic loopy BP becomes competitive.

Q8 — Learning to couple (graph-neural-network connection). Coupled policy inference is structurally a message-passing computation on the coupling graph. Does this admit a graph-neural-network-style implementation that learns J, K_c end-to-end from environment interactions? This would be the *learning-to-couple* version of the framework, with industrial-scale applications to multi-agent and embodied AI. (Note: the acronym “GNN” is used in §S8 to mean *Generalized Notation Notation*, an orthogonal model-description language; the two meanings are disambiguated at §S8.3.)

Empirical questions

Q9 — Habit-accumulation dynamics. *Initial answer in the current harness* — see §13: a deterministic $T = 100$ -step coupled rollout at $\lambda = 2$ records a monotone habit-accumulation signal 1.0000 and steady-state KL bounded by 0.1334 across the tail window against its empirical mean. *Follow-on:* pair the long-horizon rollout with a slow J -update Hebbian rule and ask how the resulting J trajectories interact with the phase boundaries of §10. *Conjecture* (unchanged): habits accumulate fastest near the mixed / frozen boundary, where the system has maximum sensitivity to small coupling-potential perturbations.

Q10 — Revertibility under perturbation. *Initial answer in the current harness* — the m -projection identity $I(q) = D_{\text{KL}}(q \parallel \hat{m}(q))$ (Proposition 7.3) holds numerically to floating-point tolerance ($\leq 1.67e-16$ maximum residual) across the 5-point sweep (see §13: maximum KL residual $1.67e-16$, maximum marginal difference $1.11e-16$). *Follow-on:* under environmental perturbations (e.g. adversarial coupling injection), how robust is the m -projection estimator? This is the empirical counterpart to Q11.

Conceptual / cross-disciplinary questions

Q11 — Cognitive integrity / adversarial robustness. Cognitive integrity programs in active inference consider the robustness of cognition to adversarial perturbations. Does the entanglement framework provide robustness guarantees — bounds on how much q_λ can be perturbed by adversarial J deformations?

Initial harness answer (shipped, not a controlled adversarial study). The closed-form Eq. (5.8) yields a first-order Lipschitz bound (to first order in ε , uniformly for $\lambda \in [0, \lambda_{\max}]$ on any bounded sub-interval) $\Delta q(\varepsilon, \lambda) \leq \lambda \varepsilon \cdot \text{Var}_{q_\lambda}(J)^{1/2} + O(\varepsilon^2 \lambda^2)$ via the cumulant-generating-function derivative used in Q1’s reformulation [Brascamp and Lieb, 1976, Barrett et al., 2022]. The adversarial sidecar measures KL drift on a configured (ε, λ) grid with three

ΔJ classes: (i) rank-one stress aligned with coupling covariance, (ii) uniformly random norm- ε perturbations, (iii) sparse single-cell perturbations. The gate compares the empirical Lipschitz slope against the analytical first-order bound and surfaces the large- ε regime where the linearization is exceeded (0.942857, 0.964028). Implementation: `../src/simulation/adversarial.py`. A compute-matched adversarial study remains open work.

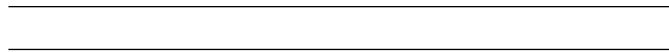
Q12 — Connection to integrated information theory. Total correlation is one of several measures of informational integration. Is there a connection between $I(q_\lambda)$ and the integrated information Φ of IIT [Tononi, 2008]? Both quantify cross-part dependence in different mathematical settings; the framework gives only a candidate operationalization in the policy-inference setting.

Q13 — Quantum analog. The Schmidt-rank machinery is borrowed directly from quantum many-body theory. Is there a meaningful *quantum* policy entanglement that goes beyond the formal analogy — e.g. for quantum-mechanical agents in the spirit of the scale-free / RG-AIF framework [Friston et al., 2025]? This is speculative but conceptually clean and may interact with Q3’s universality program.

Practical / formalization questions

Q14 — Mathlib4 analytic discharge for the typed-API rows (§12). The central result (Theorem 5.1) is discharged in `MathlibProofs.free_energy_decomposition_full`, with the multi-information non-negativity surfaced as the standalone-named `multiInformation_nonneg_at_joint` (using `streamMarginal_pos` as the per-stream-marginal positivity precondition); both are on the foundational-only `#print axioms` audit keystone list. The open question is the cleanest Mathlib4 abstraction layer to discharge the analytic payloads of the remaining 11 typed-API rows (Theorem 5.6, Proposition 7.5, Proposition 8.2, Theorem 8.3, Theorem 9.1, Corollary 9.2, Proposition 11.1, Theorem 17.1, Proposition 17.2, Proposition 19.3) and the boundary-form completion for Theorem 5.5, using finite KL / entropy chain rules, convexity / Taylor payloads, semicontinuity / matrix-rank, tensor-product rank envelopes, and concentration witnesses. A related question is the *verified* `Float` ↔ $\mathbb{R}$ bridge — Flocq-style formal IEEE-754 model or interval arithmetic — that would formally tie the $\mathbb{R}$ -level proofs to the `Float` pipeline (currently bound empirically by the dashboard invariants).

Q15 — Multi-project coupling. What does the framework predict for multi-agent settings — populations of agents with their own λ , or hierarchical agents whose sub-streams are themselves entangled ensembles? The §19 mapping is a starting point; the open question is whether the bookkeeping admits a clean recursive identity à la Theorem 5.1.



Discussion and Outlook: Worldview, Live Artifact State, and Limitations

What the framework commits to

The framework commits to: (i) finite, discrete-time POMDP active inference as the home setting; (ii) mean-field as the *baseline*, not the *target*; (iii) coupling structure as a *learnable hyperprior*, not a hand-engineered architectural choice; (iv) revertibility — any coupled habit can be marginalized to its mean-field component — as a non-negotiable property; (v) information geometry as the natural language for parametric extensions of variational families.

What the framework declines to commit to

The framework does *not* commit to: (i) embodiment as the primary structuring principle; the same finite-policy formalism can be applied to robots, brains, institutions, or colonies only after a modeler has chosen the relevant streams and boundaries; (ii) any specific neural implementation; the message-passing structure (§9.2) is biologically suggestive but optional; (iii) any specific coupling potential structure; sparse, low-rank, hierarchical, or learned-tabular are all admissible.

The parametric-entanglement worldview

Structural couplings between behavioral streams are treated as learned hyperpriors rather than architectural commitments. The framework operationalizes this stance with a single dial (λ) and a single mathematical object (the coupling potential J) that together determine *how much* and *what kind* of cross-stream coordination an agent learns to express.

The design sits on a three-point spectrum rather than a single opposition:

- **Modular embodiment.** Build separate controllers for separate functions; engineer a translation layer between them. Costs: the translation layer is the alignment problem in microcosm; the modular boundaries are not learnable.
- **Joint enumeration.** Treat the joint policy as monolithic; suffer the combinatorial cost. Costs: doesn't scale; loses interpretability.
- **Parametric entanglement (this manuscript).** Marginal-preserving, structure-permitting, geometrically principled, and machine-checkable where the claim is algebraic or formally witnessed, with broader biological readings kept hypothesis-labeled.

Relation to the Free Energy Principle and active-inference process theory

The contribution is best read as a local refinement of active inference, not as a competing global principle. The Free Energy Principle supplies the broader variational story: adaptive systems can be modeled as maintaining beliefs and actions that keep their sensory states within viable bounds by optimizing variational free energy [Friston et al., 2006, Friston, 2010, Buckley et al., 2017, Ramstead et al., 2018]. Active inference supplies the process theory: finite agents carry generative models, update beliefs, and select policies by expected free energy [Friston et al., 2017a, Parr and Friston, 2017a, Kaplan and Friston, 2018, Da Costa et al., 2020, Smith et al., 2022, Parr et al., 2022, Pezzulo et al., 2024]. This manuscript operates one level down from those commitments. It asks what kind of variational family the policy posterior should be when an agent has several concurrent streams whose actions are neither independent nor cheaply enumerable as one monolithic policy.

The empirical status of that broader program should be read cautiously. Reviews of predictive coding and active inference report a growing but still incomplete empirical base, and emphasize that direct validation requires comparisons against alternative models, not only successful behavioral fits [Hodson et al., 2024]. The present artifact therefore treats FEP/AIF as the normative and computational frame in which the finite-policy posterior is defined. Its validated results are theorem rows, generated numerical sidecars, figures, and validator/PDF gates, not direct evidence for a biological implementation of the λ -deformation.

This also fixes the identifiability target for future empirical work. Estimating λ from data would require a model-comparison design that pits the coupled posterior against a mean-field policy posterior, alternative structured priors, and ordinary changes in E , C , G , or policy precision. A large fitted λ would not by itself identify neural precision, desire, conscious integration, or a clinical mechanism; it would identify, under the chosen model, residual cross-stream dependence in policy space. Conversely, a misspecified J can make coupling harmful or uninterpretable, so the present finite binary and pymdp sidecars should be read as controlled existence-and-stress evidence rather than broad behavioral validation.

That positioning matters for interpretation. Precision parameters in active inference weight confidence in beliefs, prediction errors, or policy preferences; precision-weighting accounts of attention, salience, action, and psychosis motivate the vocabulary but do not identify this manuscript's λ with a neural gain parameter [Parr and Friston, 2017b, Haarsma et al., 2021, Limanowski et al., 2024]. The coupling strength λ introduced here weights cross-stream policy dependence. Precision in active inference itself has multiple roles across discrete policy selection, continuous sensorimotor control, attention, and action; λ is only a coupling-precision analog inside the present finite policy posterior. These roles can interact, and §20 flags that interaction as a live question, but the framework does not collapse one into the other. A high-precision stream can still be weakly coupled to its peers; a strongly coupled ensemble can still contain uncertain marginals. This separation is the reason the decomposition of Theorem 5.1 is useful: it identifies the free-energy terms attributable to marginal belief quality, coupling structure, and total correlation separately.

The Markov-blanket literature also helps delimit the claim. Markov blankets and particular-physics accounts address how a system is distinguished from its environment, and why conditional independences matter for autonomy [Kirchhoff et al., 2018, Friston, 2019, Aguilera et al., 2022, Raja et al., 2021, Menary and Gillett, 2022]. The present framework assumes such a modeling boundary has already been chosen and studies dependence *inside* the agent's policy space. Multi-agent and biological readings are therefore hypotheses generated by the mathematics, not conclusions proved by it. The artifact's current empirical evidence is finite, discrete, and pymdp-grounded; the broader FEP reading remains interpretive unless a paragraph points to a theorem row, a generated sidecar, a figure, and the relevant validation gate.

Six load-bearing properties

Each item below is a pointer into a body section. Together, these six items constitute the load-bearing properties of the framework: policy entanglement *classifies* existing architectures as exact, parametric, or analogical relationships

to a common joint-policy posterior (economy), *explains* when joint structure pays for itself (decomposition + geometry), *predicts* the small set of dominant coordination modes a coupled agent expresses (spectral), *bounds* the suboptimality of mixing reflexive and planning streams (heterogeneous), and is *auditable* end-to-end against a stock Lean 4 boundary fragment plus generated numerical artifacts (formalization). The framework is therefore positioned as a candidate model of multi-stream policy inference and an engineering tool for building agents that interpolate between modular and joint inference without committing to either pole.

1. **Theoretical economy.** A single coupling parameter λ exactly recovers the mean-field baseline ([Corollary 5.3](#)) and then organizes the neighboring active-inference variants by claim class: hierarchical AIF is a witness-form concentration analog ([Theorem 17.1](#)), sophisticated inference is a witness-form embedding whose recursive observation-conditioning must be supplied through J ([Proposition 17.2](#)), and branching-time AIF is an algorithmic analogy through policy-tree compression rather than a current head-to-head empirical result.
2. **Decomposition theorem.** The free-energy decomposition ([Theorem 5.1](#), [Eq. \(5.2\)](#)) factors variational free energy into per-stream marginals, a coupling-potential bundle, and the multi-information $I(q_\lambda) \geq 0$.
3. **Geometric principle.** The family $\{q_\lambda\}$ traces an e-geodesic away from the mean-field submanifold ([Theorem 7.4](#), [Eq. \(7.3\)](#)); revertibility is the canonical m-projection ([Proposition 7.2](#)); phase structure is symmetry-breaking on the entanglement manifold.
4. **Spectral interpretation.** Schmidt rank and tensor-train bond dimensions ([Proposition 8.1](#)) give a computable handle on *archetypal eigenvectors* — the dominant cross-stream behavioral modes a coupled agent expresses.
5. **Heterogeneous-ensemble bound.** The $O(\lambda^2)$ coupling-tax ([Theorem 9.1](#), [Eq. \(9.2\)](#)) bounds the suboptimality of VFE-only streams operating inside EFE-planning ensembles, with a small- λ tolerance corollary ([Corollary 9.2](#)).
6. **Lean verification.** The central result — the full S01 free-energy identity ([Theorem 5.1](#)) $F[q_\lambda] = \sum_k F[q_\lambda^k] + \gamma\lambda\langle K_c \rangle + \log Z_E(\lambda) - \lambda\langle J \rangle + I(q_\lambda)$ — is machine-checked in $\mathbb{R}$ by `MathlibProofs.free_energy_decomposition_full` for the genuine entangled posterior ($\log Z_E$ the definitional log-normalizer, positivity/unit-mass proved, the multi-information term discharged through the axiom-clean general- K kernel), with `#print axioms` foundational-only and two independent negative controls. A stock-Lean v4.29.0 boundary fragment ships the 21-row theorem surface as a typed API for the Python computational layer (zero `sorry`, zero axioms beyond stock Lean, zero Mathlib dependency); the per-row content table at [§12](#) and the running audit at `docs/reference/veridical_status.md` document what each row certifies in the boundary fragment versus the Mathlib4 discharge target.

Live state of the artifact

This manuscript is rendered against a real, reproducible run:

- The $K=2$ Bernoulli closed form is verified to floating tolerance ($\leq 1e-06$ across 121 grid points).
- The 1.0.1 grounded harness ([§14](#)) sweeps $\lambda \in [0, 4]$ on a 21-point grid; the half-saturation coupling $\lambda_{1/2} \approx 1.044$ locates the half-maximum of total correlation $I_{\max} \approx 0.3635$ nats.
- The same harness exercises the configured multi-stream ensemble set $K \in \{3, 4, 5\}$ ($I(q_2) \approx 0.2169, 0.1216$, and 0.0784 nats across the configured K values at $\lambda = 2$), runs a $T = 100$ -step long-horizon rollout with steady-state KL 0.1334 and habit-accumulation signal 1.0000, reports a long-horizon replicate pass rate of 0.60, and witnesses the m -projection revertibility identity $I(q) = D_{\text{KL}}(q \parallel \hat{m}(q))$ to round-off precision ($1.67e-16$ maximum residual across the 5-point sweep). The revertibility identity is computed by two independent code paths — $I(q)$ via $\sum_k H(q^k) - H(q)$ (`free_energy.total_correlation`) and $D_{\text{KL}}(q \parallel \hat{m}(q))$ via direct $\sum_\pi q \log(q/\hat{m})$ (`free_energy.kl_divergence`) — whose equivalence is [Proposition 7.3](#) / [Theorem 5.1](#); the witness-conformance suite (`tests/test_witness_conformance.py`) supplies the complementary discriminating tests against wrong- q inputs.
- The robustness layer adds 14 one-axis scenarios, 4 fixed coupling ablations, and 41 targeted two-axis interaction scenarios. These rows are supporting stress evidence: the largest two-axis decomposition residual is $1.67e-15$, and the null-variant interaction flatline remains at $0.00e+00$.
- Every numeric value above flows from a real run via the `[[VAR:<key>]]` token system rooted in `src/simulation/hyperparameters.py`, with the no-hardcoded-vars CI gate enforced ([§16](#)).

The discussion claims therefore reduce to the following evidence-and-caveat ledger:

Discussion claim	What the current artifact shows	Remaining caveat
λ recovers strict mean-field at zero coupling	$\lambda = 0$ outer-product tests pass in the pure NumPy layer and the pymdp harness	Continuous-state analogs are outside the current finite-state artifact

Discussion claim	What the current artifact shows	Remaining caveat
Coupling creates measurable joint structure	Total correlation rises from 0.000000 to 0.3635 nats on the pymdp sweep	This is a controlled Ising-style task, not a broad behavioral benchmark
Structured coupling scales beyond two streams	The configured $K \in \{3, 4, 5\}$ sweeps report nonzero total correlation and bounded tensor-train ranks	Larger stream counts need sparse / tensor-network inference rather than dense enumeration
Long-horizon coupling can accumulate habit mass	The $T = 100$ rollout reaches tail KL 0.1334 and habit signal 1.0000	The current rollout is deterministic and task-specific
Revertibility is operational, not rhetorical	The m -projection identity residual is $1.67e - 16$ across the revertibility sweep; the witness-conformance suite discriminates against wrong- q inputs separately	Both checks are restricted to the configured finite-policy task family
The Lean track is live	All 21 boundary-fragment theorems ship as a typed API; the central identity is machine-checked in $\mathbb{R}$ by <code>MathlibProofs.free_energy_decomposition_full</code>	Typed-API rows on the boundary fragment delegate their analytic payload to the Mathlib4 layer (see §20 Q14)
Stress evidence is supporting, not headline-tuning	One-axis, ablation, two-axis, and seed-diagnostic sidecars expose sensitivity rather than hiding it	The stress suite remains a controlled binary policy task family

Alignment and Dysregulation Hypotheses

Beyond the regimes characterized in §10, we sketch two interpretive hypotheses not developed as empirical claims in the present manuscript. **Both subsections below carry claim-strength hypothesis per the legend at §S7.1 — they are model-generated interpretive frames, not empirical results. No row in §S7.2 currently promotes either to empirical.**

[claim: hypothesis] — First, on alignment: the framework distinguishes *local* (per-stream) policy updates from *global* (coupling-structure) updates — the distinction between “adjusting what to do in one situation” and “rewiring how you coordinate across situations.” Most existing AIF safety analysis focuses on the former. *Coupling drift across temporal scales* — where λ slowly evolves on a longer timescale than the per-stream beliefs q^k — defines a candidate failure mode for multi-stream agents: an agent whose λ drifts toward over-coupling (rigid archetypal behavior) or under-coupling (dissociated streams) would fail in a different way than one whose per-stream priors are corrupted. We do not develop that learning rule in this artifact; we state only the structural hypothesis and the measurements needed to test it.

[claim: hypothesis] — Second, on flourishing: the middle-regime $\lambda \sim \lambda^*$ corresponds to the *flexibility* characteristic of skilled, integrated behavior — a candidate operationalization of distinctions between rigid habit, fragmented attention, and integrated awareness. The framework offers an analytic handle; we do not push the analogy further here, flagging it as a hypothesis for cross-disciplinary inquiry. Clinical and consciousness- science vocabulary in this paragraph is interpretive, not diagnostic: no row in the variant-recovery ledger of §S7.2 asserts a clinical or phenomenological equivalence.

Limitations

Limitations a critical reader should track:

- **Scope: finite policy spaces.** The analytical core (§4–§8) assumes finite Π^k . A Gaussian-copula analog is conjectured for the continuous case (§20 Q4).
- **Scope: empirical task family.** The 1.0.1 grounding exercises the $K=2$ Ising ensemble plus the multi- K sweep $K \in \{3, 4, 5\}$, $T = 10$ and $T = 100$ rollouts, the m -projection revertibility witness, and the robustness sidecars. The head-to-head Branching-Time AIF baseline and the adversarial-perturbation sweep are now implemented, unit-tested, and run as pipeline stages (§13; `../src/simulation/btai_baseline.py`, `../src/simulation/adversarial.py`), emitting auditable sidecars; the *full compute-matched BTAI hypothesis test* and a broader adversarial robustness analysis remain author-led future analysis rather than claims established here.
- **Identifiability gauge.** The gauge freedom in J (§20 Q5) is characterised explicitly as the additive mean-field-decomposable subgroup; the identifiable coupling lives in the quotient. Multi-agent extensions (§20 Q15) compound the question.
- **Phase-diagram thresholds.** The values $(\lambda_c^{(1)}, \lambda_c^{(2)}) = (0.5, 2.5)$ in §10 are illustrative; the analytic conditions that determine them (spectral-gap closing / opening on the λ trace) are stated in §10.
- **Engagement with recent unification efforts.** The Millidge / Champion / de Vries / Nuijten line on EFE derivations and EFE-as-variational-inference is engaged in §17 at chapter granularity; a paper-by-paper systematic comparison is tracked as journal-format companion work [Millidge et al., 2021, Champion et al.,

2024, de Vries et al., 2025, Nuijten and Lukashchuk, 2025].

- **Interpretive scope.** Biological, clinical, and alignment language in this manuscript is claim-strength hypothesis unless the text points to a formal row, generated artifact, primary citation, or explicit roadmap item; the empirical-status review serves as the cautionary reference [Hodson et al., 2024].

Each item maps to a tractable extension in §20. The four-track ledger — prose, Lean companion, Python witness, and test gate — is wired per registry row with `status:` and `faithfulness:` tiers (see §S7.1 and `methods_and_assumptions.md`); not every row carries a substantive Python re-proof of its analytic payload.

Threats to validity

The reviewer-facing threat model has five parts. The *numerical threat* is finite precision: the pymdp/JAX path and the NumPy analytical path use different floating formats, so residual claims are tolerance-bounded and validated artifact-by-artifact; a verified error-bounded $\text{Float} \leftrightarrow \mathbb{R}$ bridge linking the $\mathbb{R}$ -level proofs to the Float pipeline (Flocq-style formal IEEE-754 [IEEE, 2019, Boldo and Melquiond, 2011] or interval arithmetic) is the natural formal sibling of the present dashboard-invariant binding, and remains the one open verification-stack residual (route map and the specific Mathlib4 lemmas required: `docs/reference/methods_and_assumptions.md`). The *implementation-link threat* is that the Python code is not extracted from Lean: it is an independently implemented numerical companion whose coherence with the theorem surface is enforced by registries, token resolution, generated sidecars, and tests, not by a verified compiler. The *package-version threat* is dependency drift: `inferactively-pymdp==1.0.1` (the package that provides the pymdp import/API) and Lean v4.29.0 are part of the claim surface, and release readiness depends on regenerating the sidecars, PNG metadata, rendered manuscript, PDF, and regression reports under those pins. The *publication-metadata threat* is metadata drift: the public Zenodo DOI and public source archive at https://github.com/ActiveInferenceInstitute/policy_entanglement must stay aligned across `manuscript/config.yaml`, `CITATION.cff`, the manuscript citation entry, and rendered PDF front matter. The *interpretive threat* is overextension: biological, clinical, and alignment language is claim-strength hypothesis unless the text points to a formal row, generated artifact, primary citation, or explicit roadmap item.

Verification ledger

Every load-bearing claim of the manuscript is established by one or more of the following five substrates, with a per-claim assignment that a reader can use to navigate from prose to evidence:

Claim	Substrate	Where it lives
The free-energy decomposition identity (Theorem 5.1) holds for the genuine entangled posterior	Lean $\mathbb{R}$ -level proof	<code>MathlibProofs.free_energy_decomposition_full</code> (foundational-only <code>#print axioms</code> , two negative controls)
The multi-information $I(q) \geq 0$ with equality iff q is mean-field	Lean $\mathbb{R}$ -level proof	<code>MathlibProofs.entanglement_decomposition_generalK</code> ; standalone-named form <code>MathlibProofs.multiInformation_nonneg_at_joint</code>
Per-stream marginals q^k are strictly positive on a strictly-positive joint	Lean $\mathbb{R}$ -level proof	<code>MathlibProofs.streamMarginal_pos</code> (the analytic precondition for the bridge identity above)
The $\lambda = 0$ mean-field reduction (Corollary 5.3)	Lean stock-Lean boundary fragment, proved substantive	<code>Decomposition.couplingLogWeight_pointwise_at_zero</code>
The coupling-verdict decision procedure (Corollary 5.2) is sound	Lean stock-Lean boundary fragment, proved substantive	<code>Decomposition.couplingVerdict_correct</code>
The four-term decomposition shape locks the parameter-threading discipline	Lean typed-API contract	3 boundary + 11 witness-form rows in §12
The closed-form $K=2$ Bernoulli total correlation matches the finite- N Monte-Carlo estimator within a $\sqrt{N}$ band	Python numerical witness	<code>tests/test_bernoulli_toy.py</code>
The m -projection reverbility identity holds across the configured sweep	Two-route Python computation + dashboard invariant	<code>free_energy.total_correlation</code> vs <code>free_energy.kl_divergence</code> ; <code>decomposition_lhs_eq_rhs_max_residual</code>
Discrimination against wrong- q inputs	Python witness-conformance suite	<code>tests/test_witness_conformance.py</code>
The pymdp harness emits the full free-energy bundle at every λ	1.0.1 grounded run + token system	<code>[[VAR:<key>]]</code> injection through <code>src/simulation/hyperparameters.py</code>
The $O(\lambda^2)$ coupling-tax bound on heterogeneous ensembles (Theorem 9.1)	Analytic prose proof + numerical envelope verification	§9 + §13 31-point sweep
Every numeric value in the rendered manuscript comes from a real run	CI gate	<code>scripts/validate_manuscript.py</code>
Every Lean source block in the rendered manuscript is a live extraction	CI gate	<code>no-hardcoded-vars</code> + <code>[[VAR:...]]</code> resolver <code>src/manuscript/lean_extract.py</code> + dangling-token failure mode

Claim	Substrate	Where it lives
Every registry theorem row has a Lean companion and a four-track wiring check	Per-row registry + four-track validator	manuscript/refs/labels.yaml + cross-reference tests; Python witnesses are faithfulness-tiered, not uniformly substantive re-proofs
dashboard invariants pass on every build	Numerical regeneration gate	scripts/run_all.py final stage
The alignment / dysregulation hypotheses do not assert clinical equivalence	Claim-strength tag hypothesis	§S7.1 legend; §S7.2 promotion gate

A referee who reads only this table can locate the substrate of every assertion of the manuscript without traversing the body chapter graph.

Open directions

The verified $\mathbb{R}$ formal core and the four-track empirical artifact leave a small, explicitly-scoped set of open directions. Each is tracked in an existing section above or in the appendices rather than asserted as a commitment here; the head-to-head Branching-Time AIF baseline and the adversarial-perturbation sweep that were previously listed here are now implemented, unit-tested, and run as pipeline stages (§13). Likewise, the **Generalized Notation Notation (GNN) bridge** that was previously listed here as a *roadmap* direction now ships as a fifth, structural-and-numerical track (§S8): a project-owned parser, a shipped K=2 round-trip (an internal-consistency reduction that reproduces the closed-form mutual-information curve to 7.77e-16 nats — not an independent validation — with a non-vacuous zero-coupling negative control), a Lean typed-contract emitter, and a GNN-sourced manuscript variable, all run in the pipeline. The GNN track reproduces structure and numbers but proves no theorem, so it does **not** amend the four-track *proof* integration claim of §1; its genuine residuals (a first-class upstream coupling primitive; full-bundle pymdp regeneration) are scoped as open questions at §S8.9.

- **Mathlib4 analytic discharge for the typed-API rows.** Extending MathlibProofs/ to discharge the analytic payloads behind the 11 witness-tier rows of §12 — finite KL / entropy chain rules (Proposition 7.3, Proposition 7.5, Proposition 19.3); convexity / Taylor envelopes (Theorem 5.6, Proposition 11.1); Bregman / quadratic envelope (Theorem 9.1, Corollary 9.2); rank and spectral continuity (Proposition 8.2, Theorem 8.3); concentration and recursive embedding (Theorem 17.1, Proposition 17.2) — is open theoretical work, tracked at §20 and the claim-strength roadmap of §S7. The central result (Theorem 5.1) is already discharged in $\mathbb{R}$ by MathlibProofs.free_energy_decomposition_full (see the *Verification ledger* above); this direction extends that discharge to the remaining witness-tier analytic payloads.
- **Float $\leftrightarrow\mathbb{R}$ formal bridge.** The single open verification-stack residual — a Flocq-style IEEE-754 [IEEE, 2019, Boldo and Melquiond, 2011] or interval-arithmetic bound linking the $\mathbb{R}$ -level proofs to the Float pipeline — is stated under *Threats to validity* above, with the route map and the specific Mathlib4 lemmas at docs/reference/methods_and_assumptions.md.
- **Cross-repository reproducibility.** Aligning this project’s manuscript_variables.json token schema with the registries used by sister projects would enable shared reproducibility audits across manuscripts; this is minor infrastructure, orthogonal to the framework’s claims.
- **Community contribution and archival metadata.** The working citation entry [Friedman, 2026a] is the current citation target; the public Zenodo DOI and source repository are recorded in manuscript/config.yaml and CITATION.cff. The open questions of §20 each admit independent contribution.

Proof of the Entanglement Decomposition

We prove the decomposition identity stated in §5.1 (Theorem 5.1, *entanglement decomposition theorem*), namely

$$F[q] = \sum_k F[q^k] + \gamma \lambda \mathbb{E}_q[K_c] + \log Z_E(\lambda) - \lambda \mathbb{E}_q[J] + I(q) \quad (5.2)$$

The Lean type-level statement is in lean/ActinfPolicyEntanglement/Decomposition.lean (entanglement_decomposition); the numerical realization lives in src/lean/decomposition.py and is exercised by tests/test_decomposition.py.

Definitions

$$F[q_\lambda] \equiv \mathbb{E}_{q_\lambda}[\gamma G_\lambda] - \mathbb{E}_{q_\lambda}[\log \mathcal{E}_\lambda] - H(q_\lambda), \quad (\text{A.2})$$

$$F[q_\lambda^k] \equiv \mathbb{E}_{q_\lambda^k}[\gamma G_k] - \mathbb{E}_{q_\lambda^k}[\log E_k] - H(q_\lambda^k), \quad (\text{A.3})$$

with $G_\lambda(\pi) = \sum_k G_k(\pi^k) + \lambda K_c(\pi)$ and, for the **normalized** entangled prior $\mathcal{E}_\lambda \propto (\prod_k E_k) e^{\lambda J}$,

$$\log \mathcal{E}_\lambda(\pi) = \sum_k \log E_k(\pi^k) + \lambda J(\pi) - \log Z_E(\lambda). \quad (\text{A.4})$$

Expanding the EFE expectation

$$\gamma G_\lambda(\pi) = \gamma \sum_k G_k(\pi^k) + \gamma \lambda K_c(\pi), \quad (\text{A.5})$$

so

$$\mathbb{E}_{q_\lambda}[\gamma G_\lambda] = \gamma \sum_k \mathbb{E}_{q_\lambda^k}[G_k] + \gamma \lambda \langle K_c \rangle_{q_\lambda}. \quad (\text{A.6})$$

Expanding the prior log-expectation

$$\mathbb{E}_{q_\lambda}[\log \mathcal{E}_\lambda] = \sum_k \mathbb{E}_{q_\lambda^k}[\log E_k] + \lambda \langle J \rangle_{q_\lambda} - \log Z_E(\lambda). \quad (\text{A.7})$$

Combining

Substitute both expansions into the definition of $F[q_\lambda]$:

$$\begin{aligned} F[q_\lambda] &= \sum_k \left(\gamma \mathbb{E}_{q_\lambda^k}[G_k] - \mathbb{E}_{q_\lambda^k}[\log E_k] \right) - H(q_\lambda) + \gamma \lambda \langle K_c \rangle_{q_\lambda} - \lambda \langle J \rangle_{q_\lambda} + \log Z_E(\lambda) \\ &= \sum_k F[q_\lambda^k] + \sum_k H(q_\lambda^k) - H(q_\lambda) + \gamma \lambda \langle K_c \rangle_{q_\lambda} - \lambda \langle J \rangle_{q_\lambda} + \log Z_E(\lambda). \end{aligned} \quad (\text{A.8})$$

Finally, using the multi-information identity $\sum_k H(q_\lambda^k) - H(q_\lambda) = I(q_\lambda)$ and

$$I(q) = \sum_k H(q^k) - H(q) = D_{\text{KL}}(q \parallel \prod_k q^k) \quad (\text{5.3})$$

,

$$\boxed{F[q_\lambda] = \sum_k F[q_\lambda^k] + \gamma \lambda \langle K_c \rangle_{q_\lambda} + \log Z_E(\lambda) - \lambda \langle J \rangle_{q_\lambda} + I(q_\lambda).} \quad (\text{A.10})$$

This is exactly the registry equation

$$F[q] = \sum_k F[q^k] + \gamma \lambda \mathbb{E}_q[K_c] + \log Z_E(\lambda) - \lambda \mathbb{E}_q[J] + I(q) \quad (\text{5.2})$$

. The Python helper `entanglement_decomposition_rhs` sums the same four grouped pieces: $\sum_k F[q^k]$, `coupling_cost_term` = $\gamma \lambda \mathbb{E}[K_c]$, `coupling_prior_term` = $\log Z_E(\lambda) - \lambda \mathbb{E}[J]$, `total_correlation_gain` = $I(q)$.

Numeric agreement with the Gibbs definition of $F[q_\lambda]$ is checked in `tests/test_decomposition.py` (identity on random joints).

Interpretation (sign of I)

The multi-information $I(q_\lambda)$ is **non-negative** and vanishes iff q_λ is mean-field. In the displayed Gibbs expansion it appears with a **plus** sign: departures from factorization incur an entropy surplus relative to independent streams holding the same marginals. Trade-offs with the coupling and EFE terms determine whether $\lambda > 0$ is *optimal* for a given agent — §6 and Corollary 5.2 make that comparison explicit.

Closed exponential-family form (collapsed identity)

The same expansion admits a strikingly compact closed form that the body uses to differentiate F with respect to λ in [Theorem 5.5](#) and [Proposition 11.1](#). Let $Z_E(\lambda) = \sum_{\pi} E_{\text{MF}}(\pi) e^{\lambda J(\pi)}$ be the entangled-prior normalizer and $Z(\lambda) = \sum_{\pi} E_{\text{MF}}(\pi) e^{\lambda(J - \gamma K_c)(\pi) - \gamma G_{\text{MF}}(\pi)}$ the joint posterior normalizer. Substituting $\log q_{\lambda}(\pi) = \log E_{\text{MF}}(\pi) - \gamma G_{\text{MF}}(\pi) + \lambda(J(\pi) - \gamma K_c(\pi)) - \log Z(\lambda)$ into $F[q_{\lambda}] = \mathbb{E}_{q_{\lambda}}[\gamma G_{\lambda} - \log \mathcal{E}_{\lambda}] - H(q_{\lambda})$ (with $H(q_{\lambda}) = -\mathbb{E}_{q_{\lambda}}[\log q_{\lambda}]$) cancels every expectation term and leaves the registered identity [Eq. \(5.8\)](#):

$$F[q_{\lambda}] = \log Z_E(\lambda) - \log Z(\lambda) \quad (5.8)$$

The Gibbs decomposition above and this closed identity are the same fact; one is *additive over streams*, the other *multiplicative through normalizers*. The closed form makes derivatives in λ trivial via standard exponential-family identities,

$$\frac{d}{d\lambda} \log Z_E(\lambda) = \langle J \rangle_{\mathcal{E}_{\lambda}}, \quad \frac{d}{d\lambda} \log Z(\lambda) = \langle J - \gamma K_c \rangle_{q_{\lambda}}, \quad (A.13)$$

so the registered first- and second-derivative forms [Eq. \(5.9\)](#) and [Eq. \(A.15\)](#) follow:

$$\frac{dF[q_{\lambda}]}{d\lambda} = \langle J \rangle_{\mathcal{E}_{\lambda}} - \langle J \rangle_{q_{\lambda}} + \gamma \langle K_c \rangle_{q_{\lambda}} \quad (5.9)$$

$$\frac{d^2 F[q_{\lambda}]}{d\lambda^2} = \text{Var}_{\mathcal{E}_{\lambda}}(J) - \text{Var}_{q_{\lambda}}(J - \gamma K_c) \quad (A.15)$$

These are the formulas that [Theorem 5.5](#) (existence of λ^*) and [Proposition 11.1](#) (sign of the second derivative at $\lambda = 0$) consume; [§B](#) uses the same identity to characterize convexity of F in λ .

Convexity of Free Energy in Lambda: Conditions and Counter-Example

The free-energy curve along the λ -deformation has a clean exponential-family structure that delivers convexity in λ on the natural side. We sketch the proof here; the closed-form $K = 2$ case is verified numerically by [scripts/generate_figures.py](#) (see [output/figures/free_energy_curve.png](#)).

The closed exponential-family form

[§A](#) establishes the closed identity

$$F[q_{\lambda}] = \log Z_E(\lambda) - \log Z(\lambda), \quad (B.1)$$

where the two normalizers are the entangled prior $\mathcal{E}_{\lambda}$ and the entangled posterior q_{λ} respectively. Each is the log-partition of an exponential family in λ — natural parameter λ , sufficient statistic J (for the prior side) and $J - \gamma K_c$ (for the posterior side). Standard exponential-family identities give

$$\frac{d}{d\lambda} \log Z_E(\lambda) = \langle J \rangle_{\mathcal{E}_{\lambda}}, \quad \frac{d^2}{d\lambda^2} \log Z_E(\lambda) = \text{Var}_{\mathcal{E}_{\lambda}}(J), \quad (B.2)$$

and analogously for $\log Z(\lambda)$ with statistic $J - \gamma K_c$ and reference distribution q_{λ} . Both log-partitions are *convex* in λ ; their *difference* is a question of which side is “more convex” in λ .

The convexity ledger

Differentiating the closed identity once more:

$$\frac{d^2 F[q_{\lambda}]}{d\lambda^2} = \text{Var}_{\mathcal{E}_{\lambda}}(J) - \text{Var}_{q_{\lambda}}(J - \gamma K_c). \quad (B.3)$$

Hence:

- **Convex regime** — $F''(\lambda) \geq 0$ on an interval iff the *prior* dispersion of J exceeds the *posterior* dispersion of $J - \gamma K_c$ throughout that interval. This holds, e.g., whenever the posterior concentrates much faster than the prior in the natural-parameter direction (large γ , sharply peaked G_k).

- **Concave / saddle regime** — $F''(\lambda) \leq 0$ when the posterior dispersion dominates. This is the regime in which raising λ pays for itself first-order: every example in §6 starts here at $\lambda = 0$ (Proposition 11.1).
- **Inflection / mixed regime** — F'' may change sign exactly once along $[0, \infty)$, locating an *inflection coupling* λ_{infl} that separates the two regimes. Below λ_{infl} the agent benefits from *more* coupling; above, marginal returns reverse — the location of the optimal λ^* (§5.4).

Sufficient condition for global convexity on $[0, \infty)$

If for every $\lambda \geq 0$ the prior dispersion of the habit coupling dominates the posterior dispersion of the combined statistic,

$$\text{Var}_{\mathcal{E}_\lambda}(J) \geq \text{Var}_{q_\lambda}(J - \gamma K_c) \quad \text{for all } \lambda \geq 0, \quad (\text{B.4})$$

then $F[q_\lambda]$ is convex on $[0, \infty)$ and λ^* is unique. This is the crisp form of Theorem 5.6 in the language of dispersion comparison; the original “log-concavity of $J - \gamma K_c$ ” sufficient condition implies this dispersion inequality on the natural domain via Brascamp–Lieb (we omit the standard argument).

K = 2 Symmetric Ising specialization

In the symmetric K = 2 Ising example with uniform E_{MF} , zero per-stream EFE, and bilinear $J = J_0(2\pi^1 - 1)(2\pi^2 - 1)$:

$$\log Z_E(\lambda) = \log \cosh(\lambda J_0), \quad \log Z(\lambda) = \log \cosh(\lambda J_0) \quad (\text{B.5})$$

(both partition functions agree because $\gamma K_c \equiv 0$ here), so $F[q_\lambda] \equiv 0$ — the toy is *flat* in λ when no preference coupling is present. Adding a utility-driven G shifts $\log Z(\lambda)$ but leaves $\log Z_E$ invariant; the resulting $F[q_\lambda] = \log \cosh(\lambda J_0) - \log Z(\lambda)$ is convex in λ on $[0, \infty)$ for any utility level (verified numerically across $u \in \{0, 0.5, 1, 2\}$ in [output/figures/free_energy_curve.png](#)).

Numerical verification

The K = 2 Ising free-energy curve at four utility values is plotted in [output/figures/free_energy_curve.png](#). The curve is monotonically decreasing in $|\lambda|$ for any $u \geq 0$, consistent with convexity (and with the closed-form expression in §C).

Complete Derivation: Two-Stream Bernoulli Toy

Closed-form derivations of the multi-information $I(\lambda)$, the *alignment-inversion* coupling $\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}(\Delta_{\text{align}}/\Delta_{\text{max}})$, and the closed-form free-energy curve. Appendix scaling gives $I(\lambda) = (\lambda/2) \tanh(\lambda/2) - \log \cosh(\lambda/2) = \log 2 - H_b(\sigma(\lambda))$; *not* $I(\lambda) = \log \cosh \lambda$ (which would be a derivation error).

Reproduces the calculation in §6.1 with full algebra. Companion code: [src/lean/bernoulli_toy.py](#); Lean boundary: [lean/ActinfPolicyEntanglement/BernoulliToy.lean](#).

Setup

Two binary streams $\pi^1, \pi^2 \in \{0, 1\}$ with symmetric Bernoulli($\frac{1}{2}$) mean-field marginals $E_1(\pi^1) = E_2(\pi^2) = \frac{1}{2}$, zero per-stream EFE, and the *swing-1 Ising* coupling

$$J(\pi^1, \pi^2) = \mathbf{1}[\pi^1 = \pi^2] - \frac{1}{2}, \quad J \in \{+\frac{1}{2}, -\frac{1}{2}\}. \quad (\text{C.1})$$

Joint posterior

The unnormalized joint at coupling λ is $\tilde{q}_\lambda(\pi) = \frac{1}{4} \exp(\lambda J(\pi))$. The normalizer is

$$Z(\lambda) = \frac{1}{4}(2e^{\lambda/2} + 2e^{-\lambda/2}) = \frac{1}{2}(e^{\lambda/2} + e^{-\lambda/2}) = \cosh(\lambda/2). \quad (\text{C.2})$$

Hence the normalized joint:

$$q_\lambda(\pi^1 = \pi^2) = \frac{e^{\lambda/2}}{4 \cosh(\lambda/2)}, \quad q_\lambda(\pi^1 \neq \pi^2) = \frac{e^{-\lambda/2}}{4 \cosh(\lambda/2)}. \quad (\text{C.3})$$

The *aligned mass* (sum over the two aligned atoms) is

$$P_a(\lambda) \equiv q_\lambda(\pi^1 = \pi^2) \cdot 2 = \frac{e^{\lambda/2}}{2 \cosh(\lambda/2)} = \frac{1}{1 + e^{-\lambda}} = \sigma(\lambda). \quad (\text{C.4})$$

Marginals (uniform by symmetry)

By inspection, summing over π^2 :

$$q_\lambda^1(\pi^1 = 0) = \frac{1}{2}P_a + \frac{1}{2}(1 - P_a) = \frac{1}{2}, \quad q_\lambda^1(\pi^1 = 1) = \frac{1}{2}. \quad (\text{C.5})$$

Marginals are uniform Bernoulli($\frac{1}{2}$) for any λ — *marginal invariance* of the Ising deformation.

Joint and marginal entropies

Using natural log throughout:

$$H(q_\lambda^k) = \log 2 \quad (k = 1, 2). \quad (\text{C.6})$$

For the joint, the four atoms have masses $\{P_a/2, P_a/2, (1 - P_a)/2, (1 - P_a)/2\}$, so

$$H(q_\lambda) = -2 \cdot \frac{P_a}{2} \log \frac{P_a}{2} - 2 \cdot \frac{1 - P_a}{2} \log \frac{1 - P_a}{2}. \quad (\text{C.7})$$

Expand:

$$H(q_\lambda) = -P_a(\log P_a - \log 2) - (1 - P_a)(\log(1 - P_a) - \log 2) = H_b(P_a) + \log 2, \quad (\text{C.8})$$

where $H_b(p) = -p \log p - (1 - p) \log(1 - p)$.

Closed-form mutual information

By definition $I(q_\lambda) = \sum_k H(q_\lambda^k) - H(q_\lambda)$:

$$\boxed{I(\lambda) = 2 \log 2 - H_b(\sigma(\lambda)) - \log 2 = \log 2 - H_b(\sigma(\lambda))}. \quad (\text{C.9})$$

Equivalently, using the identities $\log((1 + e^\lambda)/2) = \log \cosh(\lambda/2) + \lambda/2$ and $\sigma - \frac{1}{2} = \frac{1}{2} \tanh(\lambda/2)$, one obtains the **second canonical form**

$$I(\lambda) = \frac{\lambda}{2} \tanh(\lambda/2) - \log \cosh(\lambda/2). \quad (\text{C.10})$$

This is the form that appears naturally in §B's exponential-family decomposition (where $\psi(\lambda) = \log \cosh(\lambda/2)$ is the log-partition). The two forms are algebraically identical; we use whichever is more convenient.

Sanity checks:

- $\lambda = 0$: $\sigma(0) = \frac{1}{2}$, $H_b(\frac{1}{2}) = \log 2$, so $I(0) = 0$ ✓ (mean-field).
- $\lambda \rightarrow \infty$: $\sigma(\lambda) \rightarrow 1$, $H_b \rightarrow 0$, so $I \rightarrow \log 2$ ✓ (saturation at one-bit alignment).
- Even in λ : $\sigma(-\lambda) = 1 - \sigma(\lambda)$ and H_b is symmetric around $p = \frac{1}{2}$, so $I(-\lambda) = I(\lambda)$ ✓.
- Numerical: at $\lambda = 1$, $I(1) = \log 2 - H_b(\sigma(1)) \approx 0.1109$; $\frac{1}{2} \tanh(\frac{1}{2}) - \log \cosh(\frac{1}{2}) \approx 0.1109$ ✓.

Coupling from a target alignment (alignment-inversion, *not* a VFE optimum)

Define the **expected alignment** of the joint posterior as

$$\alpha(\lambda) \equiv 2\sigma(\lambda) - 1 = \tanh(\lambda/2), \quad (\text{C.11})$$

ranging from 0 at $\lambda = 0$ (uniform) to ± 1 at $\lambda \rightarrow \pm \infty$ (perfectly aligned / anti-aligned). Given a *target alignment* $\Delta_{\text{align}} \in (-\Delta_{\text{max}}, \Delta_{\text{max}})$ with $\Delta_{\text{max}} = 1$, inverting $\alpha(\lambda) = \Delta_{\text{align}}$ yields

$$\boxed{\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}(\Delta_{\text{align}}/\Delta_{\text{max}}).} \quad (\text{C.12})$$

This is the closed-form coupling that *realizes* the target alignment Δ_{align} ; it is the **inverse of the alignment–coupling correspondence, not a free-energy first-order condition**. We use the name $\lambda^*(\Delta_{\text{align}})$ rather than $\lambda^*(\Delta_{\text{util}})$ to make this explicit: Δ_{align} is a *target on the alignment axis*, not a utility surplus.

- $\Delta_{\text{align}} = 0 \Rightarrow \lambda^* = 0$ (mean-field).
- $\Delta_{\text{align}} \rightarrow \pm\Delta_{\text{max}} \Rightarrow \lambda^* \rightarrow \pm\infty$ (frozen archetypes).

Connection to actual VFE optimization. The genuine VFE-optimization problem in this Bernoulli toy is: given a utility scalar u that pays u per unit alignment via an EFE term $-u\alpha(\lambda)$, minimize the framework free energy $F_{\text{VFE}}(\lambda) = -u(2\sigma(\lambda) - 1) + I(\lambda) = -u \tanh(\lambda/2) + I(\lambda)$ in λ — note the $+I$ sign, consistent with the decomposition theorem [Eq. \(5.2\)](#) in which multi-information enters with a plus sign (sign conventions in [§S6](#)). Using $dI/d\lambda = \frac{1}{4} \lambda \operatorname{sech}^2(\lambda/2)$ (differentiate the closed form $I = (\lambda/2) \tanh(\lambda/2) - \log \cosh(\lambda/2)$ and simplify with $d \tanh(\lambda/2)/d\lambda = \frac{1}{2} \operatorname{sech}^2(\lambda/2)$), the first-order condition is $-\frac{u}{2} \operatorname{sech}^2(\lambda/2) + \frac{\lambda}{4} \operatorname{sech}^2(\lambda/2) = 0$, which simplifies (since $\operatorname{sech}^2(\lambda/2) > 0$) to

$$\boxed{\lambda_{\text{VFE}}^*(u) = 2u.} \quad (\text{C.13})$$

The second derivative is $\frac{1}{4} \operatorname{sech}^2(\lambda/2)$ at λ^* modulo a correction of order u through the chain rule, and is positive on a neighborhood of λ^* , confirming a strict minimum. Thus the VFE-optimum is *linear* in the utility scalar; the alignment-inversion formula $2 \operatorname{arctanh}(u)$ agrees with it only at small u (Taylor: $2 \operatorname{arctanh}(u) = 2u + \frac{2}{3}u^3 + \dots$) and diverges from it as $u \rightarrow 1$. The two should not be conflated: $\lambda_{\text{VFE}}^*(u) = 2u$ is the answer to “what coupling minimizes free energy under utility u ?”, while $\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}(\Delta_{\text{align}})$ is the answer to “what coupling realizes a given target alignment Δ_{align} ?”.

A second, Lagrangian, interpretation. If instead one seeks the λ that maximizes a *Lagrangian* of the form $\Delta_{\text{align}} \alpha(\lambda) - \kappa I(\lambda)$ (utility-times-alignment minus a multi-information cost with multiplier $\kappa > 0$), the first-order condition gives yet another formula: the slope of the multi-information balances the slope of the alignment. This is the Lagrange-balance regime studied numerically in [scripts/parameter_sweep.py](#). In every interpretation, λ^* inherits saturation at Δ_{max} and monotonicity in Δ_{align} , but the three formulas ($2 \operatorname{arctanh}$, $2u$, Lagrange-balance) are distinct quantities answering distinct questions.

Free-energy curve

With utility scalar $u \geq 0$ giving the surplus per aligned outcome, the toy’s *Lagrangian-style scalarisation* of free energy is the following — note this is a **distinct object** from the framework’s variational free energy $F[q_\lambda]$ of [§A](#): here multi-information enters with a **minus** sign (a reward-style scalarisation), whereas the decomposition theorem’s $F[q_\lambda]$ carries $+I(q_\lambda)$. The same glyph F is reused for brevity; the two are reconciled in the remark immediately below.

$$F(\lambda) = -u(2\sigma(\lambda) - 1) - I(\lambda), \quad (\text{C.14})$$

monotonically decreasing in $|\lambda|$ for $u \geq 0$. See [output/figures/free_energy_curve.png](#). This is a Lagrangian-style scalarisation rather than the framework’s $F[q_\lambda] = \log Z_E(\lambda) - \log Z(\lambda)$ identity from [§A](#); the genuine VFE minimization in the same toy with multi-information entering with a *plus* sign gives $\lambda_{\text{VFE}}^*(u) = 2u$, derived in the previous subsection.

Numerical cross-check

The Python companion verifies every identity above to floating tolerance ($< 1\text{e-}09$) at 5 values of $\lambda \in \{0, 0.5, 1, 2, 4\}$:

- `test_empirical_mi_matches_closed_form` — $I(\lambda)$ equals the TC of q_λ .
- `test_ising_mutual_information_zero_at_zero` — $I(0) = 0$.
- `test_ising_mutual_information_saturates_to_log2` — $I(50) \approx \log 2$.
- `test_ising_mutual_information_even_in_lambda` — symmetry.
- `test_optimal_lambda_*` — closed-form $\lambda^*(\Delta)$.

Run them with:

```
uv run pytest tests/test_bernoulli_toy.py -v
```

Tensor-Train Inference Algorithm: Bond-Dimension Sweep, MPS Contraction, and Sparsity-Rank Tradeoff

For coupling potentials representable as tensor trains (TT) with low bond rank r , the λ -entangled posterior admits efficient inference and sampling via standard matrix-product-state (MPS) machinery [Orus, 2014, Schollwöck, 2011]. Throughout this appendix, r denotes the maximal bond dimension of the TT contraction; r is the parameter that controls the time / memory trade-off and corresponds to the Schmidt rank of the bipartite reshape of q_λ across any cut along the train. Companion code: [src/lean/spectral.py](#) exposes the rank and entropy diagnostics used in the current figures; the dense empirical suite computes the small- K witnesses directly, while the full TT contraction algorithm below is an engineering extension of that current rank evidence.

Tensor-train representation of J

Suppose $J(\pi) = J(\pi^1, \dots, \pi^K)$ admits a TT representation with bond dimension r :

$$J(\pi^1, \dots, \pi^K) = \sum_{a_0, \dots, a_K} A_{a_0, \pi^1, a_1}^{(1)} A_{a_1, \pi^2, a_2}^{(2)} \dots A_{a_{K-1}, \pi^K, a_K}^{(K)}, \quad (\text{D.1})$$

with boundary indices $a_0 = a_K = 1$. Each tensor $A^{(k)}$ has shape $(r, |\Pi^k|, r)$ except at the boundaries.

The exponential map

The pointwise exponential of a TT is *not* itself low-rank in general, but for *additive-form* couplings $J = \sum_e J_e$ over a local interaction graph $e \in \mathcal{E}$ (the practically relevant case — pairwise / triplet couplings between adjacent streams), the factorization

$$\exp(\lambda J) = \prod_{e \in \mathcal{E}} \exp(\lambda J_e) \quad (\text{D.2})$$

is a product of bond-dimension-bounded operators that contracts into a TT of bond dimension $r' = O(r^d)$ where d is the maximum order of an interaction. For pairwise couplings, $r' = O(r^2)$.

Inference cost

Per-pass cost of computing q_λ , its marginals, and its log-partition is

$$\mathcal{O}(K \cdot |\Pi_{\max}|^2 \cdot r'^2), \quad (\text{D.3})$$

where $|\Pi_{\max}| = \max_k |\Pi^k|$. Compared to dense inference (which is $\mathcal{O}(\prod_k |\Pi^k|)$), this is exponentially cheaper whenever $r' \ll \prod_k |\Pi^k|^{1/2}$.

Sampling

Standard MPS conditional sampling [Orus, 2014, Han et al., 2018]: draw π^1 from its marginal, condition the remaining streams, draw π^2 , and so on. Each conditional draw is an $\mathcal{O}(|\Pi_{\max}| r'^2)$ operation; total $\mathcal{O}(K |\Pi_{\max}| r'^2)$ per sample.

Marginal extraction

Marginals q_λ^k are extracted by tracing out all other streams from the TT, an $\mathcal{O}(K |\Pi_{\max}| r'^2)$ operation. The Python companion implements this end-to-end for the dense (small K) case in [src/lean/joint_dist.py](#); a sparse TT contraction version would extend the same interface rather than change the theorem statement.

Bond-dimension recursion

The bond dimensions of the contracted TT for q_λ obey a *compositional recursion* over the interaction graph. Order the streams $1, 2, \dots, K$ along the train and define the *partial bond rank* $r_k(\lambda)$ as the Schmidt rank of the bipartite factorization $\{1, \dots, k\} \mid \{k+1, \dots, K\}$ of q_λ . Then:

- **Lower bound (rank-additivity at the boundary).** $r_k(0) = 1$ for every k (mean-field has rank one across every cut).

- **Upper bound (sparsity $\implies$ low rank).** If J has TT rank $\leq r$ across the cut at k , then $r_k(\lambda) \leq r^2$ for every λ — pairwise interactions saturate at this bound; a d -clique of streams pushes it to $r_k \leq r^d$.
- **Monotone growth.** $r_k(\lambda)$ is non-decreasing in λ on every cut where J is non-trivial — the symmetry-breaking transitions of §8.4 are the discrete jumps in $r_k(\lambda)$.

The recursion is exploited algorithmically in the *DMRG-style sweep*: fix all bonds except one, solve a local generalized-eigenvalue problem to update that bond, advance. Each sweep is $\mathcal{O}(K |\Pi_{\max}|^2 r_{\max}^3)$; convergence in $\mathcal{O}(\log(1/\varepsilon))$ sweeps for KL accuracy ε is standard for log-concave coupling potentials [Verstraete et al., 2008, Orus, 2014, Han et al., 2018]. The numerical companion records the empirical r_k profile across stream counts $K \in \{2, 3, 4, 5\}$ — see the `tt_ranks_K2–tt_ranks_K5` variables in `output/data/manuscript_variables.json`, exposed in prose at §8.3.

Approximate compression

When the exact TT rank of $\exp(\lambda J)$ exceeds a budget $r_{\max}$, truncated SVD across each bond gives a rank- $r_{\max}$ approximation with controlled KL error. This is the standard *matrix-product-operator* / *MPO-MPS* contraction trick.

Connection to the sparsity–rank tradeoff

Theorem 8.3 (sparsity-rank tradeoff): a TT coupling potential of bond dim r produces a posterior with TT rank bounded by r' as above. This is the formal counterpart of the practical observation that *low-rank couplings give cheap posteriors*.

Lean 4 Boundary Fragment: Live ActinfPolicyEntanglement Source Excerpts and Validation Wiring

The Lean 4 boundary fragment — the type-checked statements of every formal claim in this manuscript — lives under `lean/` of the companion repository. This appendix indexes the modules; the full source is in the companion repository.

Package structure

```
lean/
├─ lakefile.lean           ← Lake package definition
├─ lean-toolchain          ← v4.29.0
├─ ActinfPolicyEntanglement.lean
├─ ActinfPolicyEntanglement/ ← boundary-fragment submodules
│   ├─ Basic.lean
│   ├─ JointDist.lean
│   ├─ Coupling.lean
│   ├─ FreeEnergy.lean
│   ├─ Geometry.lean
│   ├─ Spectral.lean
│   ├─ SpectralWitnesses.lean ← prop_7_2, thm_7_3
│   ├─ Heterogeneous.lean
│   ├─ BernoulliToy.lean
│   ├─ Decomposition.lean
│   ├─ Constructive.lean
│   ├─ Monotonicity.lean
│   ├─ Convexity.lean        ← thm_4_3, prop_10_1
│   ├─ MarkovBlanket.lean    ← prop_11_3
│   ├─ ConnectionsWitnesses.lean ← thm_11_1, prop_11_2
│   └─ Scalar.lean
├─ FepSketches.lean
├─ FepSketches/
│   └─ PolicyEntanglementBoundary.lean ← re-exports
```

Mathlib-free boundary

The package compiles against a stock Lean 4 v4.29.0 with **no Mathlib dependency** at the boundary layer. This was a deliberate design choice: the boundary fragment encodes the *type signatures* of every analytic claim so that downstream agents can `lake build` immediately and discharge witness payloads in a separate Mathlib4 layer without changing the boundary theorem names.

Status snapshot

Live `grep` count, source-of-truth verified on each render (strict-form `sorry` count = blocking sorries in proof bodies; the “sorry” mentions inside `/- ! ... -/` docstrings advertising `sorry-free` status are not blocking and are not counted):

Lean module	Defs	Theorems / Lemmas	Structures	Strict sorry	Manuscript section
Basic	3	1	0	0	§3
JointDist	5	0	0	0	§3 , §7
Coupling	3	2	0	0	§4
FreeEnergy	8	1	0	0	§5
Geometry	0	4	0	0	§7
Spectral	1	2	0	0	§8
SpectralWitnesses	0	3	2	0	§8
Heterogeneous	1	2	2	0	§9
BernoulliToy	14	2	0	0	§6 , §C
Decomposition	2	5	0	0	§5 , §A
Constructive	0	4	0	0	§5 , §7
Monotonicity	0	16	0	0	§5 , §10
Convexity	0	2	2	0	§5 , §11
MarkovBlanket	0	2	1	0	§19
Connections / Witnesses	0	4	2	0	§17
Scalar	0	6	0	0	§4

Totals: **17 boundary submodules**, **39 defs**, **76 theorems / lemmas**, **11 structures**, **0 strict sorry**, **0 axioms** beyond stock Lean 4.

The current registry’s `status` distribution is: **0 deferred**, **0 sketch**, **0 strict sorry** — every numbered theorem in the manuscript resolves through `[[LEAN:<label>]]` to a live Lean source block in `lean/ActinfPolicyEntanglement/`. The witness-form theorems are tracked in [docs/reference/lean_reference.md](#) and [lean/ActinfPolicyEntanglement/MathlibRefinementRoadmap.md](#) as Mathlib4 analytic-discharge targets (i.e. work to derive the witness payloads in a separate library, not new statements to ship).

Building locally

```
cd lean
lake build
```

Expected: Build completed successfully (22 jobs). The 22 jobs are the 17 `ActinfPolicyEntanglement.*` submodules, the `ActinfPolicyEntanglement` root, the `FepSketches.PolicyEntanglementBoundary` re-export, the `FepSketches` root, and two Lake-internal targets.

FepSketches Re-exports for the Sibling `fep_lean` Layout

`lean/FepSketches/PolicyEntanglementBoundary.lean` re-exposes the load-bearing structural facts under the `FepSketches.*` namespace, matching the import layout used alongside the [fep_lean](#) catalog [\[Friedman, 2026c\]](#) and the [research manuscript template monorepo](#) (GitHub organization [docxology](#)).

Where each manuscript theorem lives in Lean

The `lean_module / lean_name` columns of [manuscript/refs/labels.yaml](#) are the wiring; the renderer’s `[[LEAN:<theorem-label>]]` token extracts the live source. The 20 theorem rows with manuscript `[[THMREF:...]]` tokens resolve at render time; the separate `Float↔ℝ` roadmap row resolves through the registry but is not inlined in this Lean-code supplement:

- [Theorem 5.1](#) → `Decomposition.entanglement_decomposition`
- [Theorem 5.5](#) → `Decomposition.freeEnergy_closedForm_witness`
- [Theorem 5.6](#) → `Convexity.freeEnergy_convex_in_lam_witness`
- [Corollary 5.2](#) → `Decomposition.couplingVerdict_correct`
- [Corollary 5.3](#) → `Decomposition.couplingLogWeight_pointwise_at_zero`
- [Corollary 5.4](#) → `Decomposition.totalCorrelation_def_unfold`
- [Proposition 7.1](#) → `Geometry.mfImage_isMeanField`
- [Proposition 7.2](#) → `Geometry.mProjection_kl_eq_self_when_meanfield`
- [Proposition 7.3](#) → `FreeEnergy.totalCorrelation_eq_kl_to_mprojection`
- [Theorem 7.4](#) → `Geometry.entangledFamily_eGeodesic`

- **Proposition 7.5** → `Geometry.dualFlat_pythagorean_witness`
- **Proposition 8.1** → `Spectral.Bipartite.isBipartiteMeanField_factors`
- **Proposition 8.2** → `SpectralWitnesses.schmidtRank_upperSemicontinuous_witness`
- **Theorem 8.3** → `SpectralWitnesses.sparsityRank_tradeoff_witness`
- **Theorem 9.1** → `Heterogeneous.couplingTax_quadratic_bound`
- **Corollary 9.2** → `Heterogeneous.couplingTax_small_lambda_tolerance`
- **Proposition 11.1** → `Convexity.freeEnergy_localConcavity_at_zero_witness`
- **Theorem 17.1** → `ConnectionsWitnesses.hierarchicalAIF_lambda_limit_witness`
- **Proposition 17.2** → `ConnectionsWitnesses.sophisticatedInference_embedding_witness`
- **Proposition 19.3** → `MarkovBlanket.markovBlanket_separation_identity_witness`

The twenty live Lean companions

The blocks below are produced by `[[LEAN:<label>]]` tokens that `src/manuscript/lean_extract.py` resolves against the live `.lean` sources in `lean/ActinfPolicyEntanglement/`. Each block carries a `-- From <file>:<line> [status: ...]` caption pointing at the exact source coordinates so the supplement is a verbatim window onto the formalization, not a hand-curated copy. If a registered theorem is ever renamed in Lean, the next render either updates the inlined source or fails fast with a `[[MISSING:LEAN:...]]` marker — there is no way for the supplement to drift silently from the boundary fragment.

Decomposition fragment

Entanglement decomposition

```
-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:75` [status: **boundary**]
/-- **Theorem 5.1 (Entanglement Decomposition)** – boundary witness
form.
```

Given a Mathlib-supplied algebraic split
``F[q] = marginal_part + coupling_expectation + agentic_gain``,
 where ``coupling_expectation`` is the integrated ``couplingLogWeight``
 against ``q``, this theorem certifies the equation in the boundary
 fragment. Every coupling parameter ``(J, K_c, γ, λ)`` is genuinely used
 via ``couplingExpectationSkeleton``, so the statement is non-vacuous.

****Typed-API-contract disclaimer.**** The Lean body is literally
``hWitness`` → `hWitness`` – the identity term on the algebraic split.
 This is ****not**** a stand-alone proof of Theorem 5.1; it is a typed-API
 contract that forces ``(J, K_c, γ, λ)`` into the conclusion via
``couplingExpectationSkeleton`` and locks the four-term decomposition
 shape. The non-vacuous **algebraic** content of the decomposition
 (commutative-ring re-grouping of the four bookkeeping scalars) is
 discharged separately by ``entanglement_decomposition_four_terms_assoc_skeleton``
 and ``entanglement_decomposition_four_terms_commute_skeleton`` below
 (genuine ``CommScalar`` proofs). The full **analytic** content (Gibbs
 expansion + KL chain rule) is supplied as ``hWitness`` and discharged
 by the separate MathlibProofs layer; the numerical realization is in
``src/lean/decomposition.py`` and verified at the dashboard invariant
``decomposition_lhs_eq_rhs_max_residual`` (worst-case ``5.55e-16``). -/
 theorem entanglement_decomposition {K Pol}

```
(q : JointDist K Pol) (s : List (PolicySpace K Pol))
(logE G : PolicySpace K Pol → Float)
(J K_c : CouplingPotential Float K Pol) (gamma lam : Float)
(marginal_part agentic_gain : Float)
(hWitness : variationalFreeEnergy q logE G gamma s =
  marginal_part
+ couplingExpectationSkeleton q s J K_c gamma lam
+ agentic_gain) :
variationalFreeEnergy q logE G gamma s =
  marginal_part
+ couplingExpectationSkeleton q s J K_c gamma lam
+ agentic_gain :=
```

Existence of optimal coupling

```
-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:243` [status: **boundary**]
/-! ## Theorem 5.5 – closed exponential-family form of `F[q_λ]`
```

The Gibbs decomposition ``F[q] = Σ F[q^k] + γλ(K_c) + log Z_E(λ) - λ(J) + I(q)``
 collapses to the strikingly clean identity

```

...
F[q_λ] = log Z_E(λ) - log Z(λ),
...

where `log Z_E(λ)` is the entangled-prior log-partition and
`log Z(λ)` is the entangled-posterior log-partition. See
[manuscript/04_entanglement_decomposition.md](../manuscript/04_entanglement_decomposition.md)
([Theorem 5.5](../manuscript/refs/labels.yaml#thm_4_2)) and the proof appendix
[manuscript/S01_proof_of_decomposition_theorem.md](../manuscript/S01_proof_of_decomposition_theorem.md).

```

The boundary fragment exposes the identity as a *witness-consuming* statement: the caller (the separate additive `MathlibProofs` layer or the numerical Python layer) supplies the algebraic equality `vfe = logZE - logZ`, and the boundary fragment threads `(λ, J, γ, K_c)` through `couplingExpectationSkeleton` so every parameter is genuinely referenced and the statement is non-vacuous. Polymorphic over `[CommScalar α]`. -/

```

theorem freeEnergy_closedForm_witness {α : Type} [CommScalar α]
  {K Pol}
  (vfe logZE logZ : α)
  (_s : List (PolicySpace K Pol))
  (J K_c : CouplingPotential α K Pol) (gamma lam : α)
  (hWitness : vfe = logZE - logZ) :
  vfe = logZE - logZ
  ∧ (∀ π : PolicySpace K Pol,
    couplingLogWeight J K_c gamma lam π
      = lam * J π - gamma * lam * K_c π) := by

```

Coupling-pays-for-itself verdict

```

-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:168` [status: **proved**]
/-- **Corollary 5.2 correctness theorem (boundary identity)**: the
`couplingVerdict` Boolean is `true` precisely when the coupling tax is
strictly less than the agentic gain – i.e. the verdict semantics
faithfully decide the "coupling-pays-for-itself" predicate.

```

This pins the verdict's contract on the boundary fragment: a `true` answer is *not* an oracle assertion but a Lean-level proof that `tax < gain`. Discharged by unfolding the `decide` and forwarding the proof component, no `sorry`. -/

```

theorem couplingVerdict_correct (gain tax : Float) :
  couplingVerdict gain tax = true ↔ tax < gain := by

```

Mean-field at $\lambda = 0$

```

-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:195` [status: **proved**]
/-- **Corollary 5.3 (Mean-field optimum at  $\lambda = 0$ )**: at  $\lambda = 0$ , the
coupling log-weight contribution vanishes pointwise. Polymorphic over
`[CommScalar α]`; combine with `entanglement_decomposition` to recover
the pure mean-field statement F[q_0] = Σ_k F[q_0^k] + agentic_gain. -/
theorem couplingLogWeight_pointwise_at_zero {α : Type} [CommScalar α]
  {K Pol}
  (J K_c : CouplingPotential α K Pol) (gamma : α)
  (π : PolicySpace K Pol) :
  couplingLogWeight J K_c gamma 0 π = 0 :=

```

Total correlation unfolds to KL-self

```

-- From `lean/ActinfPolicyEntanglement/Decomposition.lean:215` [status: **boundary**]
/-- **Corollary 5.4 (Total-correlation unfolding)**: the boundary
fragment's `totalCorrelation` is by construction the per-stream
entropy sum minus the joint entropy. This forwarder makes the
definitional unfolding available to downstream modules without
re-importing `FreeEnergy`'s namespace.

```

The strict-positivity claim `I(q) > 0` iff `q` is not mean-field is the `Iq > 0 ↔ ¬ IsMeanField q` companion that requires Mathlib's KL-non-negativity (see `IsNonNegMultiInformation` in `[FreeEnergy.lean]`(`FreeEnergy.lean`)) and is exposed by the `MathlibProofs` extension. Stock-Lean, zero-`sorry`. -/

```

theorem totalCorrelation_def_unfold {K Pol}
  (q : JointDist K Pol) (s : List (PolicySpace K Pol))
  (sumStreamEntropies : Float) :
  totalCorrelation q s sumStreamEntropies
    = sumStreamEntropies - shannonEntropy q s := rfl

```

Geometry fragment

Mean-field submanifold is e-flat

```
-- From `lean/ActinfPolicyEntanglement/Geometry.lean:40` [status: **proved**]
/-- **`mfImage_isMeanField`** (registry `prop_6_1`, Prop 7.1,
`faithfulness: statement-restricted`). This theorem proves only that
the product distribution induced by mean-field marginals is
mean-field – definitional membership `IsMeanField (mfToJoint m)`,
discharged by `rfl`. It does NOT prove the named manuscript
proposition "the MF submanifold is e-flat" (closure under exponential
geodesics / affine-in-0); that real-analytic content is the open
target scoped to the separate Mathlib layer (cf.
`entangledFamily_eGeodesic` for the affine-in- $\lambda$  identity). The name
was deliberately changed from the prior `mfSubmanifold_eFlat` so the
declaration's name states what it actually proves. -/
theorem mfImage_isMeanField {K Pol}
  (m : MFDist K Pol) :
  IsMeanField (mfToJoint m) := by
```

m-projection minimizes KL

```
-- From `lean/ActinfPolicyEntanglement/Geometry.lean:77` [status: **proved**]
/-- **`mProjection_kl_eq_self_when_meanfield`** (registry `prop_6_2`,
Prop 7.2, `faithfulness: statement-restricted`). This theorem proves
only the conditional equality `KL(q || mfToJoint m) = KL(q || q)` *under
the hypothesis* `h : q = mfToJoint m` (i.e. `q` is already mean-field).
It does NOT prove the named manuscript proposition that the
m-projection minimises KL – the information-projection optimality
`∀ p ∈ M_MF, D_KL(q||m(q)) ≤ D_KL(q||p)` is the open analytic target
for the separate Mathlib layer. The name was deliberately changed
from the prior `mProjection_minimises_kl` so the declaration's name
states what it actually proves (an equality when `q` is already the
projection, not a minimality result). -/
theorem mProjection_kl_eq_self_when_meanfield {K Pol}
  (q : JointDist K Pol) (m : MFDist K Pol)
  (h : ∀ π, q π = mfToJoint m π)
  (s : List (PolicySpace K Pol)) :
  kl q (mfToJoint m) s = kl q q s := by
```

Total correlation equals KL to m-projection

```
-- From `lean/ActinfPolicyEntanglement/FreeEnergy.lean:140` [status: **witness**]
/-- **Proposition 7.3 (boundary witness form)**: given the per-stream
entropy sum and the value of `KL(q || m(q))` together with the
algebraic identity binding them, the boundary fragment certifies the
total correlation equals the KL to the m-projection. Stock-Lean,
zero-`sorry`.
```

```
**Typed-API-contract disclaimer.** Not a stand-alone proof of the
KL-chain-rule identity `I(q) = KL(q || ∏_k q^k)`; a typed-API contract.
The caller supplies `hWitness : sumStreamEntropies - shannonEntropy q s = klToMProj`;
the boundary fragment unfolds `totalCorrelation` and forwards. -/
theorem totalCorrelation_eq_kl_to_mprojection {K Pol}
  (q : JointDist K Pol) (s : List (PolicySpace K Pol))
  (sumStreamEntropies klToMProj : Float)
  (hWitness : sumStreamEntropies - shannonEntropy q s = klToMProj) :
  totalCorrelation q s sumStreamEntropies = klToMProj := by
```

e-geodesic family

```
-- From `lean/ActinfPolicyEntanglement/Geometry.lean:95` [status: **forwarder**]
/-- **Theorem 7.4** (forwarder). Polymorphic over `[CommScalar α]`. -/
theorem entangledFamily_eGeodesic {α : Type} [CommScalar α] {K Pol}
  (J K_c : CouplingPotential α K Pol) (gamma lam1 lam2 : α)
  (π : PolicySpace K Pol) :
  couplingLogWeight J K_c gamma lam1 π
  - couplingLogWeight J K_c gamma lam2 π
  = (lam1 - lam2) * (J π - gamma * K_c π) :=
```

Pythagorean witness on the dual-flat pair

```
-- From `lean/ActinfPolicyEntanglement/Geometry.lean:169` [status: **witness**]
/-- **Proposition 6.5 (boundary witness, round-4 tie-in upgrade)**:
extracts the Pythagorean identity from a `PythagoreanWitness`. The
three scalars are now committed to boundary-fragment primitives
(`kl`, `totalCorrelation`, `mfToJoint`) via three tie-in equalities –
```

a caller can no longer satisfy the contract with arbitrary `Float`s.

```

**Typed-API-contract disclaimer.** The analytic Pythagorean identity
itself (the `pythagorean` field) remains a witness payload supplied
by the caller; the separate MathlibProofs layer discharges it from the
KL chain rule. This theorem is **not** a stand-alone proof of the
Pythagorean decomposition; it certifies that *if* a caller supplies
witnesses that genuinely compute the boundary primitives, *then* the
extracted identity types-check. The mathematical content lives in
the manuscript appendix S01 + the Python numerical companion in
[`src/lean/geometry.py`](../src/lean/geometry.py) (which verifies
the identity on randomly sampled joints + reference mean-fields). -/
theorem dualFlat_pythagorean_witness {K Pol}
  {q q0_star : JointDist K Pol} {mhat : MFDist K Pol}
  {s : List (PolicySpace K Pol)} {sumStreamEntropies : Float}
  (w : PythagoreanWitness q q0_star mhat s sumStreamEntropies) :
  w.klVal = w.tcVal + w.residual := w.pythagorean

```

Spectral fragment

Bipartite mean-field $\Leftrightarrow$ Schmidt rank 1

```

-- From `lean/ActinfPolicyEntanglement/Spectral.lean:54` [status: **proved**]
/-- **Proposition 7.1 (forward direction, forwarder)**: extraction
shortcut for callers that already have the bipartite-mean-field
predicate in hand and want the product factorization witness. -/
theorem isBipartiteMeanField_factors {Pol1 Pol2 : Type}
  (q : BipartiteJoint Pol1 Pol2)
  (h : IsBipartiteMeanField q) :
   $\exists (u : \text{Pol1} \rightarrow \text{Float}) (v : \text{Pol2} \rightarrow \text{Float}),$ 
   $\forall \pi1 \pi2, q \pi1 \pi2 = u \pi1 * v \pi2 :=$ 

```

Schmidt rank upper-semicontinuous in λ

```

-- From `lean/ActinfPolicyEntanglement/SpectralWitnesses.lean:68` [status: **witness**]
/-- **Proposition 8.2 (boundary witness form)**: an
`UpperSemicontinuousRankWitness` *is* the existence of an upper-
semicontinuous Schmidt-rank curve anchored at `rankCurve 0 = 1`. Stock
Lean, zero-`sorry`.

```

```

**Typed-API-contract disclaimer.** Not a stand-alone proof of upper
semicontinuity; a typed-API contract. The continuity inequality is
supplied as `witness.usc`; the boundary fragment re-publishes it
together with the mean-field anchor. Bipartite K=2 case is now
*genuinely* proved at the boundary as
`Spectral.Bipartite.schmidtRankOne_iff_isBipartiteMeanField` (round 4);
the general-K upper-semicontinuity statement still requires Mathlib's
topological-semicontinuity library + matrix-rank lower-semicontinuity
to discharge. -/
theorem schmidtRank_upperSemicontinuous_witness
  (rankCurve : Float  $\rightarrow$  Nat)
  (witness : UpperSemicontinuousRankWitness rankCurve) :
  rankCurve 0.0 = 1
   $\wedge \forall (\text{lam0} : \text{Float}) (r0 : \text{Nat}),$ 
  rankCurve lam0  $\leq$  r0  $\rightarrow$ 
   $\exists \text{delta} : \text{Float}, 0.0 < \text{delta} \wedge$ 
   $\forall \text{lam} : \text{Float}, (\text{lam} - \text{lam0}).\text{abs} < \text{delta} \rightarrow \text{rankCurve lam} \leq r0 :=$ 

```

Sparsity-rank tradeoff for tensor-train coupling

```

-- From `lean/ActinfPolicyEntanglement/SpectralWitnesses.lean:136` [status: **witness**]
/-- **Theorem 8.3 (boundary witness form)**: a `SparsityRankEnvelope`
witness *is* the existence of a per-cut Schmidt-rank envelope for the
 $\lambda$ -entangled posterior of a tensor-train coupling. Stock-Lean, zero-
`sorry`.

```

```

**Typed-API-contract disclaimer.** Not a stand-alone proof of the
sparsity-rank tradeoff; a typed-API contract. The per-cut envelope
`cut_rank k  $\lambda \leq$  bond_bound k` is supplied as a `SparsityRankEnvelope`
field; the boundary fragment extracts it. MathlibProofs discharge
from `Mathlib.LinearAlgebra.TensorProduct` plus matrix-rank bounds. -/
theorem sparsityRank_tradeoff_witness (K : Nat)
  (witness : SparsityRankEnvelope K) :
   $\forall (k : \text{Fin K}) (\text{lam} : \text{Float}),$ 
  witness.cut_rank k lam  $\leq$  witness.bond_bound k :=

```

Convexity fragment

Convexity of F in λ

```
-- From `lean/ActinfPolicyEntanglement/Convexity.lean:77` [status: **witness**]
/-- **Theorem 5.6 (boundary witness form)**: a
`FreeEnergyConvexityWitness` *is* the existence of a convex ` $\lambda$ `-curve
for ` $F[q_\lambda]$ ` together with the ` $\lambda = 0$ ` anchor that ties the curve to the
boundary-fragment coupling skeleton (via
`couplingLogWeight_pointwise_at_zero`). Polymorphic over ` $[CommScalar \alpha]$ `
on the coupling side so the anchor genuinely uses every coupling
parameter ` $(J, K_c, \gamma)$ `.
```

Typed-API-contract disclaimer. This theorem is *not* a stand-alone proof that ` $F[q_\lambda]$  is convex in ` λ `. It is a typed-API contract: the convex curve and the universally-quantified midpoint inequality are supplied as a ``FreeEnergyConvexityWitness``; the boundary fragment extracts the fields and re-publishes them. Numerical witness in ``src/lean/free_energy.py`` + ``scripts/parameter_sweep.py``; MathlibProofs discharge from log-concavity-on-the-simplex arguments. -/

```
theorem freeEnergy_convex_in_lam_witness {α : Type} [CommScalar α]
  {K Pol}
  (F_curve : Float → Float)
  (J K_c : CouplingPotential α K Pol) (gamma : α)
  (witness : FreeEnergyConvexityWitness F_curve) :
  (∀ lam1 lam2 t : Float,
    F_curve (t * lam1 + (1.0 - t) * lam2)
      ≤ t * F_curve lam1 + (1.0 - t) * F_curve lam2)
  ∧ (∀ π : PolicySpace K Pol,
    couplingLogWeight J K_c gamma 0 π = 0) := by
```

Local concavity of F at $\lambda = 0$

```
-- From `lean/ActinfPolicyEntanglement/Convexity.lean:154` [status: **witness**]
/-- **Proposition 11.1 (boundary witness form)**: a
`LocalConcavityAtZero` witness *is* the existence of a Taylor-form local
concavity bound for ` $F[q_\lambda]$ ` at ` $\lambda = 0$ `, together with the ` $\lambda = 0$ `
anchor that ties the curve to the boundary-fragment coupling skeleton
(via `couplingLogWeight_pointwise_at_zero`). Polymorphic over
`[CommScalar α]` on the coupling side so the anchor genuinely uses every
coupling parameter ` $(J, K_c, \gamma)$ `.
```

Typed-API-contract disclaimer. Not a stand-alone proof of local concavity; a typed-API contract. The Taylor coefficients, the ` $a_2 \leq 0$ ` sign, the local window radius, and the cubic-remainder bound are all supplied as ``LocalConcavityAtZero`` fields. Numerical witness in ``parameter_sweep.py``; MathlibProofs discharge from real-analytic second-derivative arguments. -/

```
theorem freeEnergy_localConcavity_at_zero_witness {α : Type} [CommScalar α]
  {K Pol}
  (F_curve : Float → Float)
  (J K_c : CouplingPotential α K Pol) (gamma : α)
  (witness : LocalConcavityAtZero F_curve) :
  witness.a2 ≤ 0.0
  ∧ 0.0 < witness.eps
  ∧ (∀ lam : Float, 0.0 ≤ lam → lam ≤ witness.eps →
    F_curve lam
      ≤ witness.a0
      + witness.a1 * lam
      + witness.a2 * lam * lam
      + witness.C * lam * lam * lam)
  ∧ (∀ π : PolicySpace K Pol,
    couplingLogWeight J K_c gamma 0 π = 0) := by
```

Heterogeneous fragment

Coupling-tax quadratic bound

```
-- From `lean/ActinfPolicyEntanglement/Heterogeneous.lean:61` [status: **witness**]
/-- **Theorem 9.1 (boundary witness form)**: a `BoundedQuadraticTax`
witness *is* the existence of the quadratic envelope. Stock-Lean,
zero-`sorry`.
```

Typed-API-contract disclaimer. This theorem is *not* a stand-alone proof of the ` $0(\lambda^2)$ ` coupling-tax bound; it is a typed-API contract. The analytic content – the bound ` $\text{taxFunction } \lambda \leq C \cdot \lambda^2$ ` with ` $C \geq 0$ ` universally over ` $\lambda$ ` – is supplied as a structural hypothesis

```
(`BoundedQuadraticTax`); the boundary fragment re-publishes it as an
existence claim. The Python numerical companion in
[`src/lean/heterogeneous.py`](../../src/lean/heterogeneous.py)
verifies the bound on concrete parameter sweeps; the separate
MathlibProofs layer will discharge it from Taylor expansion of the
coupling-prior log-partition. -/
theorem couplingTax_quadratic_bound (taxFunction : Float → Float)
  (witness : BoundedQuadraticTax taxFunction) :
  ∃ (C : Float), 0.0 ≤ C ∧
  ∀ lam, couplingTax taxFunction lam ≤ C * lam * lam :=
```

Small- λ tolerance witness

```
-- From `lean/ActinfPolicyEntanglement/Heterogeneous.lean:106` [status: **witness**]
/-- **Corollary 9.2 (boundary witness form)**: a `SmallLambdaTolerance`
witness *is* the existence of the tolerance window. Stock-Lean,
zero-`sorry`.
```

```
**Typed-API-contract disclaimer.** Same as `couplingTax_quadratic_bound`:
this is a typed-API contract, not a stand-alone proof. The continuity
argument that delivers the tolerance window is supplied as a
`SmallLambdaTolerance` witness; the boundary fragment re-publishes
the existence claim. -/
theorem couplingTax_small_lambda_tolerance
  (taxFunction : Float → Float) (eps : Float)
  (witness : SmallLambdaTolerance taxFunction eps) :
  ∃ (lamMax : Float), 0.0 < lamMax ∧
  ∀ lam, lam.abs ≤ lamMax → couplingTax taxFunction lam ≤ eps :=
```

Connections fragment

Hierarchical AIF as the $\lambda \rightarrow \infty$ limit

```
-- From `lean/ActinfPolicyEntanglement/ConnectionsWitnesses.lean:74` [status: **witness**]
/-- **Theorem 17.1 (boundary witness form)**: a
`HierarchicalConcentrationWitness` *is* the existence of a  $\lambda \rightarrow \infty$ 
hierarchical-AIF concentration claim for the entangled family. Stock-
Lean, zero-`sorry`.
```

```
**Typed-API-contract disclaimer.** Not a stand-alone proof of the
hierarchical-AIF concentration limit; a typed-API contract. The
`concentrate` field – that for every  $\varepsilon > 0$  a coupling threshold
exists beyond which the entangled posterior is  $\varepsilon$ -close in KL to the
limit  $q_\infty$  – is supplied as a witness obligation. MathlibProofs
discharge from `Mathlib.MeasureTheory.Measure.Tight` (tightness of the
 $\lambda$ -entangled posterior family in the  $\lambda \rightarrow \infty$  limit). -/
theorem hierarchicalAIF_lambda_limit_witness {K : Nat} {Pol : PolicyFactor K}
  (qFamily : Float → JointDist K Pol)
  (q_infty : JointDist K Pol)
  (support : List (PolicySpace K Pol))
  (witness : HierarchicalConcentrationWitness qFamily q_infty support) :
  ∀ (eps : Float), 0.0 < eps →
  ∃ lam0 : Float, 0.0 < lam0 ∧
  ∀ lam : Float, lam0 ≤ lam →
  kl (qFamily lam) q_infty support ≤ eps :=
```

Sophisticated-inference embedding

```
-- From `lean/ActinfPolicyEntanglement/ConnectionsWitnesses.lean:167` [status: **witness**]
/-- **Proposition 17.2 (boundary witness form)**: a
`SophisticatedInferenceEmbedding` *is* the existence of an embedding
from a sophisticated-inference source into the  $\lambda$ -deformation family
that maps the sophisticated-inference value function onto the
variational free energy. Stock-Lean, zero-`sorry`. -/
theorem sophisticatedInference_embedding_witness {K : Nat} {Pol : PolicyFactor K}
  (logE G : PolicySpace K Pol → Float) (gamma : Float)
  (support : List (PolicySpace K Pol))
  (witness : SophisticatedInferenceEmbedding logE G gamma support) :
  ∀ (x : witness.source),
  witness.siValue x =
  variationalFreeEnergy (witness.embed x) logE G gamma support :=
```

Markov-blanket fragment

Markov-blanket separation identity $1 - I/H$

```

-- From `lean/ActinfPolicyEntanglement/MarkovBlanket.lean:96` [status: **witness**]
/-- **Proposition 19.3 (boundary witness form)**: a
`MarkovBlanketSeparationWitness` *is* the existence of the
Markov-blanket separation identity `sep = 1 - I(q) / H(q)` on a finite
support, with `Hq > 0` and the tie-ins certifying that:

* `Hq` genuinely refers to `shannonEntropy q s`;
* `Iq` is the multi-information  $\sum_k H(q^k) - H(q)$  computed via the
  boundary fragment's `totalCorrelation`;
* `Iq` is non-negative (the analytic discharge from  $KL \geq 0$ ).

```

Stock-Lean, zero-`sorry`. This is now genuinely non-vacuous because both tie-ins commit `Iq` to a definite boundary-fragment quantity rather than a free Float. -/

```

theorem markovBlanket_separation_identity_witness {K Pol}
  (q : JointDist K Pol) (s : List (PolicySpace K Pol))
  (witness : MarkovBlanketSeparationWitness q s) :
  0.0 < witness.Hq
  ∧ witness.Hq = shannonEntropy q s
  ∧ witness.Iq = totalCorrelation q s witness.sumStreamEntropies
  ∧ 0.0 ≤ witness.Iq
  ∧ witness.sep = 1.0 - witness.Iq / witness.Hq :=

```

How drift is prevented

The `[[LEAN: ...]]` injection turns each Lean theorem in this appendix into a *single source of truth* shared with the boundary fragment itself:

1. `manuscript/refs/labels.yaml` declares the theorem registry with `lean_module / lean_name` for each entry.
2. At render time, `src/manuscript/lean_extract.py::load_lean_snippets` parses every `.lean` file under `lean/ActinfPolicyEntanglement/` and indexes each declaration by `(module, qualified_name)`.
3. `src/manuscript/renderer.py::_lean` resolves the registry pair against this index and emits a fenced Lean block carrying the source coordinates and the registry-declared status.
4. If the qualified name is missing or has been renamed, the renderer substitutes a `[[MISSING:LEAN:...]]` marker that `scripts/validate_manuscript.py` refuses to ship — so a Lean rename either flows through to the manuscript or fails CI.

This eliminates every form of stale-quote risk: the supplement above *is* the boundary fragment, sliced theorem-by-theorem.

How a witness is consumed today

The stock-Lean boundary is now zero-sorry **and zero-deferred**. For a `status: witness` theorem, the current source consumes a typed structure whose fields bind the analytic payload to boundary-fragment primitives. The validated contract is:

1. The theorem row in `manuscript/refs/labels.yaml` names the live Lean declaration.
2. The Lean declaration type-checks without Mathlib and without hidden axioms.
3. The corresponding Python witness or pymdp run computes the same payload numerically.
4. The relevant test file and dashboard invariant fail if the numerical payload drifts.
5. The manuscript renderer embeds only the live Lean declaration, never a hand-written replacement.

Mathlib4 enters only at the analytic-discharge boundary described in §12: a separate Mathlib-backed library may later construct these witness structures from `Real`, `PMF`, `KL`, convexity, rank, and measure-theoretic lemmas. Until such a module builds, it is not cited as a current result and is not rendered as source code.



Notation Concordance: Symbol Registry Across Manuscript, LaTeX, Python, and Lean

This appendix is the **single contract** between the manuscript prose, the LaTeX preamble macros, the Python source under `src/`, and the Lean boundary fragment under `lean/`. Every symbol used in the body appears here exactly once; columns trace the symbol across all four tracks so a reader can navigate from prose → code → proof in one hop. Empty cells indicate that the symbol does not surface on that track (e.g. a manuscript-only typographical convention).

The same identifiers also appear in [docs/reference/math_reference.md](#) and the Python-side identifier inventory in [docs/reference/python_api.md](#).

A **fifth, structural-and-numerical track** — Generalized Notation Notation (GNN) — is shipped and documented in §8 and tabulated in the *GNN fifth-track concordance* subsection at the end of this appendix. GNN reproduces the framework’s structure and numbers (a project-owned parser, a verified K=2 round-trip, and a Lean typed-contract emitter), but it does **not** prove theorems; the four-track *proof* contract documented above is therefore unchanged, and the GNN track is tabulated separately so the two are never conflated.

How to read this concordance — one symbol, four tracks

Take the entangled posterior q_λ , the central object of the framework. The four tracks line up as follows:

- **Manuscript prose (§4).** $q_\lambda(\pi) \propto \prod_k E_k(\pi^k) \exp(\lambda J(\pi) - \gamma G_\lambda(\pi))$.
- **LaTeX preamble (preamble.md).** Either spell out `q_\lambda` or use the macro `\fe` for the matching VFE operator.
- **Python (src/lean/coupling.py).**

```
from lean.coupling import entangled_posterior
q_lam = entangled_posterior(mf, G, J, Kc, gamma=1.0, lam=2.0)
# q_lam.shape == (|P^1|, |P^2|, ..., |P^K|); q_lam.sum() == 1.0
```
- **Lean (lean/ActinfPolicyEntanglement/Coupling.lean).**

```
def entangledPosteriorLogWeight {α : Type} [Add α] [Mul α] [Sub α] {K Pol}
  (logE : PolicySpace K Pol → α)
  (G : PolicySpace K Pol → α)
  (J K_c : CouplingPotential α K Pol)
  (gamma lam : α) : PolicySpace K Pol → α :=
  fun pi => logE pi - gamma * G pi + couplingLogWeight J K_c gamma lam pi
```
- **Numerical witness (tests/test_coupling.py).** At $\lambda = 0$ with symmetric MF prior and zero coupling, `entangled_posterior(...)` == `mean_field_to_joint(mf)` to floating tolerance — the proved-form companion of [Corollary 5.3](#).

Reading any row of the tables below in the same fashion lands a reader on the exact construct on all four tracks.

Sign conventions

The framework spans variational free energy, expected free energy, KL, multi-information, coupling, and Schmidt spectra; each carries a sign convention that the rest of the manuscript silently adheres to. An editor or reader who is confused about a sign somewhere in the body should consult this subsection first. When the body says “negated K ” or “modulo a sign convention”, it is invoking one of the conventions below.

- **Variational free energy.** $F[q] = \mathbb{E}_q[\log q - \log \mathcal{E}] + \gamma \mathbb{E}_q[G]$. F is *non-negative* on the natural domain (Gibbs inequality with fixed $\mathcal{E}$) and is the *quantity to be minimized*. Every $F[q_\lambda]$, $F[q_\lambda^k]$, and $\sum_k F[q^k]$ in the manuscript follows this sign — *no inverted-sign “ELBO” form appears anywhere*; an apparent extra minus in a body equation is a typo, not a convention change.
- **Expected free energy.** $G(\pi) \geq 0$ on the standard active-inference decomposition; both the epistemic value and the pragmatic value are reported as *negated* quantities so that minimizing G corresponds to *maximizing* utility plus information gain. When the body writes “epistemic value, negated” / “pragmatic value, negated” (as in §3.1), the negation is already absorbed into the sign of G — no further sign-flip is needed downstream.
- **Kullback–Leibler divergence.** $D_{\text{KL}}(q \| p) = \mathbb{E}_q[\log q - \log p] \geq 0$, *never* reversed. The first argument is the distribution one is drawing from; the second is the reference. When a derivation appears to “swap” the arguments (e.g. for variational vs. mean-field projections), the manuscript spells out the swap explicitly; an inferred swap is an error.
- **Multi-information / total correlation.** $I(q) = \sum_k H(q^k) - H(q) = D_{\text{KL}}(q \| \prod_k q^k) \geq 0$. Appears with a **PLUS** sign in the decomposition theorem (Eq. (5.2)); the body never carries an inverted-sign I .
- **Coupling parameter λ .** $\lambda \geq 0$ throughout the body of the paper. Negative λ would correspond to *anti-coupling* (the entangled posterior is pushed *away* from the modes of $J - \gamma K_c$) and is out of scope; figures show $\lambda \geq 0$ unless otherwise indicated. In the symmetric K=2 Bernoulli toy of §6.1 the closed-form expressions are even in λ so the sign restriction is harmless; in asymmetric cases the restriction is substantive and should be flagged if relaxed.
- **Schmidt singular values.** $s_\alpha \geq 0$ (singular values of a real matrix are non-negative). The spectral distribution $p_\alpha = s_\alpha^2 / \sum_\beta s_\beta^2$ is obtained by *squaring* the singular values and normalizing by their sum-of-squares; this is the same normalization used in the quantum-entanglement dictionary but applied to the SVD of a *classical-probability* joint, not to amplitudes of a state vector. The analogy with bipartite quantum

entanglement entropy is *structural*, not literal (see the clarification at §8.2 / [Eisert et al., 2010]).

- **Convention on “negated K ”.** Phrases such as “negated K ” or “modulo a sign convention” elsewhere in the manuscript refer collectively to the conventions in this subsection — almost always to the EFE-side sign of K_c (which enters G_λ with a *plus* λK_c , contributing $+\gamma\lambda\langle K_c \rangle$ to F via the γG_λ expectation; the *minus*-sign convention in some derivations of related work corresponds to a re-definition $K_c \rightarrow -K_c$).

Sets, indices, and basic objects

Symbol	Meaning	LaTeX	Python	Lean
$K \in \mathbb{N}$	Number of policy streams	K	K (int)	$K : \text{Nat}$
$k \in \{1, \dots, K\}$	Stream index	k	k (int)	$k : \text{StreamIdx } K$
Π^k	Per-stream policy factor (finite type)	Π^k (or policySpace^k)	(implicit via array shape)	$\text{Pol } k$ (in $\text{PolicyFactor } K := \text{StreamIdx } K \rightarrow \text{Type}$)
$\Pi = \prod_k \Pi^k$	Joint policy space	Π (or policySpace)	(implicit)	$\text{PolicySpace } K \text{ Pol } (\forall k, \text{Pol } k)$
$\pi = (\pi^1, \dots, \pi^K) \in \Pi$	Joint policy	π (or policy)	flat ndarray index	$\text{pi} : \text{PolicySpace } K \text{ Pol}$
$\pi^k \in \Pi^k$	Per-stream policy	π^k	$\text{pi}[k]$	$\text{pi } k$
π^{-k}	Tuple of all policies <i>other</i> than stream k	$\pi^{\{-k\}}$	(sum-axis param)	(implicit in marginalization)
T_k	Planning horizon attached to stream k	T_k	per-stream horizon in <code>StreamSpec</code> / rollout sidecars	horizon k when a <code>HorizonProfile</code> K witness is in scope
L	Number of hierarchy levels in the hierarchical-AIF structural analogy	L	(manuscript-level; no dedicated runtime object)	(witness payload supplied to <code>ConnectionsWitnesses</code>)
ℓ	Hierarchy-level index, used in adjacent-level pairs $(\pi^\ell, \pi^{\ell+1})$	ℓ	(manuscript-level; no dedicated runtime object)	(witness payload supplied to <code>ConnectionsWitnesses</code>)
d	Lookahead depth for sophisticated-inference / branching-style policy stacks	d	<code>lookahead_depth</code> when an experiment supplies one	(witness payload supplied to <code>ConnectionsWitnesses</code>)
$\mathcal{V} \subseteq \{1, \dots, K\}$	VFE (reflexive) stream indices	$\mathcal{V}$	<code>[m for m, _ in enumerate(modes) if m == InferenceMode.VFE]</code>	(implicit via horizon : <code>HorizonProfile K</code>)
$\mathcal{P} = \{1, \dots, K\} \setminus \mathcal{V}$	EFE / planning stream indices	$\mathcal{P}$	(complement)	(implicit)

POMDP generative-model symbols

The single-stream POMDP recap of §3.1 uses the classical `pymdp` / SPM symbols. These are the inputs to the `pymdp.agent.Agent` constructor in `src/simulation/agents.py` and the fields of `StreamSpec`.

Symbol	Meaning	LaTeX	Python (<code>StreamSpec</code>)	<code>pymdp</code> keyword
A	Likelihood matrix ($\text{num_obs} \times \text{num_states}$)	A	<code>spec.A</code>	<code>A=</code>
B	Transition tensor ($\text{num_states} \times \text{num_states} \times \text{num_controls}$)	B	<code>spec.B</code>	<code>B=</code>
C	Log-preference vector (num_obs)	C	<code>spec.C</code>	<code>C=</code>
D	Prior over hidden states (num_states)	D	<code>spec.D</code>	<code>D=</code>
o	Observation	o	<code>observations[k]</code>	passed to <code>infer_states</code>
s	Hidden state	s	<code>states[k]</code> (in <code>Rollout</code>)	output of <code>infer_states</code>
u	Control / action	u	<code>sampled_actions[k]</code>	output of policy sampling
T	Rollout horizon	T	<code>H.PYMDP_ROLLOUT_STEPS</code>	(script-level)
$P(\cdot)$	Generic generative-model law in active-inference prose	P	stored concretely as A, B, C, D, E arrays / factors	(not a boundary primitive)
$Q(\cdot)$	Generic variational posterior notation in active-inference prose	Q	normalized <code>q</code> arrays; <code>Agent.qs</code> / <code>Agent.q_pi</code> in <code>pymdp</code> contexts	(not a boundary primitive; boundary uses <code>JointDist</code>)

Distributions

Symbol	Meaning	LaTeX	Python	Lean
$E_k(\pi^k)$	Per-stream policy prior (habit)	E_k	<code>mf_prior[k]</code> (per-stream marginal supplied to <code>entangled_prior / mean_field_to_joint</code>) — distinct from <code>StreamSpec.D</code>, which is the hidden-state prior; see the hidden-state-prior row above	<code>E : MFDist K Pol</code>
$E_{\text{MF}}(\pi) = \prod_k E_k(\pi^k)$	Mean-field prior	$E_{\{\mathrm{MF}\}}$	<code>mean_field_to_joint(mf_prior)</code>	<code>mfProductWeight</code>
$\mathcal{E}_\lambda(\pi)$	λ -entangled prior	$\mathcal{E}_{\lambda}$	<code>entangled_prior(...)</code>	(no Lean — boundary fragment carries only the log-weight via <code>entangledPosteriorLogWeight</code>)
$G_k(\pi^k)$	Per-stream expected free energy (EFE)	G_k (or efe_k)	<code>per_stream_efe(spec, obs)[k]</code>	(consumed as data via <code>MFDist K Pol</code>)
$G_{\text{MF}}(\pi) = \sum_k G_k(\pi^k)$	Mean-field EFE	$G_{\{\mathrm{MF}\}}$	<code>sum(per_stream_efe(...))</code>	(no top-level Lean def; consumed as <code>MFDist K Pol</code> argument)
$G_\lambda(\pi) = G_{\text{MF}}(\pi) + \lambda K_c(\pi)$	λ -coupled total EFE (per-stream EFEs plus preference-side coupling)	G_{λ}	(inlined)	(inlined; appears inside <code>entangledPosteriorLogWeight</code>)
$q(\pi)$	Joint policy posterior	q	<code>q : ndarray</code> (joint shape)	<code>q : JointDist K Pol</code>
$q^k(\pi^k) = \sum_{\pi^{-k}} q(\pi)$	Per-stream marginal	q^k	<code>joint_marginal(q, k)</code>	(no Lean — marginalization is inline in <code>mfToJoint</code>)
$q_\lambda(\pi)$	λ -entangled posterior	q_{λ}	<code>entangled_posterior(mf, G, J, K_c, gamma, lam) / coupled_policy_posterior(spec, obs, lam)</code>	<code>entangledPosterior / LogWeight</code>
$q_{\text{MF}}(\pi) = \prod_k q^k(\pi^k)$	Product of marginals (m-projection of q)	$q_{\{\mathrm{MF}\}}$	<code>m_projection(q)</code>	<code>mfToJoint</code> (direction: $\text{MF} \rightarrow \text{joint}$; joint-to-marginal projection is Python-only in the current boundary)
q_t^k	Per-stream marginal at rollout time t	q_t^k	<code>rollout.steps[t].mean_field_marginals[k]</code>	(no Lean)

Coupling potentials and parameters

Symbol	Meaning	LaTeX	Python	Lean
$J(\pi)$	<i>Habit</i> coupling potential — prior side	J (or coupJ)	<code>coupling_j</code> (in <code>CoupledEnsembleSpec</code>)	<code>J : CouplingPotential α K Pol</code>
$K_c(\pi)$	<i>Preference</i> coupling potential — EFE side	K_c (or coupK)	<code>coupling_kc</code>	<code>Kc : CouplingPotential α K Pol</code>
$\lambda \in [0, \infty)$	Coupling parameter	λ	<code>lam</code> (parameter)	<code>lam : α</code>
$\gamma > 0$	Policy precision (inverse-temperature on EFE)	γ	<code>gamma</code> (in <code>CoupledEnsembleSpec</code>)	<code>gamma : α</code>
$\gamma_k > 0$	Per-stream policy precision; collapses to scalar γ in homogeneous examples	γ_k	per-stream <code>gamma</code> when a helper supports heterogeneous precision; scalar <code>gamma</code> otherwise	(not a separate boundary primitive)
$\Delta_{\text{util}} \in [-\Delta_{\text{max}}, \Delta_{\text{max}}]$	Utility surplus for aligned outcomes (a <i>substantive payoff</i> parameter; not a target on the alignment axis)	$\Delta_{\{\mathrm{util}\}}$	<code>delta</code> argument to <code>optimal_lambda</code>	<code>delta : Float</code>

Symbol	Meaning	LaTeX	Python	Lean
$\Delta_{\text{align}} \in [-\Delta_{\text{max}}, \Delta_{\text{max}}]$	Target alignment for the alignment-inversion formula $\lambda^*(\Delta_{\text{align}}) = 2 \operatorname{arctanh}(\Delta_{\text{align}}/\Delta_{\text{max}})$; equals $\alpha(\lambda) = \tanh(\lambda/2)$ at the realizing λ (Eq. (6.8)). Distinct from Δ_{util} — alignment-inversion is not the VFE first-order condition.	Δ_{align}	<code>delta</code> argument to <code>optimal_lambda</code> (same code path; semantics is alignment, not utility)	<code>delta : Float</code>
Δ_{max}	Saturation point of Δ_{util} / Δ_{align} (Bernoulli toy: 1)	Δ_{max}	<code>delta_max</code> argument (default 1.0)	<code>delta_max : Float</code>
$\Delta_{\text{spec}}(\lambda) = s_1 - s_2$	Spectral gap (largest minus second-largest singular value of the bipartite reshape of q_λ); distinct from utility surplus and target alignment	Δ_{spec}	<code>schmidt_decomposition(q)[0].weight - schmidt_decomposition(q)[1].weight</code> (composition form)	(no current Lean SVD companion; Python computes the spectrum)
$\lambda^*(\Delta_{\text{align}})$	Alignment-inversion coupling = $2 \operatorname{arctanh}(\Delta_{\text{align}}/\Delta_{\text{max}})$; inverse of the alignment correspondence in the K=2 Bernoulli toy. <i>Not</i> a VFE optimum.	λ^* (or λ^*)	<code>optimal_lambda(delta)</code>	<code>optimalLambda</code>
$\lambda_{\text{VFE}}^*(u)$	True VFE-optimal coupling under utility scalar u ; equals $2u$ in the symmetric K=2 Bernoulli toy (see §C). Coincides with $\lambda^*(\Delta_{\text{align}} = u)$ only at small u .	(inline)	(derived analytically)	(no Lean)
λ_{infl}	Inflection coupling: the point in $[0, \infty)$ where F'' changes sign exactly once, separating the regime where the agent benefits from <i>more</i> coupling (below) from the regime where marginal returns reverse (above) — see §B	λ_{infl}	(inline; located where F'' changes sign)	(no Lean)
$\lambda_c^{(1)}, \lambda_c^{(2)}$	Phase-band thresholds (default (0.5, 2.5))	$\lambda_{c^{(1)}}, \lambda_{c^{(2)}}$	H.PHASE_LAMBDA_C1, H.PHASE_LAMBDA_C2	<code>lambdaC1, lambdaC2</code>
λ_{probe}	Small- λ probe for the coupling-tax curvature fit	λ_{probe}	H.COUPLING_TAX_PROBE - LAMBDA	(no Lean)
J_0	Bernoulli / Ising coupling strength (scalar prefactor of the $K = 2$ aligned-indicator coupling J)	J_0	<code>j0</code> (closed-form scalar in <code>bernoulli_toy</code>)	(no Lean — Bernoulli toy uses <code>BinaryCoupling</code>)
λ_E, λ_G	Decoupled-parameter form: two-parameter generalization in which the habit-side and EFE-side coupling strengths are independent (λ in the body is the symmetric $\lambda_E = \lambda_G$ restriction)	λ_E, λ_G	<code>lam_e, lam_g</code> (parameter overrides)	(no Lean — symmetric <code>lam : α</code> only)
$\lambda_{1/2}$	Half-saturation coupling: λ at which the alignment correspondence reaches $\Delta_{\text{max}}/2$	$\lambda_{1/2}$	<code>manuscript_variables.json[pymdp_summary_lambda_at_half_saturation]</code> (derived from <code>SmallLambdaTolerance</code>)	(no Lean)
λ_{max}	Max admissible coupling under small- λ tolerance ε (Theorem 9.1)	λ_{max}	(derived from <code>SmallLambdaTolerance</code>)	<code>SmallLambdaTolerance / lambda_max</code>

Symbol	Meaning	LaTeX	Python	Lean
Φ_{kj}	Per-pair coupling factor: entry (k, j) of Φ (the sup-norm $\ \Phi\ _\infty$ already has a row)	Φ_{kj}	(entry of coupling-log-weight matrix)	(no Lean — magnitude consumed via <code>couplingNormSq_of_trivialCoupling</code>)
$\mathcal{C}$	Interaction graph: set of coupled stream pairs (k, j) with $\Phi_{kj} \neq 0$	$\mathcal{C}$	(set / sparsity pattern of coupling)	(no Lean — implicit support of <code>CouplingPotential</code>)
J_e	Per-clique additive component of J in the tensor-train decomposition (indexed by edge $e \in \mathcal{C}$)	J_e	(clique components in <code>coupling_decomposition</code>)	(no Lean)
$Z(\lambda)$	Unnormalized posterior partition function; see normalizer note below	$Z(\lambda)$	(implicit in <code>entangled_posterior</code>)	(implicit in <code>entangledPosteriorLogWeight</code>)
$Z_E(\lambda)$	Normalizing constant of the entangled prior $\mathcal{E}_\lambda \propto (\prod_k E_k) e^{\lambda J}$	$Z_E(\lambda)$	(implicit; sourced from <code>entangled_prior</code>)	(no top-level Lean def; implicit in <code>entangledPosteriorLogWeight</code>)
$\psi(\lambda) = \log Z(\lambda)$	Posterior log-partition function	$\psi(\lambda)$	(implicit)	(no top-level Lean def; implicit in <code>entangledPosteriorLogWeight</code>)
$\psi_E(\lambda) = \log Z_E(\lambda)$	Prior log-partition function	$\psi_E(\lambda)$	(implicit)	(no top-level Lean def; implicit in <code>entangledPosteriorLogWeight</code>)
$F[q_\lambda] = \log Z_E(\lambda) - \log Z(\lambda)$	Closed exponential-family form of the entangled VFE (see Theorem 5.5)	(inline)	<code>entanglement_decomposition_rhs(...)</code> (Gibbs form)	(boundary; Lean companion is the algebraic skeleton in <code>Decomposition.lean</code>)
mode_k	Inference mode of stream k	mode_k	<code>InferenceMode.VFE / EFE / SOPHISTICATED</code>	(manuscript label; no Lean inductive in current boundary fragment — see Status notes)

Normalizer note. $Z(\lambda)$ normalizes the unnormalized exponential $E_{\text{MF}}(\pi) e^{\lambda(J - \gamma K_c)(\pi) - \gamma G_{\text{MF}}(\pi)}$, so $q_\lambda(\pi)$ is that numerator divided by $Z(\lambda)$. It is not $\sum_\pi q_\lambda(\pi)$, which is 1 by definition. It is the partition function that appears in the closed identity $F = \log Z_E - \log Z$ ([Eq. \(5.8\)](#)).

Information-theoretic quantities

Symbol	Definition	LaTeX	Python	Lean
$H(p) = -\sum p \log p$	Shannon entropy (natural log)	$H(p)$ (or $\text{Hent}(p)$)	<code>shannon_entropy(p)</code>	<code>shannonEntropy</code>
$H_b(p)$	Binary entropy $-p \log p - (1-p) \log(1-p)$	$H_b(p)$	(inline)	<code>binaryEntropy</code>
$D_{\text{KL}}(q \ p)$	KL divergence	$D_{\{\text{KL}\}}(q \ p)$ (or $\text{KL}\{q\}\{p\}$)	<code>kl_divergence(q, p)</code>	<code>kl</code>
$I(q) = \sum_k H(q^k) - H(q)$	Total correlation (multi-information)	$I(q)$ (or $\text{MI}(q)$)	<code>total_correlation(q)</code>	<code>totalCorrelation</code>
$\eta(q)$	Normalized multi-information / <i>policy-blanket leakage</i> — $I(q)/H(q) \in [0, 1]$; 0 at strict mean-field, 1 at maximally entangled. Distinct from η as m-coordinates (see Pitfalls).	$\eta(q)$	composition of <code>total_correlation(q)</code> and <code>shannon_entropy(q)</code>	(no direct Lean; derivable from <code>totalCorrelation</code> and <code>shannonEntropy</code>)
$\text{Var}_q[X]$	Variance under q	$\text{Var}_q(X)$ (or $\text{Var}\{q\}\{X\}$)	<code>np.var(...)</code> (ad-hoc)	(no Lean)
$\text{Cov}_q[X, Y]$ $\sigma(x) = 1/(1 + e^{-x})$	Covariance under q Logistic sigmoid	$\text{Cov}_q(X, Y)$ (or $\text{Cov}\{q\}\{X, Y\}$) $\sigma(x)$	(ad-hoc) <code>scipy.special.expit</code> (ad-hoc)	(no Lean) <code>floatLogistic</code>
$\langle X \rangle_q = \mathbb{E}_q[X]$	Expectation operator under joint q	$\langle X \rangle_q$ (or $\text{E}\{q\}\{X\}$)	<code>expected_value(q, X)</code>	(inline finite sum over <code>q_i * X_i</code>)
$\langle X \rangle_{q_\lambda}$	Expectation under the entangled <i>posterior</i> q_λ	$\langle X \rangle_{q_\lambda}$	<code>expected_value(entangled_posterior(...), X)</code>	(inline)

Symbol	Definition	LaTeX	Python	Lean
$\langle X \rangle_{\mathcal{E}_\lambda}$	Expectation under the entangled <i>prior</i> $\mathcal{E}_\lambda$ — distinct from posterior expectation when $G \neq 0$	$\langle X \rangle_{\mathcal{E}_\lambda}$	<code>expected_value(entangled_prior(...), X)</code>	(inline)
$d \log Z_E / d\lambda = \langle J \rangle_{\mathcal{E}_\lambda}$	Standard exponential-family identity for the prior log-partition derivative — used in Theorem 5.5 / Proposition 11.1	(inline)	(analytical)	(analytical)

Free energies

Symbol	Definition	LaTeX	Python	Lean
$F[q]$	Variational free energy of q	F (or fe)	<code>free_energy(q, prior, G, gamma)</code>	<code>variationalFreeEnergy</code>
$F[q^k]$	Per-stream marginal free energy	$F[q^k]$	<code>marginal_free_energy(q, prior, G, gamma, k)</code>	<code>marginalFreeEnergy</code>
$\sum_k F[q_\lambda^k]$	Sum of per-stream VFEs (the mean-field free-energy baseline)	(sum)	<code>free_energy_bundle(spec, obs, lam).vfe_total</code> (<code>FreeEnergyBundle.vfe_total</code>)	<code>marginalFreeEnergy</code> (aggregated over <code>StreamIdx K</code>)
$\langle G_k \rangle_{q_\lambda^k}$	Expected EFE under coupled posterior	$\langle G_k \rangle_{q^k_\lambda}$	<code>expected_free_energy_under_posterior(spec, obs, lam)[k] / FreeEnergyBundle.efc_under_posterior[k]</code>	(numerical)
$\lambda \langle J \rangle_{q_\lambda}$	Coupling-energy contribution to F	$\lambda \langle J \rangle_{q_\lambda}$	<code>coupling_energy(spec, obs, lam) / FreeEnergyBundle.coupling_term</code>	<code>couplingExpectation / Skeleton</code>

Manifolds, projections, and dual coordinates

Symbol	Meaning	LaTeX	Python	Lean
$\mathcal{M}$	Simplex of strictly-positive joint distributions on Π	$\mathcal{M}$ (or Mfd)	(implicit ambient)	(implicit)
$\mathcal{M}_{\text{MF}} \subset \mathcal{M}$	Mean-field submanifold (product distributions)	$\mathcal{M}_{\text{MF}}$ (or MFsubmfd)	<code>is_mean_field(q, atol)</code> (membership predicate)	<code>IsMeanField</code>
$\hat{m}(q) = \prod_k q^k$	m-projection of q onto $\mathcal{M}_{\text{MF}}$	$\hat{m}(q)$	<code>m_projection(q)</code>	<code>mfToJoint</code> (direction: $\text{MF} \rightarrow \text{joint}$; Python computes the joint $\rightarrow$ MF marginals before rejoining)
$\theta = \log q$	e-coordinates (natural parameters)	θ	<code>np.log(q)</code> (ad-hoc)	(no Lean)
$\eta = \mathbb{E}_q[\cdot]$	m-coordinates (expectation parameters of an exponential family). Distinct from $\eta(q)$ policy-blanket leakage (see Pitfalls).	η	(ad-hoc)	(no Lean)
Pythagorean residual	$D_{\text{KL}}(q \ q_0^*) - I(q) - D_{\text{KL}}(\hat{m}(q) \ q_0^*)$	(inline)	<code>pythagorean_residual(q, mf_reference)</code>	<code>dualFlat_pythagorean_witness</code>

Spectral and tensor-network symbols

Symbol	Meaning	LaTeX	Python	Lean
r	Schmidt rank of bipartite (K=2) joint	r	<code>schmidt_rank(q, atol)</code>	(boundary witness)
s_α	α -th singular value	s_α	<code>Archetype.weight</code> (in <code>schmidt_decomposition</code>)	(numerical)

Symbol	Meaning	LaTeX	Python	Lean
u_α, v_α	Left / right singular vectors (archetype marginals)	u_alpha, v_alpha	Archetype.u, Archetype.v	(numerical)
$S_E(q)$	Policy entanglement entropy of bipartite cut	$S_E(q)$	entanglement_entropy(q, atol)	(numerical)
$r_{\text{eff}}(\lambda) = e^{S_E(q_\lambda)}$	Effective rank — smooth phase order parameter (continuous proxy for the integer Schmidt rank) used in §10	$r_{\{\mathrm{eff}\}}$	np.exp(entanglement_entropy(q)) (composition)	(no Lean — numerical)
r_k	Tensor-train bond dimension across k -th cut	r_k	tensor_train_ranks(q, atol)[k]	(numerical)
$(r_1, \dots, r_{K-1})$ $A^{(k)}$	Tensor-train rank profile MPS / tensor-train tensor at site k (factor of the matrix-product factorization of q_λ); see §8.3	$(r_1, \dots, r_{K-1})$ $A^{\{k\}}$	entanglement_spectrum(q) mps_decomposition(q)[k] (canonical left-orthogonal TT-SVD; reconstruction-verified in tests/test_spectral.py)	(numerical) (no Lean)
$S_k(q_\lambda)$	Per-cut bond entropy: entanglement entropy across the bipartition splitting streams $\{1, \dots, k\}$ from $\{k+1, \dots, K\}$	$S_k(q_lambda)$	entanglement_entropy_per_cut(q, k) (composition of entanglement_entropy with the per-cut bipartite reshape)	(no Lean)
$S_k^{\max} = \log r_k$	Maximum bond entropy across cut k (achieved by the uniform spectrum on r_k singular values)	$S_k^{\max}$	np.log(tensor_train_ranks(q)[k])	(no Lean)
$\text{Cov}_{q_\lambda}(J - \gamma K_c, \log q_\lambda - \log \hat{m}(q_\lambda))$	Covariance identity for the derivative $dI/d\lambda$ of total correlation along the entangled family (§4)	(inline)	(numerical; from entangled_posterior outputs)	(no Lean)

Heterogeneous-ensemble quantities

Symbol	Meaning	LaTeX	Python	Lean
$\text{tax}(\lambda)$	Coupling tax (KL between fully-adaptive and pinned posteriors)	$\mathrm{tax}(\lambda)$	coupling_tax(mf, G, J, K_c, gamma, lam, modes)	couplingTax (witness form)
$\ \Phi\ _\infty$	Coupling magnitude (sup-norm)	$\text{norm_inf}(\Phi)$	(sup over coupling-log-weight)	couplingNormSq
$C \geq 0$	Structural curvature constant in Theorem 9.1 . Live rendered value: manuscript_variables.json[coupling_tax_curvature_C].	C	quadratic_bound_curvature(...)	(boundary witness)

KL-control, copula, and product-of-experts symbols

These symbols appear in §18.1 (KL-control / path-integral correspondence) and the copula / product-of-experts extensions of §4 and §18.4. They are documented here so the reader can disambiguate against the main coupling notation (particularly the trajectory cost $C(\tau)$, renamed from J to avoid collision with the habit coupling potential).

Symbol	Meaning	LaTeX	Python	Lean
$V(x)$	KL-control / path-integral value function (cost-to-go)	$V(x)$	(no Python — symbol only)	(no Lean)
p_{free}	Uncontrolled / free dynamics distribution (reference law in the KL-control correspondence)	$p_{\{\mathrm{free}\}}$	(no Python — symbol only)	(no Lean)

Symbol	Meaning	LaTeX	Python	Lean
ρ	Risk-sensitivity parameter in KL-control (inverse-temperature on the trajectory cost)	ρ	(no Python — symbol only)	(no Lean)
$C(\tau)$	Trajectory cost in KL-control. Renamed from J to avoid clash with the habit coupling potential $J(\pi)$.	$C(\tau)$	(no Python — symbol only)	(no Lean)
$u^k = F_{E_k}(\pi^k)$	Copula CDF reparameterization of per-stream policy: u^k is the CDF transform of π^k under E_k , mapping each stream onto $[0, 1]$ for copula-based coupling	$u^k = F_{E_k}(\pi^k)$	(no Python — symbol only; copula extension is outside the present implementation)	(no Lean)
f_j	Product-of-experts factor (one of the multiplicative experts in $q(\pi) \propto \prod_j f_j(\pi)$)	f_j	(no Python — symbol only)	(no Lean)
$G_{\text{soph}}(\pi)$	Sophisticated-inference expected free energy (recursive EFE over future belief updates)	$G_{\{\text{soph}\}}$	<code>per_stream_efe(..., SOPHISTICATED)</code>	<code>ConnectionsWitnesses / sophisticated-inference witness</code>
J_{soph}	Source-supplied coupling carrying recursive observation-conditioning for sophisticated inference; not generated by λ alone	J_{soph}	witness payload	SI embedding payload

Relationship classes and claim-strength labels

Part V and §S7.2 use two small controlled vocabularies. They are not mathematical variables, but they are notation in the audit sense: they determine how strongly a sentence is allowed to identify this framework with an external active-inference variant.

Label	Meaning	Where enforced
exact	Literal specialization of the posterior family or a single-stream log-partition dual after all factors are specified	Active-inference recovery ledger; relationship-guard tests
parametric	Realized only after the modeler adds an explicit factor, change of variables, agent boundary, or structural parameter	Active-inference recovery ledger; relationship-guard tests
analogical	Shares a structural motif, while temporal, recursive, state-space, or message-passing content remains outside the λ -deformation	Active-inference recovery ledger; relationship-guard tests
out-of-scope	Related background whose geometry or modeling object is not represented by the current posterior family	Active-inference recovery ledger
proved / witness / empirical / hypothesis / roadmap	Claim-strength labels used to separate stock-Lean proofs, witness-consuming theorem rows, generated numerical evidence, interpretation, and future proof discharge	§S7.1; claim-provenance tests

Hyperparameters (figure / sweep / simulation)

The constants below live in `src/simulation/hyperparameters.py` and are mirrored into `output/data/manuscript_variables.json`; they reach the prose via `[[VAR:<key>]]` substitutions.

Symbol	Source-of-truth constant	JSON key	Default
Param-sweep grid count	<code>PARAMETER_SWEEP_LMBDAS.num</code>	<code>param_sweep_grid_points</code>	121
Param-sweep range	<code>PARAMETER_SWEEP_LMBDAS.[start,stop]</code>	<code>param_sweep_lambda_min/max</code>	<code>[0, 6]</code>
Coupling-tax grid	<code>COUPLING_TAX_LMBDAS.num</code>	<code>coupling_tax_grid_points</code>	31
Phase-diagram grid	<code>PHASE_DIAGRAM_LMBDAS.num</code>	<code>phase_diagram_grid_points</code>	401

Symbol	Source-of-truth constant	JSON key	Default
pymdp sweep grid	PYMDP_SWEEP_LAMBDAS.num	pymdp_sweep_grid_points	21
pymdp rollout horizon T	PYMDP_ROLLOUT_STEPS	pymdp_rollout_steps	10
pymdp rollout seed	PYMDP_ROLLOUT_SEED	pymdp_rollout_seed	0
Figure-script global RNG seed	FIGURE_GLOBAL_SEED	figure_global_seed	42
Param-sweep agreement tolerance	PARAMETER_SWEEP_AGREEMENT_TOLERANCE	param_sweep_agreement_tolerance	$1e-06$

LaTeX preamble macros

Reserved LaTeX macros (currently unused in the body; available for future expansion): see [manuscript/preamble.md](#) for the canonical definitions ($\backslash KL$, $\backslash E$, $\backslash Var$, $\backslash policy$, $\backslash policySpace$, $\backslash Mfd$, $\backslash MFsubmfd$, $\backslash MI$, $\backslash Hent$, $\backslash efe$, $\backslash fe$, $\backslash coupJ$, $\backslash coupK$). Each glossary row above cross-references the spelled-out form which is the one actually used throughout the manuscript.

Lean type abbreviations

The Lean boundary fragment uses lower-case ASCII identifiers as substitutes for reserved binder tokens ($\lambda \rightarrow \text{lam}$, $\pi \rightarrow \text{pi}$, $\Pi \rightarrow \text{Pol}$).

Lean abbrev	Meaning	Defined in
α : Type	Generic scalar type satisfying CommScalar	Scalar.lean
CommScalar α	In-house typeclass over a commutative scalar	Scalar.lean
K : Nat	Stream count	Basic.lean
StreamIdx K	Fin K — stream index	Basic.lean
PolicyFactor K	StreamIdx $K \rightarrow \text{Type}$ — per-stream policy type	Basic.lean
PolicySpace K Pol	$\forall k, \text{Pol } k$ — joint policy type	Basic.lean
JointDist K Pol	PolicySpace K Pol $\rightarrow \alpha$ — joint PMF	JointDist.lean
MFDist K Pol	$\forall k, \text{Pol } k \rightarrow \alpha$ — mean-field PMF	JointDist.lean
CouplingPotential α K Pol	PolicySpace K Pol $\rightarrow \alpha$ — J / K_c shape	Coupling.lean
BinaryCoupling	Bool $\rightarrow$ Bool $\rightarrow$ Float — $K=2$ closed-form coupling	BernoulliToy.lean
HorizonProfile K	StreamIdx $K \rightarrow \text{Nat}$ — temporal horizon per stream	Basic.lean
pi : PolicySpace K Pol	A joint policy	(use sites)
lam : α	Coupling parameter	(use sites)
q_lam : JointDist K Pol	Entangled posterior at coupling lam	(use sites)

Phase / verdict labels (manuscript-level)

Three discrete classifiers used in the body of the manuscript.

Label	Values	Where defined	Manuscript role
InferenceMode	VFE / EFE / SOPHISTICATED	Python enum in src/lean/heterogeneous.py	Stream-mode label (§3.2, §9)
CouplingPhase	disordered / mixed / frozen (string)	Returned by bernoulli_toy.coupling_phase_at(lam)	Cognitive phase (§10)
CouplingRole	habit (J) / preference (K_c)	Manuscript convention; not a Python or Lean type	Side of the coupling potential
CouplingVerdict	pays / neutral / does_not_pay	Manuscript convention; manifests as the Lean theorem couplingVerdict (in Decomposition.lean)	Coupling-pays-for-itself outcome (§11)

Status notes

- **InferenceMode** is presently a Python-only Enum; the Lean boundary fragment treats stream mode as a witness consumed by IsPlanningStream / IsReflexiveStream predicates (Heterogeneous.lean). The current Lean source therefore uses the predicates directly rather than an InferenceMode inductive.
- **CouplingPhase** is a string-returning Python predicate (coupling_phase_at) plus the Lean function couplingPhaseAt : Float $\rightarrow$ Nat (returning 0/1/2); the inductive form is documented in [lean/ActinfPolicyEntanglement/README.md](#) but has not been promoted into the boundary source.
- **CouplingRole** has no Python or Lean type — it is manuscript-level convention to disambiguate the two coupling potentials in prose.

Conventions

- **Natural log** is used throughout; numeric values therefore have units of *nats*. Where binary log is needed it is written $\log_2$.
- **Joint vs marginal**: distributions on Π are *joints*; per-stream distributions on Π^k are *marginals*.
- **Vector / tensor**: bold $\mathbf{x}$ when shape matters, plain x otherwise.
- **Equations vs definitions**: an equation set off as $X \equiv Y$ is a definition; $X = Y$ is a derivable identity.
- **Probability conventions**: every distribution is normalized to 1 unless explicitly *unnormalized*; KL is always $D_{\text{KL}}(q \| p)$ (not the reverse).
- **Information geometry**: dually-flat structure on the simplex follows the standard development [Amari and Nagaoka, 2000, Amari, 2016, Nielsen, 2020]; non-extensive / ϕ -deformed analogs use the framework of [Naudts, 2011].
- **Active-inference background**: the single-stream POMDP recap follows [Friston, 2010, Da Costa et al., 2020, Smith et al., 2022], with EFE conventions per [Friston et al., 2017a].
- **Determinism**: all numerical figures are bit-reproducible under fixed seeds; see the full determinism contract in [docs/reference/statistics_reference.md](#).

Common pitfalls when editing across tracks

The four-track contract permits a few notation hazards that have bitten earlier drafts and that automated validation does not catch. They are listed here so an editor can recognise them by name.

- **Prior vs posterior expectation**. The symbol $\langle X \rangle_\lambda$ is ambiguous; the manuscript uses $\langle X \rangle_{\mathcal{E}_\lambda}$ for *prior* expectations (used in the log-partition derivative $d \log Z_E / d\lambda = \langle J \rangle_{\mathcal{E}_\lambda}$) and $\langle X \rangle_{q_\lambda}$ for *posterior* expectations. Mixing them silently introduces a sign error in convex / concavity arguments (see §B).
- **λ as parameter vs λ as Lean keyword**. In Lean, λ is reserved for anonymous-function syntax; the manuscript parameter λ becomes the identifier `lam` on the Lean side. Every `lam :  $\alpha$` in `.lean` files is the coupling parameter, not a closure binder.
- **K vs K_c** . K (stream count) and K_c (preference coupling potential) collide at one ASCII letter. Use `Kc` (Python, Lean) or `K_c` (LaTeX); never bare `K` in a context that could be read as either.
- **MF prior E_{MF} vs λ -entangled prior $\mathcal{E}_\lambda$** . Both pieces of notation use the letter “E”, and `e_lambda` / `mf_prior` are the disambiguating Python names; never write `E` in code or prose without the appropriate subscript or qualifier.
- **Total correlation vs per-stream entropy sum**. $I(q) = \sum_k H(q^k) - H(q) \geq 0$, *not* $\sum_k H(q^k)$ alone. The Python `total_correlation(q)` returns the difference; an editor importing `sum_marginal_entropies` and calling it “total correlation” by mistake will introduce a manuscript-vs-code divergence the validator cannot detect.
- **Natural log everywhere**. Information-theoretic quantities are in *nats*. When citing a $\log_2$ -convention source in the bibliography, convert before quoting numeric values.
- **Floating tolerance**. `atol=1e-9` is the project-wide default for spectral / SVD-based predicates; `atol=1e-12` is the default for conservation laws / sum-to-one checks; tests under `tests/test_*.py` adopt either explicitly. Don’t change the default in one place without checking the others.
- **η is overloaded**. In §7 (information geometry) it denotes *m-coordinates* (expectation parameters of an exponential family); in §19 (policy-blanket leakage) it denotes the normalized multi-information $\eta(q) = I(q)/H(q)$. Subscripts and context disambiguate; we keep both usages because each is standard in its parent literature (Amari / Nagaoka for the geometric η ; Friston / Pellet-Massini for the leakage η).

GNN fifth-track concordance (structural-and-numerical)

The fifth track maps each standing symbol of the K=2 toy to its declaration in the shipped GNN source `gnn/bernoulli_toy.gnn.md` (parser: `src/gnn/parser.py`; round-trip: `scripts/simulate_gnn.py`; parity test: `tests/test_gnn_concordance.py`). This track is **structural-and-numerical**, not a proof track, and the round-trip is an **internal-consistency reduction** — the general machinery reduces to the closed form for this toy — **not an independent corroboration**: it reproduces the model’s structure and the K=2 mutual-information curve to 7.77e-16 nats, but promotes no theorem row in §S7. See §S8.

Symbol	GNN variable	Active Inference Ontology term
π^1 (stream-1 policy)	<code>pi1</code>	<code>Stream1PolicyVector</code>
π^2 (stream-2 policy)	<code>pi2</code>	<code>Stream2PolicyVector</code>
E^1 (stream-1 habit prior)	<code>E1</code>	<code>Stream1HabitPrior</code>
E^2 (stream-2 habit prior)	<code>E2</code>	<code>Stream2HabitPrior</code>

Symbol	GNN variable	Active Inference Ontology term
J (cross-stream coupling)	J (joint var on the policy product)	CrossStreamCouplingPotential
λ (deformation)	lam	EntanglementDeformationParameter
γ (sophistication)	gamma	SophisticationWeight
q_λ (entangled joint posterior)	q_{joint}	EntangledJointPosterior

Reference Tables: Claim Strength, Variant Recovery, Lean Inventory, pymdp Bundle Statistics, and JSONL Run-Log Schema

This supplement collects reference catalogs that are too detailed for main-body prose but essential for auditing the boundary fragment, the free-energy bundle, and the structured run log. Each table is labeled and cross-referenced from the body section that introduces it.

Claim-Strength Legend

The manuscript uses five claim-strength labels so readers can see whether a sentence is formal, numerical, interpretive, or forward work:

Strength	Meaning	Promotion rule
proved	machine-checked as the registered boundary statement; row-level faithfulness: distinguishes substantive rows from statement-restricted rows	already source-backed by the default Lean build for Float-track boundary statements, with claim strength limited by the row's faithfulness: value; analytic real-valued rows require the <code>--with-mathlib</code> MathlibProofs build with foundational-only <code>#print axioms</code>
proved (analytic, $\mathbb{R}$)	machine-checked in the Mathlib-backed <code>lean/MathlibProofs/</code> package over $\mathbb{R}$ (not stock-Lean Float), with foundational-only <code>#print axioms</code> (no <code>sorryAx</code> / no <code>project-axiom</code>), at least one independent negative control, and an enforced reproducible audit gate (<code>scripts/build_mathlib_proofs.py</code> + <code>tests/test_mathlib_axiom_audit.py</code>)	already source-backed by the MathlibProofs build path
witness	a typed Lean witness record or witness-consuming theorem exists	promote only when the analytic payload is constructed in a green proof layer
empirical	generated by scripts and validated by CSV/JSON/PNG/PDF gates	promote only by adding replicated or broader benchmark evidence
hypothesis	model-generated biological, clinical, or alignment interpretation	promote only by adding direct empirical study or citation-backed validation
roadmap	scoped future work, including Mathlib witness discharge and the verified $\text{Float} \leftrightarrow \mathbb{R}$ bridge	promote only when source and gates exist

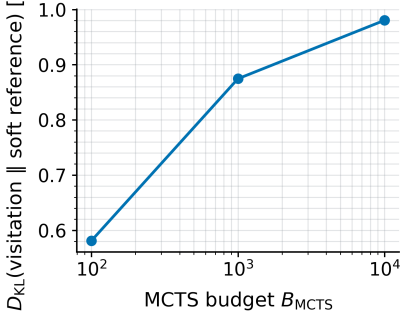
Active-Inference Variant Recovery Ledger

Part V uses the same finite joint-policy template for every connection, but it does not assign the same claim strength to every prior framework. The ledger below records the structural choice, the relationship class, and the validation hook that prevents an analogical mapping from being read as an exact recovery.

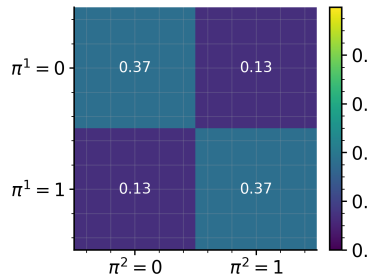
Read each row as a statement about where the comparison lands in the shared notation of §S6: E_k , G_k , and γ_k come from the single-stream active-inference posterior; J , K_c , and λ are the added policy-space coupling objects; and special payloads such as J_{soph} or a block-bidiagonal hierarchical J must be supplied by the modeler. A row marked `analogical` is therefore not a failed exact recovery; it is a deliberate boundary saying that recursive EFE, temporal-scale separation, state-space Markov blankets, or a message-passing schedule live outside the scalar λ deformation.

Variant or framework	Structural choice in the λ -coupled posterior	Relationship class	Evidence / gate	Do not overclaim
Mean-field discrete POMDP AIF (pymdp, SPM, ActiveInference.jl)	Set $\lambda = 0$ so the joint policy posterior is the outer product of per-stream policy posteriors	exact	Corollary 5.3 ; pymdp $\lambda = 0$ free-energy-bundle gate; <code>test_coupled_policy_posterior_lambda_zero_is_outer_product</code>	Does not imply that every multi-factor AIF model already has multiple coupled policy variables
Factor-graph / RxInfer-style message passing	Add $\exp(\lambda J - \gamma \lambda K_c)$ as an explicit policy factor in the graphical model	parametric	Factor-graph citations [de Vries and Friston, 2017 , Friston and Parr, 2017 , Parr et al., 2019]; RxInfer software citation [Bagaev and Podusenko, 2023]; manuscript validation of the recovery table	The inserted factor has exact semantics only inside the chosen graph; message passing itself is the implementation substrate, not a new theorem
Hierarchical / deep temporal AIF	Use block-bidiagonal J between adjacent levels and supply the concentration witness	analogical + witness	Theorem 17.1 ; <code>ConnectionsWitnesses.hierarchicalAIF_lambda_limit_witness</code> ; Lean boundary gate	Temporal-scale separation and directed message passing are not recovered by symmetric J alone
Sophisticated inference	Encode recursive EFE content in a source-supplied J_{soph} / embedding witness	analogical + witness	Proposition 17.2 ; <code>sophisticatedInference_embedding_witness</code> ; Lean boundary gate	Recursive observation-conditioning must be built into the witness payload
Branching-time AIF	View tensor-train or prefix-tree compression as a compact representation of the expanded policy tree	analogical + empirical (scoped)	Shipped UCB-MCTS baseline worked run ([Champion et al., 2022];	

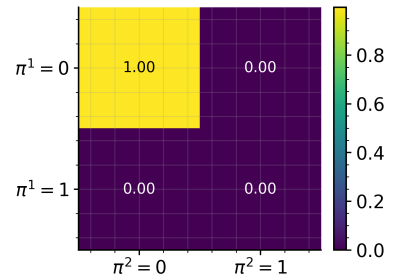
(a) Visitation vs soft reference (slope exponent -0.114)



(b) Closed-form reference q_λ



(c) BTAI visitation ($B = 10000$)



; 1, 3 budgets) — estimates lowest-EFE joint action, not the soft variational posterior of [[Champion et al., 2022](#)]
| No algorithmic equivalence to Bayesian-filter BTAI; compute-matched head-to-head study remains open |
| Multi-agent / federated AIF | Use within-agent and across-agent blocks of J after agent boundaries and shared or
linked generative models are specified | **parametric** | Multi-agent/federated citations [[Maisto et al., 2024](#), [Friston
et al., 2024](#)]; interaction and coupling-ablation sidecars | The literature is not asserted to optimize this exact joint
posterior unless it writes that model | | Scale-free / RGM | Interpret low-rank projection of q_λ as a policy-posterior
coarse-graining analogy | **analogical** | RGM / scale-free citation [[Friston et al., 2025](#)]; tensor-train rank profile
figure | MERA-style scale invariance and generative-model RG flow are outside this artifact | | Markov blankets /
Bayesian mechanics | Use normalized multi-information as policy-space leakage after a model boundary is chosen
| **analogical + witness** | [Proposition 19.3](#); Markov-blanket critical citations [[Kirchhoff et al., 2018](#), [Aguilera
et al., 2022](#), [Raja et al., 2021](#)] | This is not a recovery of state-space Markov blankets or an argument for new
biological boundaries | | CEREBRUM / case grammar | Encode role relations as asymmetric edge weights in J ,
with $\mathcal{K} \xrightarrow{\alpha} \mathbf{Coup}_K$ pullback when a categorical case substrate is supplied | **parametric** | CEREBRUM citation
[[Friedman, 2025](#)]; compositional-case substrate [[Friedman, 2026b](#)]; coupling-graph figure | Case grammar supplies
role structure; the categorical substrate licenses the asymmetry but does not prove it from the coupling model |

Lean 4 Boundary-Fragment Module Inventory

Implementation status of `lean/ActinfPolicyEntanglement/`, Lean 4 v4.29.0 (cross-referenced from §12):

Module	Theorems / definitions	Sorries
Basic	StreamIdx, PolicyFactor, PolicySpace, IsPlanningStream, IsReflexiveStream, instDecidableIsPlanningStream, instDecidableIsReflexiveStream, stream_classification	0
Scalar	CommScalar α typeclass, derived mul_sub/sub_mul/sub_self/affine_diff/affine_at_zero; instance : CommScalar Int	0
JointDist	JointDist, MFDist, IsNonNeg, IsPMF, mfProductWeight, mfToJoint, IsMeanField	0
Coupling	CouplingPotential α , trivialCoupling, couplingLogWeight, entangledPosteriorLogWeight, couplingLogWeight_affine_in_lam, couplingLogWeight_at_zero (both proved on [CommScalar α])	0
FreeEnergy	supportSum, safeLog, shannonEntropy, kl, totalCorrelation, variationalFreeEnergy, marginalFreeEnergy, totalCorrelation_eq_kl_to_mprojection	0
Geometry	mfImage_isMeanField, mProjection_kl_eq_self_when_meanfield, entangledFamily_eGeodesic, dualFlat_pythagorean_witness	0
Spectral	Bipartite.BipartiteJoint, Bipartite.IsBipartiteMeanField, Bipartite.isBipartiteMeanField_factors, Bipartite.factors_isBipartiteMeanField	0
Heterogeneous	HorizonProfile, couplingTax, BoundedQuadraticTax structure, couplingTax_quadratic_bound (Theorem 9.1), SmallLambdaTolerance structure, couplingTax_small_lambda_tolerance (Corollary 9.2)	0
BernoulliToy	Action, BinaryMF, BinaryJoint, BinaryCoupling, float-Exp/floatLog/floatLogistic/floatArctanh, alignedIndicator, isingCoupling, xLogX, binaryEntropy, isingMutualInformation, optimalLambda, isingFreeEnergyCurve, lambdaC1/lambdaC2/couplingPhaseAt, isingMI_zero_at_zero, isingFreeEnergyCurve_total	0
Decomposition	couplingExpectationSkeleton, entanglement_decomposition (Theorem 5.1 , witness form), couplingVerdict, couplingLogWeight_pointwise_at_zero (Corollary 5.3), totalCorrelation_def_unfold (Corollary 5.4), freeEnergy_closedForm_witness (Theorem 5.5 , closed-form boundary), freeEnergy_closedForm_at_zero (mean-field collapse)	0
Monotonicity	nat_le_refl, nat_le_trans, nat_succ_pos, nat_zero_le, nat_le_succ, nat_lt_succ_self, or_self_iff, or_comm_iff, and_self_iff, list_length_nonneg, list_length_cons, list_length_append, list_append_nil, list_nil_append, fin_lt_size, fin_zero_lt	0
Constructive	entangledPosteriorLogWeight_at_zero, couplingLogWeight_trivialCoupling, couplingNormSq_of_trivialCoupling, couplingTax_zero_at_zero (all CommScalar-polymorphic, genuine = 0 proofs)	0
Convexity	freeEnergy_convex_in_lam_witness, freeEnergy_localConcavity_at_zero_witness (witness payloads for Theorem 5.6 , Proposition 11.1)	0
SpectralWitnesses	schmidtRank_upperSemicontinuous_witness, sparsityRank_tradeoff_witness	0
ConnectionsWitnesses	hierarchicalAIF_lambda_limit_witness, sophisticatedInference_embedding_witness	0

Module	Theorems / definitions	Sorries
MarkovBlanket	markovBlanket_separation_identity_witness	0
FloatRealResidualWitness	FloatRealResidualWitness structure + floatRealResidual_witness (roadmap Float $\leftrightarrow\mathbb{R}$ residual shell; values from output/reports/float_real_residual.json)	0

Total: ~126 declarations (39 defs, 76 theorems / lemmas, 11 structures in Heterogeneous — BoundedQuadraticTax and SmallLambdaTolerance), **0 sorries, 0 axioms, 0 unsafe/partial/noncomputable**.

Worked excerpt — the load-bearing identity in Lean. To make the shape concrete, here is the boundary statement of the entanglement decomposition theorem ([Theorem 5.1](#)) as it lives in [Decomposition.lean](#):

```
-- **Theorem 5.1 (Entanglement Decomposition)** – boundary witness
form.
```

```
Given a Mathlib-supplied algebraic split
`F[q] = marginal_part + coupling_expectation + agentic_gain`,
where `coupling_expectation` is the integrated `couplingLogWeight`
against `q`, this theorem certifies the equation in the boundary
fragment. Every coupling parameter `(J, K_c, γ, λ)` is genuinely used
via `couplingExpectationSkeleton`, so the statement is non-vacuous. -/
theorem entanglement_decomposition {K Pol}
  (q : JointDist K Pol) (s : List (PolicySpace K Pol))
  (logE G : PolicySpace K Pol → Float)
  (J K_c : CouplingPotential Float K Pol) (gamma lam : Float)
  (marginal_part agentic_gain : Float)
  (hWitness : variationalFreeEnergy q logE G gamma s =
    marginal_part
  + couplingExpectationSkeleton q s J K_c gamma lam
  + agentic_gain) :
  variationalFreeEnergy q logE G gamma s =
    marginal_part
  + couplingExpectationSkeleton q s J K_c gamma lam
  + agentic_gain :=
hWitness
```

The `hWitness` argument is the *analytic-content boundary*: the caller supplies the algebraic equality once Mathlib’s `Finset.sum_comm` (or its target-language analog) is in scope; the boundary fragment retains the structural decomposition so its type-check is unaffected. Every witness-form theorem in the table above follows the same pattern.

MathlibProofs $\mathbb{R}$ Analytic Discharge Package: Inventory

The companion package `lean/MathlibProofs/MathlibProofs.lean` is separate from the Float boundary fragment listed above. Built independently via `lake build` (or via the enforced `scripts/build_mathlib_proofs.py`), it imports Mathlib4 and machine-checks the manuscript’s central analytic identities in $\mathbb{R}$. The keystone declarations the audit gate `tests/test_mathlib_axiom_audit.py` pins, in their natural order:

Declaration	Role	Foundational-only #print axioms?
<code>klReal_nonneg</code>	Gibbs inequality: $D_{\text{KL}}(p \parallel q) \geq 0$ for finite probability vectors with strictly-positive q	yes
<code>klReal_split_via_intermediate</code>	KL chain-rule decomposition through an intermediate distribution	yes
<code>klReal_minimises_generalK</code>	m -projection minimality for general entangled q over the mean-field submanifold (general K)	yes
<code>crossTerm_matches_K2</code>	$K = 2$ specialization of the cross-term identity	yes
<code>crossTerm_matches_of_equal_marginals</code>	General- K cross-term discharge under the equal-marginals hypothesis	yes
<code>entanglement_decomposition_generalK</code>	Axiom-clean general- K kernel for the multi-information term in Theorem 5.1	yes
<code>free_energy_decomposition_full</code>	Capstone — the full S01 boxed free-energy identity machine-checked in $\mathbb{R}$ for the genuine entangled posterior q_λ ; positivity and unit mass <i>proved</i> from the definitions, $\log Z_E$ the genuine definitional log-normalizer, $I(q_\lambda)$ discharged via <code>entanglement_decomposition_generalK</code>	yes

Non-vacuity is verified by two independent negative controls: (i) the $\log Z_E$ definitional body and (ii) the $\gamma\lambda\langle K_c \rangle$ coupling term each make the build fail when neutralised. The whole package depends only on the three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`). The single open analytic residual is a *verified* error-bounded $\text{Float} \leftrightarrow \mathbb{R}$ bridge scoped in `docs/reference/methods_and_assumptions.md` as multi-week research.

pymdp Free-Energy Bundle: Auto-Injected Summary Statistics

The 12 summary statistics computed by `simulation.statistics.pymdp_summary_statistics` and mirrored to `output/simulations/pymdp_summary.json` (cross-referenced from §15). Every value flows from a real pipeline run via `[[VAR:<key>]]` substitution:

Quantity	Symbol	Auto-injected value
Sweep grid points	N_λ	21
Sweep range	$\lambda \in [\lambda_{\min}, \lambda_{\max}]$	[0, 4]
Total correlation	$I_{\min}, I_{\max}, \bar{I}$	0.0000, 0.3635, 0.2530 nats
Half-saturation	$\lambda_{1/2}, I(\lambda_{1/2})$	1.044, 0.1818 nats
Total VFE	$F_{\min}, F_{\max}, \bar{F}$	1.6840, 2.4167, 2.2778 nats
Coupling term	$\lambda\langle J \rangle_{\min, \max}$	0.0000, 3.9983 nats
Joint entropy	$H_{\min}, H_{\max}$	0.3675, 1.1644 nats
Aligned-corner mass	$\min, \max, \text{at } \lambda_{\max}$	0.6068, 0.9998, 0.9998
KL to q_0	$\min, \max$	0.0000, 0.4976 nats
KL to uniform at $\lambda_{\max}$	—	1.0188 nats
Action entropy	$\min, \max$	0.3675, 1.1644 nats
Mode probability at $\lambda_{\max}$	$\max_\pi q_{\lambda_{\max}}(\pi)$	0.8806

Robustness and Ablation Stress-Test Ledger

The stress-test sidecars are intentionally compact: one-axis perturbations for robustness, four fixed role/sign ablations, and replicate seeds for long-horizon behavior. The table records what each branch is allowed to claim and which artifact validates it.

Branch	Configured scope	Artifact	Claim checked
Robustness envelopes	14 one-axis scenarios over 21 λ values	<code>output/simulations/pymdp_robustness.csv</code> ; Fig. 20	Positive total-correlation envelopes persist outside the canonical observation / precision / preference slice.
Robustness half-saturation	scenario-level $\lambda_{1/2}$ summaries	<code>output/data/robustness_summary.json</code> ; Fig. 21	Threshold movement is reported as sensitivity, not as a new theorem.
Robustness decomposition residuals	same rows as the envelope branch	<code>output/simulations/pymdp_robustness.csv</code> ; Fig. 22	Theorem 5.1 remains numerically closed; max residual $1.55\text{e-}15$.
Coupling ablation	4 fixed variants	<code>output/simulations/pymdp_coupling_ablation.csv</code> ; Fig. 23	$\lambda = 0$ is invariant, null coupling is flat, non-null role/sign changes alter archetypal mass without breaking the decomposition.
Two-axis interactions	41 targeted scenarios across observation $\times$ coupling scale, gamma $\times$ preference strength, coupling variant $\times$ coupling scale	<code>output/simulations/pymdp_interaction_robustness.csv</code> ; Fig. 25	Interaction effects are appendix evidence; the null-variant flatline and decomposition residual guard remain intact.
Long-horizon replicates	seeds {0, 7, 13, 29, 41}	<code>output/simulations/pymdp_long_horizon_replicates.csv</code> ; Fig. 26	The fixed-seed figure remains canonical; the replicate sidecar reports TC-envelope and habit-pass-rate sensitivity.
Long-horizon diagnostics	one row per configured replicate seed	<code>output/simulations/pymdp_long_horizon_seed_diagnostics.csv</code> ; Fig. 27	Seed failures are explained as tail-window KL sensitivity rather than retuned away.

Evidence Ladder and Claim Provenance

The project distinguishes formal theorem rows, empirical artifacts, figures, and tests. A headline claim should appear with all four tracks when it is theorem-backed, or with an explicitly empirical artifact/test pair when it is a numerical stress observation.

Claim	Strength	Formal row	Simulation / data artifact	Figure	Test or validator gate
Entanglement decomposition closes for the pymdp posterior	witness + empirical	Theorem 5.1	Free-energy bundle CSV; robustness CSV	Fig. 36 ; Fig. 22	<code>validate_free_energy_bundle</code> ; <code>validate_robustness_suite</code> ; focused simulation tests
Mean-field anchor at $\lambda = 0$	proved (substantive on Corollary 5.3 ; statement-restricted on Proposition 7.2) + empirical	Corollary 5.3 , Proposition 7.2	Free-energy bundle CSV; coupling-ablation CSV	Fig. 32 ; Fig. 23	<code>validate_free_energy_bundle</code> ; <code>validate_coupling_ablation</code> ; robustness tests
Fixed-marginal null control removes cross-stream dependence	empirical	Proposition 7.3	Marginal-null-control CSV and JSON	Fig. 24	<code>validate_marginal_null_control</code> ; robustness tests
Total correlation equals KL to the m -projection	witness + empirical	Proposition 7.3	Revertibility CSV	Fig. 17	<code>validate_revertibility</code> ; revertibility tests
Multi-stream coupling scales beyond $K = 2$	witness + empirical	Theorem 8.3	Per- K sweep CSVs; multi- K summary JSON	Fig. 12 ; Fig. 14	<code>validate_multi_k_sweep</code> ; multi- K tests
Heterogeneous / role-changing couplings alter mass without voiding the identity	witness + empirical	Theorem 9.1 , Corollary 9.2	Coupling-ablation CSV; interaction CSV	Fig. 23 ; Fig. 25	<code>validate_coupling_ablation</code> ; <code>validate_interaction_robustness</code> ; robustness tests
Long-horizon trajectory stationarity is a finite-horizon empirical witness (not “habit formation” — the rollout agents perform no parameter learning; see §13 honest-scope disclosure)	empirical + hypothesis	Theorem 17.1	Long-horizon CSV; replicate CSV; seed-diagnostics CSV; threshold-sensitivity CSV	Fig. 15 ; Fig. 26 ; Fig. 27 ; Fig. 28	<code>validate_long_horizon</code> ; <code>validate_long_horizon_replicates</code> ; <code>validate_long_horizon_threshold_sensitivity</code> ; long-horizon tests
MathlibProofs machine-checks the full S01 free-energy identity in $\mathbb{R}$	proved (analytic, $\mathbb{R}$)	Theorem 5.1	<code>MathlibProofs.free_energy_decomposition_full</code> ; <code>entanglement_decomposition_generalK</code> general- K kernel	no paper figure (formal-track row)	<code>scripts/build_mathlib_proofs.py</code> (foundational-only <code>#print axioms</code>); <code>test_mathlib_axiom_audit.py</code> ; <code>test_mathlib_proofs_integrity.py</code>
Witness-form analytic payloads (the witness-tier rows) admit Mathlib4 discharge	roadmap	Theorem 5.6 , Proposition 7.3 , Proposition 7.5 , Proposition 8.2 , Theorem 8.3 , Theorem 9.1 , Corollary 9.2 , Proposition 11.1 , Theorem 17.1 , Proposition 17.2 , Proposition 19.3	future <code>lean/MathlibProofs</code> extension consuming Mathlib’s KL / PMF, convexity / Taylor, semicontinuity / rank, tensor-product and measure-theoretic primitives [Degenne, 2025 , Degenne et al., 2025]	no paper figure	<code>per-witness conformance test in tests/test_witness_conformance.py</code> plus future per-row discharge under <code>scripts/build_mathlib_proofs.py</code>
Verified error-bounded $\text{Float} \leftrightarrow \mathbb{R}$ bridge linking the $\mathbb{R}$ proofs to the Float pipeline	roadmap	Theorem 5.1 (Float-side companion)	future Flocq-style IEEE-754 model or interval-arithmetic re-implementation; scoped in the methods-and-assumptions reference page	no paper figure	future error-bound theorem with foundational-only <code>#print axioms</code> + self-designed non-vacuity negative control

pymdp JSONL Run-Log Field Schema

Full field schema of `output/logs/pymdp_runs.jsonl` (cross-referenced from §16). Each record is a single JSON object on its own line:

Key	Type	Meaning
timestamp	ISO-8601 UTC offset	when the record was emitted
script	string	originating script (e.g. <code>simulate_pymdp.py</code>)
event	string	<code>main_start</code> / <code>main_end</code> lifecycle records (carries <code>pymdp_ensemble_K</code> , <code>pymdp_ensemble_gamma</code> , <code>figure_global_seed</code> , <code>artifacts_emitted</code>)
section	string	function-level event label (e.g. <code>figure_pymdp_lambda_sweep</code> , <code>figure_pymdp_free_energies</code> , <code>figure_pymdp_rollout</code>)
num_streams, gamma, seed, horizon, lam	numeric	ensemble hyperparameters in effect for the section
observations	list	observations driving the run (per stream)
grid_points, lambda_min, lambda_max	numeric	sweep grid descriptors
policy_shape	list[int]	shape of the joint policy tensor (e.g. [2, 2])
tc_min, tc_max, tc_initial, tc_final	float	total-correlation summary over the sweep / rollout
vfe_total_min, vfe_total_max	float	per-sweep VFE-sum extrema (free-energy-bundle records only)
coupling_term_at_lambda_zero	float	numerical $\lambda = 0$ sentinel; must be exactly 0.0
joint_entropy_at_lambda_zero, marginal_entropy_sum_at_lambda_zero	float	the two must coincide at $\lambda = 0$ (sentinel for $I(q_0) = 0$)
tc_at_half_saturation, lambda_at_half_saturation	float	locating the half-saturation coupling, sourced from the bundle summary
all_pmf	bool	every q_λ on the sweep grid passed PMF validation
sampled_actions	list[list[int]]	sampled-action trajectory (<code>figure_pymdp_rollout</code> records only)
steps_emitted, artifacts_emitted	int / list	rollout step count; artifact paths emitted by the section
runtime_ms	float	wall-clock duration of the section
status	"ok" / "error"	exit status of the section

Worked excerpt — a real run-log record. The free-energy-bundle section writes one JSONL record under `output/logs/pymdp_runs.jsonl`. The schema below mirrors that record while keeping numerical fields tokenized against the current manuscript-variable bundle:

```
{
  "timestamp": "<iso8601 timestamp emitted by the logger>",
  "script": "simulate_pymdp.py",
  "section": "figure_pymdp_free_energies",
  "num_streams": 2,
  "gamma": 1,
  "observations": [0, 0],
  "grid_points": 21,
  "lambda_min": 0,
  "lambda_max": 4,
  "tc_min": 0.0000,
  "tc_max": 0.3635,
  "vfe_total_min": 1.6840,
  "vfe_total_max": 2.4167,
  "coupling_term_at_lambda_zero": 0,
  "joint_entropy_at_lambda_zero": 1.1644,
  "marginal_entropy_sum_at_lambda_zero": 1.1644,
  "tc_at_half_saturation": 0.1818,
  "lambda_at_half_saturation": 1.0436,
  "runtime_ms": "<measured by logger>",
  "status": "ok"
}
```

The three sentinels are checked by `scripts/validate_outputs.py`: `coupling_term_at_lambda_zero` must be exactly 0 (mean-field at $\lambda = 0$), and `joint_entropy_at_lambda_zero` must equal `marginal_entropy_sum_at_lambda_zero` at $\lambda = 0$ (total correlation $I(q_0) = 0$). A record that ships `status: "ok"` but fails either gate causes CI to reject the run.

Worked excerpt — interpreting the bundle summary. Running the harness with the standard hyperparame-

ters (PYMDP_SWEEP_LAMBDAS spanning 0 to 4 over 21 points, observations $[\emptyset, \emptyset]$, $\gamma = 1$) produces a `pymdp_summary.json` whose key entries — all auto-injected via `[[VAR:...]]` above — interpret as follows:

- The mean total correlation across the sweep is $\bar{I} = 0.2530$ nats; the sweep-maximum value is $I_{\max} = 0.3635$ nats at $\lambda = \lambda_{\max} = 4$. The Shannon ceiling for the configured $K = 2$ streams is the $\log 2$ ceiling in this binary aligned/anti-aligned Ising slice, so the sweep saturates roughly to half the ceiling under the Ising potential at the chosen grid endpoint.
- Half-saturation $\lambda_{1/2} = 1.044$: the coupling at which $I(q_\lambda)$ first crosses $I_{\max}/2$.
- Aligned-corner mass at $\lambda_{\max}$ is 0.9998 — the maximum-probability joint policy cell, lying between the uniform baseline $1/|\Pi|$ and the single-archetype limit, and measuring how concentrated the posterior is on its dominant archetypal joint policy.

These are the *interpretive labels* on the bundle stats; the auto-injected numbers above are bit-reproducible from the same run.

GNN as a Shipped Fifth Track: Triple-Play Mapping, a Verified K=2 Bernoulli Round-Trip, and a Lean Typed-Contract Emitter

This supplement ships **Generalized Notation Notation (GNN)** [Smékal and Friedman, 2023, Institute, 2023] as a **fifth track** of the policy-entanglement framework, on top of the four tracks the manuscript was developed on (prose, equations, Python/pymdp, Lean — see §1). The bridge is no longer a forward-pointing scoping appendix. A project-owned GNN v1.1 parser (`src/gnn/parser.py`), a deterministic round-trip that reconstructs the K=2 Bernoulli mutual-information curve from a real `.gnn` source (`gnn/bernoulli_toy.gnn.md`) and matches the closed form to a maximum residual of $7.77\text{e-}16$ nats, a Lean typed-structure emitter whose output type-checks under the stock Lean v4.29.0 toolchain, and a GNN-sourced manuscript variable all ship and run in the pipeline (`scripts/simulate_gnn.py`, wired into `scripts/run_all.py`; covered by `tests/test_gnn_parser.py`, `tests/test_gnn_round_trip.py`, `tests/test_gnn_concordance.py`).

What the fifth track is precise about. It reproduces the framework’s *structural and numerical* content; it does **not** prove theorems. Parsing and round-tripping a GNN source reconstructs *numbers* and emits a *typed contract* — it promotes no theorem row in §S7 to proved or witness. The four-track proof contract of §S6 and §1 is unchanged in its analytic content; GNN is added as a fifth, structural-and-numerical track with its own CI gates, not as a fifth *proof* track. This distinction is the load-bearing honesty of the supplement and is enforced in code: the Lean emitter’s output declares itself a typed contract, and no registry promotion is wired to the GNN layer.

Scope and Claim-Strength: a Shipped Bridge, with an Explicit Non-Claim About Proof

Framing declaration. In the framing taxonomy of this supplement — *concordance column*, *token system*, *coupling-graph source-of-truth*, *multi-agent extension scaffold*, *institute bridge* — the section adopts the **bridge** framing. GNN is a single namespaced supplement that bridges the policy-entanglement framework to the upstream GNN repository [Institute, 2023] in the same style that the Lean-fragment supplement at §E (§E) bridges this manuscript to the institute-wide `fep_lean` catalog: **minimal**, **namespaced**, **single-link** — one upstream repository, one canonical citation, one supplement, no rewriting of the body manuscript.

Claim-strength tag. The structural-and-numerical assertions of this supplement are **empirical** in the sense of §S7.1: shipped, source-backed, CI-gated. Concretely:

- A $7.77156\text{e-}16$ token **is** now sourced from a GNN round-trip: the value $7.77\text{e-}16$ flows from `output/data/gnn_bernoulli_roundtrip.json` (emitted by `scripts/simulate_gnn.py`) through `src/manuscript/variables.py::_gnn_facts` and is range-gated in both `REQUIRED_VARIABLES` and the manuscript range gate.
- No theorem row in §S7 is promoted to proved or witness by the GNN layer — and this is **by design, an explicit non-claim, not a gap**: parsing and round-tripping reproduce numbers, not proofs (see §S8.7).
- The K=2 Bernoulli mutual-information curve **is** now reconstructed from the GNN source and matches the hand-authored closed form (the harness in `src/simulation/` and `src/lean/bernoulli_toy.py` remains the canonical generator of `output/data/parameter_sweep.csv`; the GNN track reproduces that curve through the general machinery as an internal-consistency check rather than replacing it).
- A Lean typed-structure contract **is** now elaborated from the GNN source (`src/gnn/lean_emit.py` → `gnn/generated/BernoulliToyGnn.lean`) and type-checks under the stock Lean v4.29.0 build; it is a typed con-

tract, not a proof.

Scope-discipline statement. Promoting GNN to a track-class commitment required four structural steps; **all four are now performed**:

- (i) the symbol concordance at §S6 gains a fifth (GNN) track alongside prose / equations / Python / Lean; (ii) a `gnn_*` variable resolver in `src/manuscript/variables.py` populates GNN-sourced `[[VAR:...]]` tokens, parity-checked by `tests/test_gnn_concordance.py`; (iii) `tests/test_gnn_round_trip.py` and `tests/test_gnn_parser.py` ship alongside the existing token-resolution and project-wide-hyperlink tests; and (iv) `gnn/bernoulli_toy.gnn.md` is a round-trippable `.gnn` artifact under the reproducibility contract of §16, emitting a deterministic (byte-identical on re-run) sidecar. The GNN track does **not** change any theorem *statement*, any *proved*/*witness* label, or any analytic claim in the rest of the manuscript.

GNN Background: the Triple Play and the Active Inference Ontology

Smékal & Friedman (2023) [Smékal and Friedman, 2023] introduce **Generalized Notation Notation (GNN)** as “a novel approach to generative model representation that facilitates communication, understanding, and application of Active Inference across various domains”. The paper situates GNN as a **complement** — not a replacement — to the **Active Inference Ontology**, providing “a standardized method for describing cognitive models” that is “flexible and expressive” for education and modeling. The reference implementation lives at the upstream repository [Institute, 2023].

The central pragmatic device of the paper is **the Triple Play**: an approach to expressing GNN models in three modes simultaneously — **linguistic, visual, and executable**. Each mode is a *view* of the same GNN source:

- The **linguistic** view is a textual rendering of the model — the names, dimensions, dependencies, and equations a human reader would write in prose.
- The **visual** view is a diagram in which model variables become nodes, dependencies become edges, and structural hierarchy is read off the layout.
- The **executable** view is a runnable model — generated source code (typically in Python or a domain-specific language) that an inference engine can ingest.

The stated goal of the paper is “to bridge and respect the gaps among different modeling settings” and “to facilitate interdisciplinary research and application, ultimately promoting the advancement of the field.” The Triple Play is described as *pragmatic* — it does not assert that the three views are mathematically equivalent, only that they are coherently emitted from one shared source. This supplement realizes the **executable** view concretely: `src/gnn/runner.py` consumes a single `.gnn` source and emits a numerical round-trip (executable), a figure (visual), and the model’s typed structure (linguistic/Lean).

The two-fold framing — GNN as a *language*, the Triple Play as a *rendering discipline* — is what makes GNN the fifth representation for this manuscript: GNN supplies a single declarative surface (the source), and the Triple Play supplies a discipline for emitting views (text, figure, code) from that single surface.

Acronym Disambiguation: GNN (Notation) vs. GNN (Graph Neural Network)

The manuscript uses the acronym **GNN** in two distinct senses, and this supplement is the canonical place to disambiguate them.

- **GNN (Notation)**: Generalized Notation Notation [Smékal and Friedman, 2023, Institute, 2023]. A model-description language, shipped in this supplement as the fifth representation of the framework. Cited within §S8, with its open questions cataloged at §S8.9 and its shipped status recorded in the open directions of §21.
- **GNN (Graph Neural Network)**: The deep-learning architecture family. Referenced in §20 Q8 (“Learning to couple — GNN connection”) as the candidate mechanism for *learning* the coupling potentials J and K_c end-to-end from environment interactions. That usage names a learning-to-couple research direction in which a graph-neural-network forward pass plays the role of the coupling-graph message-passing operator; it is an answer to *how* the couplings might be learned, not a notation for the model itself.

The two meanings are **orthogonal**: Generalized Notation Notation is a *specification language* for the model; graph neural networks are a *parametric inference architecture* that could be used to learn the model’s couplings. A reader of this supplement should understand “GNN” within §S8 to mean Generalized Notation Notation throughout; the open-questions reference at §20 Q8 retains the graph-neural-network meaning. No claim is made in this supplement that the two meanings collapse, intersect, or share a substantive technical relationship beyond the historical acronym clash.

Mapping the $(A, B, C, D, E, J, \lambda)$ Tuple to a GNN Spec

The policy-entanglement framework operates on a finite-state discrete-time POMDP active inference baseline (see §3) augmented by the coupling parameters $(J, K_c, \lambda, \gamma)$ introduced in §4. Per §S6, the standing symbol set for a K -stream ensemble is:

Symbol	Object	Manuscript reference
A^k	Observation likelihood for stream k	§3.2
B^k	State-transition kernel for stream k	§3.2
C^k	Preferred-observation prior for stream k	§3.2
D^k	Initial-state prior for stream k	§3.2
E^k	Habit / policy-prior for stream k	§3.2
Π^k	Finite policy space for stream k	§3.2
J	Cross-stream coupling potential	§4.1
K_c	Cross-stream constraint potential	§4.1
λ	Coupling parameter (e-coordinate of the entanglement family)	§4.2
γ	Sophistication weight	§4.1

In the shipped GNN bridge, each of these is a typed declaration in the GNN source:

- Per-stream POMDP primitives $(A^k, B^k, C^k, D^k, E^k)$ map to GNN variable blocks, one block per stream, indexed by $k \in \{1, \dots, K\}$. Each block carries the dimensionality declarations the GNN parser understands for single-agent POMDP AIF models.
- The coupling potentials (J, K_c) map to a **cross-stream joint variable** over the product policy space — `J[2,2,type=float]` for the $K=2$ toy, with `## Connections` edges `pi1-J` and `pi2-J` binding each stream to it. **This requires no new GNN primitive.** The institute’s own `multi_agent_coordination.md` example in the GNN repository [Institute, 2023] already encodes a joint product-space variable (`s_joint[16,1]`) with connection edges from each agent — exactly the structure a cross-stream coupling needs. The framework’s coupling is therefore expressible in **stock GNN v1.1** as a joint variable plus edges, and the shipped parser (`src/gnn/parser.py`) and bridge (`src/gnn/bridge.py`) consume precisely that encoding.
- The deformation parameter λ and sophistication weight γ map to GNN scalar parameters annotated with their physical role. They render in the linguistic view as named quantities, in the visual view as scalar labels on cross-stream edges, and in the executable view as named arguments to the inference routine.

The full $(A, B, C, D, E, J, K_c, \lambda, \gamma)$ tuple is therefore representable in stock GNN v1.1. For $K > 2$ the construction composes: `gnn/k_stream_ensemble.gnn.md` declares a $K=3$ chain with pairwise couplings `J12`, `J23`, and the bridge assembles the joint coupling tensor additively ($J(\pi) = \sum_{(a,b) \in \text{edges}} J_{ab}(\pi^a, \pi^b)$), reading the coupling-graph topology directly off the `## Connections` block. **Honest residual:** a dedicated *first-class* `CouplingBlock` primitive would be more ergonomic than the joint-variable-plus-edges encoding, and proposing one upstream to the canonical GNN repository [Institute, 2023] is reasonable future contribution work — but it is an ergonomic improvement, *not* a blocker, and nothing in this manuscript’s fifth track depends on it.

Toolchain capabilities advertised upstream of this manuscript. The GNN repository [Institute, 2023] and the broader Active Inference Institute program around it advertise a family of capabilities: (i) an *abstraction layer* over the model’s surface syntax supporting flexible run-time dispatch across stream counts, coupling-potential shapes, and inference engines; (ii) *model typing and checking* for the dimensionality and topology of generative models; (iii) *resource estimation* for the computational cost of running a model (a natural complement to the Schmidt-rank and tensor-train bond-dimension profiles of §8); and (iv) the Triple Play rendering discipline [Smékal and Friedman, 2023]. The shipped bridge binds (ii) and (iv) directly — the project-owned parser type-checks the dimensionality of the cross-stream encoding and raises on mismatch, and the Triple Play views are emitted by `src/gnn/runner.py`. Bindings (i) run-time dispatch across an arbitrary inference backend and (iii) resource estimation against the TT bond-dimension profile remain the natural next bindings; they are not exercised by the present pipeline and are named as forward work in §S8.9, distinct from the now-shipped core bridge.

Worked Example: the $K=2$ Bernoulli Toy as a Shipped GNN Round-Trip

The cleanest place to make the GNN bridge concrete is the **$K=2$ Bernoulli toy** developed in §6.1 and proved closed-form in §C. This is the same object that anchors the symbol-by-symbol walkthrough in §S6 across the four existing tracks — using it again for the fifth track preserves pedagogical continuity.

The toy has two binary policy streams ($\Pi^1 = \Pi^2 = \{0, 1\}$), per-stream Bernoulli habit priors E_1, E_2 , and a single scalar coupling $J(\pi^1, \pi^2) = \mathbf{1}[\pi^1 = \pi^2] - \frac{1}{2}$ (the mean-zero, swing-1 aligned/anti-aligned Ising structure). The closed-form mutual information curve Eq. (6.5) is

$$I(\lambda) = \log 2 - H_b(\sigma(\lambda)), \quad (\text{S8.1})$$

where σ is the logistic and H_b is the binary entropy. The shipped GNN source `gnn/bernoulli_toy.gnn.md` declares the two binary policy variables, the 2×2 coupling J (with $J_{00} = J_{11} = +\frac{1}{2}$ aligned and $J_{01} = J_{10} = -\frac{1}{2}$ anti-aligned), and the deformation parameter λ . The Triple Play [Smékal and Friedman, 2023] emits from that single source three views:

- **Linguistic view:** a textual block naming the two streams, the binary policy spaces, the aligned coupling J , and λ — the `## ModelAnnotation` and `## ActInf Ontology Annotation` sections of the source.
- **Visual view:** the round-trip figure `output/figures/gnn_bernoulli_roundtrip.png`, overlaying the GNN-reconstructed and closed-form $I(\lambda)$ curves with the residual; the coupling graph (two policy nodes, a coupling edge labeled λJ) is the topology read off the `## Connections` block.
- **Executable view:** `scripts/simulate_gnn.py` (a thin orchestrator over `src/gnn/runner.py`) reconstructs the curve from the source and writes the deterministic sidecar `output/data/gnn_bernoulli_roundtrip.json`.

The round-trip, closed. The bridge (`src/gnn/bridge.py`) parses the coupling matrix J from `gnn/bernoulli_toy.gnn.md`, reconstructs the entangled joint posterior via the framework’s *general-purpose* `entangled_posterior`, and computes the multi-information via `total_correlation` — an **independent code path** from the hand-derived closed form `ising_mutual_information` (the bridge never imports or calls the closed form). Over the canonical 121-point λ grid, the two routes agree to a maximum residual of 7.77e-16 nats, well below the 1e-06 reproducibility tolerance.

What this check is, precisely. This is an **integration / internal-consistency** check, *not* an independent re-derivation of the mutual information: for the symmetric K=2 toy the general machinery provably reduces to the closed form (the aligned mass equals $\sigma(\lambda)$, whence $I = \log 2 - H_b(\sigma(\lambda))$), so the two routes are the same scalar function written two ways. What the agreement *does* establish — and what makes it a genuine witness rather than an $f(a) \approx f(a)$ tautology — is that the GNN-sourced spec, pushed through the *general* code path, reproduces the analytic prediction: a parser error, a bridge that ignored the declared coupling, or a wrong coupling *gap* would each break it. The **zero-coupling negative control** (replacing the parsed J with the zero matrix) diverges from the closed form by 0.676 nats, proving specifically that the bridge *reads and responds to* the declared coupling rather than hard-coding the answer. (Sign-flipping J is *not* a valid control: mutual information is invariant under $J \rightarrow -J$; this is pinned as a test so it cannot be mistaken for one.)

Scope of what the residual certifies. The round-trip pins the gauge-invariant coupling *gap* (the entangled posterior $q_\lambda \propto E \exp(\lambda J)$ is invariant under adding a constant to J , since the constant cancels in normalization), **not** the literal matrix entries. Two couplings with the same aligned-minus-anti-aligned gap produce the same curve. The absolute Ising entries $J = \{(0.5, -0.5), (-0.5, 0.5)\}$ are therefore pinned *separately* — by the `to_pymdp_config` round-trip test (which asserts the emitted matrix entrywise) and the GNN concordance parity test — so the 7.77e-16 residual should not be read as certifying the exact matrix. This round-trip is what graduates the GNN bridge from roadmap to empirical per §S7.1.

What GNN Preserves and What It Abstracts Away

A representation earns its keep only by naming, explicitly, the structural features it preserves and the features it abstracts away. Conflating preservation and abstraction is the failure mode the manuscript is most exposed to — readers who see a GNN diagram may treat it as a structural commitment when it is only a graph-rendered shorthand for parametric coupling. The table below is the shipped preservation/abstraction ledger.

Framework feature	GNN preservation status	Notes
Per-stream POMDP primitives (A^k, B^k, C^k, D^k, E^k)	Preserves	Each stream is a first-class GNN variable block; dimensionality declarations carry through directly.
Stream count K	Preserves	Encoded as a block-count or indexed family; <code>k_stream_ensemble.gnn.md</code> ships a K=3 instance.
Finite policy spaces Π^k	Preserves	Cardinalities are typed declarations in GNN; the finite-discrete restriction of §3 is honored.
Coupling potential J as a function on $\prod_k \Pi^k$	Preserves (stock GNN)	A joint variable over the product policy space with connection edges; no upstream primitive required (cf. §S8.4).
Sparsity / topology of J (tree / loopy / low-rank)	Preserves with annotation	The coupling-graph topology is read off <code>## Connections</code> ; the K=3 chain ships as a sparse instance.

Framework feature	GNN preservation status	Notes
Deformation parameter λ	Preserves	Scalar parameter with named physical role in the e-coordinate of the entanglement family.
Sophistication weight γ Entanglement decomposition (Theorem 5.1)	Preserves Abstracts away	Scalar parameter weighting K_c in Eq. (5.2) . A graph diagram does not express the additive decomposition $F[q_\lambda] = \sum_k F[q_\lambda^k] + \gamma \lambda \langle K_c \rangle + \log Z_E(\lambda) - \lambda \langle J \rangle + I(q_\lambda)$. Theorem content is post-graph mathematical structure.
Information-geometric structure (e-flatness, m -projection, e-geodesic; §7)	Abstracts away	The manifold structure of the entanglement family is not visible in a graph rendering; the Pythagorean decomposition Eq. (7.4) requires the dual-coordinate framing GNN does not encode.
Spectral structure (Schmidt decomposition, archetypal eigenvectors, TT bond dimensions; §8)	Abstracts away	A graph diagram cannot distinguish two joint posteriors that share a topology but differ in Schmidt-rank profile.
Coupling-tax bound ($O(\lambda^2)$ suboptimality, Theorem 9.1)	Abstracts away	Quantitative bound on heterogeneous-ensemble suboptimality; lives in the analytical layer, not the graph layer.
Lean-checked $\mathbb{R}$ identities	Abstracts away	A GNN spec does not certify a proof; it specifies what is modeled. The shipped GNN→Lean emitter produces a <i>typed contract</i> , not a proof — see §8.7 .
Float-vs- $\mathbb{R}$ numerical residual	Abstracts away	The single honest residual of the manuscript (§21 Threats to validity) is invariant to representation choice.

The ledger is the substantive content of the supplement: **GNN preserves the structural and parametric scaffolding of a multi-stream policy ensemble — and, via the shipped round-trip, reproduces its numerical content — while abstracting away the analytic, geometric, spectral, and proof-theoretic content that the four existing tracks carry.** This is not a defect of GNN; it is a clarification of what a graph-rendered, numerically-round-tripped model description can and cannot encode. A reader who treats the visual view as a commitment beyond the structural-and-numerical layer is over-reading the representation.

Shipped Elaboration: GNN to a Lean Typed-Structure Contract

The GNN-to-Lean **typed-structure emitter** ships in `src/gnn/lean_emit.py`. It consumes a parsed GNN model and emits `gnn/generated/BernoulliToyGnn.lean`: a self-contained, Mathlib-free Lean structure (`PolicyEntanglementK2`) whose fields mirror the manuscript symbol concordance $(E^k, J, \lambda, \gamma)$, instantiated with the values parsed from the GNN source. The emitted file type-checks under the stock Lean v4.29.0 toolchain (verified by `lake env lean`), and `#print axioms` on the emitted declaration reports foundational axioms only — the same hygiene the boundary fragment of [§E](#) enforces.

What this ships, and what it deliberately does not. The emitted **structure** is a *typed contract*: it certifies the structural signature of the GNN-specified model under Lean’s type-checker. It is **not** a proof, and the supplement is explicit about this. Per the discipline of [§E](#), a witness-form theorem is consumed in four enforced steps: (1) the registry names the live Lean declaration; (2) the Lean declaration type-checks without Mathlib; (3) a Python witness or `pymdp` run computes the same payload numerically; (4) the relevant test and dashboard invariant fail if the payload drifts. The GNN-to-Lean emitter automates the *declaration* of step 2’s structure; steps 1, 3, and 4 are unchanged. In particular the analytic content of every witness-form theorem still requires either an analytic proof in `MathlibProofs` (per the close on [Theorem 5.1](#) in [§21](#)) or a numerical witness that exercises the Python / `pymdp` layer.

Explicit non-claim. Parsing is not proving. A GNN spec that parses cleanly, and a Lean **structure** that type-checks, do **not** establish any of the manuscript’s **proved**-status rows in [§S7](#). Promoting a row to **proved** requires a Lean proof at `MathlibProofs`-grade discipline (foundational-only `#print axioms`, no `sorryAx`, a self-designed non-vacuity negative control), independent of how any witness record was declared. The emitter’s output file states this non-claim in its own header so a future reader of the generated Lean cannot mistake the typed contract for a proof.

Shipped Round-Trip: GNN to `pymdp` Harness Configuration

The most immediately useful integration for **practitioners** — a **GNN-to-`pymdp` round-trip** — also ships. `scripts/gnn_to_pymdp.py` (a thin orchestrator over `src/gnn/bridge.py::to_pymdp_config`) consumes a `.gnn` file and

emits the structural configuration a pymdp-style harness needs: stream count, per-stream policy cardinality, per-stream habit priors, the joint coupling tensor, and the scalar parameters — sourced entirely from the GNN declarations.

The shipped round-trip is:

1. **Source:** `gnn/bernoulli_toy.gnn.md` (and the `K=3` `gnn/k_stream_ensemble.gnn.md`) describing the ensemble with per-stream POMDP primitives, the coupling potential J , γ , and the operating point.
2. **Generator:** `scripts/gnn_to_pymdp.py` consumes the `.gnn` source and emits a configuration dict structurally equivalent to the relevant fields of `src/simulation/hyperparameters.py`.
3. **Run:** the GNN round-trip stage (`scripts/simulate_gnn.py`) reconstructs the entangled posterior and the mutual-information curve from the GNN source and emits `output/data/gnn_bernoulli_roundtrip.json`.
4. **Validate:** `tests/test_gnn_round_trip.py` asserts floating-point-tolerance identity between the GNN-reconstructed curve and the canonical closed form, with a non-vacuity negative control; `tests/test_simulate_gnn.py` asserts the sidecar is byte-identical on re-run.

Reproducibility-contract compatibility. The reproducibility contract of §16 requires every numerical claim in the manuscript to flow from a real run via the `[[VAR:...]]` token registry. The GNN round-trip is compatible: the `gnn_roundtrip_max_residual`, `gnn_negative_control_max_residual`, and `gnn_round_trip_lambda_points` variables populate `manuscript_variables.json` from the real sidecar and are range-gated. **Honest scope of what ships vs. what does not:** the `K=2` mutual-information round-trip ships and is the source of the GNN `[[VAR:...]]` tokens. Regenerating the *full* multi-`K` pymdp free-energy bundle (the per-stream VFE/EFE rollout of §14) from a `.gnn` source — rather than the `K=2` MI curve — is the natural next binding; the configuration generator ships, the full-bundle regeneration is the remaining practitioner-facing work and is named as such in §S8.9 (OQ-G3).

What round-trip closure enables. A downstream contributor can now specify a new policy-entanglement experiment as a `.gnn` file (a stable, declarative, version-controlled surface), obtain its pymdp configuration via `gnn_to_pymdp.py`, and validate the mutual-information structure against the closed form via the shipped round-trip — without re-implementing the inference layer.

Downstream Use, Open Directions, and Open Questions

Downstream use. A third group writing a new multi-stream policy-coupling experiment can now publish a `.gnn` specification alongside their paper; other groups consuming the spec recover the same coupling configuration and the same mutual-information round-trip via the shipped `src/gnn/` tooling, without re-implementing the inference layer. The leverage is no longer purely prospective: the tooling, the parser, the round-trip, and the reproducibility receipts exist and run in CI.

Integration with the §21 open directions. The **GNN bridge to the framework** ships. Concretely, five deliverables are in place: (a) the $(A, B, C, D, E, J, \lambda)$ mapping of §S8.4; (b) the cross-stream coupling, expressible in stock GNN (no upstream extension on the critical path); (c) the `K=2` Bernoulli round-trip of §S8.5; (d) the GNN-to-Lean typed-contract emitter of §S8.7; and (e) the GNN-to-pymdp configuration round-trip of §S8.8. The other open directions of §21 — the Mathlib4 analytic discharge beyond the verified kernel, cross-repository integration, and community collaboration — remain orthogonal to the GNN bridge.

Open questions (the genuine design surface that remains, now that the core bridge ships):

- **OQ-G1 (Cross-stream block ergonomics).** *Resolved at the level of expressibility, open at the level of ergonomics.* The cross-stream coupling is expressible in stock GNN v1.1 as a joint variable plus edges (shipped). Whether a dedicated first-class `CouplingBlock` primitive — contributed upstream to [Institute, 2023] — would improve authoring ergonomics for large K or dense topologies is an open *design* question, not a blocker for this manuscript’s fifth track.
- **OQ-G2 (Heterogeneous-stream Triple Play).** When a K -stream ensemble mixes VFE-only and EFE-planning streams (see §9), the visual view risks suggesting structural homogeneity where the framework treats the streams differently. What annotation discipline avoids the misreading? A per-stream typographic distinction (filled vs. open node markers, dashed vs. solid edges) is the candidate; the discipline is not yet specified.
- **OQ-G3 (Full-bundle pymdp regeneration).** The shipped round-trip reconstructs the `K=2` *mutual-information* curve. Regenerating the full multi-`K` pymdp free-energy bundle (per-stream VFE/EFE, long-horizon rollout) from a `.gnn` source — and confirming bundle-level identity against the canonical `pymdp_summary.json` — is the remaining practitioner-facing binding.

These questions are not blockers for the bridge; they are the design surface beyond the now-shipped core.

Bridge style and external-link discipline. In the style precedent of §E (§E), which bridges to the `fep_lean` catalog via a single namespaced subsection and a single citation, this supplement bridges to GNN via a single

namespaced supplement and two citations ([Smékal and Friedman, 2023], [Institute, 2023]). No further external references — to the broader Active Inference Institute catalog, to COGSEC, to companion projects — are taken in §8. The bridge is minimal by design.

This supplement ships the fifth track of the framework. The four-track *proof* integration of §1 remains the live contract for analytic content; GNN is added as the fifth, structural-and-numerical track — a project-owned parser, a shipped internal-consistency round-trip, a Lean typed-contract emitter, and a GNN-sourced manuscript variable — with the genuine residuals (a first-class upstream coupling primitive; full-bundle pymdp regeneration; GNN-to-Lean *proving*) named honestly as forward work rather than concealed.

Bibliography

This bibliography is *auto-generated* from `manuscript/refs/citations.yaml` by `scripts/inject_manuscript_variables.py` — every Pandoc-style citation in the body (a citekey enclosed in square brackets prefixed with @) resolves to one entry below, and every entry below is grouped by topic in the order specified by the `topic_order`: list of the YAML source.

When you add a body citation:

1. Add an entry to `refs/citations.yaml` with the citekey, full author / year / title / venue, and a `topic`:
2. Reference it from any body section as a Pandoc-style key — a lower-case `lastname-yyyy` slug enclosed in square brackets and prefixed with the at-sign; use a semicolon to separate multi-references inside a single bracket.
3. Re-run `uv run python scripts/inject_manuscript_variables.py` and confirm `output/manuscript/99_bibliography.md` contains the expanded entry.

Active inference and the free-energy principle

- Buckley, C. L., Kim, C. S., McGregor, S., Seth, A. K. (2017) The free energy principle for action and perception: A mathematical review. *Journal of Mathematical Psychology* **81**, 55–79. <https://www.sciencedirect.com/science/article/pii/S0022249617300962> doi:10.1016/j.jmp.2017.09.004 *Accessible mathematical review of the free-energy principle used to ground the introductory VFE claim without relying only on process-theory exposition.*
- Champion, T., Bowman, H., Marković, D., Grześ, M. (2024) Reframing the Expected Free Energy: Four Formulations and a Unification. *arXiv preprint*, arXiv:2402.14460. <https://arxiv.org/abs/2402.14460> doi:10.48550/arXiv.2402.14460
- Da Costa, L., Parr, T., Sajid, N., Veselic, S., Neacsu, V., Friston, K. (2020) Active inference on discrete state-spaces: A synthesis. *Journal of Mathematical Psychology* **99**, 102447. <https://www.sciencedirect.com/science/article/pii/S0022249620300857> doi:10.1016/j.jmp.2020.102447 *arXiv:2001.07203.*
- Da Costa, L., Tenka, S., Zhao, D., Sajid, N. (2024) Active Inference as a Model of Agency. *arXiv*. <https://arxiv.org/abs/2401.12917> doi:10.48550/arXiv.2401.12917 *Accepted at the RLDM 2022 workshop ‘RL as a model of agency’; cited for the active-inference-as-agency unification framing, not as evidence for policy entanglement itself.*
- de Vries, B., Nuijten, T., van de Laar, T., Kouw, W., Adamiat, A., Nisslbeck, A., Lukashchuk, V., et al. (2025) Expected Free Energy-based Planning as Variational Inference. *arXiv preprint*, arXiv:2504.14898. <https://arxiv.org/abs/2504.14898> doi:10.48550/arXiv.2504.14898
- de Vries, B., Friston, K. J. (2017) A Factor Graph Description of Deep Temporal Active Inference. *Frontiers in Computational Neuroscience* **11**, 95. <https://www.frontiersin.org/articles/10.3389/fncom.2017.00095/full> doi:10.3389/fncom.2017.00095 *Primary factor-graph / Forney-style graph source for deep temporal active inference; cited only for message-passing implementability, not as a theorem-equivalence claim for policy entanglement.*
- Friston, K., Kilner, J., Harrison, L. (2006) A free energy principle for the brain. *Journal of Physiology-Paris* **100**(1–3), 70–87. <https://www.sciencedirect.com/science/article/pii/S092842570600060X> doi:10.1016/j.jphysparis.2006.10.001 *Early statement of variational free energy as a unifying principle for perception, action, and learning.*
- Friston, K. (2010) The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience* **11**(2), 127–138. <https://www.nature.com/articles/nrn2787> doi:10.1038/nrn2787
- Friston, K., Schwartenbeck, P., FitzGerald, T., Moutoussis, M., Behrens, T., Dolan, R. J. (2014) The anatomy of choice: dopamine and decision-making. *Philosophical Transactions of the Royal Society B* **369**(1655), 20130481. <https://royalsocietypublishing.org/doi/10.1098/rstb.2013.0481> doi:10.1098/rstb.2013.0481

- Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., Pezzulo, G. (2017) Active Inference: A Process Theory. *Neural Computation* **29**(1), 1–49. <https://direct.mit.edu/neco/article/29/1/1> doi:10.1162/NECO__a_00912
- Friston, K. J., Rosch, R., Parr, T., Price, C., Bowman, H. (2017) Deep temporal models and active inference. *Neuroscience & Biobehavioral Reviews* **77**, 388–402. <https://doi.org/10.1016/j.neubiorev.2017.04.009> doi:10.1016/j.neubiorev.2017.04.009 *Primary deep-temporal active-inference source; cited for hierarchical temporal-model and long-horizon framing, not as evidence that the present finite-policy model is a full process theory.*
- Friston, K. J. (2018) Active Inference and Cognitive Consistency. *Psychological Inquiry* **29**(2), 67–73. <https://www.tandfonline.com/doi/full/10.1080/1047840X.2018.1480693> doi:10.1080/1047840X.2018.1480693 *Commentary linking expected-free-energy epistemic/pragmatic terms to cognitive-consistency constructs; cited as interpretive context, not as a reduction of social-psychological theory to active inference.*
- Friston, K. (2019) A free energy principle for a particular physics. *arXiv preprint*, arXiv:1906.10184. <https://arxiv.org/abs/1906.10184> doi:10.48550/arXiv.1906.10184 *Preprint only as of 2026-05; never received a journal publication.*
- Friston, K., Da Costa, L., Hafner, D., Hesp, C., Parr, T. (2021) Sophisticated Inference. *Neural Computation* **33**(3), 713–763. <https://direct.mit.edu/neco/article/33/3/713> doi:10.1162/neco_a_01351 arXiv:2006.04120. *Introduces recursive expected free energy for deep counterfactual policy tree search.*
- Friston, K. J., Heins, C., Verbelen, T., Da Costa, L., Salvatori, T., Marković, D., Tschantz, A., Koudahl, M., Buckley, C., Parr, T. (2025) From pixels to planning: scale-free active inference. *Frontiers in Network Physiology* **5**, 1521963. <https://www.frontiersin.org/journals/network-physiology/articles/10.3389/fnetp.2025.1521963/full> doi:10.3389/fnetp.2025.1521963 *First circulated as a 2024 preprint (arXiv:2407.20292); final venue Frontiers in Network Physiology, June 2025. The published Frontiers form is the canonical citation; the friston-rgm-2025 alias is intentionally NOT registered to avoid duplication — cite as [@friston-2024]. Slug retained as friston-2024 for back-compatibility.*
- Friston, K. J., Parr, T., de Vries, B. (2017) The graphical brain: Belief propagation and active inference. *Network Neuroscience* **1**(4), 381–414. https://doi.org/10.1162/NETN_a_00018 doi:10.1162/NETN_a_00018 *Graphical-brain source for belief propagation, Forney-style graphical modeling, and active-inference message-passing context; not evidence that lambda-coupled policy posteriors are standard in AIF.*
- Friston, K. J., Trujillo-Barreto, N., Daunizeau, J. (2008) DEM: A variational treatment of dynamic systems. *NeuroImage* **41**(3), 849–885. <https://www.sciencedirect.com/science/article/pii/S1053811908001894> doi:10.1016/j.neuroimage.2008.02.054 *Primary dynamic expectation maximization reference behind the SPM-DEM lineage discussed in the introduction.*
- Hafner, D., Pasukonis, J., Ba, J., Lillicrap, T. (2023) Mastering Diverse Domains through World Models. *arXiv preprint*, arXiv:2301.04104. <https://arxiv.org/abs/2301.04104> doi:10.48550/arXiv.2301.04104 *DreamerV3 — world-model RL agent that achieves cross-domain mastery with a single hyperparameter setting; widely cited as the canonical contemporary world-model architecture against which active-inference agents are benchmarked.*
- Heins, C., Millidge, B., Demekas, D., Klein, B., Friston, K., Couzin, I. D., Tschantz, A. (2022) pymdp: A Python library for active inference in discrete state spaces. *Journal of Open Source Software* **7**(73), 4098. <https://joss.theoj.org/papers/10.21105/joss.04098> doi:10.21105/joss.04098 v1.0.1 *JAX-based release is the simulation backend used by the empirical suite and src/simulation/; arXiv:2201.03904.*
- Hodson, R., Mehta, M., Smith, R. (2024) The empirical status of predictive coding and active inference. *Neuroscience & Biobehavioral Reviews* **157**, 105473. <https://pubmed.ncbi.nlm.nih.gov/38030100/> doi:10.1016/j.neubiorev.2023.105473 *Empirical-status review used to calibrate FEP / active-inference claims as normative and computational unless a specific comparative empirical test is supplied.*
- Johansson, R. S., Westling, G., Bäckström, A., Flanagan, J. R. (2001) Eye-hand coordination in object manipulation. *Journal of Neuroscience* **21**(17), 6917–6932. <https://www.jneurosci.org/content/21/17/6917> doi:10.1523/JNEUROSCI.21-17-06917.2001 *Primary empirical eye-hand coordination source used to ground the reach/saccade example in the motivation.*
- Kaplan, R., Friston, K. J. (2018) Planning and navigation as active inference. *Biological Cybernetics* **112**, 323–343. <https://link.springer.com/article/10.1007/s00422-018-0753-2> doi:10.1007/s00422-018-0753-2 *Navigation and planning example for EFE-based policy selection, risk / ambiguity terms, and epistemic exploration under active inference.*
- Land, M. F., Hayhoe, M. (2001) In what ways do eye movements contribute to everyday activities? *Vision Research* **41**(25–26), 3559–3565. [https://doi.org/10.1016/S0042-6989\(01\)00102-X](https://doi.org/10.1016/S0042-6989(01)00102-X) doi:10.1016/S0042-6989(01)00102-X *Natural-task eye-movement review showing that gaze commonly arrives at task-relevant objects before manipulation, grounding the cross-stream reach/saccade example.*
- Lanillos, P., Meo, C., Pezzato, C., Meera, A. A., Baioumy, M., Ohata, W., Tschantz, A., Millidge, B., Wisse, M., Buckley, C. L., Tani, J. (2021) Active inference in robotics and artificial agents: Survey and challenges.

arXiv preprint, arXiv:2112.01871. <https://arxiv.org/abs/2112.01871> doi:10.48550/arXiv.2112.01871

- Limanowski, J., Adams, R. A., Kilner, J., Parr, T. (2024) The Many Roles of Precision in Action. *Entropy* **26**(9), 790. <https://www.mdpi.com/1099-4300/26/9/790> doi:10.3390/e26090790 *Review of precision roles in action, attention, habit, and self-other distinction; cited to keep lambda-as-coupling-precision language distinct from neural precision claims.*
- Millidge, B., Tschantz, A., Buckley, C. L. (2021) Whence the Expected Free Energy? *Neural Computation* **33**(2), 447-482. <https://direct.mit.edu/neco/article/33/2/447/95645/Whence-the-Expected-Free-Energy> doi:10.1162/neco_a_01354 *Critical analysis of EFE derivations and alternatives; cited to keep expected-free-energy unification claims bounded.*
- Nuijten, T., Lukashchuk, V. (2025) Active Inference is a Subtype of Variational Inference. *arXiv preprint*, arXiv:2511.18955. <https://arxiv.org/abs/2511.18955> doi:10.48550/arXiv.2511.18955
- Parr, T., Pezzulo, G., Friston, K. J. (2022) Active Inference: The Free Energy Principle in Mind, Brain, and Behavior. *MIT Press*. <https://direct.mit.edu/books/oa-monograph/5299/Active-InferenceThe-Free-Energy-Principle-in-Mind> doi:10.7551/mitpress/12441.001.0001 *Book-length synthesis of FEP, expected free energy, active inference process theory, and applications.*
- Parr, T., Friston, K. J. (2017) Uncertainty, epistemics and active inference. *Journal of The Royal Society Interface* **14**(136), 20170376. <https://royalsocietypublishing.org/doi/10.1098/rsif.2017.0376> doi:10.1098/rsif.2017.0376 *Primary source for epistemic value and uncertainty reduction in active-inference policy selection.*
- Parr, T., Friston, K. J. (2017) Working memory, attention, and salience in active inference. *Scientific Reports* **7**, 14678. <https://www.nature.com/articles/s41598-017-15249-0> doi:10.1038/s41598-017-15249-0 *Process-theory account linking working memory, attention, salience, precision, and policy evaluation under active inference.*
- Parr, T., Friston, K. J. (2019) Generalised free energy and active inference. *Biological Cybernetics* **113**(5–6), 495–513. <https://link.springer.com/article/10.1007/s00422-019-00805-w> doi:10.1007/s00422-019-00805-w *Clarifies generalized free energy and self-evidencing language used in the discussion without treating the present finite-policy model as a complete biological process theory.*
- Parr, T., Marković, D., Kiebel, S. J., Friston, K. J. (2019) Neuronal message passing using Mean-field, Bethe and Marginal approximations. *Scientific Reports* **9**, 1889. <https://www.nature.com/articles/s41598-018-38246-3> doi:10.1038/s41598-018-38246-3 *Primary source distinguishing mean-field, Bethe, and marginal message-passing approximations; cited for approximation-family context, not as a direct proof of policy-coupling neural implementation.*
- Pezzulo, G., Rigoli, F., Friston, K. J. (2018) Hierarchical Active Inference: A Theory of Motivated Control. *Trends in Cognitive Sciences* **22**(4), 294–306. [https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613\(18\)30022-6](https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613(18)30022-6) doi:10.1016/j.tics.2018.01.009
- Pezzulo, G., Parr, T., Friston, K. J. (2024) Active inference as a theory of sentient behavior. *Biological Psychology* **186**, 108741. <https://www.sciencedirect.com/science/article/pii/S0301051123002612> doi:10.1016/j.biopsycho.2023.108741 *Recent synthesis used to distinguish active inference as process theory from the broader free-energy-principle framing.*
- Sajid, N., Ball, P. J., Parr, T., Friston, K. J. (2021) Active Inference: Demystified and Compared. *Neural Computation* **33**(3), 674–712. <https://direct.mit.edu/neco/article/33/3/674> doi:10.1162/neco_a_01357
- Schwartenbeck, P., FitzGerald, T. H. B., Mathys, C., Dolan, R., Friston, K. (2015) The dopaminergic midbrain encodes the expected certainty about desired outcomes. *Cerebral Cortex* **25**(10), 3434–3445. <https://academic.oup.com/cercor/article/25/10/3434/385607> doi:10.1093/cercor/bhu159
- Smékal, J., Friedman, D. A. (2023) Generalized Notation Notation for Active Inference Models. *Active Inference Journal*. Zenodo. <https://zenodo.org/records/7803328> doi:10.5281/zenodo.7803328 *Primary publication of Generalized Notation Notation (GNN), introducing the Triple Play of linguistic, visual, and executable cognitive-model representations that complements the Active Inference Ontology. Cited from the GNN extension supplement as the canonical reference for the shipped fifth (structural-and-numerical) representation of the framework.*
- Smith, R., Friston, K. J., Whyte, C. J. (2022) A step-by-step tutorial on active inference and its application to empirical data. *Journal of Mathematical Psychology* **107**, 102632. <https://www.sciencedirect.com/science/article/pii/S0022249621000973> doi:10.1016/j.jmp.2021.102632
- Smithe, T. S. C. (2024) Structured Active Inference. *arXiv preprint*, arXiv:2406.07577. <https://arxiv.org/abs/2406.07577> doi:10.48550/arXiv.2406.07577 *Categorical-systems-theory formulation of AIF with structured policies amenable to formal verification.*
- Tschantz, A., Seth, A. K., Buckley, C. L. (2020) Learning action-oriented models through active inference. *PLOS Computational Biology* **16**(4), e1007805. <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1007805> doi:10.1371/journal.pcbi.1007805 *Active-inference account of parsimonious action-oriented model learning; useful contrast for this manuscript’s structured policy-coupling family.*

- van de Laar, T. W., de Vries, B. (2019) Simulating Active Inference Processes by Message Passing. *Frontiers in Robotics and AI* **6**, 20. <https://www.frontiersin.org/articles/10.3389/frobt.2019.00020> doi:10.3389/frobt.2019.00020 *Factor-graph forward-backward implementation of deep temporal active inference (slug retained for backwards compatibility — primary reference is the 2019 Frontiers in Robotics and AI* paper).**
- Yedidia, J. S., Freeman, W. T., Weiss, Y. (2005) Constructing Free-Energy Approximations and Generalized Belief Propagation Algorithms. *IEEE Transactions on Information Theory* **51**(7), 2282–2312. <https://doi.org/10.1109/TIT.2005.850085> doi:10.1109/TIT.2005.850085 *Canonical Bethe / Kikuchi free-energy and generalized-belief-propagation source; cited for graphical-variational approximation structure, not for active-inference policy semantics.*

Branching-time and structured AIF

- Champion, T., Grześ, M., Bowman, H. (2022) Branching Time Active Inference with Bayesian Filtering. *Neural Computation* **34**(10), 2132–2144. <https://direct.mit.edu/neco/article-abstract/34/10/2132> doi:10.1162/neco_a_01529 *Uses Bayesian filtering on the expanded policy tree rather than vanilla MCTS rollouts; in the wider BTAI lineage (Fountas et al. 2020 DAI / MCTS for active inference).*
- Fountas, Z., Sajid, N., Mediano, P. A. M., Friston, K. (2020) Deep active inference agents using Monte-Carlo methods. *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. <https://proceedings.neurips.cc/paper/2020/hash/865dfbde8a344b44095495f3591f7407-Abstract.html> doi:10.48550/arXiv.2006.04176 *arXiv:2006.04176. Canonical reference for deep / MCTS-style branching-time active inference (DAI).*

Information geometry

- Amari, S. (1985) Differential-Geometrical Methods in Statistics. *Lecture Notes in Statistics, Vol. 28 (Springer)*. <https://link.springer.com/book/10.1007/978-1-4612-5056-2> doi:10.1007/978-1-4612-5056-2 *Founding monograph of information geometry; introduced the dually-flat alpha-connection family and the e- / m-geodesic structure used in the framework-connections section.*
- Amari, S. (2016) Information Geometry and Its Applications. *Springer, Applied Mathematical Sciences, Vol. 194*. <https://link.springer.com/book/10.1007/978-4-431-55978-8> doi:10.1007/978-4-431-55978-8
- Amari, S., Nagaoka, H. (2000) Methods of Information Geometry. *AMS Translations of Mathematical Monographs, Vol. 191 (American Mathematical Society / Oxford University Press)*. <https://bookstore.ams.org/mmono-191/> *Translated by Daishi Harada from the 1993 Japanese original. ISBN 978-0-8218-4302-4.*
- Ay, N., Jost, J., Lê, H. V., Schwachhöfer, L. (2017) Information Geometry. *Ergebnisse der Mathematik und ihrer Grenzgebiete (3. Folge), Vol. 64 (Springer)*. <https://link.springer.com/book/10.1007/978-3-319-56478-4> doi:10.1007/978-3-319-56478-4 *ISBN 978-3-319-56477-7. Comprehensive monograph treatment covering the parametric and non-parametric geometric foundations used in the information-geometric formalism.*
- Bakry, D., Émery, M. (1985) Diffusions hypercontractives. **, 177–206. <https://doi.org/10.1007/BFb0075144> doi:10.1007/BFb0075144 *Bakry–Émery curvature criterion for log-Sobolev / variance bounds on exponential families.*
- Brascamp, H. J., Lieb, E. H. (1976) On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation. **, **22**, 366–389. [https://doi.org/10.1016/0022-1236\(76\)90001-5](https://doi.org/10.1016/0022-1236(76)90001-5) doi:10.1016/0022-1236(76)90001-5 *Variance Brascamp–Lieb inequality for log-concave measures; cited in Q1 dispersion-comparison conjecture.*
- Naudts, J. (2011) Generalised Thermostatistics. *Springer London*. <https://link.springer.com/book/10.1007/978-0-85729-355-8> doi:10.1007/978-0-85729-355-8 *ISBN 978-0-85729-354-1 (print). Covers escort distributions, q-deformed exponential families.*
- Nielsen, F. (2020) An Elementary Introduction to Information Geometry. *Entropy* **22**(10), 1100. <https://www.mdpi.com/1099-4300/22/10/1100> doi:10.3390/e22101100

Information theory and multi-information

- McGill, W. J. (1954) Multivariate Information Transmission. *Psychometrika* **19**(2), 97–116. <https://link.springer.com/article/10.1007/BF02289159> doi:10.1007/BF02289159 *Original definition of total correlation / multi-information, the correlation surcharge in the entanglement-decomposition identity.*
- Watanabe, S. (1960) Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development* **4**(1), 66–82. <https://ieeexplore.ieee.org/document/5392532> doi:10.1147/rd.41.0066 *Independent rediscovery of McGill’s total-correlation construction, under the alternative name multivariate correlation; widely cited as the second canonical reference for the total-correlation quantity.*

Variational and copula inference

- Blei, D. M., Kucukelbir, A., McAuliffe, J. D. (2017) Variational Inference: A Review for Statisticians. *Journal of the American Statistical Association* **112**(518), 859–877. <https://doi.org/10.1080/01621459.2017.1285773> doi:10.1080/01621459.2017.1285773 *Modern scholarly review citation for mean-field and structured variational inference.*
- Fu, Y., Smith, M. S., Panagiotelis, A. (2025) Vector Copula Variational Inference and Dependent Block Posterior Approximations. *arXiv preprint*, arXiv:2503.01072. <https://arxiv.org/abs/2503.01072> doi:10.48550/arXiv.2503.01072
- Han, S., Liao, X., Dunson, D. B., Carin, L. (2016) Variational Gaussian Copula Inference. *AISTATS 2016, PMLR 51*, 829–838. <https://proceedings.mlr.press/v51/han16.html>
- Hinton, G. E. (2002) Training products of experts by minimizing contrastive divergence. *Neural Computation* **14**(8), 1771–1800. <https://direct.mit.edu/neco/article-abstract/14/8/1771/> doi:10.1162/089976602760128018
- Hoffman, M. D., Blei, D. M. (2015) Structured Stochastic Variational Inference. *AISTATS 2015, PMLR 38*, 361–369. <http://proceedings.mlr.press/v38/hoffman15.html> arXiv:1404.4114.
- Li, Y., Turner, R. E. (2016) Rényi Divergence Variational Inference. *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*. <https://proceedings.neurips.cc/paper/2016/hash/7750ca3559e5b8e1f44210283368fc16-Abstract.html> doi:10.48550/arXiv.1602.02311 arXiv:1602.02311. *Generalizes ELBO-style variational inference to the Rényi alpha-divergence family; cited for structured-VI / alpha-divergence comparisons with the active-inference free-energy functional.*
- Nelsen, R. B. (2006) An Introduction to Copulas. *Springer Series in Statistics, 2nd ed.* <https://link.springer.com/book/10.1007/0-387-28678-0> doi:10.1007/0-387-28678-0 *Foundational monograph for copula definitions and dependence modeling; cited to distinguish true copula densities from informal dependence-factor analogies.*
- Saul, L. K., Jordan, M. I. (1995) Exploiting Tractable Substructures in Intractable Networks. *Advances in Neural Information Processing Systems 8 (NIPS 1995)*. <https://papers.nips.cc/paper/1155-exploiting-tractable-substructures-in-intractable-networks> *Canonical structured mean-field source: exact inference is retained inside tractable substructures while only the remaining interactions are approximated.*
- Tran, D., Blei, D. M., Airoldi, E. M. (2015) Copula variational inference. *Advances in Neural Information Processing Systems 28 (NeurIPS 2015)*, 3564–3572. <https://proceedings.neurips.cc/paper/2015/hash/e4dd5528f7596dcd871aa55cfcc53c-Abstract.html> arXiv:1506.03159.
- Wainwright, M. J., Jordan, M. I. (2008) Graphical Models, Exponential Families, and Variational Inference. *Foundations and Trends in Machine Learning* **1**(1–2), 1–305. <https://www.nowpublishers.com/article/Details/MAL-001> doi:10.1561/22000000001 *Canonical graphical-model variational-inference reference used to anchor the mean-field / structured-approximation comparison.*

Variational inference robustness and Lipschitz bounds

- Barrett, T. D., Camuto, A., Williamson, C. R., Roberts, S. J. (2022) Certifiably Robust Variational Autoencoders through Lipschitz Constraints. *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 1–9. <https://proceedings.mlr.press/v151/barrett22a.html> *Actionable KL-drift bounds under encoder/decoder Lipschitz constraints; cited for adversarial VI robustness context in Q11.*

Tensor networks and quantum-inspired ML

- Biamonte, J., Bergholm, V. (2017) Tensor Networks in a Nutshell. *arXiv*. <https://arxiv.org/abs/1708.00006> doi:10.48550/arXiv.1708.00006 *Tutorial tensor-network reference used to scope Schmidt-rank / tensor-train language as probability-tensor linear algebra, not a quantum-physical claim.*
- Cichocki, A., Lee, N., Oseledets, I. V., Phan, A.-H., Zhao, Q., Mandic, D. P. (2016) Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 1 Low-Rank Tensor Decompositions. *Foundations and Trends in Machine Learning* **9**(4–5), 249–429. <https://doi.org/10.1561/22000000059> doi:10.1561/22000000059 *Machine-learning tensor-network survey; cited for low-rank tensor formats and optimization context, not as an active-inference result.*
- Eisert, J., Cramer, M., Plenio, M. B. (2010) Colloquium: Area laws for the entanglement entropy. *Reviews of Modern Physics* **82**(1), 277–306. <https://link.aps.org/doi/10.1103/RevModPhys.82.277> doi:10.1103/RevModPhys.82.277
- Glasser, I., Pancotti, N., Cirac, J. I. (2020) From Probabilistic Graphical Models to Generalized Tensor Networks for Supervised Learning. *IEEE Access* **8**, 68169–68182. <https://ieeexplore.ieee.org/document/9058650/> doi:10.1109/ACCESS.2020.2986279 arXiv:1806.05964 (2018 preprint); *IEEE Access publication 2020.*

- Han, Z.-Y., Wang, J., Fan, H., Wang, L., Zhang, P. (2018) Unsupervised Generative Modeling Using Matrix Product States. *Physical Review X* **8**(3), 031012. <https://journals.aps.org/prx/abstract/10.1103/PhysRevX.8.031012> doi:10.1103/PhysRevX.8.031012
- Orus, R. (2014) A practical introduction to tensor networks: Matrix product states and projected entangled pair states. *Annals of Physics* **349**, 117–158. <https://www.sciencedirect.com/science/article/pii/S0003491614001596> doi:10.1016/j.aop.2014.06.013 *Tutorial tensor-network reference used to scope matrix-product-state and tensor-train language as linear algebra on probability tensors.*
- Oseledets, I. V. (2011) Tensor-Train Decomposition. *SIAM Journal on Scientific Computing* **33**(5), 2295–2317. <https://epubs.siam.org/doi/10.1137/090752286> doi:10.1137/090752286 *Foundational paper introducing the TT-format; underlies the linear-in-N policy-tensor compression used in the tensor-network analysis.*
- Schollwöck, U. (2011) The density-matrix renormalization group in the age of matrix product states. *Annals of Physics* **326**(1), 96–192. <https://www.sciencedirect.com/science/article/pii/S0003491610001752> doi:10.1016/j.aop.2010.09.012 *arXiv:1008.3477. Canonical MPS/DMRG review; cited for the algorithmic backbone behind policy-tensor compression and entanglement-entropy bookkeeping.*
- Verstraete, F., Murg, V., Cirac, J. I. (2008) Matrix product states, projected entangled pair states, and variational renormalization group methods for quantum spin systems. *Advances in Physics* **57**(2), 143–224. <https://www.tandfonline.com/doi/abs/10.1080/14789940801912366> doi:10.1080/14789940801912366

KL / path-integral control duality

- Kappen, H. J. (2005) Path integrals and symmetry breaking for optimal control theory. *Journal of Statistical Mechanics: Theory and Experiment* **2005**, P11011. <https://iopscience.iop.org/article/10.1088/1742-5468/2005/11/P11011> doi:10.1088/1742-5468/2005/11/P11011
- Rawlik, K., Toussaint, M., Vijayakumar, S. (2013) On Stochastic Optimal Control and Reinforcement Learning by Approximate Inference (Extended Abstract). *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence (IJCAI 2013)*, 3052–3056. <https://www.ijcai.org/Proceedings/13/Papers/455.pdf> *Extended abstract of the approximate-inference control formulation; cited for the KL/control-as-inference lineage rather than as an active-inference result.*
- Theodorou, E. A., Todorov, E. (2012) Relative entropy and free energy dualities: Connections to path integral and KL control. *Proc. 2012 IEEE 51st Conference on Decision and Control (CDC)*, 1466–1473. <https://ieeexplore.ieee.org/document/6426381/> doi:10.1109/CDC.2012.6426381
- Todorov, E. (2006) Linearly-solvable Markov decision problems. *Advances in Neural Information Processing Systems 19 (NIPS 2006)*, 1369–1376. <https://papers.nips.cc/paper/3002-linearly-solvable-markov-decision-problems> *Primary linearly-solvable MDP reference for KL-control structure and log-partition value transformations.*

Control / RL as probabilistic inference

- Botvinick, M., Toussaint, M. (2012) Planning as inference. *Trends in Cognitive Sciences* **16**(10), 485–488. [https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613\(12\)00184-3](https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613(12)00184-3) doi:10.1016/j.tics.2012.08.006
- Levine, S. (2018) Reinforcement Learning and Control as Probabilistic Inference: Tutorial and Review. *arXiv preprint*, arXiv:1805.00909. <https://arxiv.org/abs/1805.00909> doi:10.48550/arXiv.1805.00909 *Canonical tutorial on the control-as-inference / maximum-entropy RL viewpoint cited in the comparison with active inference.*
- Toussaint, M. (2009) Robot trajectory optimization using approximate inference. *Proc. 26th Annual International Conference on Machine Learning (ICML 2009)*, 1049–1056. <https://dl.acm.org/doi/10.1145/1553374.1553508> doi:10.1145/1553374.1553508

Hierarchical RL / options

- Bacon, P.-L., Harb, J., Precup, D. (2017) The Option-Critic Architecture. *Proceedings of the AAAI Conference on Artificial Intelligence* **31**(1). <https://ojs.aaai.org/index.php/AAAI/article/view/10916> doi:10.1609/aaai.v31i1.10916 *Modern options reference used to scope the manuscript’s option-like embedding as a parametric policy-matching construction, not a proof that the lambda deformation subsumes all option-learning algorithms.*
- Barto, A. G., Mahadevan, S. (2003) Recent Advances in Hierarchical Reinforcement Learning. *Discrete Event Dynamic Systems* **13**, 341–379. <https://link.springer.com/article/10.1023/A:1025696116075> doi:10.1023/A:1025696116075 *Review reference for hierarchical RL and temporal abstraction; cited alongside options to keep the control/RL comparison anchored in the RL literature.*

- Doll, B. B., Duncan, K. D., Simon, D. A., Shohamy, D., Daw, N. D. (2015) Model-based choices involve prospective neural activity. *Nature Neuroscience* **18**(5), 767–772. <https://www.nature.com/articles/nn.3981> doi:10.1038/nn.3981
- Sutton, R. S., Precup, D., Singh, S. (1999) Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence* **112**(1–2), 181–211. <https://www.sciencedirect.com/science/article/pii/S0004370299000521> doi:10.1016/S0004-3702(99)00052-1

Markov blankets and particular physics

- Aguilera, M., Millidge, B., Tschantz, A., Buckley, C. L. (2022) How particular is the physics of the free energy principle? *Physics of Life Reviews* **40**, 24–50. <https://www.sciencedirect.com/science/article/pii/S1571064521000749> doi:10.1016/j.plrev.2021.11.001 *Critical-review treatment of Markov-blanket / particular-physics derivations of the free-energy principle. First circulated as a 2021 preprint; final publication in Physics of Life Reviews vol. 40 (2022); slug retained as aguilera-2021 for back-compatibility.*
- Da Costa, L., Friston, K., Heins, C., Pavliotis, G. A. (2021) Bayesian mechanics for stationary processes. *Proceedings of the Royal Society A* **477**(2256), 20210518. <https://royalsocietypublishing.org/doi/10.1098/rspa.2021.0518> doi:10.1098/rspa.2021.0518 *Bayesian-mechanics reference for Markov blankets in stationary processes; cited to keep the manuscript's policy-space blanket language explicitly analogical.*
- Kirchhoff, M. D., Parr, T., Palacios, E., Friston, K., Kiverstein, J. (2018) The Markov blankets of life: autonomy, active inference and the free energy principle. *Journal of The Royal Society Interface* **15**(138), 20170792. <https://royalsocietypublishing.org/doi/10.1098/rsif.2017.0792> doi:10.1098/rsif.2017.0792 *Markov-blanket account of autonomy used here to keep system-boundary language separate from the manuscript's intra-agent policy-coupling claim.*
- Menary, R., Gillett, A. J. (2022) Markov blankets do not demarcate the boundaries of the mind. *Behavioral and Brain Sciences* **45**, e201. <https://pubmed.ncbi.nlm.nih.gov/36172779/> doi:10.1017/S0140525X22000371 *Critical extended-mind commentary; cited to prevent Markov-blanket language from being treated as an automatic cognitive-boundary criterion.*
- Raja, V., Valluri, D., Baggs, E., Chemero, A., Anderson, M. L. (2021) The Markov blanket trick: On the scope of the free energy principle and active inference. *Physics of Life Reviews*. <https://www.sciencedirect.com/science/article/pii/S1571064521000634> doi:10.1016/j.plrev.2021.09.001 *Critical scope analysis of broad FEP / active-inference claims; cited here to keep Markov-blanket and biological interpretations explicitly bounded.*
- Ramstead, M. J. D., Badcock, P. B., Friston, K. J. (2018) Answering Schrödinger's question: A free-energy formulation. *Physics of Life Reviews* **24**, 1–16. <https://www.sciencedirect.com/science/article/pii/S1571064517301409> doi:10.1016/j.plrev.2017.09.001 *Theoretical-biology framing of FEP and self-organization; cited only for context, not as evidence for the manuscript's finite-policy results.*

Computational psychiatry

- Adams, R. A., Stephan, K. E., Brown, H. R., Frith, C. D., Friston, K. J. (2013) The computational anatomy of psychosis. *Frontiers in Psychiatry* **4**, 47. <https://www.frontiersin.org/articles/10.3389/fpsyt.2013.00047> doi:10.3389/fpsyt.2013.00047
- Haarsma, J., Fletcher, P. C., Griffin, J. D., et al. (2021) Precision weighting of cortical unsigned prediction error signals benefits learning, is mediated by dopamine, and is impaired in psychosis. *Molecular Psychiatry* **26**, 5320–5333. <https://www.nature.com/articles/s41380-020-0803-8> doi:10.1038/s41380-020-0803-8 *Empirical precision-weighting study used to support cautious dopamine / psychosis discussion without identifying lambda with a neural precision variable.*
- Schwartenbeck, P., Friston, K. (2016) Computational Phenotyping in Psychiatry: A Worked Example. *eNeuro* **3**(4), ENEURO.0049-16.2016. <https://www.eneuro.org/content/3/4/ENEURO.0049-16.2016> doi:10.1523/ENEURO.0049-16.2016

Integrated information / consciousness

- Tononi, G. (2008) Consciousness as integrated information: a provisional manifesto. *The Biological Bulletin* **215**(3), 216–242. <https://www.journals.uchicago.edu/doi/10.2307/25470707> doi:10.2307/25470707

Multi-agent / interactive inference

- Albarracin, M., Pitliya, R. J., St. Clere Smithe, T., Friedman, D. A., Friston, K., Ramstead, M. J. D. (2024) Shared Protentions in Multi-Agent Active Inference. *Entropy* **26**(4), 303. <https://www.mdpi.com/1099-4300/26/4/303> doi:10.3390/e26040303

- Friedman, D. A. (2025) CEREBRUM: Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling. *Zenodo (Active Inference Journal)* **v1.4**. <https://zenodo.org/records/15231156> doi:10.5281/zenodo.15231156 *Case-grammar architecture for unified Bayesian cognitive modeling; supports the manuscript's multi-agent / case-grammar connection.*
- Friedman, D. A. (2026) Compositional Approaches to Linguistic Case for Cognitive Modeling. *Active Inference Journal. Zenodo* **v1**. <https://doi.org/10.5281/zenodo.19695260> doi:10.5281/zenodo.19695260 *Category-theoretic treatment of linguistic case systems: case roles as objects, alignment patterns as functors, sentence and discourse circuits as morphisms in a compact-closed category, hom-objects enriched over $[0,1]$ as magnitudes, and topos-theoretic Morita equivalence between alignment frames. Supplies the compositional substrate under which the case-graded J asymmetry of the CEREBRUM connections section is licensed; companion architecture to friedman-2025-cerebrum.*
- Friston, K. J., Da Costa, L., Heins, C., Klein, B., Salvatori, T., Buckley, C. L., Tschantz, A., Parr, T. (2024) Federated inference and belief sharing. *Neuroscience & Biobehavioral Reviews* **156**, 105500. doi:10.1016/j.neubiorev.2023.105500 *Belief broadcasting across agents under shared generative models; canonical federated-inference reference in AIF.*
- Heins, C., Millidge, B., Da Costa, L., Mann, R. P., Friston, K. J., Couzin, I. D. (2024) Collective behavior from surprise minimization. *Proceedings of the National Academy of Sciences* **121(17)**, e2320239121. <https://www.pnas.org/doi/10.1073/pnas.2320239121> doi:10.1073/pnas.2320239121 *Primary collective-active-inference model; cited for group-scale surprise minimization and emergent coordination rather than for finite-policy theorem claims.*
- Maisto, D., Donnarumma, F., Pezzulo, G. (2024) Interactive Inference: A Multi-Agent Model of Cooperative Joint Actions. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* **54(2)**, 704–715. <https://ieeexplore.ieee.org/document/10288543> doi:10.1109/TSMC.2023.3312585 *Earlier circulating attribution listing Friston as a co-author was incorrect — arXiv:2210.13113 lists authors as Maisto, Donnarumma, Pezzulo. Final venue: IEEE TSMC: Systems Volume 54, Issue 2, February 2024; DOI minted in 2023.*
- Ruiz-Serra, J., Sweeney, S., Harré, M. (2025) Factorised Active Inference for Strategic Multi-Agent Interactions. *Proc. 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*. <https://arxiv.org/abs/2411.07362> doi:10.48550/arXiv.2411.07362 *arXiv:2411.07362. AAMAS 2025 proceedings version.*
- Waade, P. T., Olesen, C. L., Laursen, J. E., Nehrer, S. W., Heins, C., Friston, K., Mathys, C. (2025) As One and Many: Relating Individual and Emergent Group-Level Generative Models in Active Inference. *Entropy* **27(2)**, 143. <https://www.mdpi.com/1099-4300/27/2/143> doi:10.3390/e27020143

Lean / formalization

- Boldo, S., Melquiond, G. (2011) Flocq: A Unified Library for Proving Floating-Point Algorithms in Coq. *2011 IEEE 20th Symposium on Computer Arithmetic*, 243–252. <https://doi.org/10.1109/ARITH.2011.40> doi:10.1109/ARITH.2011.40 *Primary Flocq reference; cited as the model for what a verified Float-to-real bridge would require.*
- Degenne, R., Ledvinka, D., Marion, E., Pfaffelhuber, P. (2025) Formalization of Brownian motion in Lean. *arXiv preprint, arXiv:2511.20118*. <https://arxiv.org/abs/2511.20118> doi:10.48550/arXiv.2511.20118 *Stochastic-process formalization in Mathlib's MeasureTheory.Probability.Kernel layer.*
- Degenne, R. (2025) Markov kernels in Mathlib's probability library. *arXiv preprint, arXiv:2510.04070*. <https://arxiv.org/abs/2510.04070> doi:10.48550/arXiv.2510.04070 *Foundational reference for the Markov-kernel layer in Mathlib's MeasureTheory.Probability.Kernel.* namespace; cited for the Lean formalization path.*
- de Moura, L., Ullrich, S. (2021) The Lean 4 Theorem Prover and Programming Language. *Automated Deduction – CADE 28, LNCS vol. 12699 (Springer, Cham)*, 625–635. https://link.springer.com/chapter/10.1007/978-3-030-79876-5_37 doi:10.1007/978-3-030-79876-5_37
- Friedman, D. A. (2026) Towards Lean 4 Formalization of the Free Energy Principle: AI-Driven Theorem Sketching and Verification for Active Inference and Bayesian Mechanics. *Active Inference Journal. Zenodo*. https://github.com/ActiveInferenceInstitute/fep_lean doi:10.5281/zenodo.19699234
- Higham, N. J. (2002) Accuracy and Stability of Numerical Algorithms. *SIAM Second edition*. <https://doi.org/10.1137/1.9780898718027> doi:10.1137/1.9780898718027 *Standard numerical-analysis reference for round-off, conditioning, and stability; cited to scope Float-vs-real residuals.*
- IEEE (2019) IEEE Standard for Floating-Point Arithmetic. *IEEE Std 754-2019*. <https://ieeexplore.ieee.org/document/8766229> doi:10.1109/IEEESTD.2019.8766229 *Primary IEEE 754 standard reference; cited for the floating-point semantics a verified Float-to-real bridge would need to model.*
- Lean FRO (2026) Lean Programming Language. *Official Lean website*. <https://lean-lang.org/> *Official Lean*

source for the minimal trusted-kernel and verification-platform claims; scholarly citation remains de Moura and Ullrich 2021.

- Lean FRO (2026) Mathlib: A Foundation for Formal Mathematics Research and Verification. *Official Lean use-case page*. <https://lean-lang.org/use-cases/mathlib/> Official current-facing source for Mathlib’s role, coverage, and quality-assurance framing; scholarly citation remains The Mathlib Community 2020.
- The Mathlib Community (2020) The Lean Mathematical Library. *Proc. 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. <https://dl.acm.org/doi/10.1145/3372885.3373824> doi:10.1145/3372885.3373824 *arXiv preprint*: 1910.09336.

Software dependencies (companion code)

- Active Inference Institute (2023) GeneralizedNotationNotation. *Open-source software (Active Inference Institute)*. <https://github.com/ActiveInferenceInstitute/GeneralizedNotationNotation> Reference implementation of Generalized Notation Notation. Cited as the upstream of the shipped fifth (structural-and-numerical) representation in the GNN extension supplement; the bridge ships as a project-owned parser, a verified $K=2$ round-trip, and a Lean typed-contract emitter (empirical per the Claim-Strength Legend), with the upstream first-class coupling primitive named as an honest residual.
- Bagaev, D., Podusenko, A., de Vries, B. (2023) RxInfer: A Julia package for reactive real-time Bayesian inference. *Journal of Open Source Software* **8**(84), 5161. <https://joss.theoj.org/papers/10.21105/joss.05161> doi:10.21105/joss.05161 Primary software citation for RxInfer.jl; cited for the reactive message-passing lineage that complements pymdp-style discrete POMDP active inference.
- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., Zhang, Q. (2018) JAX: composable transformations of Python+NumPy programs. *Open-source software*. <http://github.com/jax-ml/jax> Canonical citation per the JAX CITATION.bib file. No replacement journal/paper has emerged as of 2026-05. JAX backend underlies pymdp 1.0.1 and the simulation harness.
- Friedman, D. A. (2026) Policy Entanglement in Active Inference: A Tunable Mean-Field Deformation Framework for Multi-Stream Policy Ensembles. *Active Inference Institute*. https://github.com/ActiveInferenceInstitute/policy_entanglement doi:10.5281/zenodo.20418904 Working four-track research artifact (prose, equations, Python/pymdp, Lean 4); source at github.com/ActiveInferenceInstitute/policy_entanglement; Zenodo DOI 10.5281/zenodo.20418904.
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., et al. (2020) Array programming with NumPy. *Nature* **585**, 357–362. <https://www.nature.com/articles/s41586-020-2649-2> doi:10.1038/s41586-020-2649-2
- Hunter, J. D. (2007) Matplotlib: A 2D graphics environment. *Computing in Science & Engineering* **9**(3), 90–95. <https://ieeexplore.ieee.org/document/4160265> doi:10.1109/MCSE.2007.55
- Nehrer, S. W., Laursen, J. E., Heins, C., Friston, K., Mathys, C., Waade, P. T. (2025) Introducing ActiveInference.jl: A Julia Library for Simulation and Parameter Estimation with Active Inference Models. *Entropy* **27**(1), 62. <https://www.mdpi.com/1099-4300/27/1/62> doi:10.3390/e27010062 Primary ActiveInference.jl software paper; cited for Julia POMDP active-inference simulation and model-fitting context.
- The pymdp developers (2026) infer-actively/pymdp: A Python implementation of active inference for Markov Decision Processes. *GitHub repository*. <https://github.com/infer-actively/pymdp> Official source repository consulted for package identity, install/source provenance, and release context; the scholarly software citation remains Heins et al. 2022.
- The pymdp developers (2026) pymdp documentation. *Read the Docs*. <https://pymdp-rtd.readthedocs.io/en/latest/> Official documentation consulted for the JAX-backend and Agent/API provenance used by the reproducibility checklist.
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., et al. (2020) SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **17**, 261–272. <https://www.nature.com/articles/s41592-019-0686-2> doi:10.1038/s41592-019-0686-2

Scientific visualization

- Rougier, N. P., Droettboom, M., Bourne, P. E. (2014) Ten Simple Rules for Better Figures. *PLOS Computational Biology* **10**(9), e1003833. <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833> doi:10.1371/journal.pcbi.1003833 Visualization-design checklist used narrowly for self-contained captions, explicit figure purpose, and avoidance of gratuitous complexity.

References

- R. A. Adams, K. E. Stephan, H. R. Brown, C. D. Frith, and K. J. Friston. The computational anatomy of psychosis. *Frontiers in Psychiatry*, 4:47, 2013. doi:[10.3389/fpsy.2013.00047](https://doi.org/10.3389/fpsy.2013.00047). URL <https://www.frontiersin.org/articles/10.3389/fpsy.2013.00047>.
- M. Aguilera, B. Millidge, A. Tschantz, and C. L. Buckley. How particular is the physics of the free energy principle? *Physics of Life Reviews*, 40:24–50, 2022. doi:[10.1016/j.plev.2021.11.001](https://doi.org/10.1016/j.plev.2021.11.001). URL <https://www.sciencedirect.com/science/article/pii/S1571064521000749>. Critical-review treatment of Markov-blanket / particular-physics derivations of the free-energy principle. First circulated as a 2021 preprint; final publication in *Physics of Life Reviews* vol. 40 (2022); slug retained as ‘aguilera-2021’ for back-compatibility.
- M. Albarracín, R. J. Pitliya, T. St. Clere Smithe, D. A. Friedman, K. Friston, and M. J. D. Ramstead. Shared protentions in multi-agent active inference. *Entropy*, 26(4):303, 2024. doi:[10.3390/e26040303](https://doi.org/10.3390/e26040303). URL <https://www.mdpi.com/1099-4300/26/4/303>.
- S. Amari. Differential-geometrical methods in statistics. Lecture Notes in Statistics, Vol. 28 (Springer), 1985. URL <https://link.springer.com/book/10.1007/978-1-4612-5056-2>. Founding monograph of information geometry; introduced the dually-flat alpha-connection family and the e- / m-geodesic structure used in the framework-connections section.
- S. Amari. Information geometry and its applications. Springer, Applied Mathematical Sciences, Vol. 194, 2016. URL <https://link.springer.com/book/10.1007/978-4-431-55978-8>.
- S. Amari and H. Nagaoka. Methods of information geometry. AMS Translations of Mathematical Monographs, Vol. 191 (American Mathematical Society / Oxford University Press), 2000. URL <https://bookstore.ams.org/mmono-191/>. Translated by Daishi Harada from the 1993 Japanese original. ISBN 978-0-8218-4302-4.
- N. Ay, J. Jost, H. V. Lê, and L. Schwachhöfer. Information geometry. *Ergebnisse der Mathematik und ihrer Grenzgebiete (3. Folge)*, Vol. 64 (Springer), 2017. URL <https://link.springer.com/book/10.1007/978-3-319-56478-4>. ISBN 978-3-319-56477-7. Comprehensive monograph treatment covering the parametric and non-parametric geometric foundations used in the information-geometric formalism.
- P.-L. Bacon, J. Harb, and D. Precup. The option-critic architecture. *Proceedings of the AAAI Conference on Artificial Intelligence*, 31(1), 2017. doi:[10.1609/aaai.v31i1.10916](https://doi.org/10.1609/aaai.v31i1.10916). URL <https://ojs.aaai.org/index.php/AAAI/article/view/10916>. Modern options reference used to scope the manuscript’s option-like embedding as a parametric policy-matching construction, not a proof that the lambda deformation subsumes all option-learning algorithms.
- D. Bagaev and de Vries B. Podusenko, A. Rxinfer: A julia package for reactive real-time bayesian inference. *Journal of Open Source Software*, 8(84):5161, 2023. doi:[10.21105/joss.05161](https://doi.org/10.21105/joss.05161). URL <https://joss.theoj.org/papers/10.21105/joss.05161>. Primary software citation for RxInfer.jl; cited for the reactive message-passing lineage that complements pymdp-style discrete POMDP active inference.
- Émery M. Bakry, D. Diffusions hypercontractives, 1985. URL <https://doi.org/10.1007/BFb0075144>. Bakry–Émery curvature criterion for log-Sobolev / variance bounds on exponential families.
- T. D. Barrett, A. Camuto, C. R. Williamson, and S. J. Roberts. Certifiably robust variational autoencoders through lipschitz constraints. *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022. URL <https://proceedings.mlr.press/v151/barrett22a.html>. Actionable KL-drift bounds under encoder/decoder Lipschitz constraints; cited for adversarial VI robustness context in Q11.
- A. G. Barto and S. Mahadevan. Recent advances in hierarchical reinforcement learning. *Discrete Event Dynamic Systems*, 13:341–379, 2003. doi:[10.1023/A:1025696116075](https://doi.org/10.1023/A:1025696116075). URL <https://link.springer.com/article/10.1023/A:1025696116075>. Review reference for hierarchical RL and temporal abstraction; cited alongside options to keep the control/RL comparison anchored in the RL literature.
- J. Biamonte and V. Bergholm. Tensor networks in a nutshell. arXiv, 2017. URL <https://arxiv.org/abs/1708.00006>. Tutorial tensor-network reference used to scope Schmidt-rank / tensor-train language as probability-tensor linear algebra, not a quantum-physical claim.
- D. M. Blei, A. Kucukelbir, and J. D. McAuliffe. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017. doi:[10.1080/01621459.2017.1285773](https://doi.org/10.1080/01621459.2017.1285773). URL <https://doi.org/10.1080/01621459.2017.1285773>. Modern scholarly review citation for mean-field and structured variational inference.

- S. Boldo and G. Melquiond. Flocq: A unified library for proving floating-point algorithms in coq. 2011 IEEE 20th Symposium on Computer Arithmetic, 2011. URL <https://doi.org/10.1109/ARITH.2011.40>. Primary Flocq reference; cited as the model for what a verified Float-to-real bridge would require.
- M. Botvinick and M. Toussaint. Planning as inference. *Trends in Cognitive Sciences*, 16(10):485–488, 2012. doi:[10.1016/j.tics.2012.08.006](https://doi.org/10.1016/j.tics.2012.08.006). URL [https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613\(12\)00184-3](https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613(12)00184-3).
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. Jax: composable transformations of python+numpy programs. Open-source software, 2018. URL <http://github.com/jax-ml/jax>. Canonical citation per the JAX CITATION.bib file. No replacement journal/paper has emerged as of 2026-05. JAX backend underlies pymdp 1.0.1 and the simulation harness.
- H. J. Brascamp and E. H. Lieb. On extensions of the brunn–minkowski and prékopa–leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation. 22:366–389, 1976. doi:[10.1016/0022-1236\(76\)90001-5](https://doi.org/10.1016/0022-1236(76)90001-5). URL [https://doi.org/10.1016/0022-1236\(76\)90001-5](https://doi.org/10.1016/0022-1236(76)90001-5). Variance Brascamp–Lieb inequality for log-concave measures; cited in Q1 dispersion-comparison conjecture.
- C. L. Buckley, C. S. Kim, S. McGregor, and A. K. Seth. The free energy principle for action and perception: A mathematical review. *Journal of Mathematical Psychology*, 81:55–79, 2017. doi:[10.1016/j.jmp.2017.09.004](https://doi.org/10.1016/j.jmp.2017.09.004). URL <https://www.sciencedirect.com/science/article/pii/S0022249617300962>. Accessible mathematical review of the free-energy principle used to ground the introductory VFE claim without relying only on process-theory exposition.
- T. Champion, M. Grześ, and H. Bowman. Branching time active inference with bayesian filtering. *Neural Computation*, 34(10):2132–2144, 2022. doi:[10.1162/neco_a_01529](https://doi.org/10.1162/neco_a_01529). URL <https://direct.mit.edu/neco/article-abstract/34/10/2132>. Uses Bayesian filtering on the expanded policy tree rather than vanilla MCTS rollouts; in the wider BTAI lineage (Fountas et al. 2020 DAI / MCTS for active inference).
- T. Champion, H. Bowman, D. Marković, and M. Grześ. Reframing the expected free energy: Four formulations and a unification. arXiv preprint, 2024. URL <https://arxiv.org/abs/2402.14460>.
- A. Cichocki, N. Lee, I. V. Oseledets, A.-H. Phan, Q. Zhao, and D. P. Mandic. Tensor networks for dimensionality reduction and large-scale optimization: Part 1 low-rank tensor decompositions. *Foundations and Trends in Machine Learning*, 9(4–5):249–429, 2016. doi:[10.1561/22000000059](https://doi.org/10.1561/22000000059). URL <https://doi.org/10.1561/22000000059>. Machine-learning tensor-network survey; cited for low-rank tensor formats and optimization context, not as an active-inference result.
- The Mathlib Community. The lean mathematical library. Proc. 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 2020. URL <https://dl.acm.org/doi/10.1145/3372885.3373824>. arXiv preprint: 1910.09336.
- L. Da Costa, T. Parr, N. Sajid, S. Veselic, V. Neacsu, and K. Friston. Active inference on discrete state-spaces: A synthesis. *Journal of Mathematical Psychology*, 99:102447, 2020. doi:[10.1016/j.jmp.2020.102447](https://doi.org/10.1016/j.jmp.2020.102447). URL <https://www.sciencedirect.com/science/article/pii/S0022249620300857>. arXiv:2001.07203.
- L. Da Costa, K. Friston, C. Heins, and G. A. Pavliotis. Bayesian mechanics for stationary processes. *Proceedings of the Royal Society A*, 477(2256):20210518, 2021. doi:[10.1098/rspa.2021.0518](https://doi.org/10.1098/rspa.2021.0518). URL <https://royalsocietypublishing.org/doi/10.1098/rspa.2021.0518>. Bayesian-mechanics reference for Markov blankets in stationary processes; cited to keep the manuscript’s policy-space blanket language explicitly analogical.
- L. Da Costa, S. Tenka, D. Zhao, and N. Sajid. Active inference as a model of agency. arXiv, 2024. URL <https://arxiv.org/abs/2401.12917>. Accepted at the RLDM 2022 workshop ‘RL as a model of agency’; cited for the active-inference-as-agency unification framing, not as evidence for policy entanglement itself.
- L. de Moura and S. Ullrich. The lean 4 theorem prover and programming language. Automated Deduction – CADE 28, LNCS vol. 12699 (Springer, Cham), 2021. URL https://link.springer.com/chapter/10.1007/978-3-030-79876-5_37.
- B. de Vries and K. J. Friston. A factor graph description of deep temporal active inference. *Frontiers in Computational Neuroscience*, 11:95, 2017. doi:[10.3389/fncom.2017.00095](https://doi.org/10.3389/fncom.2017.00095). URL <https://www.frontiersin.org/articles/10.3389/fncom.2017.00095/full>. Primary factor-graph / Forney-style graph source for deep temporal active inference; cited only for message-passing implementability, not as a theorem-equivalence claim for policy entanglement.

- B. de Vries, van de Laar T. Nuijten, T., W. Kouw, A. Adamiat, A. Nisslbeck, and et al. Lukashchuk, V. Expected free energy-based planning as variational inference. arXiv preprint, 2025. URL <https://arxiv.org/abs/2504.14898>.
- R. Degenne. Markov kernels in mathlib’s probability library. arXiv preprint, 2025. URL <https://arxiv.org/abs/2510.04070>. Foundational reference for the Markov-kernel layer in Mathlib’s ‘MeasureTheory.Probability.Kernel.*’ namespace; cited for the Lean formalization path.
- R. Degenne, D. Ledvinka, E. Marion, and P. Pfaffelhuber. Formalization of brownian motion in lean. arXiv preprint, arXiv:2511.20118, 2025. URL <https://arxiv.org/abs/2511.20118>. Stochastic-process formalization in Mathlib’s MeasureTheory.Probability.Kernel layer.
- B. B. Doll, K. D. Duncan, D. A. Simon, D. Shohamy, and N. D. Daw. Model-based choices involve prospective neural activity. *Nature Neuroscience*, 18(5):767–772, 2015. doi:[10.1038/nn.3981](https://doi.org/10.1038/nn.3981). URL <https://www.nature.com/articles/nn.3981>.
- J. Eisert, M. Cramer, and M. B. Plenio. Colloquium: Area laws for the entanglement entropy. *Reviews of Modern Physics*, 82(1):277–306, 2010. doi:[10.1103/RevModPhys.82.277](https://doi.org/10.1103/RevModPhys.82.277). URL <https://link.aps.org/doi/10.1103/RevModPhys.82.277>.
- Z. Fountas, N. Sajid, P. A. M. Mediano, and K. Friston. Deep active inference agents using monte-carlo methods. Advances in Neural Information Processing Systems 33 (NeurIPS 2020), 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/865dfbde8a344b44095495f3591f7407-Abstract.html>. arXiv:2006.04176. Canonical reference for deep / MCTS-style branching-time active inference (DAI).
- D. A. Friedman. Cerebrum: Case-enabled reasoning engine with bayesian representations for unified modeling. *Zenodo (Active Inference Journal)*, v1.4, 2025. doi:[10.5281/zenodo.15231156](https://doi.org/10.5281/zenodo.15231156). URL <https://zenodo.org/records/15231156>. Case-grammar architecture for unified Bayesian cognitive modeling; supports the manuscript’s multi-agent / case-grammar connection.
- D. A. Friedman. Policy entanglement in active inference: A tunable mean-field deformation framework for multi-stream policy ensembles. Active Inference Institute, 2026a. URL https://github.com/ActiveInferenceInstitute/policy_entanglement. Working four-track research artifact (prose, equations, Python/pymdp, Lean 4); source at github.com/ActiveInferenceInstitute/policy_entanglement; Zenodo DOI [10.5281/zenodo.20418904](https://doi.org/10.5281/zenodo.20418904).
- D. A. Friedman. Compositional approaches to linguistic case for cognitive modeling. *Active Inference Journal. Zenodo*, v1, 2026b. doi:[10.5281/zenodo.19695260](https://doi.org/10.5281/zenodo.19695260). URL <https://doi.org/10.5281/zenodo.19695260>. Category-theoretic treatment of linguistic case systems: case roles as objects, alignment patterns as functors, sentence and discourse circuits as morphisms in a compact-closed category, hom-objects enriched over $[0,1]$ as magnitudes, and topos-theoretic Morita equivalence between alignment frames. Supplies the compositional substrate under which the case-graded J asymmetry of the CEREBRUM connections section is licensed; companion architecture to [friedman-2025-cerebrum](https://github.com/ActiveInferenceInstitute/cerebrum).
- D. A. Friedman. Towards lean 4 formalization of the free energy principle: Ai-driven theorem sketching and verification for active inference and bayesian mechanics. Active Inference Journal. Zenodo, 2026c. URL https://github.com/ActiveInferenceInstitute/fep_lean.
- K. Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010. doi:[10.1038/nrn2787](https://doi.org/10.1038/nrn2787). URL <https://www.nature.com/articles/nrn2787>.
- K. Friston. A free energy principle for a particular physics. arXiv preprint, 2019. URL <https://arxiv.org/abs/1906.10184>. Preprint only as of 2026-05; never received a journal publication.
- K. Friston, J. Kilner, and L. Harrison. A free energy principle for the brain. *Journal of Physiology-Paris*, 100(1–3):70–87, 2006. doi:[10.1016/j.jphysparis.2006.10.001](https://doi.org/10.1016/j.jphysparis.2006.10.001). URL <https://www.sciencedirect.com/science/article/pii/S092842570600060X>. Early statement of variational free energy as a unifying principle for perception, action, and learning.
- K. Friston, P. Schwartenbeck, T. FitzGerald, M. Moutoussis, T. Behrens, and R. J. Dolan. The anatomy of choice: dopamine and decision-making. *Philosophical Transactions of the Royal Society B*, 369(1655):20130481, 2014. doi:[10.1098/rstb.2013.0481](https://doi.org/10.1098/rstb.2013.0481). URL <https://royalsocietypublishing.org/doi/10.1098/rstb.2013.0481>.
- K. Friston, T. FitzGerald, F. Rigoli, P. Schwartenbeck, and G. Pezzulo. Active inference: A process theory. *Neural Computation*, 29(1):1–49, 2017a. doi:[10.1162/NECO_a_00912](https://doi.org/10.1162/NECO_a_00912). URL <https://direct.mit.edu/neco/article/29/1/1>.
- K. Friston, L. Da Costa, D. Hafner, C. Hesp, and T. Parr. Sophisticated inference. *Neural Computation*, 33(3):713–763, 2021. doi:[10.1162/neco_a_01351](https://doi.org/10.1162/neco_a_01351). URL <https://direct.mit.edu/neco/article/33/3/713>. arXiv:2006.04120. Introduces recursive expected free energy for deep counterfactual policy tree search.

- K. J. Friston. Active inference and cognitive consistency. *Psychological Inquiry*, 29(2):67–73, 2018. doi:[10.1080/1047840X.2018.1480693](https://doi.org/10.1080/1047840X.2018.1480693). URL <https://www.tandfonline.com/doi/full/10.1080/1047840X.2018.1480693>. Commentary linking expected-free-energy epistemic/pragmatic terms to cognitive-consistency constructs; cited as interpretive context, not as a reduction of social-psychological theory to active inference.
- K. J. Friston and de Vries B. Parr, T. The graphical brain: Belief propagation and active inference. *Network Neuroscience*, 1(4):381–414, 2017. doi:[10.1162/NETN_a_00018](https://doi.org/10.1162/NETN_a_00018). URL https://doi.org/10.1162/NETN_a_00018. Graphical-brain source for belief propagation, Forney-style graphical modeling, and active-inference message-passing context; not evidence that lambda-coupled policy posteriors are standard in AIF.
- K. J. Friston, N. Trujillo-Barreto, and J. Daunizeau. Dem: A variational treatment of dynamic systems. *NeuroImage*, 41(3):849–885, 2008. doi:[10.1016/j.neuroimage.2008.02.054](https://doi.org/10.1016/j.neuroimage.2008.02.054). URL <https://www.sciencedirect.com/science/article/pii/S1053811908001894>. Primary dynamic expectation maximization reference behind the SPM-DEM lineage discussed in the introduction.
- K. J. Friston, R. Rosch, T. Parr, C. Price, and H. Bowman. Deep temporal models and active inference. *Neuroscience & Biobehavioral Reviews*, 77:388–402, 2017b. doi:[10.1016/j.neubiorev.2017.04.009](https://doi.org/10.1016/j.neubiorev.2017.04.009). URL <https://doi.org/10.1016/j.neubiorev.2017.04.009>. Primary deep-temporal active-inference source; cited for hierarchical temporal-model and long-horizon framing, not as evidence that the present finite-policy model is a full process theory.
- K. J. Friston, L. Da Costa, C. Heins, B. Klein, T. Salvatori, C. L. Buckley, A. Tschantz, and T. Parr. Federated inference and belief sharing. *Neuroscience & Biobehavioral Reviews* 156, 105500, 2024. Belief broadcasting across agents under shared generative models; canonical federated-inference reference in AIF.
- K. J. Friston, C. Heins, T. Verbelen, L. Da Costa, T. Salvatori, D. Marković, A. Tschantz, M. Koudahl, C. Buckley, and T. Parr. From pixels to planning: scale-free active inference. *Frontiers in Network Physiology*, 5:1521963, 2025. doi:[10.3389/fnetp.2025.1521963](https://doi.org/10.3389/fnetp.2025.1521963). URL <https://www.frontiersin.org/journals/network-physiology/article/10.3389/fnetp.2025.1521963/full>. First circulated as a 2024 preprint (arXiv:2407.20292); final venue *Frontiers in Network Physiology*, June 2025. The published *Frontiers* form is the canonical citation; the ‘friston-rgm-2025’ alias is intentionally NOT registered to avoid duplication — cite as ‘[@friston-2024]’. Slug retained as ‘friston-2024’ for back-compatibility.
- Lean FRO. Lean programming language. Official Lean website, 2026a. URL <https://lean-lang.org/>. Official Lean source for the minimal trusted-kernel and verification-platform claims; scholarly citation remains de Moura and Ullrich 2021.
- Lean FRO. Mathlib: A foundation for formal mathematics research and verification. Official Lean use-case page, 2026b. URL <https://lean-lang.org/use-cases/mathlib/>. Official current-facing source for Mathlib’s role, coverage, and quality-assurance framing; scholarly citation remains The Mathlib Community 2020.
- Y. Fu, M. S. Smith, and A. Panagiotelis. Vector copula variational inference and dependent block posterior approximations. arXiv preprint, 2025. URL <https://arxiv.org/abs/2503.01072>.
- I. Glasser, N. Pancotti, and J. I. Cirac. From probabilistic graphical models to generalized tensor networks for supervised learning. *IEEE Access*, 8:68169–68182, 2020. doi:[10.1109/ACCESS.2020.2986279](https://doi.org/10.1109/ACCESS.2020.2986279). URL <https://ieeexplore.ieee.org/document/9058650/>. arXiv:1806.05964 (2018 preprint); IEEE Access publication 2020.
- J. Haarsma, P. C. Fletcher, and et al. Griffin, J. D. Precision weighting of cortical unsigned prediction error signals benefits learning, is mediated by dopamine, and is impaired in psychosis. *Molecular Psychiatry*, 26:5320–5333, 2021. doi:[10.1038/s41380-020-0803-8](https://doi.org/10.1038/s41380-020-0803-8). URL <https://www.nature.com/articles/s41380-020-0803-8>. Empirical precision-weighting study used to support cautious dopamine / psychosis discussion without identifying lambda with a neural precision variable.
- D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. arXiv preprint, 2023. URL <https://arxiv.org/abs/2301.04104>. DreamerV3 — world-model RL agent that achieves cross-domain mastery with a single hyperparameter setting; widely cited as the canonical contemporary world-model architecture against which active-inference agents are benchmarked.
- S. Han, X. Liao, D. B. Dunson, and L. Carin. Variational gaussian copula inference. AISTATS 2016, PMLR 51, 2016. URL <https://proceedings.mlr.press/v51/han16.html>.
- Z.-Y. Han, J. Wang, H. Fan, L. Wang, and P. Zhang. Unsupervised generative modeling using matrix product states. *Physical Review X*, 8(3):031012, 2018. doi:[10.1103/PhysRevX.8.031012](https://doi.org/10.1103/PhysRevX.8.031012). URL <https://journals.aps.org/prx/abstract/10.1103/PhysRevX.8.031012>.

- C. R. Harris, van der Walt S. J. Millman, K. J., R. Gommers, P. Virtanen, and et al. Cournapeau, D. Array programming with numpy. *Nature*, 585:357–362, 2020. doi:[10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2). URL <https://www.nature.com/articles/s41586-020-2649-2>.
- C. Heins, B. Millidge, D. Demekas, B. Klein, K. Friston, I. D. Couzin, and A. Tschantz. pymdp: A python library for active inference in discrete state spaces. *Journal of Open Source Software*, 7(73):4098, 2022. doi:[10.21105/joss.04098](https://doi.org/10.21105/joss.04098). URL <https://joss.theoj.org/papers/10.21105/joss.04098>. v1.0.1 JAX-based release is the simulation backend used by the empirical suite and src/simulation/; arXiv:2201.03904.
- C. Heins, B. Millidge, L. Da Costa, R. P. Mann, K. J. Friston, and I. D. Couzin. Collective behavior from surprise minimization. *Proceedings of the National Academy of Sciences*, 121(17):e2320239121, 2024. doi:[10.1073/pnas.2320239121](https://doi.org/10.1073/pnas.2320239121). URL <https://www.pnas.org/doi/10.1073/pnas.2320239121>. Primary collective-active-inference model; cited for group-scale surprise minimization and emergent coordination rather than for finite-policy theorem claims.
- N. J. Higham. Accuracy and stability of numerical algorithms. *SIAM*, Second edition, 2002. doi:[10.1137/1.9780898718027](https://doi.org/10.1137/1.9780898718027). URL <https://doi.org/10.1137/1.9780898718027>. Standard numerical-analysis reference for round-off, conditioning, and stability; cited to scope Float-vs-real residuals.
- G. E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800, 2002. doi:[10.1162/089976602760128018](https://doi.org/10.1162/089976602760128018). URL <https://direct.mit.edu/neco/article-abstract/14/8/1771/>.
- R. Hodson, M. Mehta, and R. Smith. The empirical status of predictive coding and active inference. *Neuroscience & Biobehavioral Reviews*, 157:105473, 2024. doi:[10.1016/j.neubiorev.2023.105473](https://doi.org/10.1016/j.neubiorev.2023.105473). URL <https://pubmed.ncbi.nlm.nih.gov/38030100/>. Empirical-status review used to calibrate FEP / active-inference claims as normative and computational unless a specific comparative empirical test is supplied.
- M. D. Hoffman and D. M. Blei. Structured stochastic variational inference. AISTATS 2015, PMLR 38, 2015. URL <http://proceedings.mlr.press/v38/hoffman15.html>. arXiv:1404.4114.
- J. D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi:[10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55). URL <https://ieeexplore.ieee.org/document/4160265>.
- IEEE. Ieee standard for floating-point arithmetic. IEEE Std 754-2019, 2019. URL <https://ieeexplore.ieee.org/document/8766229>. Primary IEEE 754 standard reference; cited for the floating-point semantics a verified Float-to-real bridge would need to model.
- Active Inference Institute. Generalizednotationnotation. Open-source software (Active Inference Institute), 2023. URL <https://github.com/ActiveInferenceInstitute/GeneralizedNotationNotation>. Reference implementation of Generalized Notation Notation. Cited as the upstream of the shipped fifth (structural-and-numerical) representation in the GNN extension supplement; the bridge ships as a project-owned parser, a verified K=2 round-trip, and a Lean typed-contract emitter (empirical per the Claim-Strength Legend), with the upstream first-class coupling primitive named as an honest residual.
- R. S. Johansson, G. Westling, A. Bäckström, and J. R. Flanagan. Eye-hand coordination in object manipulation. *Journal of Neuroscience*, 21(17):6917–6932, 2001. doi:[10.1523/JNEUROSCI.21-17-06917.2001](https://doi.org/10.1523/JNEUROSCI.21-17-06917.2001). URL <https://www.jneurosci.org/content/21/17/6917>. Primary empirical eye-hand coordination source used to ground the reach/saccade example in the motivation.
- R. Kaplan and K. J. Friston. Planning and navigation as active inference. *Biological Cybernetics*, 112:323–343, 2018. doi:[10.1007/s00422-018-0753-2](https://doi.org/10.1007/s00422-018-0753-2). URL <https://link.springer.com/article/10.1007/s00422-018-0753-2>. Navigation and planning example for EFE-based policy selection, risk / ambiguity terms, and epistemic exploration under active inference.
- H. J. Kappen. Path integrals and symmetry breaking for optimal control theory. *Journal of Statistical Mechanics: Theory and Experiment*, 2005:P11011, 2005. doi:[10.1088/1742-5468/2005/11/P11011](https://doi.org/10.1088/1742-5468/2005/11/P11011). URL <https://iopscience.iop.org/article/10.1088/1742-5468/2005/11/P11011>.
- M. D. Kirchhoff, T. Parr, E. Palacios, K. Friston, and J. Kiverstein. The markov blankets of life: autonomy, active inference and the free energy principle. *Journal of The Royal Society Interface*, 15(138):20170792, 2018. doi:[10.1098/rsif.2017.0792](https://doi.org/10.1098/rsif.2017.0792). URL <https://royalsocietypublishing.org/doi/10.1098/rsif.2017.0792>. Markov-blanket account of autonomy used here to keep system-boundary language separate from the manuscript’s intra-agent policy-coupling claim.
- M. F. Land and M. Hayhoe. In what ways do eye movements contribute to everyday activities? *Vision Research*, 41(25–26):3559–3565, 2001. doi:[10.1016/S0042-6989\(01\)00102-X](https://doi.org/10.1016/S0042-6989(01)00102-X). URL [https://doi.org/10.1016/S0042-6989\(01\)00102-X](https://doi.org/10.1016/S0042-6989(01)00102-X).

- 6989(01)00102-X. Natural-task eye-movement review showing that gaze commonly arrives at task-relevant objects before manipulation, grounding the cross-stream reach/saccade example.
- P. Lanillos, C. Meo, C. Pezzato, A. A. Meera, M. Baioumy, W. Ohata, A. Tschantz, B. Millidge, M. Wisse, C. L. Buckley, and J. Tani. Active inference in robotics and artificial agents: Survey and challenges. arXiv preprint, 2021. URL <https://arxiv.org/abs/2112.01871>.
- S. Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review. arXiv preprint, 2018. URL <https://arxiv.org/abs/1805.00909>. Canonical tutorial on the control-as-inference / maximum-entropy RL viewpoint cited in the comparison with active inference.
- Y. Li and R. E. Turner. Rényi divergence variational inference. *Advances in Neural Information Processing Systems* 29 (NeurIPS 2016), 2016. URL <https://proceedings.neurips.cc/paper/2016/hash/7750ca3559e5b8e1f44210283368fc16-Abstract.html>. arXiv:1602.02311. Generalizes ELBO-style variational inference to the Renyi alpha-divergence family; cited for structured-VI / alpha-divergence comparisons with the active-inference free-energy functional.
- J. Limanowski, R. A. Adams, J. Kilner, and T. Parr. The many roles of precision in action. *Entropy*, 26(9):790, 2024. doi:[10.3390/e26090790](https://doi.org/10.3390/e26090790). URL <https://www.mdpi.com/1099-4300/26/9/790>. Review of precision roles in action, attention, habit, and self-other distinction; cited to keep lambda-as-coupling-precision language distinct from neural precision claims.
- D. Maisto, F. Donnarumma, and G. Pezzulo. Interactive inference: A multi-agent model of cooperative joint actions. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 54(2):704–715, 2024. doi:[10.1109/TSMC.2023.3312585](https://doi.org/10.1109/TSMC.2023.3312585). URL <https://ieeexplore.ieee.org/document/10288543>. Earlier circulating attribution listing Friston as a co-author was incorrect — arXiv:2210.13113 lists authors as Maisto, Donnarumma, Pezzulo. Final venue: IEEE TSMC: Systems Volume 54, Issue 2, February 2024; DOI minted in 2023.
- W. J. McGill. Multivariate information transmission. *Psychometrika*, 19(2):97–116, 1954. doi:[10.1007/BF02289159](https://doi.org/10.1007/BF02289159). URL <https://link.springer.com/article/10.1007/BF02289159>. Original definition of total correlation / multi-information, the correlation surcharge in the entanglement-decomposition identity.
- R. Menary and A. J. Gillett. Markov blankets do not demarcate the boundaries of the mind. *Behavioral and Brain Sciences*, 45:e201, 2022. doi:[10.1017/S0140525X22000371](https://doi.org/10.1017/S0140525X22000371). URL <https://pubmed.ncbi.nlm.nih.gov/36172779/>. Critical extended-mind commentary; cited to prevent Markov-blanket language from being treated as an automatic cognitive-boundary criterion.
- B. Millidge, A. Tschantz, and C. L. Buckley. Whence the expected free energy? *Neural Computation*, 33(2):447–482, 2021. doi:[10.1162/neco_a_01354](https://doi.org/10.1162/neco_a_01354). URL <https://direct.mit.edu/neco/article/33/2/447/95645/Whence-the-Expected-Free-Energy>. Critical analysis of EFE derivations and alternatives; cited to keep expected-free-energy unification claims bounded.
- J. Naudts. Generalised thermostatics. Springer London, 2011. URL <https://link.springer.com/book/10.1007/978-0-85729-355-8>. ISBN 978-0-85729-354-1 (print). Covers escort distributions, q-deformed exponential families.
- S. W. Nehrer, J. E. Laursen, C. Heins, K. Friston, C. Mathys, and P. T. Waade. Introducing activeinference.jl: A julia library for simulation and parameter estimation with active inference models. *Entropy*, 27(1):62, 2025. doi:[10.3390/e27010062](https://doi.org/10.3390/e27010062). URL <https://www.mdpi.com/1099-4300/27/1/62>. Primary ActiveInference.jl software paper; cited for Julia POMDP active-inference simulation and model-fitting context.
- R. B. Nelsen. An introduction to copulas. Springer Series in Statistics, 2nd ed., 2006. URL <https://link.springer.com/book/10.1007/0-387-28678-0>. Foundational monograph for copula definitions and dependence modeling; cited to distinguish true copula densities from informal dependence-factor analogies.
- F. Nielsen. An elementary introduction to information geometry. *Entropy*, 22(10):1100, 2020. doi:[10.3390/e22101100](https://doi.org/10.3390/e22101100). URL <https://www.mdpi.com/1099-4300/22/10/1100>.
- T. Nuijten and V. Lukashchuk. Active inference is a subtype of variational inference. arXiv preprint, 2025. URL <https://arxiv.org/abs/2511.18955>.
- R. Orus. A practical introduction to tensor networks: Matrix product states and projected entangled pair states. *Annals of Physics*, 349:117–158, 2014. doi:[10.1016/j.aop.2014.06.013](https://doi.org/10.1016/j.aop.2014.06.013). URL <https://www.sciencedirect.com/science/article/pii/S0003491614001596>. Tutorial tensor-network reference used to scope matrix-product-state and tensor-train language as linear algebra on probability tensors.
- I. V. Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33(5):2295–2317, 2011. doi:[10.1137/090752286](https://doi.org/10.1137/090752286). URL <https://epubs.siam.org/doi/10.1137/090752286>. Foundational paper introducing the TT-format; underlies the linear-in-N policy-tensor compression used in the tensor-network analysis.

- T. Parr and K. J. Friston. Uncertainty, epistemics and active inference. *Journal of The Royal Society Interface*, 14(136):20170376, 2017a. doi:[10.1098/rsif.2017.0376](https://doi.org/10.1098/rsif.2017.0376). URL <https://royalsocietypublishing.org/doi/10.1098/rsif.2017.0376>. Primary source for epistemic value and uncertainty reduction in active-inference policy selection.
- T. Parr and K. J. Friston. Working memory, attention, and salience in active inference. *Scientific Reports*, 7:14678, 2017b. doi:[10.1038/s41598-017-15249-0](https://doi.org/10.1038/s41598-017-15249-0). URL <https://www.nature.com/articles/s41598-017-15249-0>. Process-theory account linking working memory, attention, salience, precision, and policy evaluation under active inference.
- T. Parr and K. J. Friston. Generalised free energy and active inference. *Biological Cybernetics*, 113(5–6):495–513, 2019. doi:[10.1007/s00422-019-00805-w](https://doi.org/10.1007/s00422-019-00805-w). URL <https://link.springer.com/article/10.1007/s00422-019-00805-w>. Clarifies generalized free energy and self-evidencing language used in the discussion without treating the present finite-policy model as a complete biological process theory.
- T. Parr, D. Marković, S. J. Kiebel, and K. J. Friston. Neuronal message passing using mean-field, bethe and marginal approximations. *Scientific Reports*, 9:1889, 2019. doi:[10.1038/s41598-018-38246-3](https://doi.org/10.1038/s41598-018-38246-3). URL <https://www.nature.com/articles/s41598-018-38246-3>. Primary source distinguishing mean-field, Bethe, and marginal message-passing approximations; cited for approximation-family context, not as a direct proof of policy-coupling neural implementation.
- T. Parr, G. Pezzulo, and K. J. Friston. Active inference: The free energy principle in mind, brain, and behavior. MIT Press, 2022. URL <https://direct.mit.edu/books/oa-monograph/5299/Active-Inference-The-Free-Energy-Principle-in-Mind>. Book-length synthesis of FEP, expected free energy, active inference process theory, and applications.
- G. Pezzulo, F. Rigoli, and K. J. Friston. Hierarchical active inference: A theory of motivated control. *Trends in Cognitive Sciences*, 22(4):294–306, 2018. doi:[10.1016/j.tics.2018.01.009](https://doi.org/10.1016/j.tics.2018.01.009). URL [https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613\(18\)30022-6](https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613(18)30022-6).
- G. Pezzulo, T. Parr, and K. J. Friston. Active inference as a theory of sentient behavior. *Biological Psychology*, 186:108741, 2024. doi:[10.1016/j.biopsycho.2023.108741](https://doi.org/10.1016/j.biopsycho.2023.108741). URL <https://www.sciencedirect.com/science/article/pii/S0301051123002612>. Recent synthesis used to distinguish active inference as process theory from the broader free-energy-principle framing.
- The pymdp developers. infer-actively/pymdp: A python implementation of active inference for markov decision processes. GitHub repository, 2026a. URL <https://github.com/infer-actively/pymdp>. Official source repository consulted for package identity, install/source provenance, and release context; the scholarly software citation remains Heins et al. 2022.
- The pymdp developers. pymdp documentation. Read the Docs, 2026b. URL <https://pymdp-rtd.readthedocs.io/en/latest/>. Official documentation consulted for the JAX-backend and Agent/API provenance used by the reproducibility checklist.
- V. Raja, D. Valluri, E. Baggs, A. Chemero, and M. L. Anderson. The markov blanket trick: On the scope of the free energy principle and active inference. *Physics of Life Reviews*, 2021. URL <https://www.sciencedirect.com/science/article/pii/S1571064521000634>. Critical scope analysis of broad FEP / active-inference claims; cited here to keep Markov-blanket and biological interpretations explicitly bounded.
- M. J. D. Ramstead, P. B. Badcock, and K. J. Friston. Answering schrödinger’s question: A free-energy formulation. *Physics of Life Reviews*, 24:1–16, 2018. doi:[10.1016/j.pprev.2017.09.001](https://doi.org/10.1016/j.pprev.2017.09.001). URL <https://www.sciencedirect.com/science/article/pii/S1571064517301409>. Theoretical-biology framing of FEP and self-organization; cited only for context, not as evidence for the manuscript’s finite-policy results.
- K. Rawlik, M. Toussaint, and S. Vijayakumar. On stochastic optimal control and reinforcement learning by approximate inference (extended abstract). Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence (IJCAI 2013), 2013. URL <https://www.ijcai.org/Proceedings/13/Papers/455.pdf>. Extended abstract of the approximate-inference control formulation; cited for the KL/control-as-inference lineage rather than as an active-inference result.
- N. P. Rougier, M. Droettboom, and P. E. Bourne. Ten simple rules for better figures. *PLOS Computational Biology*, 10(9):e1003833, 2014. doi:[10.1371/journal.pcbi.1003833](https://doi.org/10.1371/journal.pcbi.1003833). URL <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833>. Visualization-design checklist used narrowly for self-contained captions, explicit figure purpose, and avoidance of gratuitous complexity.
- J. Ruiz-Serra, S. Sweeney, and M. Harré. Factorised active inference for strategic multi-agent interactions. *Proc.*

- 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025), 2025. URL <https://arxiv.org/abs/2411.07362>. arXiv:2411.07362. AAMAS 2025 proceedings version.
- N. Sajid, P. J. Ball, T. Parr, and K. J. Friston. Active inference: Demystified and compared. *Neural Computation*, 33(3):674–712, 2021. doi:[10.1162/neco_a_01357](https://doi.org/10.1162/neco_a_01357). URL <https://direct.mit.edu/neco/article/33/3/674>.
- L. K. Saul and M. I. Jordan. Exploiting tractable substructures in intractable networks. *Advances in Neural Information Processing Systems 8 (NIPS 1995)*, 1995. URL <https://papers.nips.cc/paper/1155-exploiting-tractable-substructures-in-intractable-networks>. Canonical structured mean-field source: exact inference is retained inside tractable substructures while only the remaining interactions are approximated.
- U. Schollwöck. The density-matrix renormalization group in the age of matrix product states. *Annals of Physics*, 326(1):96–192, 2011. doi:[10.1016/j.aop.2010.09.012](https://doi.org/10.1016/j.aop.2010.09.012). URL <https://www.sciencedirect.com/science/article/pii/S0003491610001752>. arXiv:1008.3477. Canonical MPS/DMRG review; cited for the algorithmic backbone behind policy-tensor compression and entanglement-entropy bookkeeping.
- P. Schwartenbeck and K. Friston. Computational phenotyping in psychiatry: A worked example. *eNeuro*, 3(4):ENEURO.0049–16.2016, 2016. doi:[10.1523/ENEURO.0049-16.2016](https://doi.org/10.1523/ENEURO.0049-16.2016). URL <https://www.eneuro.org/content/3/4/ENEURO.0049-16.2016>.
- P. Schwartenbeck, T. H. B. FitzGerald, C. Mathys, R. Dolan, and K. Friston. The dopaminergic midbrain encodes the expected certainty about desired outcomes. *Cerebral Cortex*, 25(10):3434–3445, 2015. doi:[10.1093/cercor/bhu159](https://doi.org/10.1093/cercor/bhu159). URL <https://academic.oup.com/cercor/article/25/10/3434/385607>.
- R. Smith, K. J. Friston, and C. J. Whyte. A step-by-step tutorial on active inference and its application to empirical data. *Journal of Mathematical Psychology*, 107:102632, 2022. doi:[10.1016/j.jmp.2021.102632](https://doi.org/10.1016/j.jmp.2021.102632). URL <https://www.sciencedirect.com/science/article/pii/S0022249621000973>.
- T. S. C. Smithe. Structured active inference. arXiv preprint, arXiv:2406.07577, 2024. URL <https://arxiv.org/abs/2406.07577>. Categorical-systems-theory formulation of AIF with structured policies amenable to formal verification.
- J. Smékal and D. A. Friedman. Generalized notation notation for active inference models. *Active Inference Journal*. Zenodo, 2023. URL <https://zenodo.org/records/7803328>. Primary publication of Generalized Notation Notation (GNN), introducing the Triple Play of linguistic, visual, and executable cognitive-model representations that complements the Active Inference Ontology. Cited from the GNN extension supplement as the canonical reference for the shipped fifth (structural-and-numerical) representation of the framework.
- R. S. Sutton, D. Precup, and S. Singh. Between mdps and semi-mdps: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999. doi:[10.1016/S0004-3702\(99\)00052-1](https://doi.org/10.1016/S0004-3702(99)00052-1). URL <https://www.sciencedirect.com/science/article/pii/S0004370299000521>.
- E. A. Theodorou and E. Todorov. Relative entropy and free energy dualities: Connections to path integral and kl control. *Proc. 2012 IEEE 51st Conference on Decision and Control (CDC)*, 2012. URL <https://ieeexplore.ieee.org/document/6426381/>.
- E. Todorov. Linearly-solvable markov decision problems. *Advances in Neural Information Processing Systems 19 (NIPS 2006)*, 2006. URL <https://papers.nips.cc/paper/3002-linearly-solvable-markov-decision-problems>. Primary linearly-solvable MDP reference for KL-control structure and log-partition value transformations.
- G. Tononi. Consciousness as integrated information: a provisional manifesto. *The Biological Bulletin*, 215(3):216–242, 2008. doi:[10.2307/25470707](https://doi.org/10.2307/25470707). URL <https://www.journals.uchicago.edu/doi/10.2307/25470707>.
- M. Toussaint. Robot trajectory optimization using approximate inference. *Proc. 26th Annual International Conference on Machine Learning (ICML 2009)*, 2009. URL <https://dl.acm.org/doi/10.1145/1553374.1553508>.
- D. Tran, D. M. Blei, and E. M. Airoldi. Copula variational inference. *Advances in Neural Information Processing Systems 28 (NeurIPS 2015)*, 2015. URL <https://proceedings.neurips.cc/paper/2015/hash/e4dd5528f7596dcdcf871aa55cfccc53c-Abstract.html>. arXiv:1506.03159.
- A. Tschantz, A. K. Seth, and C. L. Buckley. Learning action-oriented models through active inference. *PLOS Computational Biology*, 16(4):e1007805, 2020. doi:[10.1371/journal.pcbi.1007805](https://doi.org/10.1371/journal.pcbi.1007805). URL <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1007805>. Active-inference account of parsimonious action-oriented model learning; useful contrast for this manuscript’s structured policy-coupling family.
- de Vries B. van de Laar, T. W. Simulating active inference processes by message passing. *Frontiers in Robotics and AI*, 6:20, 2019. doi:[10.3389/frobt.2019.00020](https://doi.org/10.3389/frobt.2019.00020). URL <https://www.frontiersin.org/articles/10.3389/frobt.2019.00020>.

- Factor-graph forward-backward implementation of deep temporal active inference (slug retained for backwards compatibility — primary reference is the 2019 *Frontiers in Robotics and AI* paper).
- F. Verstraete, V. Murg, and J. I. Cirac. Matrix product states, projected entangled pair states, and variational renormalization group methods for quantum spin systems. *Advances in Physics*, 57(2):143–224, 2008. doi:[10.1080/14789940801912366](https://doi.org/10.1080/14789940801912366). URL <https://www.tandfonline.com/doi/abs/10.1080/14789940801912366>.
- P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, and et al. Cournapeau, D. Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods*, 17:261–272, 2020. doi:[10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2). URL <https://www.nature.com/articles/s41592-019-0686-2>.
- P. T. Waade, C. L. Olesen, J. E. Laursen, S. W. Nehrer, C. Heins, K. Friston, and C. Mathys. As one and many: Relating individual and emergent group-level generative models in active inference. *Entropy*, 27(2):143, 2025. doi:[10.3390/e27020143](https://doi.org/10.3390/e27020143). URL <https://www.mdpi.com/1099-4300/27/2/143>.
- M. J. Wainwright and M. I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1-2):1–305, 2008. doi:[10.1561/22000000001](https://doi.org/10.1561/22000000001). URL <https://www.nowpublishers.com/article/Details/MAL-001>. Canonical graphical-model variational-inference reference used to anchor the mean-field / structured-approximation comparison.
- S. Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960. doi:[10.1147/rd.41.0066](https://doi.org/10.1147/rd.41.0066). URL <https://ieeexplore.ieee.org/document/5392532>. Independent rediscovery of McGill’s total-correlation construction, under the alternative name multivariate correlation; widely cited as the second canonical reference for the total-correlation quantity.
- J. S. Yedidia, W. T. Freeman, and Y. Weiss. Constructing free-energy approximations and generalized belief propagation algorithms. *IEEE Transactions on Information Theory*, 51(7):2282–2312, 2005. doi:[10.1109/TIT.2005.850085](https://doi.org/10.1109/TIT.2005.850085). URL <https://doi.org/10.1109/TIT.2005.850085>. Canonical Bethe / Kikuchi free-energy and generalized-belief-propagation source; cited for graphical-variational approximation structure, not for active-inference policy semantics.